42. CAN with Flexible Data-rate (CANFD)

42.1 Overview

The CAN with Flexible Data-rate (CANFD) supports the following functions:

- CAN with Flexible Data-rate.*1

Note 1. This feature is not available in the classical CAN function.

The CANFD module has a flexible message buffer and FIFO structure that meet the requirements of various applications. It also provides test modes to achieve high testability of the module that can be useful for power-on testing.

This specification describes of the CANFD module.

The CANFD mode is only available in certain products that support it.

42.1.1 CANFD Module

Table 42.1 CANFD module specifications (1 of 2)

Parameter		Specifications
Communication		CAN functionality conforms to CANFD ISO 11898-1 (2015)
Protocol engine version		RS-CANFD_PE V3.0
Data transfer rate	CANFD*1	Up to 1 Mbps for arbitration phase and up to 8 Mbps for data phase, individually for CAN channel
	Classical CAN	Up to 1 Mbps
Operation frequency of Peripheral module clock		100 MHz (PCLKA) RAM clock: 200 MHz (PCLKE)
Data Link Layer (DLL) clock		8 MHz to 80MHz The following clocks can be selected. ● CANMCLK : External Oscillator Clock ● CANFDCLK : CANFD Core Clock
Input/Output pins		CRXn/CTXn (n = 0, 1)
CAN channels		2 channels
Selectable ID type		11-bit Standard ID
		11-bit Standard ID + 18-bit Extended ID
Selectable frame type		Data frame (RTR = 0) (CAN and CANFD frames)
		Remote frame (RTR = 1) (only CAN frames)
Variable data byte count for data frames		DLC range: 0 to F
Message buffer		Up to 16 reception message buffers
		4 transmit message buffers per channel
		1 transmission queue per channel Automatic message transfer into transmission queues supported
FIFO number		2 reception FIFO buffers 1 COMMON FIFO individually configurable as: ● Reception FIFO ● Transmission FIFO
Automatic delay interval timer for transmission		The delay timer can be applied to: ● Transmission FIFO

Table 42.1 CANFD module specifications (2 of 2)

Parameter	Specifications
Enhanced reception filtering	Support of 11 bits and 29 bits CAN identifier
	Programmable 29 bits CAN identifier acceptance filter mask for each entry
	Programmable routing capability for each FIFO and reception message buffers (up to 2 routing destinations)
	RTR and IDE masking
	Data Length Code (DLC) filter
	Message buffer payload overload protection
	Payload filter
	Updating Acceptance Filter List (AFL) entry during communication
General software support	Automatic label information added to receive message (for upper software layer support)
Timer	TX and RX Time Stamp function
Power down function	Module start stop function for each CAN node (Channel and Global Sleep mode)
RAM	RAM ECC protected (2 bits error detection, 1-bit error correction)
Module-stop function	Module-stop state can be set for each channels to reduce power consumption.
TrustZone Filter	Security and Privilege attribution can be set for each channels.

Note 1. The CANFD mode is available only for CANFD supported product.

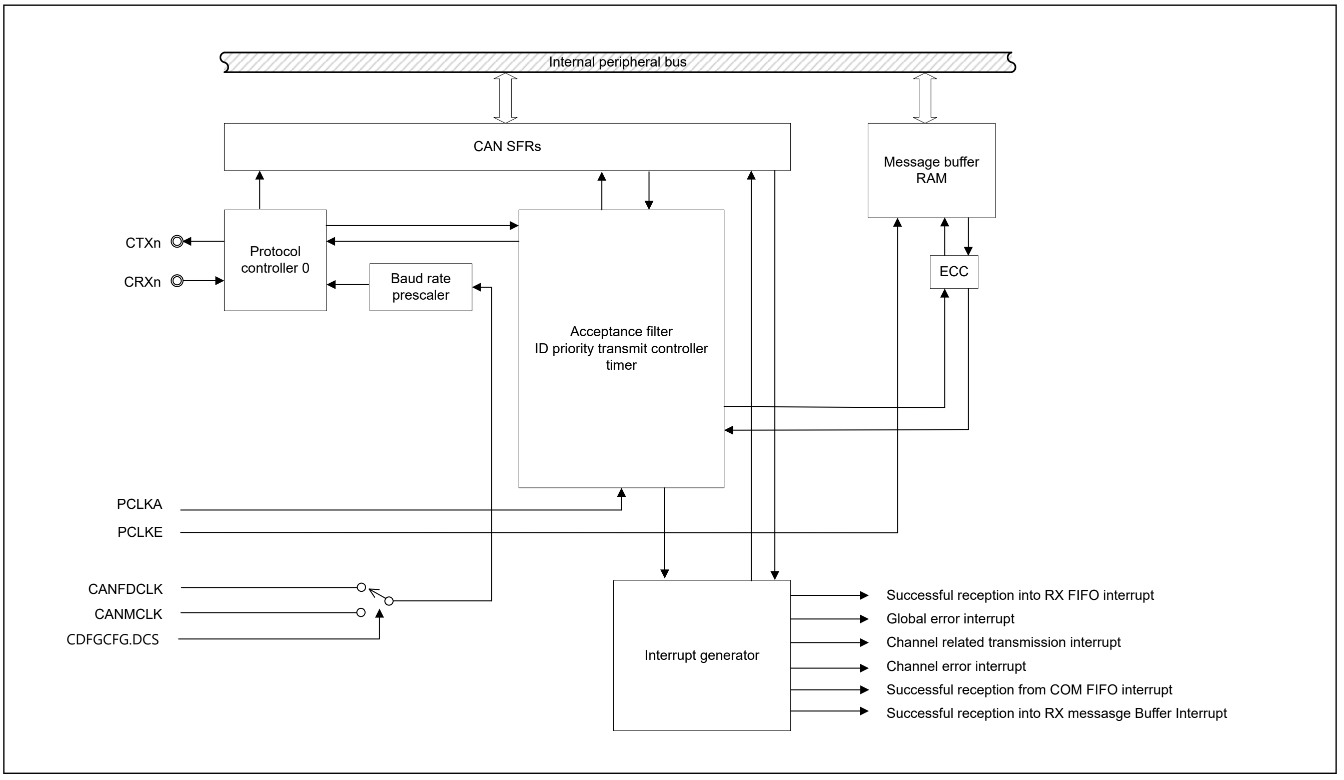


Figure 42.1 Overview of the CANFD module

- CRXn/CTXn:
  Input/Output pins of the CANFD module
- Protocol controller:
  Handles CAN protocol processing such as bus arbitration, bit timing at transmission and reception, stuffing, error handling

- Message buffer RAM (MBRAM):
  This RAM is used to store messages after reception or for transmission using a normal message buffer or a FIFO. Each message entry has an individual ID, data length code, data field, message pointer for upper layer application usage and a time stamp.
  This RAM is used to store the message acceptance filtering entries. Each acceptance filter entry has an individual ID, data length code, data field, message pointer for upper layer application usage and message direction pointer.
- Acceptance filter list RAM (AFLRAM):
  Performs filtering of received messages. The entries in the Acceptance filter list RAM are used for the filtering process.
- Two timers:
  - Reception Timestamp function
  - Transmission separation time for FIFO buffers
- Interrupt generator:
  Generates several types of global and channel interrupts
- CAN Special Function Registers (SFRs):
  Registers associated with CAN. See section 42.2. Register Descriptions.

42.1.2 Clock Restriction

For the CAN communication the following restriction for the clocks should be satisfied:

- PCLKE / 2 = PCLKA ≥ CANFDCLK
- PCLKE / 2 = PCLKA ≥ CANMCLK

To avoid missing events the CAN engine clock (CANFDCLK or CANMCLK) frequency must be less than the PCLKA clock frequency.

To avoid loss of CAN message, the PCLKA should be set to a clock with a frequency depend on the CAN communication baud rate. The constraint of a baud rate and a PCLKA clock is shown in Table 42.2.

Table 42.2 Clock restriction

	Baud rate	PCLKA
CANFD	1 Mbps Nominal 8 Mbps Data	PCLKA ≥ 40 MHz
	500 Kbps Nominal 5 Mbps Data	PCLKA ≥ 32 MHz
Classical CAN	1 Mbps Data	PCLKA ≥ 32 MHz

The frequency of CANFDCLK and CANMCLK depend on the required baud rate. For information how to configure the baud rate, see section 42.4.1.3. Baud Rate.

42.2 Register Descriptions

42.2.1 Register Table

The reset value shown for the RAM area, consisting of CFDGAFLIDr, CFDGAFLMr, CFDGAFLP0r, CFDGAFLP1r, CFDRMBCPb, CFDRFMBCPb, CFDCFMBCP0, CFDTMBCPb, CFDTHLACC0, CFDTHLACC1 and CFDRPGACCk is valid after initialization of a hardware reset. See section 42.4.2. CAN Module Configuration after Hardware Reset for details of the initialization process.

If a write access with a size of 8 or 16 bits is performed for the RAM area, then the CANFD module does a read-modify write-access to the RAM location, because the RAM requires a 32-bit access through the ECC module.

For single bit error, the correct data is written back. For multiple bit errors, unknown data is written back.

Do not access the space where the register is not assigned.

The read data from the space where the register is not assigned is unknown.

42.2.2 CFDC0NCFG : Nominal Bitrate Configuration Register

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x0000

Bit position:	31–25	24–17	16–10	9–0
Bit field:	NTSEG2[6:0]	NTSEG1[7:0]	NSJW[6:0]	NBRP[9:0]
Value after reset:	0 0 0 0 0 0 0	0 0 0 0 0 0 0 0	0 0 0 0 0 0 0	0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
9:0	NBRP[9:0]	Channel Nominal Baud Rate Prescaler Nominal baud rate prescaler division ratio	R/W
16:10	NSJW[6:0]	Resynchronization Jump Width 0x00: 1 Tq 0x01: 2 Tq : 0x7E: 127 Tq 0x7F: 128 Tq	R/W
24:17	NTSEG1[7:0]	Timing Segment 1 0x00: Reserved 0x01: 2 Tq 0x02: 3 Tq 0x03: 4 Tq : 0xFE: 255 Tq 0xFF: 256 Tq	R/W
31:25	NTSEG2[6:0]	Timing Segment 2 0x00: Reserved 0x01: 2 Tq : 0x7E: 127 Tq 0x7F: 128 Tq	R/W

Note: S-TYPE-3, P-TYPE-3
Note: Tq means time quantum.

This register configures the transmission/reception nominal baud rate parameters of the channels.

NBRP[9:0] bits (Channel Nominal Baud Rate Prescaler)

The NBRP[9:0] bits are used to define the peripheral bus clock periods contained in a time quantum.

Do not write to these bits in CH_OPERATION or CH_SLEEP mode.

Only write to these bits when the CANFD channel is in CH_RESET or CH_HALT mode.

NSJW[6:0] bits (Resynchronization Jump Width)

The NSJW[6:0] bits set the synchronization jump width. A value from 1 to 128 time quanta can be set.

Do not write to these bits in CH_OPERATION or CH_SLEEP mode.

Only write to these bits when the CANFD channel is in CH_RESET or CH_HALT mode.

NTSEG1[7:0] bits (Timing Segment 1)

The NTSEG1[7:0] bits set the segment TSEG1 to compensate for edges on the CAN bus with a positive phase error. These bits contain the propagation segment.

Do not write to these bits in CH_OPERATION or CH_SLEEP mode.

Only write to these bits when the CANFD channel is in CH_RESET or CH_HALT mode.

Additionally, configure a Tq value only between 2 and 256, inclusive. See section 42.4.1.2. CAN Bit Timing for more details.

NTSEG2[6:0] bits (Timing Segment 2)

The NTSEG2[6:0] bits set the segment TSEG2 to compensate for edges on the CAN bus with a negative phase error.

Do not write to these bits in CH_OPERATION or CH_SLEEP mode.

Only write to these bits when the CANFD channel is in CH_RESET or CH_HALT mode.

Additionally, configure a Tq value only between 2 and 128, inclusive.

42.2.3 CFDC0CTR : Control Register

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x0004

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	ROM	BFT	—	—	—	CTMS[1:0]		CTME	ERRD	BOM[1:0]		—	TDCVFIE	SOCOIE	EOCOIE	TAIE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	ALIE	BLIE	OLIE	BORIE	BOEIE	EPIE	EWIE	BEIE	—	—	—	—	RTBO	CSLPR	CHMDC[1:0]	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1

Bit	Symbol	Function	R/W
1:0	CHMDC[1:0]	Channel Mode Control 0 0: Channel operation mode request 0 1: Channel reset request 1 0: Channel halt request 1 1: Keep current value	R/W
2	CSLPR	Channel Sleep Request 0: Channel sleep request disabled 1: Channel sleep request enabled	R/W
3	RTBO	Return from Bus-Off 0: Channel is not forced to return from bus-off 1: Channel is forced to return from bus-off	R/W
7:4	—	These bits are read as 0. The write value should be 0.	R/W
8	BEIE	Bus Error Interrupt Enable 0: Bus error interrupt disabled 1: Bus error interrupt enabled	R/W
9	EWIE	Error Warning Interrupt Enable 0: Error warning interrupt disabled 1: Error warning interrupt enabled	R/W
10	EPIE	Error Passive Interrupt Enable 0: Error passive interrupt disabled 1: Error passive interrupt enabled	R/W
11	BOEIE	Bus-Off Entry Interrupt Enable 0: Bus-off entry interrupt disabled 1: Bus-off entry interrupt enabled	R/W
12	BORIE	Bus-Off Recovery Interrupt Enable 0: Bus-off recovery interrupt disabled 1: Bus-off recovery interrupt enabled	R/W
13	OLIE	Overload Interrupt Enable 0: Overload interrupt disabled 1: Overload interrupt enabled	R/W
14	BLIE	Bus Lock Interrupt Enable 0: Bus lock interrupt disabled 1: Bus lock interrupt enabled	R/W

Bit	Symbol	Function	R/W
15	ALIE	Arbitration Lost Interrupt Enable 0: Arbitration lost interrupt disabled 1: Arbitration lost interrupt enabled	R/W
16	TAIE	Transmission Abort Interrupt Enable 0: TX abort interrupt disabled 1: TX abort interrupt enabled	R/W
17	EOCOIE	Error Occurrence Counter Overflow Interrupt Enable 0: Error occurrence counter overflow interrupt disabled 1: Error occurrence counter overflow interrupt enabled	R/W
18	SOCOIE	Successful Occurrence Counter Overflow Interrupt Enable 0: Successful occurrence counter overflow interrupt disabled 1: Successful occurrence counter overflow interrupt enabled	R/W
19	TDCVFIE*1	Transceiver Delay Compensation Violation Interrupt Enable 0: Transceiver delay compensation violation interrupt disabled 1: Transceiver delay compensation violation interrupt enabled	R/W
20	—	This bit is read as 0. The write value should be 0.	R/W
22:21	BOM[1:0]	Channel Bus-Off Mode 0 0: Normal mode (comply with ISO 11898-1) 0 1: Entry to Halt mode automatically at bus-off start 1 0: Entry to Halt mode automatically at bus-off end 1 1: Entry to Halt mode (during bus-off recovery period) by software	R/W
23	ERRD	Channel Error Display 0: Only the first set of error codes displayed 1: Accumulated error codes displayed	R/W
24	CTME	Channel Test Mode Enable 0: Channel test mode disabled 1: Channel test mode enabled	R/W
26:25	CTMS[1:0]	Channel Test Mode Select 0 0: Basic test mode 0 1: Listen-only mode 1 0: Self-test mode 0 (External loopback mode) 1 1: Self-test mode 1 (Internal loopback mode)	R/W
29:27	—	These bits are read as 0. The write value should be 0.	R/W
30	BFT	Bit Flip Test 0: First data bit of reception stream not inverted 1: First data bit of reception stream inverted	R/W
31	ROM*1	Restricted Operation Mode 0: Restricted operation mode disabled 1: Restricted operation mode enabled	R/W

Note: S-TYPE-3, P-TYPE-3

Note 1. These bits are not available in the classical CAN function.

Channel Control register controls the modes of the related channel. It is used to enable generation of interrupts if errors are detected on the CAN bus connected to this channel. It is also used to configure the channel in test mode.

CHMDC[1:0] bits (Channel Mode Control)

The CHMDC[1:0] bits can be used to configure modes of the CAN channel.

CAN mode transitions are described in more details in section 42.3.3. Channel Modes.

Setting CHMDC[1:0] bits to 11b has no effect. When the CANFD module is in GL_HALT mode, these bits can only be set to 10b or 01b. These bits cannot be set in CH_SLEEP mode.

These bits can change automatically when transitioning to Halt mode by the CFDC0CTR.BOM settings.

If CPU write access to CFDC0CTR.CHMDC occurs at the same time when the CAN channel enters Halt mode (at the start of bus-off when CFDC0CTR.BOM = 01b, or at the end of bus-off when CFDC0CTR.BOM = 10b), then the CPU write access has the highest priority.

The CAN channel changes the value of CFDC0CTR.CHMDC within the Channel Control Registers for the specified cases only if the CFDC0CTR.CHMDC value is 00b (Operation mode).

CSLPR bit (Channel Sleep Request)

When the CSLPR bit is 1, a Sleep mode request is generated for the corresponding CAN channel

When this bit is 0, a request to exit Sleep mode is generated for the related CANFD channel.

Only write to this bit when the related CANFD channel is in CH_RESET or CH_SLEEP mode.

RTBO bit (Return from Bus-Off)

When the protocol controller of the CAN channel enters bus-off state, you can force a recovery from bus-off state by setting the RTBO bit in the Channel Control Register to 1.

The error state changes from bus-off state to integrating with a maximum delay of 1 CAN bit time.

When the RTBO bit is set to 1, the REC and TEC registers are initialized and the Bus-Off Status bit (Channel Bus-off Status, CFDC0STS.BOSTS) is set to 0.

Registers other than the REC and TEC registers are not initialized by this command. Even if CFDC0CTR.BORIE is set, a bus-off recovery interrupt is not generated by this recovery from the bus-off state.

The RTBO bit cannot be set in CH_SLEEP mode. Setting this bit in any state other than bus-off state has no effect and the bit is cleared immediately. The read value is always 0.

Return from the Bus-Off command should be used only when CFDC0CTR.BOM is set to 00b.

Only write to this bit when the related CANFD channel is in CH_OPERATION mode. This bit is automatically cleared when set by software.

BEIE bit (Bus Error Interrupt Enable)

When the BEIE and the CFDC0ERFL.BEF bits are both 1, an error interrupt request is generated.

This bit cannot be set in CH_SLEEP mode. Only write to this bit when the related CANFD channel is in CH_RESET mode.

EWIE bit (Error Warning Interrupt Enable)

When the EWIE and the CFDC0ERFL.EWF bits are both 1, an error interrupt request is generated.

The EWIE bit cannot be set in CH_SLEEP mode. Only write to this bit when the related CANFD channel is in CH_RESET mode.

EPIE bit (Error Passive Interrupt Enable)

An error interrupt request is generated when the EPIE bit and the CFDC0ERFL.EPF are both 1.

The EPIE bit cannot be set in CH_SLEEP mode. Only write to this bit when the related CANFD channel is in CH_RESET mode.

BOEIE bit (Bus-Off Entry Interrupt Enable)

When the BOEIE and the CFDC0ERFL.BOEF bits are both 1, an error interrupt request is generated.

The BOEIE bit cannot be set in CH_SLEEP mode. Only write to this bit when the related CANFD channel is in CH_RESET mode.

BORIE bit (Bus-Off Recovery Interrupt Enable)

When the BORIE and the CFDC0ERFL.BORF bits are both 1, an error interrupt request is generated.

The BORIE bit cannot be set in CH_SLEEP mode. Only write to this bit when the related CANFD channel is in CH_RESET mode.

OLIE bit (Overload Interrupt Enable)

When the OLIE and the CFDC0ERFL.OVLF bits are both 1, an error interrupt request is generated.

Do not write to this bit in CH_SLEEP mode. Only write to this bit when the related CANFD channel is in CH_RESET mode.

BLIE bit (Bus Lock Interrupt Enable)

When the BLIE and the CFDC0ERFL.BLF bits are both 1, an error interrupt request is generated.

Do not write to this bit in CH_SLEEP mode. Only write to this bit when the related CANFD channel is in CH_RESET mode.

ALIE bit (Arbitration Lost Interrupt Enable)

When the ALIE and the CFDC0ERFL.ALF bits are both 1, an error interrupt request is generated.

Do not write to this bit in CH_SLEEP mode. Only write to this bit when the related CANFD channel is in CH_RESET mode.

TAIE bit (Transmission Abort Interrupt Enable)

When the TAIE bit is 1 and a transmission is successfully aborted from a TX MB belonging to the corresponding CAN channel, an interrupt request is generated.

Do not write to this bit in CH_SLEEP mode. Only write to this bit when the related CANFD channel is in CH_RESET mode.

EOCOIE bit (Error Occurrence Counter Overflow Interrupt Enable)

When the EOCOIE bit is 1 and the CFDC0FDSTS.EOCO bit belonging to the corresponding CAN channel is 1, an error interrupt request is generated.

The EOCOIE bit cannot be set in CH_SLEEP mode. Only write to this bit when the related CANFD channel is in CH_RESET mode.

SOCOIE bit (Successful Occurrence Counter Overflow Interrupt Enable)

When the SOCOIE bit is 1 and the CFDC0FDSTS.SOCO bit belonging to the corresponding CAN channel is 1, an error interrupt request is generated.

The SOCOIE bit cannot be set in CH_SLEEP mode. Only write to this bit when the related CANFD channel is in CH_RESET mode.

TDCVFIE bit (Transceiver Delay Compensation Violation Interrupt Enable)

When the TDCVFIE bit is 1 and the CFDC0FDSTS.TDCVF bit belonging to the corresponding CAN channel is 1, an error interrupt request is generated.

The TDCVFIE bit cannot be set in CH_SLEEP mode.

Only write to this bit when the related CANFD channel is in CH_RESET mode. Do not set this bit when in Classical CAN mode.

Note: This bit is not available in the classical CAN function.

BOM[1:0] bits (Channel Bus-Off Mode)

The BOM[1:0]bits control the timing of the recovery from Bus-Off mode of the CANFD Channel.

Do not write to these bits in CH_SLEEP mode. Only write to these bits when the related CANFD channel is in CH_RESET mode.

Only write to these bits when the related CANFD channel is in CH_RESET mode.

ERRD bit (Channel Error Display)

The ERRD bit controls the display mode of the error flag bits [14:8] in the Channel Error Flag Register (CFDC0ERFL).

If the ERRD bit is 0 and more than one error occur at the same time, the error flag bits are set for all the errors that occurred at the same time. No further errors are flagged until CFDC0ERFL[14:8] is cleared.

Do not write to the ERRD bit in CH_SLEEP mode. Only write to this bit when the related CANFD channel is in CH_RESET or CH_HALT mode.

CTME bit (Channel Test Mode Enable)

The CTME bit enables the channel test modes.

Do not write to this bit in CH_SLEEP mode. Only write to this bit when the related CANFD channel is in CH_HALT mode.

CTMS[1:0] bits (Channel Test Mode Select)

The CTMS[1:0] bits are used to select the required test mode.

Do not write to these bits in CH_SLEEP or CH_RESET mode. Only write to these bits when the related CANFD channel is in CH_HALT mode.

These bits are cleared automatically when the related CANFD channel is in CH_RESET mode.

BFT bit (Bit Flip Test)

The BFT bit checks the internal CRC generator logic of the protocol controller.

It inverts the first bit (ID bit) of the CAN message data stream being received, so that the internal generated CRC result will not match the received CRC value of the frame. Refer to the bit stuffing rule, when using this feature, as there is the possibility of receiving a stuff error (due to the inversion) rather than a CRC error.

The internal generated CRC value is always observed in the following registers:

- CFDC0ERFL.CRCREG (Classical CAN frames)
- CFDC0FDCRC.CRCREG (CANFD frames).*1

Note 1. This feature is not available in the classical CAN function.

Some restrictions exist when using this bit:

Other CAN node will send a reference message and the receiver node(s) can invert one bit of incoming bit stream.

Note: The transmitter and receiver modes share the same CRC generator, therefore it is not necessary to consider the modes separately when testing.

The Bit Flip test mode is enabled if the BFT (new control signal that inverts the first bit of the bit stream) and CTME bits are both 1 and CFDC0CTR.CTMS is 0x00.

If this function is used by a transmitting node, a bit error or an arbitration lost will occur.

Do not write to the BFT bit in CH_SLEEP mode. Users should not use this function when the Self test mode 1 (Internal Loop back mode). Only write to this bit when the related CANFD channel is in CH_HALT mode.

This bit is cleared automatically when the related CANFD channel is in CH_RESET mode.

ROM bit (Restricted Operation Mode)

When the ROM and CTME bits are both 1, the restricted operation mode is enabled. This mode should only be used in basic test mode (CFDC0CTR.CTMS[1:0] = 00b).

The ROM bit cannot be set in CH_SLEEP mode. Only write to this bit when the related CANFD channel is in CH_HALT mode.

This bit is cleared automatically when the related CANFD channel is in CH_RESET mode. Do not set this bit when in Classical CAN mode.

Note: This bit is not available in the classical CAN function.

42.2.4 CFDC0STS : Status Register

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x0008

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	TEC[7:0]								REC[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	ESIF	COMSTS	RECSTS	TRMSTS	BOSTS	EPSTS	CSLPSTS	CHLTSTS	CRSTSTS
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1

Bit	Symbol	Function	R/W
0	CRSTSTS	Channel Reset Status 0: Channel not in Reset mode 1: Channel in Reset mode	R
1	CHLTSTS	Channel Halt Status 0: Channel not in Halt mode 1: Channel in Halt mode	R
2	CSLPSTS	Channel Sleep Status 0: Channel not in Sleep mode 1: Channel in Sleep mode	R
3	EPSTS	Channel Error Passive Status 0: Channel not in error passive state 1: Channel in error passive state	R
4	BOSTS	Channel Bus-Off Status 0: Channel not in bus-off state 1: Channel in bus-off state	R
5	TRMSTS	Channel Transmit Status 0: Channel is not transmitting 1: Channel is transmitting	R
6	RECSTS	Channel Receive Status 0: Channel is not receiving 1: Channel is receiving	R
7	COMSTS	Channel Communication Status 0: Channel is not ready for communication 1: Channel is ready for communication	R
8	ESIF*1	Error State Indication Flag 0: No CANFD message has been received when the ESI flag was set 1: At least one CANFD message was received when the ESI flag was set	R/W
15:9	—	These bits are read as 0. The write value should be 0.	R/W
23:16	REC[7:0]	Reception Error Count These bits increment or decrement the counter value according to error status of the CAN channel during reception.	R
31:24	TEC[7:0]	Transmission Error Count These bits increment or decrement the counter value according to error status of the CAN channel during transmission.	R

Note: S-TYPE-3, P-TYPE-3
Note 1. This bit is not available in the classical CAN function.

Channel Status Register shows the mode, error and transmission or reception status of the related channel together with its reception and transmission error count values.

CRSTSTS bit (Channel Reset Status)

The CRSTSTS bit indicates whether the related CAN channel is in Reset mode.

This bit is set automatically when the related CAN channel enters Channel Reset mode. When the mode is changed from Reset mode to Sleep mode, the CRSTSTS bit remains 1.

This bit is cleared automatically when the related CAN channel exits the Channel Reset mode, except when changing to Sleep mode.

CHLTSTS bit (Channel Halt Status)

The CHLTSTS bit indicates whether the related CAN channel is in Halt mode.

This bit is set automatically when the related CAN module enters Halt mode, and is cleared automatically when the related CAN module exits Halt mode.

CSLPSTS bit (Channel Sleep Status)

The CSLPSTS bit indicates whether the related CAN channel is in Sleep mode.

This bit is set automatically when the related CANFD channel enters Sleep mode, and is cleared automatically when the related CANFD channel exits Sleep mode.

EPSTS bit (Channel Error Passive Status)

The EPSTS bit indicates whether the related CANFD channel has entered the error passive state.

This bit is set automatically when the value of the CAN Transmission or Reception Counter Register exceeds the value of 0x7F.

This bit is cleared automatically when the related CANFD channel exits the error passive state or enters Reset mode.

BOSTS bit (Channel Bus-Off Status)

The BOSTS bit indicates whether the related CANFD channel has entered the error bus-off state.

This bit is set automatically when the value of the related CAN Transmission Error Count Register exceeds 0xFF and the related CANFD channel is in the bus-off state (CAN Transmission Error Count Register > 0xFF).

This bit is cleared automatically when the related CANFD channel exits bus-off state.

TRMSTS bit (Channel Transmit Status)

The TRMSTS bit indicates whether the related CANFD channel is transmitting a message.

This bit is set automatically when the related CANFD channel is operating as a transmitter node or is in the bus-off state.

This bit is cleared automatically when the related CANFD channel is in the bus-idle state or starts operating as a receiver node.

RECSTS bit (Channel Receive Status)

The RECSTS bit indicates whether the related CANFD channel is receiving a message.

This bit is set automatically when the related CANFD channel is operating as a receiver node.

This bit is cleared automatically when the related CANFD channel is in the bus-idle state or starts operating as a transmitter node.

COMSTS bit (Channel Communication Status)

The COMSTS bit indicates whether the related CANFD channel is ready for communication.

This bit is set automatically when the related CANFD channel is ready to perform communication following the detection of 11 consecutive recessive bits after exiting the Reset or Halt mode.

This bit is cleared automatically when the related CANFD channel is in CH_RESET or CD_HALT mode.

Note: This bit is 1 during bus-off state.

ESIF bit (Error State Indication Flag)

The ESIF bit is set when the ESI bit is sampled recessively for a reception CAN message without any error. When in Loopback or Mirror mode, the self-transmitted messages are considered reception messages.

If a set from the CANFD channel occurs simultaneously with a clear by a write access, then the bit is set.

This bit is cleared by writing 0 to it. This bit is cleared automatically when the related CANFD channel is in CH_RESET mode.

Only write to this bit when the related CANFD channel is in CH_HALT or CH_OPERATION mode.

Do not use the bit clear instruction to clear this bit. Use the MOV instruction to ensure that only the specified bit is cleared. Other bits remain 1.

Note: This bit is not available in the classical CAN function.

REC[7:0] bits (Reception Error Count)

The REC[7:0] bits increment or decrement the counter value according to error status of the CANFD channel during reception, and display the value of the REC error counter.

The value in bus-off state is indeterminate.

These bits are cleared automatically when the CANFD module enters GL_RESET or the CANFD channel is in CH_RESET mode.

TEC[7:0] bits (Transmission Error Count)

The TEC[7:0] bits increment or decrement the counter value according to error status of the CANFD channel during transmission, and display the value of the TEC error counter.

Only write to these bits when in test mode and CANFD channel is in CH_HALT mode.

These bits are cleared automatically when CANFD module is in GL_RESET or CANFD channel is in CH_RESET mode.

42.2.5 CFDC0ERFL : Error Flag Register

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x000C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	CRCREG[14:0]														
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	ADERR	B0ERR	B1ERR	CERR	AERR	FERR	SERR	ALF	BLF	OVLF	BORF	BOEF	EPF	EWF	BEF
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	BEF	Bus Error Flag 0: Channel bus error not detected 1: Channel bus error detected	R/W
1	EWF	Error Warning Flag 0: Channel error warning not detected 1: Channel error warning detected	R/W
2	EPF	Error Passive Flag 0: Channel error passive not detected 1: Channel error passive detected	R/W
3	BOEF	Bus-Off Entry Flag 0: Channel bus-off entry not detected 1: Channel bus-off entry detected	R/W
4	BORF	Bus-Off Recovery Flag 0: Channel bus-off recovery not detected 1: Channel bus-off recovery detected	R/W

Bit	Symbol	Function	R/W
5	OVLF	Overload Flag 0: Channel overload not detected 1: Channel overload detected	R/W
6	BLF	Bus Lock Flag 0: Channel bus lock not detected 1: Channel bus lock detected	R/W
7	ALF	Arbitration Lost Flag 0: Channel arbitration lost not detected 1: Channel arbitration lost detected	R/W
8	SERR	Stuff Error 0: Channel stuff error not detected 1: Channel stuff error detected	R/W
9	FERR	Form Error 0: Channel form error not detected 1: Channel form error detected	R/W
10	AERR	Acknowledge Error 0: Channel acknowledge error not detected 1: Channel acknowledge error detected	R/W
11	CERR	CRC Error 0: Channel CRC error not detected 1: Channel CRC error detected	R/W
12	B1ERR	Bit 1 Error 0: Channel bit 1 error not detected 1: Channel bit 1 error detected	R/W
13	B0ERR	Bit 0 Error 0: Channel bit 0 error not detected 1: Channel bit 0 error detected	R/W
14	ADERR	Acknowledge Delimiter Error 0: Channel acknowledge delimiter error not detected 1: Channel acknowledge delimiter error detected	R/W
15	—	This bit is read as 0. The write value should be 0.	R/W
30:16	CRCREG[14:0]	CRC Register value These bits show the CRC value calculated for the CAN2.0 CAN frame.	R
31	—	This bit is read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Channel Error Flag register shows the status of various error conditions detectable regardless of the setting of the related CAN Channel Error Interrupt Enable Register. It also shows the status of the various bus errors detectable by the CAN channel. Refer to the CAN specification (ISO 11898-1) to check when each error condition occurs.

For this register, only a single bit can be cleared by software. Do not use the bit clear instruction to clear the bit. Use the MOV instruction to ensure that only the specified bit is cleared. Other bits remain 1.

Example in assembler language to clear the CFDC0ERFL.BEF bit:

```
mov.b #0x0FE, CFDC0ERFL ;
```

BEF bit (Bus Error Flag)

The BEF bit indicates a detection of a CAN channel bus error state, flagged by bits [14:8] in this register.

This bit is cleared by writing 0 to it, and can only be set by CANFD module logic. Writing 1 has no effect.

This bit is set automatically when a bus error is detected, and is cleared automatically when the related CANFD channel is in CH_RESET mode.

If a set from the CAN channel occurs simultaneously with a clear by a write access, then the bit is set.

Only write to this bit when the related CANFD channel is in CH_HALT or CH_OPERATION mode.

EWF bit (Error Warning Flag)

The EWF bit indicates whether an error warning condition has been detected for the CAN channel.

This bit is cleared by writing 0 to it, and can only be set by CANFD module logic. Writing 1 has no effect.

This bit is set automatically when either TEC or REC exceeds 0x5F.

The setting of this bit only occurs when the TEC or REC initially exceeds 0x5F. Therefore, if the TEC or REC remains > 0x5F and the EWF bit is cleared by software, it is not set again until both the TEC and REC go below 0x60 and either TEC or REC crosses over again from a value 0x5F to a value > 0x5F.

If a set condition occurs simultaneously with a clear condition, then the bit is set. It is cleared automatically when the related CANFD channel is in CH_RESET mode.

Only write to this bit when the related CANFD channel is in CH_HALT or CH_OPERATION mode.

EPF bit (Error Passive Flag)

The EPF bit indicates a detection of a CAN channel error passive state.

This bit is cleared by writing 0 to it, and can only be set by CANFD module logic. Writing 1 has no effect.

This bit is set automatically when the CAN error state becomes error passive state.

The setting of this bit only occurs when the TEC or REC initially exceeds 0x7F. Therefore, if the TEC or REC remains > 0x7F and the bit is cleared by software, it is not set again until both the TEC and REC go below 0x80 and either TEC or REC crosses over again from a value ≤ 0x7F to a value > 0x7F.

If a set condition occurs simultaneously with a clear condition, then the bit is set. It is cleared automatically when the related CANFD channel is in CH_RESET mode.

Only write to this bit when the related CANFD channel is in CH_HALT or CH_OPERATION mode.

BOEF bit (Bus-Off Entry Flag)

The BOEF bit indicates a detection of a CAN channel bus-off entry state.

This bit is cleared by writing 0 to it, and can only be set by CANFD module logic. Writing 1 has no effect.

This bit is set automatically when the CAN error state enters the bus-off state.

This bit is cleared automatically when the related CANFD channel is in CH_RESET mode. If a set condition occurs simultaneously with a clear condition, then the bit is set.

Only write to this bit when the related CANFD channel is in CH_HALT or CH_OPERATION mode.

BORF bit (Bus-Off Recovery Flag)

The BORF bit indicates a detection of a CAN channel bus-off recovery state.

This bit is cleared by writing 0 to it, and can only be set by CANFD module logic. Writing 1 has no effect.

This bit is set automatically if CAN channel recovers from bus-off state in the following conditions:

- When CFDC0CTR.BOM is 00b and normal recovery (11 consecutive recessive bits x 128 times detected) occurs
- When CFDC0CTR.BOM is 10b and normal recovery (11 consecutive recessive bits x 128 times detected) occurs
- When CFDC0CTR.BOM is 11b and normal recovery (11 consecutive recessive bits x 128 times detected) occurs.

The bit is not set if CAN channel recovers from bus-off state in the following conditions:

- When CAN Reset mode is requested
- When CFDC0CTR.RTBO is set to 1 (the CAN channel returns to error active)
- When CFDC0CTR.BOM is 01b
- When CFDC0CTR.BOM is 11b and a halt request is asserted before the CAN channel reaches the end of the bus-off state.

This bit is cleared automatically when the related CANFD channel is in CH_RESET mode. If a set condition occurs simultaneously with a clear condition, the flag is set.

Only write to this bit when the related CANFD channel is in CH_HALT or CH_OPERATION mode.

OVLF bit (Overload Flag)

The OVLF flag indicates a detection of a CAN channel overload state.

The OVLF bit is cleared by writing 0 to it, and can only be set by CANFD module logic. Writing 1 has no effect.

This bit is set automatically when an overload condition is detected. If a set condition occurs simultaneously with a clear condition, then the bit is set.

This bit is cleared automatically when the related CANFD channel is in CH_RESET mode.

Only write to this bit when the related CANFD channel is in CH_HALT or CH_OPERATION mode.

BLF bit (Bus Lock Flag)

The BLF bit indicates a detection of a CAN channel bus lock condition.

This bit is cleared by writing 0 to it, and can only be set by CANFD module logic. Writing 1 has no effect.

This bit is set automatically when 32 consecutive dominant bits are detected on the CAN bus while the CAN channel is in Operation mode.

If a set condition occurs simultaneously with a clear condition, then the bit is set. It is cleared automatically when the related CANFD channel is in CH_RESET mode.

Only write to this bit when the related CANFD channel is in CH_HALT or CH_OPERATION mode.

It is cleared automatically when the related CANFD channel is in CH_RESET mode.

ALF bit (Arbitration Lost Flag)

The ALF bit indicates a detection of a CAN channel bus arbitration lost condition.

This bit is cleared by writing 0 to it, and can only be set by CANFD module logic. Writing 1 has no effect.

The bit is set automatically when an arbitration lost condition is detected on the CAN bus while the CAN channel is in Operation mode.

If a set condition occurs simultaneously with a clear condition, then the bit is set. It is cleared automatically when the related CANFD channel is in CH_RESET mode.

Only write to this bit when the related CANFD channel is in CH_HALT or CH_OPERATION mode.

SERR bit (Stuff Error)

The SERR bit indicates a detection of a CAN stuff error.

This bit is cleared by writing 0 to it, and can only be set by CANFD module logic. Writing 1 has no effect.

To clear this bit, use the following sequence:

1. Clear the corresponding flag bit.
2. Read if the flag bit is cleared.
3. If yes, continue, else go back to step 1.

This bit is set automatically when a stuff error is detected. If CFDC0CTR.ERRD bit is 1 and if the set and clear conditions occur at the same time for this bit, then this bit is set.

This bit is cleared automatically when the related CANFD channel is in CH_RESET mode. If CFDC0CTR.ERRD bit is 0 and the set and clear conditions occur at the same time for this bit, then it is cleared if a bit at CFDC0ERFL[14:8] is already set. Otherwise, this bit is set if CFDC0ERFL[14:8] is 0000000b.

Only write to this bit when the related CANFD channel is in CH_HALT or CH_OPERATION mode.

FERR bit (Form Error)

The FERR bit indicates a detection of a CAN form error.

This bit is cleared by writing 0 to it, and can only be set by CANFD module logic. Writing 1 has no effect.

To clear this bit, use the following sequence:

1. Clear the corresponding flag bit.
2. Read if the flag bit is cleared.

3. If yes, continue, else go back to step 1.

This bit is set automatically when a form error is detected. If CFDC0CTR.ERRD bit is 1 and if the set and clear conditions occur at the same time for this bit, then this bit is set.

This bit is cleared automatically when the related CANFD channel is in CH_RESET mode. If CFDC0CTR.ERRD bit is 0 and the set and clear conditions occur at the same time for this bit, then it is cleared if a bit at CFDC0ERFL[14:8] is already set. Otherwise, this bit is set if CFDC0ERFL[14:8] is 0000000b.

Only write to this bit when the related CANFD channel is in CH_HALT or CH_OPERATION mode.

AERR bit (Acknowledge Error)

The AERR bit indicates a detection of a CAN acknowledge error.

This bit is cleared by writing 0 to it, and can only be set by CANFD module logic. Writing 1 has no effect.

To clear this bit, use the following sequence:

1. Clear the corresponding flag bit.
2. Read if the flag bit is cleared.
3. If yes, continue, else go back to step 1.

This bit is set automatically when an acknowledge error is detected. If CFDC0CTR.ERRD bit is 1 and if the set and clear conditions occur at the same time for this bit, then this bit is set.

This bit is cleared automatically when the related CANFD channel is in CH_RESET mode. If CFDC0CTR.ERRD bit is 0 and the set and clear conditions occur at the same time for this bit, then it is cleared if a bit at CFDC0ERFL[14:8] is already set. Otherwise, this bit is set if CFDC0ERFL[14:8] is 0000000b.

Only write to this bit when the related CANFD channel is in CH_HALT or CH_OPERATION.

CERR bit (CRC Error)

The CERR bit indicates a detection of a CAN CRC error.

This bit is cleared by writing 0 to it, and can only be set by CANFD module logic. Writing 1 has no effect.

To clear this bit, use the following sequence:

1. Clear the corresponding flag bit.
2. Read if the flag bit is cleared.
3. If yes, continue, else go back to step 1.

This bit is set automatically when a CRC error is detected. If CFDC0CTR.ERRD bit is 1 and if the set and clear conditions occur at the same time for this bit, then this bit is set.

This bit is cleared automatically when the related CANFD channel is in CH_RESET mode. If CFDC0CTR.ERRD bit is 0 and the set and clear conditions occur at the same time for this bit, then it is cleared if a bit at CFDC0ERFL[14:8] is already set. Otherwise, this bit is set if CFDC0ERFL[14:8] is 0000000b.

Only write to this bit when the related CANFD channel is in CH_HALT or CH_OPERATION mode.

B1ERR bit (Bit 1 Error)

The B1ERR bit indicates a detection of a recessive bit error.

This bit is cleared by writing 0 to it, and can only be set by CANFD module logic. Writing 1 has no effect.

To clear this bit, use the following sequence:

1. Clear the corresponding flag bit.
2. Read if the flag bit is cleared.
3. If yes, continue, else go back to step 1.

This bit is set automatically when a recessive bit error (expected recessive bit, sampled as dominant bit) is detected. If CFDC0CTR.ERRD bit is 1 and if the set and clear conditions occur at the same time for this bit, then this bit is set.

This bit is cleared automatically when the related CANFD channel is in CH_RESET mode. If CFDC0CTR.ERRD bit is 0 and the set and clear conditions occur at the same time for this bit, then it is cleared if a bit at CFDC0ERFL[14:8] is already set. Otherwise, this bit is set if CFDC0ERFL[14:8] is 0000000b.

Only write to this bit when the related CANFD channel is in CH_HALT or CH_OPERATION mode.

B0ERR bit (Bit 0 Error)

The B0ERR bit indicates a detection of a dominant bit error.

This bit is cleared by writing 0 to it, and can only be set by CANFD module logic. Writing 1 has no effect.

To clear this bit, use the following sequence:

1. Clear the corresponding flag bit.
2. Read if the flag bit is cleared.
3. If yes, continue, else go back to step 1.

This bit is set automatically when a dominant bit error (expected dominant bit, sampled as recessive bit) is detected. If CFDC0CTR.ERRD bit is 1 and if the set and clear conditions occur at the same time for this bit, then this bit is set.

This bit is cleared automatically when the related CANFD channel is in CH_RESET mode. If CFDC0CTR.ERRD bit is 0 and the set and clear conditions occur at the same time for this bit, then it is cleared if a bit at CFDC0ERFL[14:8] is already set. Otherwise, this bit is set if CFDC0ERFL[14:8] is 0000000b.

Only write to this bit when the related CANFD channel is in CH_HALT or CH_OPERATION mode.

ADERR bit (Acknowledge Delimiter Error)

The ADERR bit indicates a detection of an acknowledge delimiter bit error.

This bit is cleared by writing 0 to it, and can only be set by CANFD module logic. Writing 1 has no effect.

To clear this bit, use the following sequence:

1. Clear the corresponding flag bit.
2. Read if the flag bit is cleared.
3. If yes, continue, else go back to step 1.

This bit is set automatically when a form error is detected during the acknowledge delimiter state of frame transmission. If CFDC0CTR.ERRD bit is 1 and if the set and clear conditions occur at the same time for this bit, then this bit is set.

This bit is cleared automatically when the related CANFD channel is in CH_RESET mode. If CFDC0CTR.ERRD bit is 0 and the set and clear conditions occur at the same time for this bit, then it is cleared if a bit at CFDC0ERFL[14:8] is already set. Otherwise, this bit is set if CFDC0ERFL[14:8] is 0000000b.

Only write to this bit when the related CANFD channel is in CH_HALT or CH_OPERATION mode.

CRCREG[14:0] bits (CRC Register value)

The CRCREG[14:0] bits read the calculated CRC value when CFDC0CTR.CTME bit is 1 for the channel.

If CFDC0CTR.CTME bit is 0, then these bits are always read as 0.

These bits show the CAN2.0 CRC value calculated by the CANFD channel logic when the CTME bit is enabled.

The CFDC0ERFL.CRCREG value is updated in the first bit of the CRC field of the CAN frame (reception and transmission).

These bits are cleared automatically when the related CANFD channel is in CH_RESET mode.

42.2.6 CFDC0DCFG : Data Bitrate Configuration Register

This register is not available in the classical CAN function.

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x0100

Bit position:	31	30	29	28	27:24	23	22	21	20	19:16	15	14	13	12:8	7:0
Bit field:	—	—	—	—	DSJW[3:0]	—	—	—	—	DTSEG2[3:0]	—	—	—	DTSEG1[4:0]	DBRP[7:0]
Value after reset:	0	0	0	0	0 0 0 0	0	0	0	0	0 0 0 0	0	0	0	0 0 0 0 0	0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
7:0	DBRP[7:0]	Channel Data Baud Rate Prescaler Data Baud Rate Prescaler division ratio	R/W
12:8	DTSEG1[4:0]	Timing Segment 1 0x00: Reserved 0x01: 2 Tq 0x02: 3 Tq 0x03: 4 Tq ⋮ 0x1E: 31 Tq 0x1F: 32 Tq	R/W
15:13	—	These bits are read as 0. The write value should be 0.	R/W
19:16	DTSEG2[3:0]	Timing Segment 2 0x0: Reserved 0x1: 2 Tq ⋮ 0xE: 15 Tq 0xF: 16 Tq	R/W
23:20	—	These bits are read as 0. The write value should be 0.	R/W
27:24	DSJW[3:0]	Resynchronization Jump Width 0x0: 1 Tq 0x1: 2 Tq ⋮ 0xF: 16 Tq	R/W
31:28	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3
Note: Tq means time quantum.

The Data Bitrate Configuration Register configures the transmission/reception data baud rate parameters of the channels.

The channel of Classical CAN mode does not perform configuration of this register.

DBRP[7:0] bits (Channel Data Baud Rate Prescaler)

The DBRP[7:0] bits define the peripheral bus clock periods contained in a time quantum.

Do not write to these bits in CH_OPERATION or CH_SLEEP mode.

Only write to these bits when the related CANFD channel is in CH_RESET or CH_HALT mode.

DTSEG1[4:0] bits (Timing Segment 1)

The DTSEG1[4:0] bits set the segment TSEG1 to compensate for edges on the CAN bus with a positive phase error. A value from 2 to 32 time quanta can be set.

The DTSEG1[4:0] bits are also used to set the propagation segment.

Do not write to these bits in CH_OPERATION or CH_SLEEP mode.

Only write to these bits when the related CANFD channel is in CH_RESET or CH_HALT mode. Do not write any other value to these bits. See section 42.4.1.2. CAN Bit Timing for more details.

DTSEG2[3:0] bits (Timing Segment 2)

The DTSEG2[3:0] bits set the segment TSEG2 to compensate for edges on the CAN bus with a negative phase error. A value from 2 to 16 time quanta can be set.

Do not write to these bits in CH_OPERATION or CH_SLEEP mode.

Only write to these bits when the related CANFD channel is in CH_RESET or CH_HALT mode. Do not write any other value to these bits.

DSJW[3:0] bits (Resynchronization Jump Width)

The DSJW[3:0] bits set the synchronization jump width. A value from 1 to 16 time quanta can be set.

Do not write to these bits in CH_OPERATION or CH_SLEEP mode.

Only write to these bits when the related CANFD channel is in CH_RESET or CH_HALT mode.

42.2.7 CFDC0FDCFG : CANFD Configuration Register

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x0104

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	CLOE	REFE	FDOE	—	—	—	—	TDCO[7:0]							
Value after reset:	0	0/1*1	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	ESIC	TDCE	TDCOC	—	—	—	—	—	EOCCFG[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
2:0	EOCCFG[2:0]	Error Occurrence Counter Configuration 0 0 0: All transmitter or receiver CAN frames 0 0 1: All transmitter CAN frames 0 1 0: All receiver CAN frames 0 1 1: Reserved 1 0 0: Only transmitter or receiver CANFD data-phase (fast bits) 1 0 1: Only transmitter CANFD data-phase (fast bits) 1 1 0: Only receiver CANFD data-phase (fast bits) 1 1 1: Reserved	R/W
7:3	—	These bits are read as 0. The write value should be 0.	R/W
8	TDCOC*2	Transceiver Delay Compensation Offset Configuration 0: Measured + offset 1: Offset-only	R/W
9	TDCE*2	Transceiver Delay Compensation Enable 0: Transceiver delay compensation disabled 1: Transceiver delay compensation enabled	R/W
10	ESIC*2	Error State Indication Configuration 0: The ESI bit in the frame represents the error state of the node itself 1: The ESI bit in the frame represents the error state of the message buffer if the node itself is not in error passive. If the node is in error passive, then the ESI bit is driven by the node itself.	R/W
11	—	This bit is read as 0. The write value should be 0.	R/W
15:12	—	These bits are read as 0. The write value should be 0.	R/W
23:16	TDCO[7:0]*2	Transceiver Delay Compensation Offset	R/W
27:24	—	These bits are read as 0. The write value should be 0.	R/W
28	FDOE*2	FD-Only Enable 0: FD-only mode disabled 1: FD-only mode enabled	R/W

Bit	Symbol	Function	R/W
29	REFE	RX Edge Filter Enable 0: RX edge filter disabled 1: RX edge filter enabled	R/W
30	CLOE*2 *3	Classical CAN Enable 0: Classical CAN mode disabled 1: Classical CAN mode enabled	R/W
31	—	This bit is read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note 1. The value after reset is 0 for products that support the CAN-FD protocol, and 1 for products that support only classical CAN protocol.

Note 2. These bits are not available in the classical CAN function.

Note 3. This bit can only be written for products that support the CAN-FD protocol. For products that support only classical CAN protocol, this bit is reserved and fixed to 1.

The CANFD Configuration Register configures which communication direction (transmitter/receiver) errors are counted.

EOCCFG[2:0] bits (Error Occurrence Counter Configuration)

The EOCCFG[2:0] bits select which type of CAN frame configuration and direction, including protocol errors are counted.

Do not write to these bits in CH_OPERATION or CH_SLEEP mode.

Only write to these bits when the related CANFD channel is in CH_RESET or CH_HALT mode.

TDCOC bit (Transceiver Delay Compensation Offset Configuration)*1

The TDCOC bit selects which offset is used when defining the position of the secondary sample point (SSP) for the CANFD channel. If the bit is set to 0, the position of the SSP is the measured transceiver delay plus the fixed offset. If the bit is 1, the position of the SSP is defined only by the offset.

Do not write to this bit in CH_OPERATION or CH_SLEEP mode.

Only write to this bit when the related CANFD channel is in CH_RESET or CH_HALT mode. Do not set this bit when in Classical CAN mode.

TDCE bit (Transceiver Delay Compensation Enable)*1

The TDCE bit enables the transceiver delay compensation for the CANFD channel.

Do not write to this bit in CH_OPERATION or CH_SLEEP mode.

Only write to this bit when the related CANFD channel is in CH_RESET or CH_HALT mode. Do not set this bit when in Classical CAN mode.

ESIC bit (Error State Indication Configuration)*1

The ESIC bit controls the transmission of either the ESI flag information or the message of ESI flag information (CFDCFFDCSTS.CFESI or CFDTMFDCTRb.TMESI).

Do not write to this bit in CH_OPERATION or CH_SLEEP mode.

Only write to this bit when the related CANFD channel is in CH_RESET or CH_HALT mode. Do not set this bit when in Classical CAN mode.

TDCO[7:0] bits (Transceiver Delay Compensation Offset)*1

The TDCO[7:0] bits set the secondary sample point offset. How this value is used, depends on the CFDC0FDCFG.TDCOC setting.

If CFDC0FDCFG.TDCOC = 0, the transceiver delay compensation result is equal to the Trv_Delay (measured delay) + the value in CFDC0FDCFG.TDCO, rounded down to the nearest integer number of time quanta. Otherwise, the result is equal to the value in CFDC0FDCFG.TDCO. See section 42.4.1.5. Transmitter Delay Compensation for details on how CFDC0FDCFG.TDCO is used.

The actual offset value is interpreted as TDCO + 1. For example, if 4 is set in TDCO, the offset is 5 clock cycles. Clock cycle is 1 cycle of CAN channel DLL clock.

Do not write to the TDCO[7:0] bits in CH_OPERATION or CH_SLEEP mode.

Only write to these bits when the related CANFD channel is in CH_RESET or CH_HALT mode. Do not set this bit when in Classical CAN mode.

FDOE bit (FD-Only Enable)*1

The FDOE bit enables the reception and transmission of CANFD-only frames. If enabled, communication in Classical CAN frame format is disabled. Transmission of Classical CAN frames is not possible because the FDF bit of the message buffer is a don't care (CFDCFFDCSTS.CFFDF/CFDTMFDCTRb.TMFDF).

If messages with Classical CAN frame format are received, the protocol controller treats them as invalid frames and response with error frames. When a Classical CAN frame is configured for transmitting, the FDF bit is sent as recessive, therefore an FD frame is sent. If the data length code (DLC) is configured of greater than 8 bytes, the remaining data bytes are padded with 0xCC.

The FDOE bit cannot be written in CH_OPERATION, CH_HALT or CH_SLEEP mode.

Do not set CFDC0FDCFG.FDOE and CFDC0FDCFG.CLOE simultaneously.

REFE bit (RX Edge Filter Enable)

The REFE bit enables the RX edge filter during the IDLE detection (bus integration). When the bit is enabled, two consecutive dominant time quanta are required to detect a synchronization edge.

The REFE bit cannot be written in CH_OPERATION, CH_HALT and CH_SLEEP mode. Do not set this bit when in Classical CAN mode.

CLOE bit (Classical CAN Enable)*1

The CLOE bit enables the Classical CAN mode. If this bit is 1, the protocol controller can only send classical frames and response with a form or CRC error on FD frames.

Do not set CFDC0FDCFG.CLOE and CFDC0FDCFG.FDOE simultaneously.

CFDC0FDCFG.CLOE	CFDC0FDCFG.FDOE	Channel mode
0	0	CANFD mode
0	1	FD-only mode
1	0	Classical CAN mode
1	1	Reserved

The CANFD mode is available only for CANFD supported product.

Do not write to this bit in CH_OPERATION, CH_HALT or CH_SLEEP mode.

Only write to these bits when the CANFD channel is in CH_RESET mode.

Note 1. These bits are not available in the classical CAN function.

42.2.8 CFDC0FDCTR : CANFD Control Register

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x0108

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	SOCCLR	EOCCLR
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	EOCCLR	Error Occurrence Counter Clear 0: No error occurrence counter clear 1: Clear error occurrence counter	R/W
1	SOCCLR	Successful Occurrence Counter Clear 0: No successful occurrence counter clear 1: Clear successful occurrence counter	R/W
31:2	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

The CANFD Control Register controls the error and successful occurrence counters.

EOCCLR bit (Error Occurrence Counter Clear)

The EOCCLR bit is used to clear the error occurrence counter.

Do not write to this bit in CH_SLEEP or CH_RESET mode. The read value is always 0.

This bit is cleared automatically by the CANFD module logic and when the related CANFD channel is in CH_RESET mode.

SOCCLR bit (Successful Occurrence Counter Clear)

The SOCCLR bit is used to clear the successful occurrence counter.

Do not write to this bit in CH_SLEEP or CH_RESET mode. The read value is always 0.

This bit is cleared automatically by the CANFD module logic and when the related CANFD channel is in CH_RESET mode.

42.2.9 CFDC0FDSTS : CANFD Status Register

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x010C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	SOC[7:0]								EOC[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	TDCVF	—	—	—	—	—	SOCO	EOCO	TDCR[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	TDCR[7:0]*1	Transceiver Delay Compensation Result	R
8	EOCO	Error Occurrence Counter Overflow 0: Error occurrence counter has not overflowed 1: Error occurrence counter has overflowed	R/W
9	SOCO	Successful Occurrence Counter Overflow 0: Successful occurrence counter has not overflowed 1: Successful occurrence counter has overflowed	R/W
14:10	—	These bits are read as 0. The write value should be 0.	R/W
15	TDCVF*1	Transceiver Delay Compensation Violation Flag 0: Transceiver delay compensation violation has not occurred 1: Transceiver delay compensation violation has occurred	R/W
23:16	EOC[7:0]	Error Occurrence Counter These bits show the error occurrence counter value.	R

Bit	Symbol	Function	R/W
31:24	SOC[7:0]	Successful occurrence counter These bits show the successful occurrence counter value.	R

Note: S-TYPE-3, P-TYPE-3

Note 1. These bits are not available in the classical CAN function.

The CANFD Status Register indicates the transceiver compensation delay result and its related FIFO message lost status.

TDCR[7:0] bits (Transceiver Delay Compensation Result)

The TDCR[7:0] bits are set when the transceiver delay has been measured.

The measured delay is a multiple of the CAN channel DLL clock. The result depends on the CFDC0FDCFG.TDCOC configuration and the offset value in CFDC0FDCFG.TDCO. See section 42.4.1.5. Transmitter Delay Compensation for details on how this value is derived.

The TDCR[7:0] bits are updated at the falling edge between FDF and the RES bit when CFDC0FDCFG.TDCOC = 0 and the transceiver delay compensation is enabled (CFDC0FDCFG.TDCE = 1).

These bits are cleared automatically when the related CANFD channel is in CH_RESET mode.

Note: These bits are not available in the classical CAN function.

EOCO bit (Error Occurrence Counter Overflow)

The EOCO bit indicates whether the related CAN channel error occurrence counter has overflowed. This bit is cleared by writing 0 to it. Writing 1 has no effect.

This bit is set automatically when CFDC0FDSTS.EOC is 0xFF and a CAN bus error is detected based on the configuration defined in CFDC0FDCFG.EOCCFG.

If the set from the CAN channel occurs simultaneously with the clear by the write access, then the bit is set.

This bit is cleared automatically when the related CANFD channel is in CH_RESET mode.

Only write to this bit when the related CANFD channel is in CH_HALT or CH_OPERATION mode.

Do not use the bit clear instruction to clear this bit. Use the MOV instruction to ensure that only the specified bit is cleared. Other bits remain 1.

SOCO bit (Successful Occurrence Counter Overflow)

The SOCO bit indicates whether the related CAN channel successful occurrence counter has overflowed. This bit is cleared by writing 0 to it. Writing 1 has no effect.

This bit is set automatically when CFDC0FDSTS.SOC is 0xFF and a successful message reception or successful message transmission occurs.

If the set from the CAN channel occurs simultaneously with the clear by the write access, then the bit is set.

This bit is cleared automatically when the related CANFD channel is in CH_RESET mode.

Write to this bit only when the related CANFD channel is in CH_HALT or CH_OPERATION mode.

Do not use the bit clear instruction to clear this bit. Use the MOV instruction to ensure that only the specified bit is cleared. Other bits remain 1.

TDCVF bit (Transceiver Delay Compensation Violation Flag)

The CANFD module captures internally the transmitted data bit-by-bit. This data is then compared against the received CAN bus level which is delayed by the transceiver loop delay.

The transceiver delay has some variations depending on the physical parameters such as temperature. The result bit CFDC0FDSTS.TDCR is updated by each message. However, temporary maximum delay violation could be missed. Therefore, the TDCVF bit captures this violation.

This bit is cleared by writing 0 to it. Writing 1 has no effect.

This bit is set automatically when the transceiver delay compensation is greater than the maximum delay compensation (6 data bit times - 2 clk_dlc) and the internal bit is overrun.

If the set from the CAN channel occurs simultaneously with the clear by the write access, then the bit is set.

This bit is cleared automatically when the related CANFD channel is in CH_RESET mode.

Only write to this bit when the related CANFD channel is in CH_HALT or CH_OPERATION mode.

Do not use the bit clear instruction to clear this bit. Use the MOV instruction to ensure that only the specified bit is cleared. Other bits remain 1.

Note: This bit is not available in the classical CAN function.

EOC[7:0] bits (Error Occurrence Counter)

The EOC[7:0] bits are used together with the SOC[7:0] bits to support an option for host-controlled fall-back to payload bit rate identical to arbitration bit rate when messages utilizing the reduced payload bit length have significant higher error rates compared to other messages.

This higher error rate can be detected depending on the configuration of the CFDC0FDCFG.EOCCFG bits.

The EOC[7:0] bits are set only by CANFD module logic. These bits are cleared by writing 1 to CFDC0FDCTR.EOCCLR. Writing any other value has no effect.

These bits are updated when an error occurs, according to the configuration of the CFDC0FDCFG.EOCCFG bits. When the counter reaches the value of 0xFF, the update stops.

These bits are cleared automatically when the related CANFD channel is in CH_RESET mode.

SOC[7:0] bits (Successful occurrence counter)

The SOC[7:0] bits are used together with the EOC[7:0] bits to support an option for host-controlled fall-back to payload bit rate identical to arbitration bit rate when messages utilizing the reduced payload bit length have significant higher error rates compared to other messages.

The SOC[7:0] bits are set only by CANFD module logic. Writing any other value has no effect.

These bits are updated when the occurrence of any error-free messages on the bus is detected through reception or transmission. When the counter reaches the value of 0xFF, the update stops.

Note: In Loopback mode, the counter is incremented twice.

These bits are cleared by writing 1 to CFDC0FDCTR.SOCCLR.

These bits are cleared automatically when the related CANFD channel is in CH_RESET mode.

42.2.10 CFDC0FDCRC : CANFD CRC Register

This register is not available in the classical CAN function.

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x0110

Bit position:	31	30	29	28	27–24	23	22	21	20–0
Bit field:	—	—	—	—	SCNT[3:0]	—	—	—	CRCREG[20:0]
Value after reset:	0	0	0	0	0 0 0 0	0	0	0	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
20:0	CRCREG[20:0]	CRC Register value These bits show the CRC value calculated for the CANFD frame.	R
23:21	—	These bits are read as 0.	R
27:24	SCNT[3:0]	Stuff bit count These bits shows the stuff bit count (mod 8) for the CANFD frame.	R
31:28	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

The CANFD CRC Register holds the CRC value calculated for the CANFD frame.

CRCREG[20:0] bits (CRC Register value)

The CRCREG[20:0] bits contain the CRC value calculated by the CANFD channel logic when the CFDC0CTR.CTME bit is enabled.

The CFDC0FDCRC.CRCREG value is updated in the first bit of the CRC field of the CANFD frame (reception and transmission).

When the CFDC0CTR.CTME bit is 0, the CRCREG[20:0] bits are always read as 0.

When bit 17th of the CRC field is used, CRCREG[20:17] are always read as 0.

These bits are cleared automatically when the related CANFD channel is in CH_RESET mode.

SCNT[3:0] bits (Stuff bit count)

The SCNT[3:0] bits contain the stuff count value of the CANFD frame. These bits indicate the number of inserted stuff bits (modulo 8, Gray coded) for a CANFD frame when the CFDC0CTR.CTME bit is enabled in CFDC0FDCRC.SCNT[3:1]. SCNT[0] is the parity bit.

When the CFDC0CTR.CTME bit is 0, the SCNT[3:0] bits are always read as 0.

The SCNT value is updated in the first bit of CRC field of the CANFD frame (reception and transmission).

These bits are cleared automatically when the related CANFD channel is in CH_RESET mode.

42.2.11 CFDGCFG : Global Configuration Register

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x0014

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	ITRCP[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	TSSS	TSP[3:0]				—	—	CMPOC	DCS	MME	DRE	DCE	TPRI
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	TPRI	Transmission Priority 0: ID priority 1: Message buffer number priority	R/W
1	DCE	DLC Check Enable 0: DLC check disabled 1: DLC check enabled	R/W
2	DRE	DLC Replacement Enable 0: DLC replacement disabled 1: DLC replacement enabled	R/W
3	MME	Mirror Mode Enable 0: Mirror mode disabled 1: Mirror mode enabled	R/W
4	DCS	Data Link Controller Clock Select 0: CANFD core clock (CANFDCLK) 1: External oscillator clock (CANMCLK)	R/W
5	CMPOC*1	CANFD Message Payload Overflow Configuration 0: Message is rejected 1: Message payload is cut to fit to configured message size	R/W
7:6	—	These bits are read as 0. The write value should be 0.	R/W

Bit	Symbol	Function	R/W
11:8	TSP[3:0]	Timestamp Prescaler 0x0: Timestamp prescaler = 1 0x1: Timestamp prescaler = 2 0x2: Timestamp prescaler = 4 0x3: Timestamp prescaler = 8 ⋮ 0xD: Timestamp prescaler = 8192 0xE: Timestamp prescaler = 16384 0xF: Timestamp prescaler = 32768	R/W
12	TSSS	Timestamp Source Select 0: Source clock for timestamp counter is peripheral clock 1: Source clock for timestamp counter is bit time clock	R/W
15:13	—	These bits are read as 0. The write value should be 0.	R/W
31:16	ITRCP[15:0]	Interval Timer Reference Clock Prescaler FIFO interval timer prescaler value	R/W

Note: S-TYPE-3, P-TYPE-3
Note 1. This bit is not available in the classical CAN function.

The Global Configuration Register is used to select the transmission priority to be used for all the TX message buffers and the clock source for the CAN protocol engine of CAN channel. The CFDGCFG register is also used to select the source for the timestamp clock and to configure the frequency for the timestamp clock and interval timer reference clock.

TPRI bit (Transmission Priority)

The TPRI bit selects the transmission priority for CAN channel.

Do not write to this bit in GL_SLEEP mode. Only write to this bit when CANFD module is in GL_RESET mode.

Message buffer number priority should not be used together with TX queue transmission.

DCE bit (DLC Check Enable)

The DCE bit enables data length code (DLC) check for CAN channel.

Do not write to this bit in GL_SLEEP mode. Only write to this bit when CANFD module is in GL_RESET mode.

DRE bit (DLC Replacement Enable)

When the DRE bit is 1 and the DCE is 1, the CANFD stores the configured value (CFDGAFLP0r.GAFLDLC) of the DLC in the destination RX message buffer or FIFO buffer if the DLC check passes. Otherwise, the DLC value in the destination RX message buffer or FIFO buffer is unchanged.

Do not write to this bit in GL_SLEEP mode. Only write to this bit when CANFD module is in GL_RESET mode.

MME bit (Mirror Mode Enable)

The MME bit enables the Mirror mode for CAN channel.

Do not write to this bit in GL_SLEEP mode. Only write to this bit when CANFD module is in GL_RESET mode.

DCS bit (Data Link Controller Clock Select)

The DCS bit selects CANFDCLK or CANMCLK as the clock source for CAN communication.

Do not write to this bit in GL_SLEEP or GL_OPERATION mode. Only write to this bit when CANFD module is in GL_RESET mode.

CMPOC bit (CANFD Message Payload Overflow Configuration)

The CMPOC bit controls the message payload acceptance mechanism when the received payload is higher than the message buffer payload size CFDRMNB.RMPLS, CFDRFCCa.RFPLS, and CFDCFCC.CFPLS. The received message payload is always compared with the available message payload size in the message buffer.

Do not write to this bit in GL_SLEEP or GL_OPERATION mode. Only write to this bit when CANFD module is in GL_RESET mode.

When this bit is set and payload overflow occurs, the DLC value is stored in the RX message buffer or FIFO buffer unchanged.

Note: This bit is not available in the classical CAN function.

TSP[3:0] bits (Timestamp Prescaler)

The value configured in the TSP[3:0] bits defines the period of the clock source used for the timestamp counter.

Do not write to this bit in GL_SLEEP mode. Only write to this bit when CANFD module is in GL_RESET mode.

TSSS bit (Timestamp Source Select)

The TSSS bit allows the selection of the clock source for the timestamp counter.

Do not write to this bit in GL_SLEEP mode. Only write to this bit when CANFD module is in GL_RESET mode. Additionally, do not set this bit to 1 when CANFD communication is used.*1

Note: The bit time clock varies depending on the nominal and data rate bit configuration.

Note 1. This feature is not available in the classical CAN function.

ITRCP[15:0] bits (Interval Timer Reference Clock Prescaler)

The ITRCP[15:0] bits allow the definition of a reference clock for the FIFO interval timer source clock.

When these bits are 0x0000, the timer is disabled.

Do not write to this bit in GL_SLEEP mode. Only write to this bit when CANFD module is in GL_RESET mode.

42.2.12 CFDGCTR : Global Control Register

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x0018

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	TSRST
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	CMPOFIE	THLEIE	MEIE	DEIE	—	—	—	—	—	GSLPR	GMDC[1:0]	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1

Bit	Symbol	Function	R/W
1:0	GMDC[1:0]	Global Mode Control 0 0: Global operation mode request 0 1: Global reset mode request 1 0: Global halt mode request 1 1: Keep current value	R/W
2	GSLPR	Global Sleep Request 0: Global sleep request disabled 1: Global sleep request enabled	R/W
7:3	—	These bits are read as 0. The write value should be 0.	R/W
8	DEIE	DLC Check Interrupt Enable 0: DLC check interrupt disabled 1: DLC check interrupt enabled	R/W
9	MEIE	Message Lost Error Interrupt Enable 0: Message lost error interrupt disabled 1: Message lost error interrupt enabled	R/W
10	THLEIE	TX History List Entry Lost Interrupt Enable 0: TX history list entry lost interrupt disabled 1: TX history list entry lost interrupt enabled	R/W

Bit	Symbol	Function	R/W
11	CMPOFIE*1	CANFD Message Payload Overflow Flag Interrupt Enable 0: CANFD message payload overflow flag interrupt disabled 1: CANFD message payload overflow flag interrupt enabled	R/W
15:12	—	These bits are read as 0. The write value should be 0.	R/W
16	TSRST	Timestamp Reset 0: Timestamp not reset 1: Timestamp reset	R/W
31:17	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note 1. This bit is not available in the classical CAN function.

The Global Control Register controls the global mode of the CANFD module and the timestamp function. The register also enables and disables the global error interrupts.

GMDC bits (Global Mode Control)

The GMDC bits can be used to configure the modes for the CANFD module. Additionally, if CFDGCTR.GSLPR bit is 1 when the CANFD module is in Reset mode, the CANFD module enters Global Sleep mode.

Setting the GMDC bits to 11b has no effect. Mode transition is described in detail in section 42.3.2. Global Modes.

Do not write to this bit when the CANFD module is in GL_SLEEP mode.

GSLPR bit (Global Sleep Request)

The GSLPR bit globally selects the sleep request for CANFD module including CAN channels. Channel sleep request is set automatically for channels.

Only write to this bit when the CANFD module is in GL_RESET or GL_SLEEP mode.

DEIE bit (DLC Check Interrupt Enable)

When the DEIE bit is 1, an interrupt is generated if a DLC error is detected in the received frames.

Do not write to this bit when the CANFD module is in GL_SLEEP mode.

MEIE bit (Message Lost Error Interrupt Enable)

When the MEIE bit is 1, an interrupt is generated if a message lost condition occurs.

Do not write to this bit when the CANFD module is in GL_SLEEP mode.

THLEIE bit (TX History List Entry Lost Interrupt Enable)

When the THLEIE bit is 1, an interrupt is generated if a TX history list entry lost condition occurs.

Do not write to this bit when the CANFD module is in GL_SLEEP mode.

CMPOFIE bit (CANFD Message Payload Overflow Flag Interrupt Enable)

When the CMPOFIE bit is 1, an interrupt is generated when a CANFD message payload overflow condition occurs.

Do not write to this bit when the CANFD module is in GL_SLEEP mode.

Note: This bit is not available in the classical CAN function

TSRST bit (Timestamp Reset)

When the TSRST bit is 1, the Global Timestamp Register is reset to 0x0000.

Do not write to this bit when the CANFD module is in GL_SLEEP or GL_RESET mode.

Read value is always 0.

This bit is cleared automatically by the CANFD module logic.

42.2.13 CFDGSTS : Global Status Register

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x001C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	GRAMINIT	GSLPSTS	GHLTSTS	GRSTSTS
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	1

Bit	Symbol	Function	R/W
0	GRSTSTS	Global Reset Status 0: Not in Reset mode 1: In Reset mode	R
1	GHLTSTS	Global Halt Status 0: Not in Halt mode 1: In Halt mode	R
2	GSLPSTS	Global Sleep Status 0: Not in Sleep mode 1: In Sleep mode	R
3	GRAMINIT	Global RAM Initialization 0: RAM initialization is complete 1: RAM initialization is ongoing	R
31:4	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

The Global Status Register indicates the global status of the CANFD module.

GRSTSTS bit (Global Reset Status)

The GRSTSTS bit indicates the status of Global CANFD module Reset mode.

This bit is set automatically when the CANFD module enters GL_RESET mode. When the mode changes from GL_RESET mode to GL_SLEEP mode, this bit remains set.

This bit is cleared automatically when the CANFD module exits the GL_RESET mode.

GHLTSTS bit (Global Halt Status)

The GHLTSTS bit indicates the status of Global CANFD module Halt mode.

This bit is set automatically when the CANFD module enters GL_HALT mode.

This bit is cleared automatically when the CANFD module exits the GL_HALT mode.

GSLPSTS bit (Global Sleep Status)

The GSLPSTS bit indicates the status of Global CANFD module Sleep mode.

This bit is set automatically when the CANFD module enters GL_SLEEP mode.

This bit is cleared automatically when the CANFD module exits the GL_SLEEP mode.

GRAMINIT bit (Global RAM Initialization)

The GRAMINIT bit indicates the status of Global CANFD module RAM initialization.

This bit is set automatically when the CANFD module enters GL_SLEEP mode after a hardware reset.

This bit is cleared automatically when the CANFD module completed RAM initialization.

This bit is cleared when the test_mode input port is set to 1.

42.2.14 CFDGERFL : Global Error Flag Register

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x0020

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	EEF0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	CMPOF	THLES	MES	DEF
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	DEF	DLC Error Flag 0: DLC error not detected 1: DLC error detected	R/W
1	MES	Message Lost Error Status 0: Message lost error not detected 1: Message lost error detected	R
2	THLES	TX History List Entry Lost Error Status 0: TX history list entry lost error not detected 1: TX history list entry lost error detected	R
3	CMPOF*1	CANFD Message Payload Overflow Flag 0: CANFD message payload overflow not detected 1: CANFD message payload overflow detected	R/W
15:4	—	These bits are read as 0. The write value should be 0.	R/W
16	EEF0	ECC Error Flag 0: ECC error not detected during TX-SCAN 1: ECC error detected during TX-SCAN	R/W
31:17	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3
Note 1. This bit is not available in the classical CAN function.

The Global Error Flag register indicates the detection of global errors.

DEF bit (DLC Error Flag)

The DEF bit indicates the error status of the DLC.

Do not write to this bit when the CANFD module is in GL_SLEEP or GL_RESET mode. Writing 1 has no effect.

Do not use the bit clear instruction to clear this bit. Use the MOV instruction to ensure that only the specified bit is cleared. Other bits remain 1.

This bit is set automatically when a DLC error is detected in a received frame.

If the set from the CAN channel occurs simultaneously with the clear by the write access, then the bit is set

The bit is cleared by writing 0 to it.

This bit is cleared automatically in GL_RESET mode.

MES bit (Message Lost Error Status)

The MES bit indicates status of the message lost error.

This bit is set automatically when a FIFO message lost error is detected.

This bit is cleared automatically when:

- All FIFO message lost flags are cleared
- The CANFD module is in GL_RESET mode.

THLES bit (TX History List Entry Lost Error Status)

The THLES bit indicates status of the TX history list entry lost error.

This bit is set automatically when a TX history list entry lost error is detected.

This bit is cleared automatically when:

- All TX history list entry lost flags are cleared
- The CANFD module is in GL_RESET mode.

CMPOF bit (CANFD Message Payload Overflow Flag)

The CMPOF bit is set automatically when a CANFD message payload overflow is detected on at least one channel.

Do not write to this bit when the CANFD module is in GL_SLEEP or GL_RESET mode.

This bit is cleared by writing 0 to it. Writing 1 to this bit has no effect.

If the set from the CAN channel occurs simultaneously with the clear by the write access, then the bit is set.

Do not use the bit clear instruction to clear this bit. Use the MOV instruction to ensure that only the specified bit is cleared. Other bits remain 1.

This bit is cleared automatically in GL_RESET mode.

Note: This bit is not available in the classical CAN function

EEF0 bit (ECC Error Flag)

The EEF0 bit specifies whether an ECC error has occurred.

Do not write to this bit when the CANFD module is in GL_SLEEP or GL_RESET mode. Writing 1 to this bit has no effect.

If the set from the CAN channel occurs simultaneously with the clear by the write access, then the bit is set.

Do not use the bit clear instruction to clear this bit. Use the MOV instruction to ensure that only the specified bit is cleared. Other bits remain 1.

The bit is cleared by writing 0 to it. This bit is cleared automatically in GL_RESET mode.

42.2.15 CFDGTINTSTS : Global TX Interrupt Status Register

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x00A4

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	THIF0	CFTIF0	TQIF0	TAI0	TSIF0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	TSIF0	TX Successful Interrupt Flag 0: Channel n TX Successful Interrupt flag not set 1: Channel n TX Successful Interrupt flag set	R
1	TAI0	TX Abort Interrupt Flag 0: Channel n TX Abort Interrupt flag not set 1: Channel n TX Abort Interrupt flag set	R

Bit	Symbol	Function	R/W
2	TQIF0	TX Queue Interrupt Flag 0: Channel n TX Queue Interrupt flag not set 1: Channel n TX Queue Interrupt flag set	R
3	CFTIF0	COM FIFO TX Mode Interrupt Flag 0: Channel n COM FIFO TX Mode Interrupt flag not set 1: Channel n COM FIFO TX Mode Interrupt flag set	R
4	THIF0	TX History List Interrupt 0: Channel n TX History List Interrupt flag not set 1: Channel n TX History List Interrupt flag set	R
31:5	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

The Global TX Interrupt Status register indicates the detection of transmit specific interrupts.

TSIF0 bit (TX Successful Interrupt Flag)

The TSIF0 bit is set to 1 when the TX Successful Interrupt flag of the related channel is set (when the interrupt is enabled). This bit is cleared automatically:

- When the related TX MB Result Status bits are cleared (when the interrupt enable is disabled)
- When in GL_RESET or CH_RESET mode.

TAI0 bit (TX Abort Interrupt Flag)

The TAIF0 bit is set to 1 when the TX Abort Interrupt flag of the related channel is set (when the interrupt is enabled).

This bit is cleared automatically:

- When the related TX MB Result Status bits are cleared (when the interrupt enable is disabled)
- When in GL_RESET or CH_RESET mode.

TQIF0 bit (TX Queue Interrupt Flag)

The TQIF0 bit is set to 1 when the TX Queue Interrupt flag of the related channel is set (when the interrupt is enabled). This bit is cleared automatically:

- When the related TX Queue Interrupt flag is cleared (when the interrupt is enable disabled)
- When in GL_RESET or CH_RESET mode.

CFTIF0 bit (COM FIFO TX Mode Interrupt Flag)

The CFTIF0 bit is set to 1 when the related COM TX FIFO Mode Interrupt flag (CFDCFSTS.CFTXIF) is set (when the interrupt is enabled).

This bit is cleared automatically:

- When the related COM TX FIFO Mode Interrupt flag (CFDCFSTS.CFTXIF) is cleared (when the interrupt enable is disabled)
- When in GL_RESET or CH_RESET mode.

THIF0 bit (TX History List Interrupt)

The THIF0 bit is set to 1 when the related TX History List Interrupt flag (CFDTHLSTS.THLIF) is set (when the interrupt is enabled).

This bit is cleared automatically:

- When the related TX History List Interrupt flag (CFDTHLSTS.THLIF) is cleared (when the interrupt enable is disabled)
- When in GL_RESET or CH_RESET mode.

42.2.16 CFDGTSC : Global Timestamp Counter Register

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x0024

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	TS[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	TS[15:0]	Timestamp value	R
31:16	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

The Global Timestamp Counter register stores the timestamp based on the selected configuration.

TS[15:0] bits (Timestamp value)

The Timestamp value is stored in the Global Timestamp Counter register based on the configuration of TSSS, TSBTCS and TSP. The accuracy of the timestamp counter cannot be guaranteed when transitioning to halt state.

The Timestamp value is stored in this register based on the configuration of TSSS, TSBTCS and TSP.

Do not write to bits TS[15:0] when the CANFD module is in GL_RESET or GL_SLEEP mode.

The TS[15:0] bits are cleared automatically in GL_RESET mode.

42.2.17 CFDGAFLECTR : Global Acceptance Filter List Entry Control Register

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x0028

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	AFLDAE	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	—	These bits are read as 0. The write value should be 0.	R/W
8	AFLDAE	Acceptance Filter List Data Access Enable 0: Acceptance Filter List data access disabled 1: Acceptance Filter List data access enabled	R/W
31:9	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

The Global Acceptance Filter List Entry Control Register is used to select the Global Acceptance Filter List page for reading or writing entries into the Global Acceptance Filter List.

AFLDAE bit (Acceptance Filter List Data Access Enable)

The AFLDAE bit prevents write access to the Acceptance Filter List when cleared after configuration of the Acceptance Filter List.

Data can be read from the Acceptance Filter List independent of the status of this bit.

Do not write to this bit when the CANFD module is in GL_SLEEP mode. Set this bit to enable write access for the Acceptance Filter List.

42.2.18 CFDGAFLCFG : Global Acceptance Filter List Configuration Register

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x002C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	RNC0[4:0]					—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	—	These bits are read as 0. The write value should be 0.	R/W
20:16	RNC0[4:0]	Rule Number Number of rules dedicated	R/W
31:21	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

The Global Acceptance Filter List Configuration Register is used to define the number of rules for entries in the Acceptance Filter List.

The total number of available entries in the Acceptance Filter List is 16.

RNC0[4:0] bits (Rule Number)

The RNC0[4:0] bits define the number of rules in the Acceptance Filter List.

Only write to these bits when the CANFD module is in GL_RESET mode. These bits can set to 5 bits for 16 rules.

42.2.19 CFDGAFLIDr : Global Acceptance Filter List ID Registers (r = 1 to 16)

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x0120 + 0x0010 × (r - 1)

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	GAFLIDE	GAFLRTR	GAFLLB	GAFLID[28:16]												
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	GAFLID[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
28:0	GAFLID[28:0]	Global Acceptance Filter List Entry ID Field ID part of the Global Acceptance Filter List entry	R/W
29	GAFLLB	Global Acceptance Filter List Entry Loopback Configuration 0: Global Acceptance Filter List entry ID for acceptance filtering with attribute RX 1: Global Acceptance Filter List entry ID for acceptance filtering with attribute TX	R/W
30	GAFLRTR	Global Acceptance Filter List Entry RTR Field 0: Data frame 1: Remote frame	R/W

Bit	Symbol	Function	R/W
31	GAFLIDE	Global Acceptance Filter List Entry IDE Field 0: Standard identifier of rule entry ID is valid for acceptance filtering 1: Extended identifier of rule entry ID is valid for acceptance filtering	R/W

Note: S-TYPE-3, P-TYPE-3

The Global Acceptance Filter List ID Registers are used to configure the ID field for the rules of entries in the Global Acceptance Filter List.

GAFLID[28:0] bits (Global Acceptance Filter List Entry ID Field)

The GAFLID[28:0] bits represent the CAN identifier (ID) field of each entry in the Global Acceptance Filter List.

Do not write to these bits when CFDGAFLECTR.AFLDAE bit is 0.

Only write to these bits when the related CANFD channel is in CH_RESET or CH_HALT mode.

GAFLLB bit (Global Acceptance Filter List Entry Loopback Configuration)

The GAFLLB bit selects whether entry in the Global Acceptance Filter List gets the attribute RX or TX.

This attribute determines the validity of the entry in Mirror mode, Loopback test mode, and during standard (non-loopback) reception. See section 42.5.5. Loopback Modes for detailed description of the validity of the Global Acceptance Filter List entry depending on transmitter/receiver case, the type of loopback mode, and RX/TX attribute.

Do not write to this bit when CFDGAFLECTR.AFLDAE bit is 0.

Only write to this bit when the related CANFD channel is in CH_RESET or CH_HALT mode.

GAFLRTR bit (Global Acceptance Filter List Entry RTR Field)

The GAFLRTR bit allows the configuration of the specified frame format (data frame or remote frame) for each entry of the Global Acceptance Filter List. For each rule entry in a CAN channel, the acceptance filter process compares this bit against the RTR bit of the received CAN message.

Do not write to this bit when CFDGAFLECTR.AFLDAE bit is 0.

Only write to this bit when the related CANFD channel is in CH_RESET or CH_HALT mode.

GAFLIDE bit (Global Acceptance Filter List Entry IDE Field)

The GAFLIDE bit allows the configuration of the ID format (standard ID or extended ID) for each entry in the Global Acceptance Filter List. For each rule entry in a CAN channel, the acceptance filter process compares this bit against the IDE bit of the received CAN message.

Do not write to this bit when CFDGAFLECTR.AFLDAE bit is 0.

Only write to this bit when the related CANFD channel is in CH_RESET or CH_HALT mode.

42.2.20 CFDGAFLMr : Global Acceptance Filter List Mask Registers (r = 1 to 16)

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x0124 + 0x0010 × (r - 1)

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	GAFLIDEM	GAFLRTRM	GAFLIFL1	GAFLIDM[28:16]												
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	GAFLIDM[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
28:0	GAFLIDM[28:0]	Global Acceptance Filter List ID Mask Field Global Acceptance Filter List Mask field bits for ID field	R/W
29	GAFLIFL1	Global Acceptance Filter List Information Label 1 Global Acceptance Filter List information label bit 1	R/W
30	GAFLRTRM	Global Acceptance Filter List Entry RTR Mask 0: RTR bit is not used for ID matching 1: RTR bit is used for ID matching	R/W
31	GAFLIDEM	Global Acceptance Filter List IDE Mask 0: IDE bit is not used for ID matching 1: IDE bit is used for ID matching	R/W

Note: S-TYPE-3, P-TYPE-3

The Global Acceptance Filter List Mask Registers are used to configure the Mask field of each rule for entries in the Global Acceptance Filter List.

GAFLIDM[28:0] bits (Global Acceptance Filter List ID Mask Field)

GAFLIDM[28:0] bits are the filter mask bits for the related bits in the CAN Identifier field of each Global Acceptance Filter List entry.

0	Corresponding STD-ID/EXT-ID bit is not used for ID matching
1	Corresponding STD-ID/EXT-ID bit is used for ID matching

Do not write to these bits when CFDGAFLECTR.AFLDAE bit is 0.

Only write to these bits when the related CANFD channel is in CH_RESET or CH_HALT mode.

GAFLIFL1 bit (Global Acceptance Filter List Information Label 1)

The GAFLIFL1 bit allows the configuration of a 2-bit information label to be attached to a received message accepted by the associated entry in the Global Acceptance Filter List. This bit is an MSB bit of an information label.

Do not write to this bit when CFDGAFLECTR.AFLDAE bit is 0.

Only write to this bit when the related CANFD channel is in CH_RESET or CH_HALT mode.

This bit is stored in the Information Label Field [1] (CFDRMFDSTSb.RMIFL [1], CFDRFFDSTSb.RFIFL [1], CFDCFFDCSTS.CFIFL [1]) of the storage location of an incoming message.

GAFLRTRM bit (Global Acceptance Filter List Entry RTR Mask)

The GAFLRTRM bit allows the configuration of the RTR mask bit for each entry in the Global Acceptance Filter List.

Do not write to this bit when CFDGAFLECTR.AFLDAE bit is 0.

Only write to this bit when the related CANFD channel is in CH_RESET or CH_HALT mode.

GAFLIDEM bit (Global Acceptance Filter List IDE Mask)

The GAFLIDEM bit allows the configuration of the IDE mask bit for each entry in the Global Acceptance Filter List.

When the IDE mask bit is 0, the ID comparison depends on the received IDE bit.

If the received IDE bit is 0, the STD-ID comparison takes place.

If the received IDE bit is 1, the EXT-ID comparison takes place.

Do not write to this bit when CFDGAFLECTR.AFLDAE bit is 0.

Only write to this bit when the related CANFD channel is in CH_RESET or CH_HALT mode.

42.2.21 CFDGAFLP0r : Global Acceptance Filter List Pointer 0 Registers (r = 1 to 16)

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x0128 + 0x0010 × (r - 1)

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	GAFLPTR[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	GAFL RMV	—	—	GAFLRMDP[4:0]					GAFLI FL0	—	—	—	GAFLDLC[3:0]			
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
3:0	GAFLDLC[3:0]	Global Acceptance Filter List DLC Field Minimum number of data bytes in a data frame required for acceptance	R/W
6:4	—	These bits are read as 0. The write value should be 0.	R/W
7	GAFLIFL0	Global Acceptance Filter List Information Label 0	R/W
12:8	GAFLRMDP[4:0]	Global Acceptance Filter List RX Message Buffer Direction Pointer RX message buffer number for storage of received messages	R/W
14:13	—	These bits are read as 0. The write value should be 0.	R/W
15	GAFLRMV	Global Acceptance Filter List RX Message Buffer Valid 0: Single message buffer direction pointer is invalid 1: Single message buffer direction pointer is valid	R/W
31:16	GAFLPTR[15:0]	Global Acceptance Filter List Pointer	R/W

Note: S-TYPE-3, P-TYPE-3

The Global Acceptance Filter List Pointer 0 Registers are used to configure the data length code (DLC), software pointer, single message buffer select, and message buffer direction pointer for each rule entry in the Global Acceptance Filter List.

GAFLDLC[3:0] bits (Global Acceptance Filter List DLC Field)

The GAFLDLC[3:0] bits allow the configuration of a minimum data length code (DLC) value for a message to be accepted by the associated entry in the Global Acceptance Filter List (automatic DLC filter function).

DLC filter process is only passed if the DLC value of the message accepted by an entry in the Global Acceptance Filter List is equal to or higher than the DLC value configured for this associated Global Acceptance Filter List entry. Automatic DLC filter function is disabled for the corresponding rule entry when this field is set to 0.

Table 42.3 shows DLC value that can be configured.

Table 42.3 Configuration of DLC value (1 of 2)

Format	DLC[3]	DLC[2]	DLC[1]	DLC[0]	Description
CAN and CANFD	0	0	0	0	DLC of received message = 0 or more (DLC filter check is disabled)
CAN and CANFD	0	0	0	1	DLC of received message = 1 or more
CAN and CANFD	0	0	1	0	DLC of received message = 2 or more
CAN and CANFD	0	0	1	1	DLC of received message = 3 or more
CAN and CANFD	0	1	0	0	DLC of received message = 4 or more
CAN and CANFD	0	1	0	1	DLC of received message = 5 or more

Table 42.3 Configuration of DLC value (2 of 2)

Format	DLC[3]	DLC[2]	DLC[1]	DLC[0]	Description
CAN and CANFD	0	1	1	0	DLC of received message = 6 or more
CAN and CANFD	0	1	1	1	DLC of received message = 7 or more
CAN	1	x	x	x	DLC of received message = 8 or more
CANFD	1	0	0	0	DLC of received message = 8 or more*1
CANFD	1	0	0	1	DLC of received message = 12 or more*1
CANFD	1	0	1	0	DLC of received message = 16 or more*1
CANFD	1	0	1	1	DLC of received message = 20 or more*1
CANFD	1	1	0	0	DLC of received message = 24 or more*1
CANFD	1	1	0	1	DLC of received message = 32 or more*1
CANFD	1	1	1	0	DLC of received message = 48 or more*1
CANFD	1	1	1	1	DLC of received message = 64*1

Note 1. This setting is not available in the classical CAN function.

Do not write to these bits when CFDGAFLECTR.AFLDAE bit is 0.

Only write to these bits when the related CANFD channel is in CH_RESET or CH_HALT mode.

GAFLIFL0 bit (Global Acceptance Filter List Information Label 0)

The GAFLIFL0 bit allows the configuration of a 2-bit information label that can be attached to a received message accepted by the related Global Acceptance Filter List entry. This bit is an LSB bit of an information label.

You cannot write to this bit when CFDGAFLECTR.AFLDAE bit is 0.

Only write to the bit when the related CANFD channel is in CH_RESET or CH_HALT mode.

This bit is stored in Information Label Field[0] (CFDRMFDSTSb.RMIFL[0], CFDRFFDSTSb.RFIFL[0], CFDCFFDCSTS.CFIFL[0]) of the storage location of an incoming message.

GAFLRMDP[4:0] bits (Global Acceptance Filter List RX Message Buffer Direction Pointer)

The GAFLRMDP[4:0] bits allow the configuration of a single reception message buffer as the destination target for a received message that passes the acceptance check of the related Global Acceptance Filter List entry. The value entered is the single destination message buffer number.

Do not write to these bits when CFDGAFLECTR.AFLDAE bit is 0.

Only write to these bits when the related CANFD channel is in CH_RESET or CH_HALT mode.

CFDRMNB.NRXMB[4:0] is the value entered in the RX Message Buffer Number Register to configure the number of RX message buffers. The value to be entered in CFDGAFLP0r.GAFLRMDP[4:0] bits should only be between 0x00 and CFDMNB.NMB[4:0] to 1 less.

If CFDRMNB.NRXMB[4:0] = 0x00, the GAFLRMV bit should be configured as 0.

GAFLRMV bit (Global Acceptance Filter List RX Message Buffer Valid)

The GAFLRMV bit allows the enabling or disabling of a single reception message buffer as the target for a received message that passes the acceptance check of the related Global Acceptance Filter List entry.

Do not write to these bits when CFDGAFLECTR.AFLDAE bit is 0.

Only write to these bits when the related CANFD channel is in CH_RESET or CH_HALT mode.

GAFLPTR[15:0] bits (Global Acceptance Filter List Pointer)

The GAFLPTR[15:0] bits allow the configuration of a 16-bit pointer to be attached to a received message accepted by the related Global Acceptance Filter List entry. The pointer is added during message storage in the Message Buffer area and can be used by the application as a support function. The pointer information can be used for example, to support PDU Identifier allocation for the received message in AUTOSAR systems.

Do not write to these bits when CFDGAFLECTR.AFLDAE bit is 0.

Only write to these bits when the related CANFD channel is in CH_RESET or CH_HALT mode.

42.2.22 CFDGAFLP1r : Global Acceptance Filter List Pointer 1 Registers (r = 1 to 16)

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x012C + 0x0010 × (r - 1)

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	GAFLFDP8	—	—	—	—	—	—	GAFLFDP1	GAFLFDP0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	GAFLFDP0	Global Acceptance Filter List FIFO Direction Pointer FIFO direction pointer bits for received message storage 0: Disable RX FIFO 0 as target for reception 1: Enable RX FIFO 0 as target for reception	R/W
1	GAFLFDP1	Global Acceptance Filter List FIFO Direction Pointer FIFO direction pointer bits for received message storage 0: Disable RX FIFO 1 as target for reception 1: Enable RX FIFO 1 as target for reception	R/W
7:2	—	These bits are read as 0. The write value should be 0.	R/W
8	GAFLFDP8	Global Acceptance Filter List FIFO Direction Pointer FIFO direction pointer bits for received message storage 0: Disable Common FIFO as target for reception 1: Enable Common FIFO as target for reception	R/W
31:9	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

The Global Acceptance Filter List Pointer 1 registers are used to configure the FIFO direction pointer fields in each Rule Entry of the Global Acceptance Filter List.

GAFLFDP8, GAFLFDP1, GAFLFDP0 bits (Global Acceptance Filter List FIFO Direction Pointer)

These bits allow the configuration of FIFO Buffers as the target for a received message passing the acceptance check of the related Global Acceptance Filter List entry. Each bit of the GAFLFDP8, GAFLFDP1, GAFLFDP0 is configuring a dedicated FIFO.

Users cannot write to these bits when CFDGAFLECTR.AFLDAE bit is 0.

For storage in Common FIFO, target for reception can only be those Common FIFO Buffers that are configured as RX FIFO.

Only write to these bits when the related CANFD channel is in CH_RESET or CH_HALT mode.

Users should only configure up to 2 destination FIFO Buffers or 1 destination FIFO Buffers plus one RX Message Buffer.

42.2.23 CFDRMNB : RX Message Buffer Number Register

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x0030

Bit position:	31																				10		8				4				0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	RMPLS[2:0]			—	—	—	NRXMB[4:0]				
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
4:0	NRXMB[4:0]	Number of RX Message Buffers	R/W
7:5	—	These bits are read as 0. The write value should be 0.	R/W
10:8	RMPLS[2:0]	Reception Message Buffer Payload Data Size 0 0 0: 8 bytes 0 0 1: 12 bytes 0 1 0: 16 bytes 0 1 1: 20 bytes 1 0 0: 24 bytes 1 0 1: 32 bytes 1 1 0: 48 bytes 1 1 1: 64 bytes	R/W
31:11	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

The RX Message Buffer Number register is used to configure the total number of RX message buffers allocated to channels.

NRXMB[4:0] bits (Number of RX Message Buffers)

The NRXMB[4:0] bits are used to configure the number of RX message buffers.

Only write to these bits when the CANFD module is in GL_RESET mode.

Enter only values between 0 and 16 inclusive, with 0x00 indicating that no RX message buffer is allocated.

RMPLS[2:0] bits (Reception Message Buffer Payload Data Size)

The RMPLS[2:0] bits are used to configure the message buffer payload data size.

Only write to these bits when the CANFD module is in GL_RESET mode.

42.2.24 CFDRMND : RX Message Buffer New Data Register

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x0034

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	RMNS[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	RMNS[15:0]	RX Message Buffer New Data Status 0: New data not stored in corresponding RX message buffer 1: New data stored in corresponding RX message buffer	R/W

Note: S-TYPE-3, P-TYPE-3

The RX Message Buffer New Data Status Register specifies the new data storage status of the RX message buffers.

RMNS[15:0] bits (RX Message Buffer New Data Status)

The RMNS[15:0] bits indicate the status of new data for the corresponding RX message buffer. RMNS bit [0] corresponds to RX message buffer [0] and so on.

The bit position of CFDRMND corresponds to the buffer number of RXMB.

Do not write to these bits when the CANFD module is in GL_RESET or GL_SLEEP mode. Writing 1 has no effect.

These bits cannot be cleared when message storage in the corresponding RX message buffer is in progress.
Do not use the bit clear instruction to clear these bits. Use the MOV instruction to ensure that only the specified bit is cleared. Other bits remain 1.

These bits are set automatically when storage of new messages are in the corresponding RX message buffer. These bits are cleared by writing 0. These bits are cleared automatically when the CANFD module is in GL_RESET mode.

When CFDRMNB.RMPLS = 000b (maximum 8 bytes payload), the duration of message storage is 6 PCLKA cycles.

When CFDRMNB.RMPLS > 000b, the duration of message storage is 6 PCLKA cycles + 1 for each 4 bytes (maximum of 20 PCLKA cycles for 64 bytes).

Note: This feature is not available in the classical CAN function.

42.2.25 CFDRFCCa : RX FIFO Configuration/Control Registers a (a = 0 to 1)

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x003C + 0x04 × a

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	RFIGCV[2:0]			RFIM	—	RFDC[2:0]			—	RFPLS[2:0]			—	—	RFIE	RFE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	RFE	RX FIFO Enable 0: FIFO disabled 1: FIFO enabled	R/W
1	RFIE	RX FIFO Interrupt Enable 0: FIFO interrupt generation disabled 1: FIFO interrupt generation enabled	R/W
3:2	—	These bits are read as 0. The write value should be 0.	R/W
6:4	RFPLS[2:0]*1	Rx FIFO Payload Data Size Configuration 0 0 0: 8 bytes 0 0 1: 12 bytes 0 1 0: 16 bytes 0 1 1: 20 bytes 1 0 0: 24 bytes 1 0 1: 32 bytes 1 1 0: 48 bytes 1 1 1: 64 bytes	R/W
7	—	This bit is read as 0. The write value should be 0.	R/W

Bit	Symbol	Function	R/W
10:8	RFDC[2:0]	RX FIFO Depth Configuration 0 0 0: FIFO Depth = 0 message 0 0 1: FIFO Depth = 4 messages 0 1 0: FIFO Depth = 8 messages 0 1 1: FIFO Depth = 16 messages 1 0 0: FIFO Depth = 32 messages 1 0 1: FIFO Depth = 48 messages 1 1 0: Reserved 1 1 1: Reserved	R/W
11	—	This bit is read as 0. The write value should be 0.	R/W
12	RFIM	RX FIFO Interrupt Mode 0: Interrupt generated when RX FIFO counter reaches RFIGCV value from values smaller than RFIGCV 1: Interrupt generated at the end of every received message storage	R/W
15:13	RFIGCV[2:0]	RX FIFO Interrupt Generation Counter Value 0 0 0: Interrupt generated when FIFO is 1/8th full 0 0 1: Interrupt generated when FIFO is 1/4th full 0 1 0: Interrupt generated when FIFO is 3/8th full 0 1 1: Interrupt generated when FIFO is 1/2 full 1 0 0: Interrupt generated when FIFO is 5/8th full 1 0 1: Interrupt generated when FIFO is 3/4th full 1 1 0: Interrupt generated when FIFO is 7/8th full 1 1 1: Interrupt generated when FIFO is full	R/W
31:16	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note 1. These bits are not available in the classical CAN function.

The RX FIFO Configuration/Control Registers are used to configure and control the two RX FIFOs.

RFE bit (RX FIFO Enable)

The RFE bit enables the FIFO. When this bit is set to 0, the RX FIFO is cleared to empty.

Only write to this bit when the CANFD module is in GL_HALT or GL_OPERATION mode.

This bit can only be set if the configured FIFO depth is greater than 0x000 (CFDRFCCa.RFDC > 0x000) and less than 0x110.

Set the RFE bit with a separate write access to the CFDRFCCa register, after all the other bits in the CFDRFCCa register are set.

This bit is cleared automatically when the CANFD module is in GL_RESET mode.

RFIE bit (RX FIFO Interrupt Enable)

The RFIE bit enables generation of the FIFO interrupt.

Do not write to this bit when the CANFD module is in GL_SLEEP mode.

RFPLS[2:0] bits (Rx FIFO Payload Data Size Configuration)

The RFPLS[2:0] bits define the message data payload allocation in the RAM.

This is the maximum number of bytes which can be received by this FIFO.

Only write to these bits when the CANFD module is in GL_RESET mode.

Note: These bits are not available in the classical CAN function.

RFDC[2:0] bits (RX FIFO Depth Configuration)

The RFDC[2:0] bits select the depth of the FIFO in terms of the number of messages. If the FIFO depth is configured to 0 messages, the FIFO cannot be used.

Only write to these bits when the CANFD module is in GL_RESET mode.

RFIM bit (RX FIFO Interrupt Mode)

The RFIM bit selects the interrupt generation condition for the FIFO.

Do not write to this bit when the CANFD module is in GL_SLEEP mode.

Only write to this bit when the CANFD module is in GL_RESET mode.

RFIGCV[2:0] bits (RX FIFO Interrupt Generation Counter Value)

The RFIGCV[2:0] bits select the counter value of the FIFO for generation of FIFO interrupts. These values represent fractions of the FIFO depth for which an interrupt is generated.

Do not write to these bits when the CANFD module is in GL_SLEEP mode.

The setting of the RFIGCV[2:0] bits should be synchronized with the RFDC[2:0] bits.

Only write to these bits when the CANFD module is in GL_RESET mode.

42.2.26 CFDRFSTSa : RX FIFO Status Registers a (a = 0 to 1)

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x0044 + 0x04 × a

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	RFMC[5:0]						—	—	—	—	RFIF	RFMLT	RFFLL	RFEMP
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1

Bit	Symbol	Function	R/W
0	RFEMP	RX FIFO Empty 0: FIFO not empty 1: FIFO empty	R
1	RFFLL	RX FIFO Full 0: FIFO not full 1: FIFO full	R
2	RFMLT	RX FIFO Message Lost 0: No message lost in FIFO 1: FIFO message lost	R/W
3	RFIF	RX FIFO Interrupt Flag 0: FIFO interrupt condition not satisfied 1: FIFO interrupt condition satisfied	R/W
7:4	—	These bits are read as 0. The write value should be 0.	R/W
13:8	RFMC[5:0]	RX FIFO Message Count Number of messages stored in FIFO	R
31:14	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

The RX FIFO Status Registers show the status of messages stored in the corresponding FIFO buffers.

RFEMP bit (RX FIFO Empty)

The RFEMP bit is set automatically when:

- The RFMC bit is 0
- RX FIFO is disabled by setting the CFDRFCCa.RFE bit to 0
- The CANFD module is in GL_RESET mode.

The RFEMP bit is cleared automatically when the first message is stored in the RX FIFO buffer.

RFFLL bit (RX FIFO Full)

The RFFLL bit is set automatically when the number of CAN messages stored in the FIFO buffer matches the configured FIFO depth.

The RFFLL is cleared automatically when:

- The number of CAN messages stored in the FIFO buffer is less than the configured FIFO depth
- RX FIFO is disabled by setting the CFDRFCCa.RFE bit to 0
- The CANFD module is in GL_RESET mode.

RFMLT bit (RX FIFO Message Lost)

Only write to the RFMLT bit when CANFD module is in GL_HALT or GL_OPERATION mode. Writing 1 has no effect.

Do not use the bit clear instruction to clear this bit. Use the MOV instruction to ensure that only the specified bit is cleared. Other bits remain 1.

This bit is set automatically whenever a message is lost due to attempted storage when the FIFO buffer is already full. If a set from the CAN channel occurs simultaneously with a clear by a write access, then the bit is set.

The bit is cleared:

- By writing 0 to it
- When the CANFD module is in GL_RESET mode.

RFIF bit (RX FIFO Interrupt Flag)

The RFIF bit is set automatically when the configured interrupt condition is satisfied. This bit is not automatically cleared when the RX FIFO buffer is disabled.

Only write to this bit when the CANFD module is in GL_HALT or GL_OPERATION mode. Writing 1 has no effect.

Do not use the bit clear instruction to clear this bit. Use the MOV instruction to ensure that only the specified bit is cleared. Other bits remain 1.

If a set from the CAN channel occurs simultaneously with a clear by a write access, then this bit is set.

The bit is cleared by writing 0 to it. The bit is also cleared when CANFD module is in GL_RESET mode.

RFMC[5:0] bits (RX FIFO Message Count)

The RFMC[5:0] bits indicate the number of CAN messages stored in the RX FIFO buffer that can be read by the CPU.

These bits are cleared automatically when the FIFO is disabled and when the CANFD module is in GL_RESET mode.

42.2.27 CFDRFPCTRa : RX FIFO Pointer Control Registers a (a = 0 to 1)

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x004C + 0x04 × a

Bit position:	31																								7							0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	RFPC[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	RFPC[7:0]	RX FIFO Pointer Control Increments read pointer of the corresponding RX FIFO buffers	W
31:8	—	The write value should be 0.	W

Note: S-TYPE-3, P-TYPE-3

The RX FIFO Pointer Control Registers can be used to increment the read pointer of the corresponding RX FIFO buffers.

RFPC bits (RX FIFO Pointer Control)

When the value 0xFF is written to the RFPC bits, the pointer of the corresponding RX FIFO buffer is moved to the next FIFO entry. Only write 0xFF to these registers when the corresponding RX FIFO buffer is enabled and not empty.

The read value from these bits is always 0x00.

Only write to these bits when the CANFD module is in GL_HALT or GL_OPERATION mode.

Do not write to the RX FIFO Pointer Control registers when DMA is enabled.

42.2.28 CFDCFCC : Common FIFO Configuration/Control Register

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x0054

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	CFITT[7:0]								CFDC[2:0]			—	—	—	CFTML[1:0]	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	CFIGCV[2:0]			CFIM	CFITR	CFITSS	—	CFM	—	CFPLS[2:0]			—	CFTXIE	CFRXIE	CFE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	CFE	Common FIFO Enable 0: FIFO disabled 1: FIFO enabled	R/W
1	CFRXIE	Common FIFO RX Interrupt Enable 0: FIFO interrupt generation disabled for Frame RX 1: FIFO interrupt generation enabled for Frame RX	R/W
2	CFTXIE	Common FIFO TX Interrupt Enable 0: FIFO interrupt generation disabled for Frame TX 1: FIFO interrupt generation enabled for Frame TX	R/W
3	—	This bit is read as 0. The write value should be 0.	R/W
6:4	CFPLS[2:0]*1	Common FIFO Payload Data Size Configuration 0 0 0: 8 bytes 0 0 1: 12 bytes 0 1 0: 16 bytes 0 1 1: 20 bytes 1 0 0: 24 bytes 1 0 1: 32 bytes 1 1 0: 48 bytes 1 1 1: 64 bytes	R/W
7	—	This bit is read as 0. The write value should be 0.	R/W
8	CFM	Common FIFO Mode 0: RX FIFO mode 1: TX FIFO mode	R/W
9	—	This bit is read as 0. The write value should be 0.	R/W
10	CFITSS	Common FIFO Interval Timer Source Select 0: Reference clock (× 1 / × 10 period) 1: Bit time clock of related channel (FIFO is linked to fixed channel)	R/W
11	CFITR	Common FIFO Interval Timer Resolution 0: Reference clock period × 1 1: Reference clock period × 10	R/W

Bit	Symbol	Function	R/W
12	CFIM	Common FIFO Interrupt Mode 0: RX FIFO mode: RX interrupt generated when Common FIFO counter reaches CFIGCV value from a lower value TX FIFO mode: TX interrupt generated when Common FIFO transmits the last message successfully 1: RX FIFO mode: RX interrupt generated at the end of every received message storage TX FIFO mode: interrupt generated for every successfully transmitted message	R/W
15:13	CFIGCV[2:0]	Common FIFO Interrupt Generation Counter Value 0 0 0: Interrupt generated when FIFO is 1/8th full 0 0 1: Interrupt generated when FIFO is 1/4th full 0 1 0: Interrupt generated when FIFO is 3/8th full 0 1 1: Interrupt generated when FIFO is 1/2 full 1 0 0: Interrupt generated when FIFO is 5/8th full 1 0 1: Interrupt generated when FIFO is 3/4th full 1 1 0: Interrupt generated when FIFO is 7/8th full 1 1 1: Interrupt generated when FIFO is full	R/W
17:16	CFTML[1:0]	Common FIFO TX Message Buffer Link Transmission scan link position of the corresponding channel	R/W
20:18	—	These bits are read as 0. The write value should be 0.	R/W
23:21	CFDC[2:0]	Common FIFO Depth Configuration 0 0 0: FIFO Depth = 0 message 0 0 1: FIFO Depth = 4 messages 0 1 0: FIFO Depth = 8 messages 0 1 1: FIFO Depth = 16 messages 1 0 0: FIFO Depth = 32 messages 1 0 1: FIFO Depth = 48 messages 1 1 0: FIFO Depth = Reserved 1 1 1: FIFO Depth = Reserved	R/W
31:24	CFITT[7:0]	Common FIFO Interval Transmission Time Delay the start of transmission from the FIFO if configured in TX mode, delay is a multiple of basic Interval Timer Clock Source unit	R/W

Note: S-TYPE-3, P-TYPE-3

Note 1. These bits are not available in the classical CAN function.

CFE bit (Common FIFO Enable)

The CFE bit enables the FIFO when set. FIFO is disabled when this bit is cleared.

This bit can also be used, by clearing it, to abort transmission from Common FIFO when configured in TX mode, or to stop reception into the Common FIFO in RX mode.

Only write to this bit when the CANFD module is in GL_HALT or GL_OPERATION mode and the related CANFD channel is not in CH_RESET mode for FIFOs configured as TX FIFO.

This bit can only be set if the configured FIFO depth is greater than 0x000 (CFDCFCC.CFDC > 0x000) and less than 0x110 (0x110 > CFDCFCC.CFDC > 0x000).

Set the CFE bit with a separate write access to the CFDCFCC register, after all the other bits in this register are set.

This bit is cleared automatically when the CANFD module is in GL_RESET mode.

This bit is also cleared automatically when the related channel is in CH_RESET mode if the FIFO is configured in TX mode.

CFRXIE bit (Common FIFO RX Interrupt Enable)

The CFRXIE bit enables generation of FIFO interrupts when the interrupt flag is set after reception of a frame in the corresponding FIFO buffer.

Do not write to this bit when the CANFD module is in GL_SLEEP mode.

CFTXIE bit (Common FIFO TX Interrupt Enable)

The CFTXIE bit enables generation of common FIFO interrupts when the interrupt flag is set after transmission of a frame from the corresponding FIFO buffer.

Do not write to this bit when the CANFD module is in GL_SLEEP mode.

CFPLS[2:0] bits (Common FIFO Payload Data Size Configuration)

The CFPLS[2:0] bits define the message data payload allocation in the RAM. This is the maximum number of bytes which can be received or transmitted by the FIFO buffer.

For details, see section 42.6. FIFO Buffers and Normal Message Buffer Configuration.

Only write to this bit when the CANFD module is in GL_RESET mode.

Note: These bits are not available in the classical CAN function.

CFM bit (Common FIFO Mode)

The CFM bit selects the mode of the FIFO. When a hardware reset is applied, all the Common FIFO buffers are configured in RX FIFO mode.

Do not write to these bits in GL_OPERATION or GL_SLEEP mode.

Only write to these bits when the CANFD module is in GL_RESET mode.

CFITSS bit (Common FIFO Interval Timer Source Select)

The CFITSS bit selects the basic clock source for the Interval Transmission Timer.

Do not write to this bit when the CANFD module is in GL_SLEEP mode. In addition, do not write to this bit when the CFE bit is set to 1.

Do not write 1 to this bit when CANFD communication is used.*1

Note: The bit time clock can vary depending on the nominal and data rate bit configuration.

Note 1. This feature is not available in the classical CAN function.

CFITR bit (Common FIFO Interval Timer Resolution)

The CFITR bit selects the resolution of the reference clock for the Interval Transmission Timer (peripheral clock is the source for the reference clock).

Do not write to this bit when the CANFD module is in GL_SLEEP mode. Also, do not write to this bit when the CFE bit is set to 1.

CFIM bit (Common FIFO Interrupt Mode)

The CFIM bit selects the interrupt generation condition for the FIFO buffer.

Do not write to this bit in GL_SLEEP mode.

Only write to this bit when the CANFD module is in GL_RESET mode.

CFIGCV[2:0] bits (Common FIFO Interrupt Generation Counter Value)

The CFIGCV[2:0] bits select the message counter value for the generation of FIFO interrupts. These values represent fractions of the FIFO depth at which the interrupt is to be generated.

Do not write to these bits when the CANFD module is in GL_SLEEP mode.

The setting of these bits should be synchronized with the CFDC[2:0] bits.

Only write to these bits when the CANFD module is in GL_RESET mode.

CFTML[1:0] bits (Common FIFO TX Message Buffer Link)

The CFTML[1:0] bits select the normal transmit message buffer position where the TX FIFO is linked to, for transmission scanning.

Do not write to these bits in GL_OPERATION or GL_SLEEP mode.

Only write to this bit when the CANFD module is in GL_RESET mode.

CFDC[2:0] bits (Common FIFO Depth Configuration)

The CFDC[2:0] bits select the depth of the common FIFO in terms of the number of messages. If the FIFO depth is configured to 0 message, the FIFO cannot be used.

Only write to these bits when the CANFD module is in GL_RESET mode.

CFITT[7:0] bits (Common FIFO Interval Transmission Time)

The CFITT[7:0] bits select the delay in the start of transmission for all messages transmitted from this FIFO buffer when configured in TX mode. The delay is a multiple of the basic interval timer clock source period (reference clock × 1, reference clock × 10, or bit time clock of the related CAN channel).

Do not write to these bits when the CANFD module is in GL_SLEEP mode.

Do not write to these bits when the CFE bit is set to 1.

When CFDGCFG.ITRCP[15:0] = 0x0000, set the CFITT[7:0] bits to 0x0000.

42.2.29 CFDCFSTS : Common FIFO Status Register

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x0058

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	CFMC[5:0]						—	—	—	CFTXIF	CFRXIF	CFMLT	CFFLL	CFEMP
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1

Bit	Symbol	Function	R/W
0	CFEMP	Common FIFO Empty 0: FIFO not empty 1: FIFO empty	R
1	CFFLL	Common FIFO Full 0: FIFO not full 1: FIFO full	R
2	CFMLT	Common FIFO Message Lost 0: Number of message lost in FIFO 1: FIFO message lost	R/W
3	CFRXIF	Common RX FIFO Interrupt Flag 0: FIFO interrupt condition not satisfied after frame reception 1: FIFO interrupt condition satisfied after frame reception	R/W
4	CFTXIF	Common TX FIFO Interrupt Flag 0: FIFO interrupt condition not satisfied after frame transmission 1: FIFO Interrupt condition satisfied after frame transmission	R/W
7:5	—	These bits are read as 0. The write value should be 0.	R/W
13:8	CFMC[5:0]	Common FIFO Message Count Number of messages stored in FIFO	R
31:14	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

CFEMP bit (Common FIFO Empty)

The CFEMP bit is set automatically when:

- The CPU has read all messages from the FIFO configured in RX mode
- All messages have been transmitted from the FIFO configured in TX mode
- The FIFO is disabled by setting the CFE bit to 0
- The CANFD module is in GL_RESET mode

- The related CANFD channel is in CH_RESET when FIFO configured in TX mode.

The CFEMP bit is cleared automatically when:

- The first reception message is stored in the FIFO buffer when configured in RX mode
- The first message to be transmitted is stored in the FIFO buffer when configured in TX mode.

CFFLL bit (Common FIFO Full)

The CFFLL bit is set automatically when the number of CAN messages stored in the FIFO matches the configured FIFO depth.

The CFFLL bit is cleared automatically when:

- The number of CAN messages stored in the FIFO is less than the configured FIFO depth
- The FIFO is disabled by setting the CFE bit to 0
- The CANFD module is in GL_RESET mode
- The related CANFD channel is in CH_RESET mode when FIFO buffer is configured in TX mode.

CFMLT bit (Common FIFO Message Lost)

The CFMLT bit is set automatically whenever a message is lost due to attempted storage of a new message when FIFO is already full in RX mode.

If a set from the CAN channel occurs simultaneously with a clear by a write access, then this bit is set.

Only write to this bit when the CANFD module is in GL_HALT or GL_OPERATION mode and the related CANFD channel is not in CH_RESET mode for FIFO configured as TX FIFO. Writing 1 has no effect.

Do not use the bit clear instruction to clear this bit. Use the MOV instruction to ensure that only the specified bit is cleared. Other bits remain 1.

The CFMLT bit is cleared:

- By writing 0 to it
- When the CANFD module is in GL_RESET mode
- When the related CANFD channel is in CH_RESET mode if the FIFO buffer is configured in TX mode.

CFRXIF bit (Common RX FIFO Interrupt Flag)

The CFRXIF bit is not cleared automatically if the Common FIFO buffer is disabled.

Only write to this bit when the CANFD module is in GL_HALT or GL_OPERATION mode and the related CANFD channel is not in CH_RESET mode for FIFO configured as TX FIFO. Writing 1 has no effect.

Do not use the bit clear instruction to clear this bit. Use the MOV instruction to ensure that only the specified bit is cleared. Other bits remain 1.

This bit is set automatically when the configured interrupt condition is satisfied for Common FIFO buffers when configured in RX mode.

If the set from the CAN channel occurs simultaneously with the clear by the write access, then the bit is set.

The CFRXIF bit is cleared:

- By writing 0 to it
- When the CANFD module is in GL_RESET mode

CFTXIF bit (Common TX FIFO Interrupt Flag)

The CFTXIF bit is not cleared automatically if the Common FIFO buffer is disabled.

Only write to this bit when the CANFD module is in GL_HALT or GL_OPERATION mode and the related CANFD channel is not in CH_RESET mode for FIFO buffer configured as TX FIFO. Writing 1 has no effect.

Do not use the bit clear instruction to clear this bit. Use the MOV instruction to ensure that only the specified bit is cleared. Other bits remain 1.

This bit is set automatically when the configured interrupt condition is satisfied for Common FIFO buffers configured in TX mode.

If the set from the CAN channel occurs simultaneously with the clear by the write access, then the bit is set.

The CFTXIF bit is cleared:

- By writing 0 to it
- When the CANFD module is in GL_RESET mode
- When the related CANFD channel is in CH_RESET mode if the FIFO buffer is configured in TX mode.

CFMC[5:0] bits (Common FIFO Message Count)

The CFMC[5:0] bits indicate the following:

- Number of CAN messages stored by the CPU in the FIFO buffer configured in TX mode pending for transmission
- Number of CAN messages stored in the FIFO buffer configured in RX mode by CANFD module to be read by the CPU

The CFMC[5:0] bits are cleared automatically when:

- The FIFO is disabled
- The CANFD module is in GL_RESET mode
- The related CANFD channel is in CH_RESET mode if the FIFO buffer is configured in TX mode.

42.2.30 CFDCFPCTR : Common FIFO Pointer Control Register

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x005C

Bit position:	31																								7							0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	CFPC[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	CFPC[7:0]	Common FIFO Pointer Control Increments read or write pointer of the corresponding Common FIFO buffers depending on the mode configuration.	W
31:8	—	The write value should be 0.	W

Note: S-TYPE-3, P-TYPE-3

The Common FIFO Pointer Control Registers can be used to increment the read or write pointer of the corresponding Common FIFO buffer.

CFPC[7:0] bits (Common FIFO Pointer Control)

When the value 0xFF is written into the CFPC[7:0] bits, the read pointer of the corresponding Common FIFO buffer (when configured in RX mode), or the write pointer of the corresponding Common FIFO buffer (when configured in TX mode) moves to the next FIFO entry.

The read value from these bits is always 0x00.

Only write to these bits when the CANFD module is in GL_HALT or GL_OPERATION mode.

Only write 0xFF to this register when:

- The Common FIFO buffer is enabled and is not empty if configured in RX mode
- The Common FIFO buffer is enabled and is not full if configured in TX mode

Do not write to the Common FIFO Pointer Control registers when DMA is enabled.

42.2.31 CFDFESTS : FIFO Empty Status Register

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x0060

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	CFEMP	—	—	—	—	—	—	RFXEMP[1:0]	
Value after reset:	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	1

Bit	Symbol	Function	R/W
1:0	RFXEMP[1:0]	RX FIFO Empty Status 0: Corresponding FIFO not empty 1: Corresponding FIFO empty	R
7:2	—	These bits are read as 0.	R
8	CFEMP	Common FIFO Empty Status 0: Corresponding FIFO not empty 1: Corresponding FIFO empty	R
31:9	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

The FIFO Empty Status register shows status of the empty bits of the FIFO buffers.

RFXEMP[1:0] bits (RX FIFO Empty Status)

The RFXEMP[1:0] bits are set when the CANFD module is in GL_RESET mode.

Each bit is set automatically when the corresponding bit is set in the RX FIFO Status Registers.

Each bit is cleared automatically when the corresponding bit is cleared in the RX FIFO Status Registers.

CFEMP bit (Common FIFO Empty Status)

The CFEMP bits are set when the CANFD module is in GL_RESET mode.

Each bit is set automatically when the corresponding bit is set in the Common FIFO Status Registers.

Each bit is cleared automatically when the corresponding bit is cleared in the Common FIFO Status Registers.

42.2.32 CFDFFSTS : FIFO Full Status Register

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x0064

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	CFFLL	—	—	—	—	—	—	RFXFLL[1:0]	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
1:0	RFXFLL[1:0]	RX FIFO Full Status 0: Corresponding FIFO not full 1: Corresponding FIFO full	R
7:2	—	These bits are read as 0.	R
8	CFFLL	Common FIFO Full Status 0: Corresponding FIFO not full 1: Corresponding FIFO full	R
31:9	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

The FIFO Full Status Register shows status of the full bits of the FIFO buffers.

RFXFLL[1:0] bits (RX FIFO Full Status)

The RFXFLL[1:0] bits are cleared when CANFD module is in GL_RESET mode.

Each bit is set automatically when the corresponding bit is set in the RX FIFO Status Registers.

Each bit is cleared automatically when the corresponding bit is cleared in the RX FIFO Status Registers.

CFFLL bits (Common FIFO Full Status)

The CFFLL bits are cleared when the CANFD module is in GL_RESET mode.

Each bit is set automatically when the corresponding bit is set in the Common FIFO Status Registers.

Each bit is cleared automatically when the corresponding bit is cleared in the Common FIFO Status Registers.

42.2.33 CFDFMSTS : FIFO Message Lost Status Register

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x0068

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	CFMLT	—	—	—	—	—	—	RFXMLT[1:0]	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
1:0	RFXMLT[1:0]	RX FIFO Message Lost Status 0: Corresponding FIFO Message Lost flag not set 1: Corresponding FIFO Message Lost flag set	R
7:2	—	These bits are read as 0.	R
8	CFMLT	Common FIFO Message Lost Status 0: Corresponding FIFO Message Lost flag not set 1: Corresponding FIFO Message Lost flag set	R
31:9	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

The FIFO Message Lost Status Register shows status of the Msg Lost bits of the FIFO buffers.

RFXMLT[1:0] bits (RX FIFO Message Lost Status)

The RFXMLT[1:0] bits are cleared when the CANFD module is in GL_RESET mode.

Each bit is set automatically when the corresponding bit is set in the RX FIFO Status Registers.

Each bit is cleared automatically when the corresponding bit is cleared in the RX FIFO Status Registers.

CFMLT bits (Common FIFO Message Lost Status)

The CFMLT bits are cleared when the CANFD module is in GL_RESET mode.

Each bit is set automatically when the corresponding bit is set in the Common FIFO Status Registers.

Each bit is cleared automatically when the corresponding bit is cleared in the Common FIFO Status Registers.

42.2.34 CFDRFISTS : RX FIFO Interrupt Flag Status Register

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x006C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	RFXIF[1:0]	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
1:0	RFXIF[1:0]	RX FIFO[x] Interrupt Flag Status 0: Corresponding RX FIFO Interrupt flag not set 1: Corresponding RX FIFO Interrupt flag set	R
31:2	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

The FIFO Interrupt Flag Status Register shows status of the interrupt flag bits of the RX FIFO buffers.

RFXIF[1:0] bits (RX FIFO[x] Interrupt Flag Status)

Each bit is set automatically when the corresponding interrupt flag bit is set in the RX FIFO Status Registers.

The RFXIF[1:0] bits are cleared when the CANFD module is in GL_RESET mode.

Each bit is cleared automatically when the corresponding interrupt flag bit is cleared in the RX FIFO Status Registers.

42.2.35 CFDCDTCT : DMA Transfer Control Register

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x00C8

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	CFDMAE	—	—	—	—	—	—	RFDMAE1	RFDMAE0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	RFDMAE0	DMA Transfer Enable for RXFIFO 0 0: DMA transfer request disabled 1: DMA transfer request enabled	R/W
1	RFDMAE1	DMA Transfer Enable for RXFIFO 1 0: DMA transfer request disabled 1: DMA transfer request enabled	R/W
7:2	—	These bits are read as 0. The write value should be 0.	R/W
8	CFDMAE	DMA Transfer Enable for Common FIFO 0 0: DMA transfer request disabled 1: DMA transfer request enabled	R/W
31:9	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

The DMA Transfer Control Register controls the start and stop of DMA transfer operation.

RFDMAEe (e = 0 to 1) bit (DMA Transfer Enable for RXFIFO e)

The RFDMAEe bit cannot be set in GL_SLEEP or GL_RESET mode.

This bit is cleared when the CANFD module is in GL_RESET mode.

CFDMAE bit (DMA Transfer Enable for Common FIFO)

The CFDMAE bit enables or disables DMA transfer request for common FIFO

The CFDMAE bit cannot be set in GL_SLEEP or GL_RESET mode.

Do not enable a DMA transfer for a Common FIFO that is configured as TX FIFO.

This bit is cleared when the CANFD module is in GL_RESET mode.

42.2.36 CFDCDTSTS : DMA Transfer Status Register

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x00CC

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	CFDMASTS	—	—	—	—	—	—	RFDMASTS1	RFDMASTS0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	RFDMASTS0	DMA Transfer Status for RX FIFO 0 0: DMA transfer stopped 1: DMA transfer on going	R
1	RFDMASTS1	DMA Transfer Status for RX FIFO 1 0: DMA transfer stopped 1: DMA transfer on going	R
7:2	—	These bits are read as 0.	R
8	CFDMASTS	DMA Transfer Status only for Common FIFO 0: DMA transfer stopped 1: DMA transfer on going	R
31:9	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

The DMA Transfer Status Register shows the status of the DMA transfer.

RFDMASTSe (e = 0 to 1) bit (DMA Transfer Status for RX FIFO e)

Each bit is set automatically when the corresponding DMA enable bit is set and the corresponding DMA FIFO is not empty.

Each bit is cleared automatically when the DMA transfer stops either because the DMA is disabled or the DMA FIFO is empty.

When CFDCDTCT.RFDMAEe (see CFDCDTCT.RFDMAEe bit in section 42.2.35. CFDCDTCT : DMA Transfer Control Register) is set to 0 while DMA transfer for the corresponding FIFO is on going, the RFDMASTSe bit becomes 0 when the DMA transfer is complete.

This bit is cleared when the CANFD module is in GL_RESET mode.

CFDMASTS bit (DMA Transfer Status only for Common FIFO)

Each bit is set automatically when the corresponding DMA enable bit is set and the corresponding DMA FIFO is not empty.

Each bit is cleared automatically when the DMA transfer stops either because the DMA is disabled or the DMA FIFO is empty.

When CFDCDTCT.CFDMAE (see CFDCDTCT.CFDMAE bit in section 42.2.35. CFDCDTCT : DMA Transfer Control Register) is set to 0 while DMA transfer for the corresponding FIFO is on going, the CFDMASTS bit becomes 0 when the DMA transfer is complete.

This bit is cleared when the CANFD module is in GL_RESET mode.

42.2.37 CFDTMCi : TX Message Buffer Control Registers i (i = 0 to 3)

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x0070 + 0x01 × i

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	TMOM	TMTAR	TMTR
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	TMTR	TX Message Buffer Transmission Request 0: TX Message buffer transmission not requested 1: TX message buffer transmission requested	R/W
1	TMTAR	TX Message Buffer Transmission Abort Request 0: TX message buffer transmission request abort not requested 1: TX message buffer transmission request abort requested	R/W
2	TMOM	TX Message Buffer One-shot Mode 0: TX message buffer not configured in one-shot mode 1: TX message buffer configured in one-shot mode	R/W
7:3	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

The TX Message Buffer Control Registers configure the TX message buffer functions.

TMTR bit (TX Message Buffer Transmission Request)

When the TMTR bit is set, the CANFD module logic tries to transmit the message stored in the corresponding message buffer.

Only write to this bit when the related CANFD module is in CH_HALT or CH_OPERATION mode.

Do not set this bit if the corresponding TX message buffer is linked to a COM FIFO in TX mode or is a part of TX Queue.

This bit cannot be directly cleared by a CPU write access.

This bit can only be set when the Transmission Result flag bits (CFDTMSTSj.TMTRF) in the CFDTMSTSj register corresponding to the message buffer are cleared to 00b.

The TMTR bit is automatically cleared by the:

- CANFD module logic at the end of a successful transmission
- CANFD module logic at the end of a transmission abort, requested by the corresponding CFDTMCi.TMTAR bit
- CANFD module logic when there is a detection of a CAN bus error or arbitration loss if CFDTMCi.TMOM bit is set for the message buffer
- CANFD module logic when the CANFD module is in GL_RESET mode or the related channel is in CH_RESET mode.

TMTAR bit (TX Message Buffer Transmission Abort Request)

When the TMTAR bit is set, the CANFD module logic tries to abort the transmission of the frame stored in the corresponding message buffer.

In most cases, transmission cannot be aborted if the internal scan for transmission is complete and the message buffer has already been selected for transmission. In this case, frame may be transmitted successfully from the message buffer. The message buffer selection is released by entering CH_HALT mode.

However, message buffer selected for transmission can be aborted by an abort request when the CAN node detects a new message on the bus (RX pin) before it starts transmission from the selected message buffer.

Only write to the TMTAR bit when the related CANFD channel is in CH_HALT or CH_OPERATION mode. This bit can only be set when the related transmit request TMTR bit is set.

The TMTAR bit cannot be cleared by a CPU write access. Clearing of this bit by CANFD has priority over setting by a CPU write access.

The TMTAR bit is automatically cleared by:

- The CANFD module logic at the end of a successful transmission
- The CANFD module logic at the end of a transmission abort
- The CANFD module logic when there is detection of a CAN bus error or arbitration loss
- The CANFD module logic when the CANFD module is in GL_RESET mode or the related channel enters CH_RESET mode.

TMOM bit (TX Message Buffer One-shot Mode)

When the TMOM bit is set, the CANFD module logic tries to transmit the message only once.

If the transmission is successful, the CFDTMSTSj.TMTRF bits are set to 10b or 11b. Otherwise, the transmission is automatically aborted and CFDTMSTSj.TMTRF bits are set to 01b due to a bus error or a bus arbitration lost.

The TMOM bit remains set if the transmission has completed successfully or aborted due to an error or a loss of arbitration.

Only write to this bit when the related CANFD channel is in CH_HALT or CH_OPERATION mode.

Set this bit at the same time as the TMTR bit. Clear this bit with a write access.

If a message has already been requested for transmission, do not write to this bit until the message has been successfully transmitted or transmission has been aborted.

The TMOM bit is automatically cleared by the CANFD module logic when the CANFD module is in GL_RESET mode or the related channel is in CH_RESET mode.

42.2.38 CFDTMSTSj : TX Message Buffer Status Registers j (j = 0 to 3)

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x0074 + 0x01 × j

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	TMTARM	TMTRM	TMTRF[1:0]		TMTSTS
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	TMTSTS	TX Message Buffer Transmission Status 0: No ongoing transmission 1: Ongoing transmission	R
2:1	TMTRF[1:0]	TX Message Buffer Transmission Result Flag 0 0: No result 0 1: Transmission aborted from the TX message buffer 1 0: Transmission successful from the TX message buffer and transmission abort was not requested 1 1: Transmission successful from the TX message buffer and transmission abort was requested	R/W
3	TMTRM	TX Message Buffer Transmission Request Mirrored 0: TX message buffer transmission not requested 1: TX message buffer transmission requested	R
4	TMTARM	TX Message Buffer Transmission Abort Request Mirrored 0: TX message buffer transmission request abort not requested 1: TX message buffer transmission request abort requested	R
7:5	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

The TX Message Buffer Status Registers show status of the transmission and transmission abort for the corresponding message buffers.

TMTSTS bit (TX Message Buffer Transmission Status)

The TMTSTS bit is set automatically at the start of the transmission from the corresponding TX message buffer.

This bit is cleared automatically when:

- Transmission stops
- The CANFD module is in GL_RESET mode
- The related CANFD channel is in CH_RESET mode.

TMTRF[1:0] bits (TX Message Buffer Transmission Result Flag)

The TMTRF[1:0] bits show the result for the corresponding TX message buffer. The status is as follows:

- 00: Transmission in progress or has not been requested
- 01: Transmission has been aborted from the corresponding TX message buffer
- 10: Transmission was successful from the corresponding TX message buffer and the CFDTMCi.TMTAR bit was not set for this TX message buffer
- 11: Transmission was successful from the corresponding TX message buffer, but the CFDTMCi.TMTAR bit was set for this TX message buffer.

Only write to these bits when the related CANFD channel is in CH_HALT or CH_OPERATION mode.

The TMTRF[1:0] bits are cleared automatically when the CANFD module is in GL_RESET mode or the related channel is in CH_RESET mode.

If the set from the CAN channel occurs simultaneously with the clear by the write access, then the bit is set.

TMTRM bit (TX Message Buffer Transmission Request Mirrored)

The TMTRM bit is set when the CFDTMCi.TMTR bit in the corresponding CFDTMCi register is set.

This bit is cleared when the CFDTMCi.TMTR bit in the corresponding CFDTMCi register is cleared.

TMTARM bit (TX Message Buffer Transmission Abort Request Mirrored)

The TMTARM bit is set when the CFDTMCi.TMTAR bit in the corresponding CFDTMCi register is set.

This bit is cleared when the CFDTMCi.TMTAR bit in the corresponding CFDTMCi register is cleared.

42.2.39 CFDTMTRSTS : TX Message Buffer Transmission Request Status Register

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x0078

Bit position:	31																												3			0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	CFDTMTRSTS[3:0]			
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
3:0	CFDTMTRSTS[3:0]	TX Message Buffer Transmission Request Status 0: Transmission not requested for corresponding TX message buffer 1: Transmission requested for corresponding TX message buffer	R
31:4	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

These bits show the TX Message Buffer Transmission Request Status for the corresponding TX Message Buffer. The bit 0 of a CFDTMTRSTS register corresponds to the TX message buffer 0.

The bit position of CFDTMTRSTS corresponds to the buffer number of TX message buffer.

CFDTMTRSTS[3:0] bits (TX Message Buffer Transmission Request Status)

The CFDTMTRSTS[3:0] bits show status of the CFDTMCi.TMTR bits of the TX Message Buffer Control Registers.

Each bit is set automatically when the corresponding bit is set in the TX Message Buffer Control Registers (CFDTMCi), and only when the message buffer does not belong to a TX Queue.

Each bit is cleared automatically when:

- The corresponding bit is cleared in the TX Message Buffer Control Registers
- The CANFD module is in GL_RESET mode
- The related CANFD channel is in CH_RESET mode.

42.2.40 CFDTMTARSTS : TX Message Buffer Transmission Abort Request Status Register

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x007C

Bit position:	31																												3			0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	CFDTMTARSTS[3:0]			
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
3:0	CFDTMTARSTS[3:0]	TX Message Buffer Transmission Abort Request Status 0: Transmission abort not requested for corresponding TX message buffer 1: Transmission abort requested for corresponding TX message buffer	R
31:4	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

These bits show the TX Message Buffer Transmission Abort Request Status for the corresponding TX Message Buffer. The bit 0 of a CFDTMTARSTS register corresponds to the TX message buffer 0.

The bit position of CFDTMTARSTS corresponds to the buffer number of TX message buffer.

CFDTMTARSTS[3:0] bits (TX Message Buffer Transmission Abort Request Status)

The CFDTMTARSTS[3:0] bits show status of the CFDTMCi.TMTAR bits of the TX Message Buffer Control Registers.

Each bit is set automatically when the corresponding bit is set in the TX Message Buffer Control Registers, and when the message buffer belongs to a TX Queue.

Each bit is cleared automatically when:

- The corresponding bit is cleared in the TX Message Buffer Control Registers
- The CANFD module is in GL_RESET mode
- The related CANFD channel is in CH_RESET mode.

42.2.41 CFDTMTCSTS : TX Message Buffer Transmission Completion Status Register

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x0080

Bit position:	31																												3			0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	CFDTMTCSTS[3:0]			
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
3:0	CFDTMTCSTS[3:0]	TX Message Buffer Transmission Completion Status 0: Transmission not complete for corresponding TX message buffer 1: Transmission completed for corresponding TX message buffer	R
31:4	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

These bits show the TX Message Buffer Transmission Completion Status for the corresponding TX Message Buffer. The bit 0 of a CFDTMTCSTS register corresponds to the TX message buffer 0.

The bit position of CFDTMTCSTS corresponds to the buffer number of TX message buffer.

CFDTMTCSTS[3:0] bits (TX Message Buffer Transmission Completion Status)

The CFDTMTCSTS[3:0] bits show status of successful completion of the TX Message Buffer Status Registers.

Each bit is set automatically when the corresponding bit is set in the TX Message Buffer Status Registers.

Each bit is cleared automatically when:

- The corresponding bit is cleared in the TX Message Buffer Status Registers
- The CANFD module is in GL_RESET mode
- The related CANFD channel is in CH_RESET mode.

If a CAN channel enters CH_RESET mode, then the bits related to that channel are cleared.

42.2.42 CFDTMTASTS : TX Message Buffer Transmission Abort Status Register

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x0084

Bit position:	31																												3			0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	CFDTMTASTS[3:0]			
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
3:0	CFDTMTASTS[3:0]	TX Message Buffer Transmission Abort Status 0: Transmission not aborted for corresponding TX message buffer 1: Transmission aborted for corresponding TX message buffer	R
31:4	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

These bits show the TX Message Buffer Transmission abort Status for the corresponding TX Message Buffer. The bit 0 of a CFDTMTASTS register corresponds to the TX message buffer 0.

The bit position of CFDTMTASTS corresponds to the buffer number of TX message buffer.

CFDTMTASTS[3:0] bits (TX Message Buffer Transmission Abort Status)

The CFDTMTASTS[3:0] bits show status of the successful transmission abort of the corresponding TX message buffer.

Each bit is set automatically when the CFDTMSTSj.TMTRF bits are set to 01b in the corresponding TX Message Buffer Status Register.

Each bit is cleared automatically when:

- The CFDTMSTSj.TMTRF bits are cleared in the corresponding TX Message Buffer Status Register
- The CANFD module is in GL_RESET mode
- The related CANFD channel is in CH_RESET mode.

42.2.43 CFDTMIEC : TX Message Buffer Interrupt Enable Configuration Register

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x0088

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	TMIEg[3:0]			
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
3:0	TMIEg[3:0]	TX Message Buffer Interrupt Enable 0: TX message buffer interrupt disabled for corresponding TX message buffer 1: TX message buffer interrupt enabled for corresponding TX message buffer	R/W
31:4	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

These bits show the TX Message Buffer Interrupt Enable for the corresponding TX Message Buffer.

The bit 0 of a CFDTMIEC register corresponds to the TX message buffer 0.

The bit position of CFDTMIEC corresponds to the buffer number of TX message buffer.

g = [0…3]

TMIEg[3:0] bits (TX Message Buffer Interrupt Enable)

If the TMIEg[3:0] bits are set, an interrupt is generated at the end of a successful transmission from the corresponding message buffer.

See section 42.7. Interrupts and DMA for TX Message Buffer Interrupt specification.

Do not write to the TMIEg[3:0] bits when:

- The CANFD module is in GL_SLEEP mode
- The related CANFD channel is in CH_SLEEP mode
- The corresponding TX message buffer is part of a TX Queue
- The corresponding TX message buffer is linked to a Common FIFO with the CFDCFCC.CFTML bits.

42.2.44 CFDTXQCC : TX Queue Configuration/Control Register

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x008C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	TXQDC[1:0]		TXQIM	—	TXQTXIE	—	—	—	—	TXQE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	TXQE	TX Queue Enable 0: TX Queue disabled 1: TX Queue enabled	R/W
4:1	—	These bits are read as 0. The write value should be 0.	R/W
5	TXQTXIE	TX Queue TX Interrupt Enable 0: TX Queue TX interrupt disabled 1: TX Queue TX interrupt enabled	R/W
6	—	This bit is read as 0. The write value should be 0.	R/W
7	TXQIM	TX Queue Interrupt Mode 0: When the last message is successfully transmitted 1: At every successful transmission	R/W
9:8	TXQDC[1:0]	TX Queue Depth Configuration 00: 0 messages 01: Reserved 10: 3 messages 11: 4 messages	R/W
31:10	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

The TX Queue Configuration/Control Registers are used to configure the TX Queue transmission.

TXQ is composed of TXMB0 to TXMB3 (at the maximum) when TXQE is enabled.

TXQE bit (TX Queue Enable)

The TXQE bit cannot be set if the configured TX Queue depth is 0x00 (CFDTXQCC.TXQDC == 0x00).

You cannot write to this bit when the CANFD module is in GL_SLEEP mode.

Do not write to this bit when the related CANFD channel is in CH_RESET or CH_SLEEP mode.

The TXQE bit is cleared automatically when the related CANFD channel is in CH_RESET mode.

TXQTXIE bit (TX Queue TX Interrupt Enable)

When the TXQTXIE bit is set, an interrupt is generated based on the setting of the TXQIM bit.

You cannot write to this bit when the CANFD module is in GL_SLEEP mode.

Do not write to this bit when the related CANFD channel is in CH_SLEEP mode.

TXQIM bit (TX Queue Interrupt Mode)

The TXQIM bit selects the interrupt generation condition for the TX Queue.

You cannot write to this bit when the CANFD module is in GL_SLEEP mode.

Do not write to this bit when the related CANFD channel is in any of the following modes:

- CH_SLEEP
- CH_HALT
- CH_OPERATION.

TXQDC[1:0] bits (TX Queue Depth Configuration)

The TXQDC[1:0] bits select the depth of the transmission queue. The message buffer selection starts from MB[0] up to MB[3] depending on the configured depth.

You cannot write to this bit when the CANFD module is in GL_SLEEP mode.

Do not write to this bit when the related CANFD channel is in any of the following modes:

- CH_SLEEP
- CH_HALT
- CH_OPERATION.

42.2.45 CFDTXQSTS : TX Queue Status Register

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x0090

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	TXQMC[2:0]			—	—	—	—	—	TXQTXIF	TXQFLL	TXQEMP
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1

Bit	Symbol	Function	R/W
0	TXQEMP	TX Queue Empty 0: TX Queue not empty 1: TX Queue empty	R
1	TXQFLL	TX Queue Full 0: TX Queue not full 1: TX Queue full	R
2	TXQTXIF	TX Queue TX Interrupt Flag 0: TX Queue interrupt condition not satisfied after a frame TX 1: TX Queue interrupt condition satisfied after a frame TX	R/W
7:3	—	These bits are read as 0. The write value should be 0.	R/W
10:8	TXQMC[2:0]	TX Queue Message Count Number of messages in the TX Queue.	R
31:11	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

The TX Queue Status Registers show the status of the TX Queue of corresponding CAN channel.

TXQEMP bit (TX Queue Empty)

The TXQEMP bit is set automatically when the TX Queue is disabled or no messages are stored in the TX Queue.

This bit is set automatically when:

- The last message is transmitted from the TX Queue
- The related CANFD channel is in CH_RESET mode.

The bit is cleared automatically when the first message to be transmitted is stored in the TX Queue.

TXQFLL bit (TX Queue Full)

The TXQFLL bit is set automatically when the number of CAN messages stored in the TX Queue matches the configured TX Queue depth.

This bit is cleared automatically when:

- The number of CAN messages stored in the TX Queue is less than the configured TX Queue depth
- The related CANFD channel is in CH_RESET mode.

TXQTXIF bit (TX Queue TX Interrupt Flag)

The TXQTXIF bit is not cleared automatically if the TX Queue is disabled.

When stopping the TX Queue, this bit should be cleared, after disabling TXQE and checking an empty state of TX Queue.

Do not use the bit clear instruction to clear this bit. Use the MOV instruction to ensure that only the specified bit is cleared. Other bits remain 1. Writing 1 has no effect.

This bit is set automatically when the configured interrupt condition is satisfied for the TX Queue.

If the set from the CAN channel occurs simultaneously with the clear by the write access, then the bit is set.

You cannot write to this bit when the related CANFD channel is in CH_SLEEP or CH_RESET mode.

The bit is cleared:

- By writing 0 to it
- When the related CANFD channel is in CH_RESET mode.

TXQMC[2:0][13:8] bits (TX Queue Message Count)

The TXQMC[2:0] bits show the number of CAN messages in the TX Queue.

These bits are cleared automatically when the related CANFD channel is in CH_RESET mode.

42.2.46 CFDTXQPCTR : TX Queue Pointer Control Register

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x0094

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	TXQPC[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	TXQPC[7:0]	TX Queue Pointer Control Increments the write pointer to the TX Queue buffer in the corresponding channel	W
31:8	—	The write value should be 0.	W

Note: S-TYPE-3, P-TYPE-3

The TX Queue Pointer Control Registers are used to confirm storage of a full message in the corresponding TX Queue buffers.

TXQPC[7:0] bits (TX Queue Pointer Control)

When the value 0xFF is written to the TXQPC[7:0] bits, the write pointer of the corresponding TX Queue buffer is updated and a transmit request is initiated for this message.

The read value from these bits is always 0x00. Do not write to the FIFO control registers when DMA is enabled.

You cannot write to these bits when the related CANFD channel is in CH_SLEEP or CH_RESET mode.

Only write 0xFF to this register when:

- The corresponding TX Queue is enabled and not full

- The Common FIFO is enabled.

42.2.47 CFDTHLCC : TX History List Configuration/Control Register

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x0098

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	THLDTE	THLIM	THLIE	—	—	—	—	—	—	—	THLE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	THLE	TX History List Enable 0: TX History List disabled 1: TX History List enabled	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W
8	THLIE	TX History List Interrupt Enable 0: TX History List Interrupt disabled 1: TX History List Interrupt enabled	R/W
9	THLIM	TX History List Interrupt Mode 0: Interrupt generated if TX History List level reaches ¾ of the TX History List depth 1: Interrupt generated for every successfully stored entry	R/W
10	THLDTE	TX History List Dedicated TX Enable 0: TX FIFO + TX Queue 1: Flat TX MB + TX FIFO + TX Queue	R/W
31:11	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

The TX History List Configuration/Control Register configures the TX History List functions.

THLE bit (TX History List Enable)

The THLE bit enables the TX History List buffer when it is set.

You cannot write to this bit when the related CANFD channel is in CH_RESET or CH_SLEEP mode.

This bit is cleared automatically when the related CANFD channel is in CH_RESET mode.

THLIE bit (TX History List Interrupt Enable)

The THLIE bit enables the generation of the TX History List interrupt when it is set.

You cannot write to this bit when the CANFD module is in GL_SLEEP mode.

THLIM bit (TX History List Interrupt Mode)

The THLIM bit selects the interrupt generation condition for the FIFO.

You cannot write to this bit when the CANFD module is in GL_SLEEP mode.

Do not write to this bit when the CANFD module is in GL_HALT or GL_OPERATION mode.

THLDTE bit (TX History List Dedicated TX Enable)

The THLDTE bit selects the condition for storing an entry in the TX History List after successful transmission.

You cannot write to this bit when the CANFD module is in GL_SLEEP mode.

Do not write to this bit when the CANFD module is in GL_HALT or GL_OPERATION mode.

42.2.48 CFDTHLSTS : TX History List Status Register

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x009C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	THLMC[3:0]				—	—	—	—	THLIF	THLELT	THLFLL	THLEMP
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1

Bit	Symbol	Function	R/W
0	THLEMP	TX History List Empty 0: TX History List not empty 1: TX History List empty	R
1	THLFLL	TX History List Full 0: TX History List not full 1: TX History List full	R
2	THLELT	TX History List Entry Lost 0: No entry lost in TX History List 1: TX History List entry Lost	R/W
3	THLIF	TX History List Interrupt Flag 0: TX History List interrupt condition not satisfied 1: TX History List interrupt condition satisfied	R/W
7:4	—	These bits are read as 0. The write value should be 0.	R/W
11:8	THLMC[3:0]	TX History List Message Count Number of messages stored in TX History List	R
31:12	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

The TX History List Status register shows the status of data stored in the TX History List buffer.

THLEMP bit (TX History List Empty)

The THLEMP bit is set automatically when the CPU has read all the entries from the TX History List buffer.

This bit is cleared automatically when the first entry is stored to the TX History List.

This bit is set automatically when:

- TX History List is disabled
- The related CANFD channel is in CH_RESET mode.

THLFLL bit (TX History List Full)

The THLFLL bit is set automatically when the number of entries in the TX History List buffer matches the TX History List depth.

Each TX History List can store up to 8 entries.

This bit is cleared automatically when:

- The number of entries in the TX History List buffer is less than the TX History List depth
- The TX History List is disabled
- The related CANFD channel is in CH_RESET mode.

THLELT bit (TX History List Entry Lost)

The THLELT bit is set when a new entry cannot be stored because the related TX History List buffer is already full.

Only write to this bit when the related CANFD channel is in CH_HALT or CH_OPERATION mode. Writing 1 has no effect.

Do not use the bit clear instruction to clear this bit. Use the MOV instruction to ensure that only the specified bit is cleared. Other bits remain 1.

If the set from the CAN channel occurs simultaneously with the clear by the write access, then the bit is set.

This bit is cleared:

- By writing 0 to it
- When the related CANFD channel is in CH_RESET mode.

THLIF bit (TX History List Interrupt Flag)

The THLIF bit is set when the configured interrupt condition is satisfied.

Only write to this bit when the related CANFD channel is in CH_HALT or CH_OPERATION mode. Writing 1 has no effect.

Do not use the bit clear instruction to clear this bit. Use the MOV instruction to ensure that only the specified bit is cleared. Other bits remain 1.

If the set from the CAN channel occurs simultaneously with the clear by the write access, then the bit is set.

This bit is cleared:

- By writing 0 to it
- When the related CANFD channel is in CH_RESET mode.

The bit is cleared by writing 0 to it.

This bit is automatically cleared in CH_RESET mode.

THLMC[3:0] bits (TX History List Message Count)

The THLMC[3:0] bits show the number of transmitted messages stored in the TX History List.

These bits are cleared automatically when the related CANFD channel is in CH_RESET mode.

42.2.49 CFDTHLACC0 : TX History List Access Register 0

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x0740

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	TMTS[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	BN[1:0]		BT[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
2:0	BT[2:0]	Buffer Type 0 0 1: Flat TX message buffer 0 1 0: TX FIFO message buffer number 1 0 0: TX Queue message buffer number	R
4:3	BN[1:0]	Buffer Number Number of the message buffer	R

Bit	Symbol	Function	R/W
15:5	—	These bits are read as 0.	R
31:16	TMTS[15:0]	Transmit Timestamp Transmit timestamp value for software drivers	R

Note: S-TYPE-3, P-TYPE-3

The TX History List Access Registers 0 provide access to the entry in the TX History List based on the read timestamp value.

BT[2:0] bits (Buffer Type)

The BT[2:0] bits indicate whether data has been stored following a transmission from a FIFO buffer, a TX Queue or a TX message buffer.

BN[1:0] bits (Buffer Number)

The BN[1:0] bits show the message buffer from which transmission was successfully completed. If a message from a Common FIFO is transmitted, then these bits show the message buffer that is linked to the Common FIFO for transmission.

TMTS[15:0] bits (Transmit Timestamp)

The TMTS[15:0] bits indicate the timestamp for use by software drivers.

42.2.50 CFDTHLACC1 : TX History List Access Register 1

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x0744

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	TIFL[1:0]	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	TID[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	TID[15:0]	Transmit ID These bits indicate that message buffer reference ID, TX FIFO reference ID, or AFL pointer field is stored for software drivers.	R
17:16	TIFL[1:0]	Transmit Information Label These bits indicate that message buffer information label, TX FIFO information label, or AFL information label is stored for software drivers.	R
31:18	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

The TX History List Access Registers 1 provide access to entry in the TX History List based on the read pointer value.

TID[15:0] bits (Transmit ID)

The TID[15:0] bits indicate whether the message buffer reference ID (CFDTMFDCTRb.TMPTR) or the TX FIFO reference ID (CFDCFFDCSTS.CFPTR) is for use by software drivers.

TIFL[1:0] bits (Transmit Information Label)

The TIFL[1:0] bits indicate whether the message buffer information label (CFDTMFDCTRb.TMIFL) or the TX FIFO information label (CFDCFFDCSTS.CFIFL) is for use by software drivers.

42.2.51 CFDTHLPCTR : TX History List Pointer Control Register

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x00A0

Bit position:	31																								7							0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	THLPC[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	THLPC[7:0]	TX History List Pointer Control Increments the write pointer to the TX History List in the corresponding channel	W
31:8	—	The write value should be 0.	W

Note: S-TYPE-3, P-TYPE-3

The TX History List Pointer Control Registers are used to increment the read pointer of the TX History List.

THLPC[7:0] bits (TX History List Pointer Control)

When 0xFF is written to the THLPC[7:0] bits, the read pointer of the TX History List is moved to the next TX History List entry address.

The read value from these bits is always 0x00. Only write to these bits when the related CANFD channel is in CH_HALT or CH_OPERATION mode.

Only write 0xFF to these registers when the corresponding TX History List is enabled and not empty.

42.2.52 CFDGRSTC : Global SW reset Register

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x00D8

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	KEY[7:0]								—	—	—	—	—	—	—	SRST
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	SRST	SW Reset 0: Normal state 1: SW reset state	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W
15:8	KEY[7:0]	Key Code These bits control the validity of rewriting of a SRST bit.	W
31:16	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

SRST bit (SW Reset)

When the SRST bit is set, the CANFD module is in the same state as hardware reset. When a reset is required, write 1 then write 0 to this bit.

This bit is cleared when the CANFD module is in GL_SLEEP mode.

When this bit is cleared, the RAM initialization sequence does not operate. The configuration of RAM is performed by software.

The RAM is not initialized when software reset is performed during the initialization of RAM. Software must perform the initialization of RAM.

KEY[7:0] bits (Key Code)

When 0xC4 is written in the KEY[15:8] bits, a write to the SRST bit is valid.

The read value from these bits is always 0x00.

CFDGRSTC.SRST bit and the CFDGRSTC.KEY bit should be written simultaneously.

42.2.53 CFDGTSTCFG : Global Test Configuration Register

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x00A8

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	RTMPS[3:0]			
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	—	These bits are read as 0. The write value should be 0.	R/W
19:16	RTMPS[3:0]	RAM Test Mode Page Select Select a RAM test mode page	R/W
31:20	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

The Global Test Configuration Register is used to configure the RAM test mode page.

RTMPS[3:0] bits (RAM Test Mode Page Select)

The RTMPS[3:0] bits select the RAM page mode for CPU read/write access when the CANFD module is configured in RAM test mode.

See section 42.9.2.1. RAM Test Mode for the RAM test mode specification.

Do not write to these bits when the CANFD module is in GL_RESET or GL_SLEEP mode.

Only enter values from 0 to 8 (0x008) for the message buffer RAM.

Only write to these bits when the CANFD module is in GL_HALT mode.

These bits are cleared automatically when the related CANFD channel is in GL_RESET mode.

42.2.54 CFDGTSTCTR : Global Test Control Register

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x00AC

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	RTME	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
1:0	—	These bits are read as 0. The write value should be 0.	R/W
2	RTME	RAM Test Mode Enable 0: RAM test mode disabled 1: RAM test mode enabled	R/W
31:3	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

The Global Test Control register is used to control the global test modes of the CANFD module.

RTME bit (RAM Test Mode Enable)

When the RTME bit is set, the CANFD module is configured in RAM test mode. See section 42.9.2.1. RAM Test Mode for RAM test mode specification.

Only write to this bit when the CANFD module is in GL_HALT mode.

Clear this bit when the CANFD module is in GL_HALT mode.

This bit is cleared automatically when the CANFD module is in GL_RESET mode.

42.2.55 CFDGFDCFG : Global FD Configuration Register

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x00B0

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	TSCCFG[1:0]		—	—	—	—	—	—	—	RPED
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	RPED	RES Bit Protocol Exception Disable 0: Protocol exception event detection enabled 1: Protocol exception event detection disabled	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W

Bit	Symbol	Function	R/W
9:8	TSCCFG[1:0]	Timestamp Capture Configuration 0 0: Timestamp capture at the sample point of SOF (start of frame) 0 1: Timestamp capture at frame valid indication 1 0: Timestamp capture at the sample point of RES bit 1 1: Reserved	R/W
31:10	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

RPED bit (RES Bit Protocol Exception Disable)

The RPED bit configures the protocol exception event handling according to ISO 11898-1.

When this bit is enabled, the protocol exception event detection is disabled, and the protocol controller transmits an error frame when the protocol exception event is detected (RES bit is sampled recessive).

Only write to this bit when the CANFD module is in GL_RESET mode.

TSCCFG[1:0] bits (Timestamp Capture Configuration)

The TSCCFG[1:0] bits configure the different capture points of the timestamp for transmission and reception.

When CFDGFDCFG.TSCCFG[1:0] = 10b, the timestamp capture is performed for CANFD frames at RES bit and for Classical frames at the start of frame.

Only write to these bits when the CANFD module is in GL_RESET mode.

42.2.56 CFDGLOCKK : Global Lock Key Register

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x00B8

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	LOCK[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	LOCK[15:0]	Lock Key Key bits for unlocking the protection of test modes	W
31:16	—	The write value should be 0.	W

Note: S-TYPE-3, P-TYPE-3

The Global Lock Key register is a write-only register that is used to unlock the protection for special test bits.

See section 42.9.2. Global Test Modes for Lock key specification.

LOCK[15:0] bits (Lock Key)

The unlock key sequence must be written in the LOCK[15:0] bits to configure the CANFD module in FIFO OTB disable and RAM test modes.

The read value from these bits is always 0x0000.

You cannot write to these bits when the CANFD module is in GL_SLEEP or GL_RESET mode.

Do not write to these bits when the CANFD module is in GL_OPERATION mode.

42.2.57 CFDRPGACCk : RAM Test Page Access Registers k (k = 0 to 63)

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x0280 + 0x0004 × k

Bit position:	31–0
Bit field:	RDTA[31:0]
Value after reset:	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
31:0	RDTA[31:0]	RAM Data Test Access RAM data bytes	R/W

Note: S-TYPE-3, P-TYPE-3

RDTA[31:0] bits (RAM Data Test Access)

Data can be read from or written into the RDTA[31:0] bits when the CANFD module is configured in RAM test mode.

Only write to this bit when the CANFD module is in GL_HALT mode and RAM test mode is enabled.

Software data should be read/written in the RAM Test Page Access registers during RAM test mode.

42.2.58 CFDGAFLIGNENT : Global AFL Ignore Entry Register

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x00C0

Bit position:	31–4	3–0
Bit field:	— —	IRN[3:0]
Value after reset:	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	0 0 0 0

Bit	Symbol	Function	R/W
3:0	IRN[3:0]	Ignore Rule Number Define rule number which ignores an AFL entry.	R/W
31:4	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

IRN[3:0] bits (Ignore Rule Number)

The IRN[3:0] bits define the rule number which updates an AFL entry.

Enter only values between 0 and 15 (0x0F) inclusive.

Only write to these bits when the CFDGAFLIGNCTR.IREN bit is 0.

You cannot write to these bits when the CANFD module is in GL_SLEEP mode.

42.2.59 CFDGAFLIGNCTR : Global AFL Ignore Control Register

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x00C4

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	KEY[7:0]								—	—	—	—	—	—	—	IREN
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	IREN	Ignore Rule Enable 0: AFL entry number is not ignored 1: AFL entry number is ignored	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W
15:8	KEY[7:0]	Key Code These bits control the validity of rewriting the IREN bit.	W
31:16	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

IREN bit (Ignore Rule Enable)

When the IREN bit is set, the entry number (selected by CFDGAFLIGNENT register) is ignored.

This bit is cleared automatically when the CANFD module is in GL_RESET mode.

KEY[7:0] bits (Key Code)

When 0xC4 is written in the KEY[7:0] bits, a write to the IREN bit is valid.

The read value from these bits is always 0x00.

CFDGAFLIGNCTR.IREN bit and the CFDGAFLIGNCTR.KEY bit should be written simultaneously

42.2.60 CFDRMIEC : RX Message Buffer Interrupt Enable Configuration Register

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x0038

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	RMIE[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	RMIE[15:0]	RX Message Buffer Interrupt Enable 0: RX Message Buffer Interrupt disabled for corresponding RX message buffer 1: RX Message Buffer Interrupt enabled for corresponding RX message buffer	R/W

Note: S-TYPE-3, P-TYPE-3

These bits show the RX Message Buffer Interrupt Enable for the corresponding RX Message Buffer. CFDRMIEC bit 0 corresponds to RX Message Buffer 0 and so on.

The bit position of CFDRMIEC corresponds to the buffer number of RXMB.

RMIE[15:0] bit (RX Message Buffer Interrupt Enable)

If this bit is set, then an interrupt will be generated at the end of a successful reception from the corresponding Message Buffer.

For details, see section 42.7.1. Interrupts.

Users cannot write to this bit when the CANFD module is in GL_SLEEP mode.

42.2.61 Message Buffer Component Structure

42.2.61.1 Start Addresses

The start address for each of the Message Buffer component is calculated using the number of related Message Buffer components.

The start addresses for each register in the Message Buffer component are depicted in Table 42.4.

Table 42.4 Message Buffer Component Register Start Addresses

b = Message buffer component index	MBCP	p	Register	Start Address
[0…15] b = [0…7]	RMBCPb[0]	x	RMID	0x0920 + b × 0x004C
		x	RMPTR	0x0924 + b × 0x004C
		x	RMFDSTSb	0x0928 + b × 0x004C
		[1…15]	RMDFbp	0x092C + b × 0x004C + p × 0x0004
[0…15] b = [8…15]	RMBCPb[0]	x	RMIDb	0x0D20 + (b-8) × 0x004C
		x	RMPTRb	0x0D24 + (b-8) × 0x004C
		x	RMFDSTSb	0x0D28 + (b-8) × 0x004C
		[1…15]	RMDFbp	0x0D2C + (b-8) × 0x004C + p × 0x0004
[0…1]	RFMBCPb[0]	x	RFIDb	0x0520 + b × 0x004C
		x	RFPTRb	0x0524 + b × 0x004C
		x	RFFDSTSb	0x0528 + b × 0x004C
		[1…15]	RFDFbp	0x052C + b × 0x004C + p × 0x0004
[0]	CFMBCPb[0]	x	CFID	0x05B8
		x	CFPTR0	0x05BC
		x	CFFDCSTS0	0x05C0
		[1…15]	CFDFp0	0x05C4 + p × 0x0004
[0…3]	TMBCPb[0]	x	TMIDb	0x0604 + b × 0x004C
		x	TMPTRb	0x0608 + b × 0x004C
		x	TMFDCTRb	0x060C + b × 0x004C
		[1…15]	TMDFbp	0x0610 + b × 0x004C + p × 0x0004

The message buffer configuration consists of four types of Message Buffer components:

- RX Message Buffer Component (CFDRMBCPb[0])
- RX FIFO Access Message Buffer Component (CFDRFMBCPb[0])
- Common FIFO Access Message Buffer Component (CFDCFMBCP0[0])
- TX Message Buffer Component (CFDTMBCPb[0]).

Where b = the Message Buffer component index that has a range that varies based on the type of Message Buffer component.

For a summary of this configuration, see Figure 42.28. For a detailed description of the number of and the different types of message buffers, see section 42.6. FIFO Buffers and Normal Message Buffer Configuration.

As described in section 42.2. Register Descriptions, each Message Buffer component consists of the following registers:

- Identifier (ID)
- Pointer (PTR)
- Data Field (DFp).

Where p = the Data Field register index that has a range that varies based on the type of message buffer component.

Rc is the Message Buffer Component register where c = Message Buffer Component register index that has a range that varies based on the type of Message Buffer component.

A description of the registers, their associated bits and their accessibility are shown below the summary and detailed figures of each component.

In each of the figures, a cell that contains '-' means reserved and has the same behavior as reserved bits for registers in section 42.2.61. Message Buffer Component Structure.

42.2.61.2 CFDRMBCPb[0] : RX Message Buffer Component b (b = 0 to 15)

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: See Table 42.4

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	Rc[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	Rc[15:0]	RX Message Buffer Component c Refer to Table 42.5, Table 42.6 and the descriptions that follow for a detailed description of each register and its related bits, contained within this message buffer component.	R/W

Note: S-TYPE-3, P-TYPE-3

Where the total number of CFDRMBCPb = 16 as shown in Figure 42.28 (c = RX Message Buffer Component Register index = [0…18])

Rc[15:0] bit (RX Message Buffer Component c)

The RX Message Buffer Component is made up of the following registers: CFDRMIDb, CFDRMPTRb, CFDRMFDSTSb, and CFDRMDFbp. Refer to Table 42.6 for details of how to interpret the structure of this buffer component and how to access the respective registers.

Table 42.5 RX Message Buffer Component Summary (1 of 2)

RX Message Buffer Component (RMBCP)	
Rc	**CANFD mode (CAN_FD_MODE = 1'b1)**
R0	RX Message Buffer (b) ID Registers
R1	RX Message Buffer (b) Pointer Registers
R2	RX Message Buffer (b) CANFD Status Registers
R3	RX Message Buffer (b) Data Field 0 Registers
R4	RX Message Buffer (b) Data Field 1 Registers
R5	RX Message Buffer (b) Data Field 2 Registers
R6	RX Message Buffer (b) Data Field 3 Registers

Table 42.5 RX Message Buffer Component Summary (2 of 2)

RX Message Buffer Component (RMBCP)	
Rc	**CANFD mode (CAN_FD_MODE = 1'b1)**
R7	RX Message Buffer (b) Data Field 4 Registers
R8	RX Message Buffer (b) Data Field 5 Registers
R9	RX Message Buffer (b) Data Field 6 Registers
R10	RX Message Buffer (b) Data Field 7 Registers
R11	RX Message Buffer (b) Data Field 8 Registers
R12	RX Message Buffer (b) Data Field 9 Registers
R13	RX Message Buffer (b) Data Field 10 Registers
R14	RX Message Buffer (b) Data Field 11 Registers
R15	RX Message Buffer (b) Data Field 12 Registers
R16	RX Message Buffer (b) Data Field 13 Registers
R17	RX Message Buffer (b) Data Field 14 Registers
R18	RX Message Buffer (b) Data Field 15 Registers
R[19…31]	—

Table 42.6 RX Message Buffer Component (RMBCP) Detailed

Rc	p	Symbol	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R0	x	CFDRMIDb	RMIDE	RMRTR	—	RMID																												
R1	x	CFDRMPTRb	RMDLC				—	—	—	—	—	—	—	—	—	—	—	—	RMTS															
R2	x	CFDRMFDSTSb	RMPTR																—	—	—	—	—	—	RMIFL		—	—	—	—	—	RMFDF	RMBRS	RMESI
R3	0	CFDRMDFbp	RMDB_HH								RMDB_HL								RMDB_LH								RMDB_LL							
R[4…18]	[1…15]	CFDRMDFbp	RMDB_HH								RMDB_HL								RMDB_LH								RMDB_LL							

42.2.61.3 CFDRMIDb : RX Message Buffer ID Registers (b = 0 to 15)

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x0920 + 0x004C × b (b = 0 to 7)
0x0D20 + 0x004C × (b - 8) (b = 8 to 15)

Bit position:	31	30	29	28–0
Bit field:	RMIDE	RMRTR	—	RMID[28:0]

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
28:0	RMID[28:0]	RX Message Buffer ID Field STD-ID/EXT-ID fields	R
29	—	This bit is read as 0.	R

Bit	Symbol	Function	R/W
30	RMRTR	RX Message Buffer RTR Bit 0: Data frame 1: Remote frame	R
31	RMIDE	RX Message Buffer IDE Bit 0: STD-ID is stored 1: EXT-ID is stored	R

Note: S-TYPE-3, P-TYPE-3

The RX Message Buffer ID Register b (b = 0 to 15) store the ID field, IDE bit, and RTR bit of the received message.

RMID[28:0] bits (RX Message Buffer ID Field)

The RMID[28:0] are the bits of the STD-ID/EXT-ID fields of the message stored in the RX message buffer.

See section 42.2.61.1. Start Addresses for details on how to interpret the structure of this buffer component.

RMRTR bit (RX Message Buffer RTR Bit)

The RMRTR bit shows whether a data frame or a remote frame was stored in the RX message buffer.

Note: There are no remote frames in CANFD format. When a CANFD frame is received, the register reflects the state of the received value (the RRS bit in FD frame format).

RMIDE bit (RX Message Buffer IDE Bit)

The RMIDE bit shows whether message with Standard Identifier or Extended Identifier was stored in the RX message buffer.

42.2.61.4 CFDRMPTRb : RX Message Buffer Pointer Registers (b = 0 to 15)

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x0924 + 0x004C × b (b = 0 to 7)
0x0D24 + 0x004C × (b - 8) (b = 8 to 15)

Bit position:	31–28	27	26	25	24	23	22	21	20	19	18	17	16	15–0
Bit field:	RMDLC[3:0]	—	—	—	—	—	—	—	—	—	—	—	—	RMTS[15:0]
Value after reset:	0 0 0 0	0	0	0	0	0	0	0	0	0	0	0	0	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
15:0	RMTS[15:0]	RX Message Buffer Timestamp Field Timestamp value stored for the message in the RX message buffer	R
27:16	—	These bits are read as 0.	R
31:28	RMDLC[3:0]	RX Message Buffer DLC Field Number of data bytes received in a CAN frame.	R

Note: S-TYPE-3, P-TYPE-3

The RX Message Buffer Pointer Register b (b = 0 to 15) store the DLC and Timestamp fields for the received message.

RMTS[15:0] bits (RX Message Buffer Timestamp Field)

The RMTS[15:0] bits store the timestamp value taken at the capture point as configured by CFDGFDCFG.TSCCFG of the received message.

RMDLC[3:0] bits (RX Message Buffer DLC Field)

The RMDLC[3:0] bits store the number of data bytes that were received in the RX message buffer.

See Table 5 in ISO 11898-1 (2015) Specification for details in defining the number of data bytes that were received.

Note: The maximum capacity of the buffer belongs to CFDRMNB.RMPLS and this is not available in the classical CAN function.

42.2.61.5 CFDRMFDSTSb : RX Message Buffer CANFD Status Registers (b = 0 to 15)

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x0928 + 0x004C × b (b = 0 to 7)
0x0D28 + 0x004C × (b - 8) (b = 8 to 15)

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	RMPTR[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	RMIFL[1:0]		—	—	—	—	—	RMFDF	RMBRS	RMESI
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	RMESI*1	Error State Indicator bit 0: CANFD frame received from error active node 1: CANFD frame received from error passive node	R
1	RMBRS*1	Bit Rate Switch bit 0: CANFD frame received with no bit rate switch 1: CANFD frame received with bit rate switch	R
2	RMFDF*1	CAN FD Format bit 0: Non CANFD frame received 1: CANFD frame received	R
7:3	—	These bits are read as 0.	R
9:8	RMIFL[1:0]	RX Message Buffer Information Label Field	R
15:10	—	These bits are read as 0.	R
31:16	RMPTR[15:0]	RX Message Buffer Pointer Field	R

Note: S-TYPE-3, P-TYPE-3
Note 1. This bit is not available in the classical CAN function.

The RX Message Buffer CANFD Status Register b (b = 0 to 15) show the status of the FDF, BRS and ESI bits, and pointer of the received CANFD frame.

RMESI bit (Error State Indicator bit)

The RMESI bit has the same value as the ESI bit of the received CANFD frame.

When the received FDF bit is 0, this means a CAN2.0 frame is received and 0 is stored to this bit.

Note: This bit is not available in the classical CAN function.

RMBRS bit (Bit Rate Switch bit)

The RMBRS bit has the same value as the BRS bit of the received CANFD frame.

When the received FDF bit is 0, this means a CAN2.0 frame is received and 0 is stored to this bit.

Note: This bit is not available in the classical CAN function.

RMFDF bit (CAN FD Format bit)

The RMFDF bit has the same value as the FDF bit of the received CANFD frame.

Note: This bit is not available in the classical CAN function.

RMIFL[1:0] bits (RX Message Buffer Information Label Field)

The RMIFL[1:0] bits store the information label value from the related Global Acceptance Filter List entry.

RMPTR[15:0] bits (RX Message Buffer Pointer Field)

The RMPTR[15:0] bits store the pointer value from the related Global Acceptance Filter List entry.

42.2.61.6 CFDRMDFb_p : RX Message Buffer Data Field p Registers (p = 0 to 15, b = 0 to 15)

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x092C + 0x004C × b + 0x0004 × p (b = 0 to 7, p = 0 to 15)
0x0D2C + 0x004C × (b - 8) + 0x0004 × p (b = 8 to 15, p = 0 to 15)

Bit position:	31–24	23–16	15–8	7–0
Bit field:	RMDB_HH[7:0]	RMDB_HL[7:0]	RMDB_LH[7:0]	RMDB_LL[7:0]
Value after reset:	0 0 0 0 0 0 0 0	0 0 0 0 0 0 0 0	0 0 0 0 0 0 0 0	0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
7:0	RMDB_LL[7:0]	RX Message Buffer Data Byte (p × 4)	R
15:8	RMDB_LH[7:0]	RX Message Buffer Data Byte ((p × 4) + 1)	R
23:16	RMDB_HL[7:0]	RX Message Buffer Data Byte ((p × 4) + 2)	R
31:24	RMDB_HH[7:0]*1	RX Message Buffer Data Byte ((p × 4) + 3)	R

Note: S-TYPE-3, P-TYPE-3
Note 1. These bits are not available in the classical CAN function.

The RX Message Buffer Data Field p Register b (p = 0 to 15, b = 0 to 15) store the data bytes (p × 4) to data bytes ((p × 4) + 3) of the received message.

RMDB_LL[7:0] bits (RX Message Buffer Data Byte (p × 4))

The RMDB_LL[7:0] bits store data bytes (p × 4) of the message in the RX message buffer.

Unused data bytes are filled with 0x00.

RMDB_LH[7:0] bits (RX Message Buffer Data Byte ((p × 4) + 1))

The RMDB_LH[7:0] bits store data bytes ((p × 4) + 1) of the message in the RX message buffer.

Unused Data Bytes will be filled with 0x00.

RMDB_HL[7:0] bits (RX Message Buffer Data Byte ((p × 4) + 2))

The RMDB_HL[7:0] bits store data bytes ((p × 4) + 2) of the message in the RX message buffer.

Unused data bytes are filled with 0x00.

RMDB_HH[7:0] bits (RX Message Buffer Data Byte ((p × 4) + 3))

The RMDB_HH[7:0] bits store data bytes ((p × 4) + 3) of the message in the RX message buffer.

Unused data bytes are filled with 0x00.

42.2.61.7 CFDRFMBCPb[0] : RX FIFO Access Message Buffer Component b (b = 0 to 1)

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: see Table 42.4

Bit position:	31–0
Bit field:	Rc[31:0]
Value after reset:	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
31:0	Rc[31:0]	RX FIFO Access Message Buffer Component c See Table 42.7, Table 42.8 and the descriptions that follow for a detailed description of each register and its related bits, contained within this message buffer component.	R

Note: S-TYPE-3, P-TYPE-3

Where the total number of CFDRFMBCPb = 2 as shown in Figure 42.28 (c = RX FIFO Access Message Buffer Component Register index = [0…18])

Rc[31:0] bits (RX FIFO Access Message Buffer Component c)

The RX FIFO Access Message Buffer component comprises the following registers:

- CFDRFIDb
- CFDRFPTRb
- CFDRFFDSTSb
- CFDRFDFbp

See Table 42.8 for details on how to interpret the structure of this buffer component and how to access the respective registers.

Table 42.7 RX FIFO Access Message Buffer component summary

Rc	
R0	RX FIFO Access ID Registers
R1	RX FIFO Access Pointer Register
R2	RX FIFO Access CANFD Status Registers
R3	RX FIFO Access Data Field 0 Registers
R4	RX FIFO Access Data Field 1 Registers
R5	RX FIFO Access Data Field 2 Registers
R6	RX FIFO Access Data Field 3 Registers
R7	RX FIFO Access Data Field 4 Registers
R8	RX FIFO Access Data Field 5 Registers
R9	RX FIFO Access Data Field 6 Registers
R10	RX FIFO Access Data Field 7 Registers
R11	RX FIFO Access Data Field 8 Registers
R12	RX FIFO Access Data Field 9 Registers
R13	RX FIFO Access Data Field 10 Registers
R14	RX FIFO Access Data Field 11 Registers
R15	RX FIFO Access Data Field 12 Registers
R16	RX FIFO Access Data Field 13 Registers
R17	RX FIFO Access Data Field 14 Registers
R18	RX FIFO Access Data Field 15 Registers
R[19...31]	—

Table 42.8 RX Message Buffer Component (RMBCP) Detailed (1 of 2)

Rc	p	Symbol	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R0	x	CFDRMIDb	RMIDE	RMRTR	—	RMID																												

Table 42.8 RX Message Buffer Component (RMBCP) Detailed (2 of 2)

Rc	p	Symbol	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R1	x	CFDRM PTRb	RMDLC				—	—	—	—	—	—	—	—	—	—	—	—	RMTS															
R2	x	CFDRM FDSTSb	RMPTR																—	—	—	—	—	—	RMIFL		—	—	—	—	—	RMFDF	RMBRS	RMESI
R3	0	CFDRM DFbp	RMDB_HH								RMDB_HL								RMDB_LH								RMDB_LL							
R[4… 18]	[1… 15]	CFDRM DFbp	RMDB_HH								RMDB_HL								RMDB_LH								RMDB_LL							

42.2.61.8 CFDRFIDb : RX FIFO Access ID Register b (b = 0 to 1)

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x0520 + 0x004C × b

Bit position:	31	30	29	28–0
Bit field:	RFIDE	RFRTR	—	RFID[28:0]
Value after reset:	0	0	0	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
28:0	RFID[28:0]	RX FIFO Buffer ID Field STD-ID/EXT-ID fields	R
29	—	This bit is read as 0.	R
30	RFRTR	RX FIFO Buffer RTR bit 0: Data frame 1: Remote frame	R
31	RFIDE	RX FIFO Buffer IDE bit 0: STD-ID has been received 1: EXT-ID has been received	R

Note: S-TYPE-3, P-TYPE-3

The RX FIFO Access ID Registers b (b = 0 to 1) store the ID field, IDE bit and RTR bit of the message.

RFID[28:0] bits (RX FIFO Buffer ID Field)

The RFID[28:0] bits are the bits of the STD-ID/EXT-ID fields of the message in the FIFO buffer.

For alignment of these bits in standard and extended frame format, see Identifier Bits Alignment.

RFRTR bit (RX FIFO Buffer RTR bit)

The RFRTR bit shows whether a data frame or a remote frame was stored in the FIFO buffer.

Note: There are no remote frames in CANFD format. When a CANFD frame was received, the register reflects the state of the received value (RRS bit in FD frame format).

RFIDE bit (RX FIFO Buffer IDE bit)

The RFIDE bit shows whether message with the Standard Identifier or Extended Identifier was received in the FIFO buffer.

42.2.61.9 CFDRFPTRb : RX FIFO Access Pointer Register b (b = 0 to 1)

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x0524 + 0x004C × b

Bit position:	31–28	27	26	25	24	23	22	21	20	19	18	17	16	15–0
Bit field:	RFDLC[3:0]	—	—	—	—	—	—	—	—	—	—	—	—	RFTS[15:0]
Value after reset:	0 0 0 0	0	0	0	0	0	0	0	0	0	0	0	0	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
15:0	RFTS[15:0]	RX FIFO Timestamp Value Timestamp value of the received CAN frame	R
27:16	—	These bits are read as 0.	R
31:28	RFDLC[3:0]	RX FIFO Buffer DLC Field Number of data bytes received in a CAN frame	R

Note: S-TYPE-3, P-TYPE-3

The FIFO Access Pointer Registers b (b = 0 to 1) store the DLC and Timestamp fields for the received message.

RFTS[15:0] bits (RX FIFO Timestamp Value)

The RFTS[15:0] bits store the timestamp value taken at the capture point as configured by the CFDGFDCFG.TSCCFG bit of the received message.

RFDLC[3:0] bits (RX FIFO Buffer DLC Field)

The RFDLC[3:0] bits store the number of data bytes that were received in the RX FIFO buffer.

See Table 5 in ISO 11898-1 (2015) Specification for details in defining the number of data bytes that were received.

42.2.61.10 CFDRFFDSTSb : RX FIFO Access CANFD Status Register b (b = 0 to 1)

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x0528 + 0x004C × b

Bit position:	31–16
Bit field:	CFDRFPTR[15:0]
Value after reset:	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit position:	15	14	13	12	11	10	9–8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	RFIFL[1:0]	—	—	—	—	—	RFFDF	RFBRS	RFESI
Value after reset:	0	0	0	0	0	0	0 0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	RFESI*1	Error State Indicator bit 0: CANFD frame received from error active node 1: CANFD frame received from error passive node	R
1	RFBRS*1	Bit Rate Switch bit 0: CANFD frame received with no bit rate switch 1: CANFD frame received with bit rate switch	R
2	RFFDF*1	CAN FD Format bit 0: Non CANFD frame received 1: CANFD frame received	R
7:3	—	These bits are read as 0.	R

Bit	Symbol	Function	R/W
9:8	RFIFL[1:0]	RX FIFO Buffer Information Label Field	R
15:10	—	These bits are read as 0.	R
31:16	CFDRFPTR[15:0]	RX FIFO Buffer Pointer Field	R

Note: S-TYPE-3, P-TYPE-3

Note 1. This bit is not available in the classical CAN function.

The RX FIFO Access CANFD Status Registers b (b = 0 to 1) show the status of the FDF, BRS, and ESI bits, including the pointer of the received CANFD frame.

RFESI bit (Error State Indicator bit)

The RFESI bit has the same value as the ESI bit of the received CANFD frame.

When the received FDF bit is 0, this means a CAN2.0 frame is received and 0 is stored to this bit.

Note: This bit is not available in the classical CAN function.

RFBRS bit (Bit Rate Switch bit)

The RFBRS bit has the same value as the BRS bit of the received CANFD frame.

When the received FDF bit is 0, this means a CAN2.0 frame is received and 0 is stored to this bit.

Note: This bit is not available in the classical CAN function.

RFFDF bit (CAN FD Format bit)

The RFFDF bit has the same value as the FDF bit of the received CANFD frame.

Note: This bit is not available in the classical CAN function.

RFIFL[1:0] bits (RX FIFO Buffer Information Label Field)

The RFIFL[1:0] bits store the information label value from the related Global Acceptance Filter List entry.

CFDRFPTR[15:0] bits (RX FIFO Buffer Pointer Field)

The CFDRFPTR[15:0] bits store the pointer value from the related Global Acceptance Filter List entry.

42.2.61.11 CFDRFDFb_p : RX FIFO Access Data Field p Register b (p = 0 to 15, b = 0 to 1)

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x052C + 0x004 × p + 0x04C × b

Bit position:	31–24	23–16	15–8	7–0
Bit field:	RFDB_HH[7:0]	RFDB_HL[7:0]	RFDB_LH[7:0]	RFDB_LL[7:0]
Value after reset:	0 0 0 0 0 0 0 0	0 0 0 0 0 0 0 0	0 0 0 0 0 0 0 0	0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
7:0	RFDB_LL[7:0]	RX FIFO Buffer Data Byte (p × 4)	R
15:8	RFDB_LH[7:0]	RX FIFO Buffer Data Byte ((p × 4) + 1)	R
23:16	RFDB_HL[7:0]	RX FIFO Buffer Data Byte ((p × 4) + 2)	R
31:24	RFDB_HH[7:0]	RX FIFO Buffer Data Byte ((p × 4) + 3)	R

Note: S-TYPE-3, P-TYPE-3

The RX FIFO Access Data Field p Registers b (p = 0 to 15, b = 0 to 1) store data bytes (p × 4) to data byte ((p × 4) + 3) of the received message.

RFDB_LL[7:0] bits (RX FIFO Buffer Data Byte (p × 4))

The RFDB_LL[7:0] bits store data bytes (p × 4) of the message present in the FIFO buffer.

Unused data bytes are filled with 0x00 according to the configured data payload size CFDRFCCa.RFPLS.

RFDB_LH[7:0] bits (RX FIFO Buffer Data Byte ((p × 4) + 1))

The RFDB_LH[7:0] bits store data bytes ((p × 4) + 1) of the message present in the FIFO buffer.

Unused data bytes are filled with 0x00.

RFDB_HL[7:0] bits (RX FIFO Buffer Data Byte ((p × 4) + 2))

The RFDB_HL[7:0] bits store data bytes ((p × 4) + 2) of the message present in the FIFO buffer.

Unused data bytes are filled with 0x00.

RFDB_HH[7:0] bits (RX FIFO Buffer Data Byte ((p × 4) + 3))

The RFDB_HH[7:0] bits store data bytes ((p × 4) + 3) of the message present in the FIFO buffer.

Unused data bytes are filled with 0x00.

42.2.61.12 CFDCFMBCP0[0] : Common FIFO Access Message Buffer Component

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: See Table 42.4

Bit position:	31 … 0
Bit field:	Rc[31:0]
Value after reset:	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
31:0	Rc[31:0]	Common FIFO Access Message Buffer Component c Refer to Table 42.9, Table 42.10 and the descriptions that follow for a detailed description of each register and its related bits, contained within this message buffer component.	R

Note: S-TYPE-3, P-TYPE-3

Where the total number of CFDCFMBCP0 = 1 as shown in Figure 42.28 (c = Common FIFO Message Buffer Component Register index = [0…18])

Rc[31:0] bit (Common FIFO Access Message Buffer Component c)

The Common FIFO Access Message Buffer Component is made up of the following registers: CFDCFID, CFDCFPTR, CFFDSTS0, and CFDCFDFp. Refer to Table 42.10 for details of how to interpret the structure of this buffer component and how to access the respective registers.

Table 42.9 Common FIFO Access Message Buffer Component Summary (1 of 2)

Common FIFO Access Message Buffer Component (CFMBCP)	
Rc	**CANFD mode (CAN_FD_MODE = 1)**
R0	Common FIFO Access ID Registers
R1	Common FIFO Access Pointer Register
R2	Common FIFO Access CANFD Status Registers
R3	Common FIFO Access Data Field 0 Registers
R4	Common FIFO Access Data Field 1 Registers
R5	Common FIFO Access Data Field 2 Registers
R6	Common FIFO Access Data Field 3 Registers
R7	Common FIFO Access Data Field 4 Registers

Table 42.9 Common FIFO Access Message Buffer Component Summary (2 of 2)

Common FIFO Access Message Buffer Component (CFMBCP)	
Rc	**CANFD mode (CAN_FD_MODE = 1)**
R8	Common FIFO Access Data Field 5 Registers
R9	Common FIFO Access Data Field 6 Registers
R10	Common FIFO Access Data Field 7 Registers
R11	Common FIFO Access Data Field 8 Registers
R12	Common FIFO Access Data Field 9 Registers
R13	Common FIFO Access Data Field 10 Registers
R14	Common FIFO Access Data Field 11 Registers
R15	Common FIFO Access Data Field 12 Registers
R16	Common FIFO Access Data Field 13 Registers
R17	Common FIFO Access Data Field 14 Registers
R18	Common FIFO Access Data Field 15 Registers
R[19…31]	—

Table 42.10 Common FIFO Access Message Buffer Component (CFMBCP) Detailed

Rc	p	Symbol	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R0	x	CFDCFID	CFIDE	CFRTR	THLEN	CFID																												
R1	x	CFDCFPTR	CFDLC				—	—	—	—	—	—	—	—	—	—	—	—	CFTS															
R2	x	CFDCFFDCSTS	CFPTR																—	—	—	—	—	—	CFIFL		—	—	—	—	—	CFFDF	CFBRS	CFESI
R3	0	CFDCFDFp	CFDB_HH								CFDB_HL								CFDB_LH								CFDB_LL							
R[4…18]	[1…15]	CFDCFDFp	CFDB_HH								CFDB_HL								CFDB_LH								CFDB_LL							
R[19…31]	x	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

42.2.61.13 CFDCFID : Common FIFO Access ID Register

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x05B8

Bit position:	31	30	29	28–0
Bit field:	CFIDE	CFRTR	THLEN	CFID[28:0]
Value after reset:	0	0	0	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
28:0	CFID[28:0]	Common FIFO Buffer ID Field STD-ID / EXT-ID fields	R/W

Bit	Symbol	Function	R/W
29	THLEN	THL Entry enable TX FIFO Mode: 0: Entry will not be stored in THL after successful TX. 1: Entry will be stored in THL after successful TX. RX FIFO Mode: Reserved, this bit is read as 0	R/W
30	CFRTR	Common FIFO Buffer RTR Bit 0: Data Frame 1: Remote Frame	R/W
31	CFIDE	Common FIFO Buffer IDE Bit 0: STD-ID will be transmitted or has been received 1: EXT-ID will be transmitted or has been received	R/W

Note: S-TYPE-3, P-TYPE-3

The Common FIFO Access ID registers store the ID field, IDE bit and RTR bit of the message.

In TX mode, users can read data from the FIFO, only for the current entry based on the write pointer value, not for the other entries.

CFID[28:0] bit (Common FIFO Buffer ID Field)

These are the bits of the STD-ID / EXT-ID fields of the message in the FIFO Buffer.

In TX mode, users can write and read from FIFO buffers.

In RX mode, users can only read data from FIFO buffers.

THLEN bit (THL Entry enable)

This bit controls the storage of an entry corresponding to the transmitted message in the TX History list at the end of a successful transmission.

In TX mode, users can write and read from FIFO buffers.

In RX mode, users can only read data from FIFO buffers.

CFRTR bit (Common FIFO Buffer RTR Bit)

This bit selects whether a Data Frame or a Remote Frame will be transmitted from or was received in the FIFO Buffer.

Note: There are no remote frames in CANFD format. In case a CANFD frame was received (RX mode) the register reflects the state of the received value (RRS bit in FD frame format). In case of CANFD transmission (TX mode CFDCFID.CFFDF =1) the bit is always transmitted dominant (Data Frame).

In TX mode, users can write and read from FIFO buffers.

In RX mode, users can only read data from FIFO buffers.

CFIDE bit (Common FIFO Buffer IDE Bit)

This bit selects whether a message with EXT-ID or STD-ID will be transmitted from or was received in the FIFO Buffer.

In TX mode, users can write and read from FIFO buffers.

In RX mode, users can only read data from FIFO buffers.

42.2.61.14 CFDCFPTR : Common FIFO Access Pointer Register

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x05BC

Bit position:	31–28	27	26	25	24	23	22	21	20	19	18	17	16	15–0
Bit field:	CFDLC[3:0]	—	—	—	—	—	—	—	—	—	—	—	—	CFTS[15:0]
Value after reset:	0 0 0 0	0	0	0	0	0	0	0	0	0	0	0	0	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
15:0	CFTS[15:0]	Common FIFO Timestamp Value Timestamp value of the received CAN frame (FIFO in RX mode).	R/W
27:16	—	These bits are read as 0. The write value should be 0.	R/W
31:28	CFDLC[3:0]	Common FIFO Buffer DLC Field Number of data bytes received in a CAN frame, or to be transmitted in a CAN frame.	R/W

Note: S-TYPE-3, P-TYPE-3

The Common FIFO Access Pointer Registers store the DLC and Timestamp fields.

In TX mode, you can read data from the FIFO buffer, only for the current entry based on the write pointer value, and not for the other entries.

CFTS[15:0] bits (Common FIFO Timestamp Value)

The CFTS[15:0] bits store the timestamp value taken at the capture point as configured by the CFDGFDCFG.TSCCFG bit of the received message (if FIFO is configured in RX mode).

In TX mode, you can read and write from FIFO buffers.

In RX mode, you can only read data from FIFO buffers.

CFDLC[3:0] bits (Common FIFO Buffer DLC Field)

The CFDLC[3:0] bits store the number of data bytes that were received in the FIFO buffer or are to be transmitted.

See Table 5 in ISO 11898-1 (2015) Specification for details in defining the number of data bytes.

In TX mode, you can read and write from the FIFO buffers. Do not read data for the other entries in the FIFO when configured in TX mode.

In RX mode, you can only read data from the FIFO buffers.

42.2.61.15 CFDCFFDCSTS : Common FIFO Access CANFD Control/Status Register

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x05C0

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	CFPTR[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	CFIFL[1:0]		—	—	—	—	—	CFFDF	CFBRS	CFESI
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	CFESI*1	Error State Indicator bit 0: CANFD frame received or to transmit by error active node 1: CANFD frame received or to transmit by error passive node	R/W
1	CFBRS*1	Bit Rate Switch bit 0: CANFD frame received or to transmit with no bit rate switch 1: CANFD frame received or to transmit with bit rate switch	R/W
2	CFFDF*1	CAN FD Format bit 0: Non CANFD frame received or to transmit 1: CANFD frame received or to transmit	R/W
7:3	—	These bits are read as 0. The write value should be 0.	R/W
9:8	CFIFL[1:0]	COMMON FIFO Buffer Information Label Field	R/W
15:10	—	These bits are read as 0. The write value should be 0.	R/W

Bit	Symbol	Function	R/W
31:16	CFPTR[15:0]	Common FIFO Buffer Pointer Field	R/W

Note: S-TYPE-3, P-TYPE-3
Note 1. This bit is not available in the classical CAN function.

The Common FIFO Access CANFD Control/Status Registers show the status of the FDF, BRS and ESI bits, including the pointer of the received CANFD frame or the CANFD frame to transmit.

In TX mode, you can read data from the FIFO, only for the current entry based on the write pointer value, and not for the other entries.

CFESI bit (Error State Indicator bit)

In TX mode, you can read and write from FIFO buffers. In this mode, when the CANFD module is not in error passive, the CFESI bit equals the write value. Otherwise, it is a don’t care, and the bit is transmitted as 1 on the CAN bus, indicating this is an error passive node.

In RX mode, you can only read data from FIFO buffers.

In RX mode, the CFESI bit is updated with the ESI bit value of the CANFD frame when it has been received, indicating the error state of the transmitting node. In RX mode, 0 is stored to this bit when the received FDF bit is 0, this means a CAN 2.0 frame is received.

Note: This bit is not available in the classical CAN function.

CFBRS bit (Bit Rate Switch bit)

In TX mode, you can read and write from FIFO buffers. In this mode, the CANFD module either transmits a 0 to indicate no bit rate switch in the frame to be transmitted or a 1 to indicate a bit rate switch in the frame to be transmitted.

In RX mode, you can only read data from FIFO buffers.

In RX mode, the CFBRS bit is updated with the BRS bit value of the CANFD frame when it has been received, indicating whether there is a bit rate switch (1) or (0) on the CANFD frame.

In RX mode, 0 is stored to the CFBRS bit when the received FDF bit is 0, this means a CAN 2.0 frame is received.

Note: This bit is not available in the classical CAN function.

CFFDF bit (CAN FD Format bit)

In TX mode, you can read and write from FIFO buffers. In this mode, the CANFD module either transmits a 0 to indicate a CAN 2.0 frame is to be transmitted or a 1 to indicate a CANFD frame is to be transmitted.

In RX mode, you can only read data from FIFO buffers.

In RX mode, the CFFDF bit is updated with the FDF bit value of the CAN frame when it has been received, indicating whether it is a CAN 2.0 frame (0) or a CANFD frame (1).

Note: This bit is not available in the classical CAN function.

CFIFL[1:0] bits (COMMON FIFO Buffer Information Label Field)

If the Common FIFO is configured in TX mode, the value programmed in CFDCFFDCSTS.CFIFL[1:0] is stored together with additional message information, to the TX History List after successful transmission of the message.

The information label value from the related Global Acceptance Filter List entry is stored in these bits (if FIFO is configured in either RX mode).

In TX mode, you can read and write from FIFO buffers.

In RX mode, you can only read data from FIFO buffers.

CFPTR[15:0] bits (Common FIFO Buffer Pointer Field)

If the Common FIFO is configured in TX mode, the value programmed in CFDCFFDCSTS.CFPTR[15:0] is stored together with additional message information, to the TX History List after successful transmission of the message.

The pointer value from the related Global Acceptance Filter List entry is stored in these bits (if FIFO is configured in either RX mode).

In TX mode, you can read and write from FIFO buffers.

In RX mode, you can only read data from FIFO buffers.

42.2.61.16 CFDCFDFp : Common FIFO Access Data Field p Registers (p = 0 to 15)

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x05C4 + 0x004 × p

Bit position:	31–24	23–16	15–8	7–0
Bit field:	CFDB_HH[7:0]	CFDB_HL[7:0]	CFDB_LH[7:0]	CFDB_LL[7:0]
Value after reset:	0 0 0 0 0 0 0 0	0 0 0 0 0 0 0 0	0 0 0 0 0 0 0 0	0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
7:0	CFDB_LL[7:0]	Common FIFO Buffer Data Bytes (p × 4)	R/W
15:8	CFDB_LH[7:0]	Common FIFO Buffer Data Bytes ((p × 4) + 1)	R/W
23:16	CFDB_HL[7:0]	Common FIFO Buffer Data Bytes ((p × 4) + 2)	R/W
31:24	CFDB_HH[7:0]	Common FIFO Buffer Data Bytes ((p × 4) + 3)	R/W

Note: S-TYPE-3, P-TYPE-3

The FIFO Access Data Field p Registers (p = 0 to 15) store data bytes (p × 4) to data bytes ((p × 4) + 3) of the message.

In TX mode, you can read data from the FIFO, only for the current entry based on the write pointer value, and not for the other entries.

CFDB_LL[7:0] bits (Common FIFO Buffer Data Bytes (p × 4))

The CFDB_LL[7:0] bits store data bytes (p × 4) of the message present in the FIFO buffer.

In TX mode, you can read and write from the FIFO buffers.

In RX mode, you can only read data from the FIFO buffers.

In RX mode, unused data bytes are filled with 0x00, according to their configured data payload size CFDCFCC.CFPLS.*1

CFDB_LH[7:0] bits (Common FIFO Buffer Data Bytes ((p × 4) + 1))

The CFDB_LH[7:0] bits store data bytes ((p × 4) + 1) of the message present in the FIFO buffer.

In TX mode, you can read and write from the FIFO buffers.

In RX mode, you can only read data from the FIFO buffers.

In RX mode, unused data bytes are filled with 0x00, according to their configured data payload size CFDCFCC.CFPLS.*1

CFDB_HL[7:0] bits (Common FIFO Buffer Data Bytes ((p × 4) + 2))

The CFDB_HL[7:0] bits store data bytes ((p × 4) + 2) of the message present in the FIFO buffer.

In TX mode, you can read and write from the FIFO buffers.

In RX mode, you can only read data from the FIFO buffers.

In RX mode, unused data bytes are filled with 0x00, according to their configured data payload size CFDCFCC.CFPLS.*1

CFDB_HH[7:0] bits (Common FIFO Buffer Data Bytes ((p × 4) + 3))

The CFDB_HH[7:0] bits store data bytes ((p × 4) + 3) of the message present in the FIFO buffer.

In TX mode, you can read and write from the FIFO buffers.

In RX mode, you can only read data from the FIFO buffers.

In RX mode, unused data bytes are filled with 0x00, according to their configured data payload size CFDCFCC.CFPLS.*1

Note 1. In RX mode, unused data bytes are filled with 0x00 according to the configured data payload size CFDCFCC.CFPLS, which is a CANFD feature not found in classical CAN.

42.2.61.17 CFDTMBCPb[0] : TX Message Buffer Component b (b = 0 to 3)

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: See Table 42.4

Bit position:	31 … 0
Bit field:	Rc[31:0]
Value after reset:	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
31:0	Rc[31:0]	TX Message Buffer Component c Refer to Table 42.11, Table 42.12 and the descriptions that follow for a detailed description of each register and its related bits, contained within this message buffer component.	R

Note: S-TYPE-3, P-TYPE-3

Where the total number of CFDTMBCPn = 4 as shown in Figure 42.28 (c = TX Message Buffer Component Register index = [0…18])

Rc[31:0] bit (TX Message Buffer Component c)

TX Message Buffer Component c

The TX Message Buffer Component is made up of the following registers: CFDTMIDb, CFDTMPTRb, CFDTMFDCTRb, and CFDTMDFbp. Refer to Table 42.12 for details of how to interpret the structure of this buffer component and how to access the respective registers.

Table 42.11 TX Message Buffer Component Summary

TX Message Buffer Component (TMBCP)	
Rc	**CANFD mode (CAN_FD_MODE = 1b)**
R0	TX Message Buffer (b) ID Registers
R1	TX Message Buffer (b) Pointer Registers
R2	TX Message Buffer (b) CANFD Status Registers
R3	TX Message Buffer (b) Data Field 0 Registers
R4	TX Message Buffer (b) Data Field 1 Registers
R5	TX Message Buffer (b) Data Field 2 Registers
R6	TX Message Buffer (b) Data Field 3 Registers
R7	TX Message Buffer (b) Data Field 4 Registers
R8	TX Message Buffer (b) Data Field 5 Registers
R9	TX Message Buffer (b) Data Field 6 Registers
R10	TX Message Buffer (b) Data Field 7 Registers
R11	TX Message Buffer (b) Data Field 8 Registers
R12	TX Message Buffer (b) Data Field 9 Registers
R13	TX Message Buffer (b) Data Field 10 Registers
R14	TX Message Buffer (b) Data Field 11 Registers
R15	TX Message Buffer (b) Data Field 12 Registers
R16	TX Message Buffer (b) Data Field 13 Registers
R17	TX Message Buffer (b) Data Field 14 Registers
R18	TX Message Buffer (b) Data Field 15 Registers
R[19…31]	—

Table 42.12 TX Message Buffer Component (TMBCP) Detailed

Rc	p	Symbol	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R0	x	CFDTMIDb	TMIDE	TMRTR	THLEN	TMID																												
R1	x	CFDTMPTRb	TMDLC				—	—	—	—	—	—	—	—	—	—	—	—	CFTS															
R2	x	CFDTMFDCTRb	TMPTR																—	—	—	—	—	—	TMIFL		—	—	—	—	—	TMFDF	TMBRS	TMESI
R3	0	CFDTMDFbp	TMDB_HH								TMDB_HL								TMDB_LH								TMDB_LL							
R[4… 18]	[1… 15]	CFDTMDFbp	TMDB_HH								TMDB_HL								TMDB_LH								TMDB_LL							

42.2.61.18 CFDTMIDb : TX Message Buffer ID Registers (b = 0 to 3)

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x0604 + 0x004C × b

Bit position:	31	30	29	28–0
Bit field:	TMIDE	TMRTR	THLEN	TMID[28:0]
Value after reset:	0	0	0	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
28:0	TMID[28:0]	TX Message Buffer ID Field STD-ID/EXT-ID fields	R/W
29	THLEN	Tx History List Entry 0: Entry not stored in THL after successful TX 1: Entry stored in THL after successful TX	R/W
30	TMRTR	TX Message Buffer RTR bit 0: Data frame 1: Remote frame	R/W
31	TMIDE	TX Message Buffer IDE bit 0: STD-ID is transmitted 1: EXT-ID is transmitted	R/W

Note: S-TYPE-3, P-TYPE-3

Each TX Message Buffer ID Register b (b = 0 to 3) are used to store the ID, IDE, RTR fields and history configuration of the message to be transmitted from the associated buffer.

TMID[28:0] bits (TX Message Buffer ID Field)

The TMID[28:0] bits are bits of the STD-ID/EXT-ID fields of the message stored in this TX message buffer.

Do not write to these bits when the related CANFD channel is in CH_SLEEP mode.

THLEN bit (Tx History List Entry)

The THLEN bit controls the storage of an entry corresponding to the transmitted message in the TX History list at the end of a successful transmission.

Do not write to these bits when the related CANFD channel is in CH_SLEEP mode.

TMRTR bit (TX Message Buffer RTR bit)

The TMRTR bit selects whether a data frame or remote frame is to be transmitted from this TX message buffer.

Note: There are no remote frames in CANFD format. For a CANFD transmission (CFDTMFDCTRb.CFFDF = 1), this bit is always transmitted dominant (data frame).

Do not write to these bits when the related CANFD channel is in CH_SLEEP mode.

TMIDE bit (TX Message Buffer IDE bit)

The TMIDE bit selects whether a message with EXT-ID or STD-ID is to be transmitted from this TX message buffer.

Do not write to these bits when the related CANFD channel is in CH_SLEEP mode.

42.2.61.19 CFDTMPTRb : TX Message Buffer Pointer Register (b = 0 to 3)

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x0608 + 0x004C × b

Bit position:	31			28																												0
Bit field:	TMDLC[3:0]				—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
27:0	—	The read values are undefined. The write value should be 0.	R/W
31:28	TMDLC[3:0]	TX Message Buffer DLC Field Number of data bytes to be transmitted in a CAN frame.	R/W

Note: S-TYPE-3, P-TYPE-3

Each TX Message Buffer Pointer Register b (b = 0 to 3) is used to store the DLC fields of the message to transmit from the associated buffer.

TMDLC[3:0] bits (TX Message Buffer DLC Field)

The TMDLC[3:0] bits select the number of data bytes to be transmitted from this TX message buffer when the corresponding TMRTR bit is configured as 0.

See Table 5 in ISO 11898-1 (2015) Specification for details in defining the number of data bytes to be transmitted.

Do not write to these bits when the related CANFD channel is in CH_SLEEP mode.

42.2.61.20 CFDTMFDCTRb : TX Message Buffer CANFD Control Register (b = 0 to 3)

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x060C + 0x004C × b

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	TMPTR[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	TMIFL[1:0]		—	—	—	—	—	TMFDF	TMBRS	TMESI
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	TMESI*1	Error State Indicator bit 0: CANFD frame to transmit by error active node 1: CANFD frame to transmit by error passive node	R/W
1	TMBRS*1	Bit Rate Switch bit 0: CANFD frame to transmit with no bit rate switch 1: CANFD frame to transmit with bit rate switch	R/W
2	TMFDF*1	CAN FD Format bit 0: Non CANFD frame to transmit 1: CANFD frame to transmit	R/W
7:3	—	The read values are undefined. The write value should be 0.	R/W
9:8	TMIFL[1:0]	TX Message Buffer Information Label Field	R/W
15:10	—	The read values are undefined. The write value should be 0.	R/W
31:16	TMPTR[15:0]	TX Message Buffer Pointer Field	R/W

Note: S-TYPE-3, P-TYPE-3

Note 1. This bit is not available in the classical CAN function.

The TX Message Buffer CANFD Control Registers b (b = 0 to 3) show the status of the FDF, BRS and ESI bits, including the pointer fields of the CANFD frame to be transmitted.

TMESI bit (Error State Indicator bit)

If the channel is not in error passive, then the TMESI bit equals the write value, otherwise it is a don’t care, and the bit is transmitted as 1 on the CAN bus, indicating this is an error passive node.

Do not write to the TMESI bit when the related CANFD channel is in CH_SLEEP mode.

Note: This bit is not available in the classical CAN function.

TMBRS bit (Bit Rate Switch bit)

Do not write to the TMBRS bit when the related CANFD channel is in CH_SLEEP mode.

Note: This bit is not available in the classical CAN function.

TMFDF bit (CAN FD Format bit)

Do not write to the TMFDF bit when the related CANFD channel is in CH_SLEEP mode.

Note: This bit is not available in the classical CAN function.

TMIFL[1:0] bits (TX Message Buffer Information Label Field)

The TMIFL[1:0] bits store the information label value to be copied, together with additional message information, in the TX History List after successful transmission of the message.

Do not write to these bits when the related CANFD channel is in CH_SLEEP mode.

TMPTR[15:0] bits (TX Message Buffer Pointer Field)

The TMPTR[15:0] bits store the pointer value to be copied, together with additional message information in the TX History List after successful transmission of the message.

Do not write to these bits when the related CANFD channel is in CH_SLEEP mode.

42.2.61.21 CFDTMDFb_p : TX Message Buffer Data Field Register (p = 0 to 15, b = 0 to 3)

Base address: CANFDn = 0x4038_0000 + 0x2000 × n (n = 0, 1)
CANFDn_NS = 0x5038_0000 + 0x2000 × n (n = 0, 1)

Offset address: 0x0610 + 0x004 × p + 0x004C × b

Bit position:	31–24	23–16	15–8	7–0
Bit field:	TMDB_HH[7:0]	TMDB_HL[7:0]	TMDB_LH[7:0]	TMDB_LL[7:0]
Value after reset:	0 0 0 0 0 0 0 0	0 0 0 0 0 0 0 0	0 0 0 0 0 0 0 0	0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
7:0	TMDB_LL[7:0]	TX Message Buffer Data Byte (p × 4)	R/W
15:8	TMDB_LH[7:0]	TX Message Buffer Data Byte ((p × 4) + 1)	R/W
23:16	TMDB_HL[7:0]	TX Message Buffer Data Byte ((p × 4) + 2)	R/W
31:24	TMDB_HH[7:0]	TX Message Buffer Data Byte ((p × 4) + 3)	R/W

Note: S-TYPE-3, P-TYPE-3

Each TX Message Buffer Data Field p Register b (p = 0 to 15, b = 0 to 3) is used to store data bytes (p × 4) to data bytes ((p × 4) + 3) of the message to transmit from the associated buffer.

TMDB_LL[7:0] bits (TX Message Buffer Data Byte (p × 4))

TMDB_LL[7:0] bits store data bytes (p × 4) of the message in the TX message buffer.

Do not write to these bits when the related CANFD channel is in CH_SLEEP mode.

TMDB_LH[7:0] bits (TX Message Buffer Data Byte ((p × 4) + 1))

TMDB_LH[7:0] bits store data bytes ((p × 4) + 1) of the message in the TX message buffer.

Do not write to these bits when the related CANFD channel is in CH_SLEEP mode.

TMDB_HL[7:0] bits (TX Message Buffer Data Byte ((p × 4) + 2))

TMDB_HL[7:0] bits store data bytes ((p × 4) + 2) of the message in the TX message buffer.

Do not write to these bits when the related CANFD channel is in CH_SLEEP mode.

TMDB_HH[7:0] bits (TX Message Buffer Data Byte ((p × 4) + 3))

TMDB_HH[7:0] bits store data bytes ((p × 4) + 3) of the message in the TX message buffer.

Do not write to these bits when the related CANFD channel is in CH_SLEEP mode.

42.3 Modes of Operation

42.3.1 Overview

The modes of the CANFD module can be classified into 2 groups:

- Global modes
- Channel modes

42.3.2 Global Modes

These modes are applicable for the complete CANFD module and therefore are called Global modes. The global modes of the CANFD module are:

- Global Sleep
- Global Reset
- Global Halt

- Global Operation.

Figure 42.2 shows the possible transitions between the Global modes.

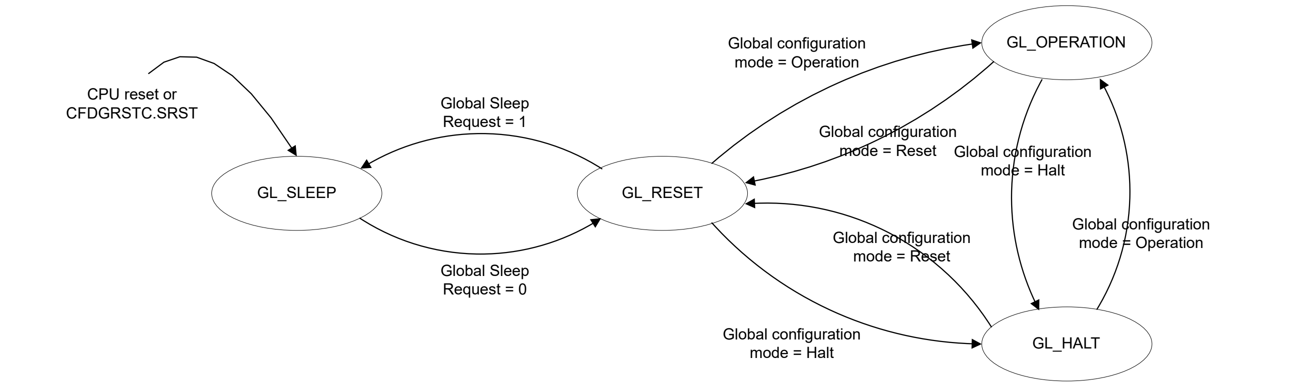


Figure 42.2 Transition between CANFD Global modes

Change in the Global mode can affect the Channel mode. Table 42.13 shows the effect of a Global mode transition on a Channel mode.

Table 42.13 Possible CANFD Channel modes and Global modes

Current Global mode	Target Global mode			
	Sleep (GL_SLEEP)	**Reset (GL_RESET)**	**Halt (GL_HALT)**	**Operation (GL_OPERATION)**
Sleep (GL_SLEEP)		Ch-Sleep: Keep Ch-Reset: N/A Ch-Halt: N/A Ch-Oper: N/A		
Reset (GL_RESET)	Ch-Sleep: Keep Ch-Reset: → Ch-Sleep Ch-Halt: N/A Ch-Oper: N/A		Ch-Sleep: Keep Ch-Reset: Keep Ch-Halt: N/A Ch-Oper: N/A	Ch-Sleep: Keep Ch-Reset: Keep Ch-Halt: N/A Ch-Oper: N/A
Halt (GL_HALT)		Ch-Sleep: Keep Ch-Reset: Keep Ch-Halt: → Ch-Reset Ch-Oper: N/A		Ch-Sleep:Keep Ch-Reset: Keep Ch-Halt: Keep Ch-Oper: N/A
Operation (GL_OPERATION)		Ch-Sleep: Keep Ch-Reset: Keep Ch-Halt: → Ch-Reset Ch-Oper: → Ch-Reset	Ch-Sleep: Keep Ch-Reset: Keep Ch-Halt: Keep Ch-Oper: → Ch-Halt	

42.3.2.1 Global Sleep Mode

After the release of a hardware reset or after setting and clearing a CFDGRSTC.SRST bit, the CANFD module automatically enters Global Sleep mode.

The CANFD module also enters the Global Sleep mode when the Global Sleep Request bit is set while it is in Global Reset mode. This control bit cannot be set in Global Halt mode or Global Operation mode.

Setting the Global Sleep Request bit sets Channel Sleep Request bit and forces the channel into the Channel Sleep mode.

Sleep mode is used for power saving purpose. When CANFD module is in Global Sleep mode, only the clock for CPU write access to the Global Sleep Mode Request bit is active. All other clocks are stopped and all other functions of the CANFD module are suspended.

Read access from all registers is still possible and all register values are preserved.

After setting the Global Sleep Request bit, it is necessary to confirm that the Global Sleep status has been updated, indicating successful transition to Global Sleep mode before the Global Sleep Request bit can be cleared again.

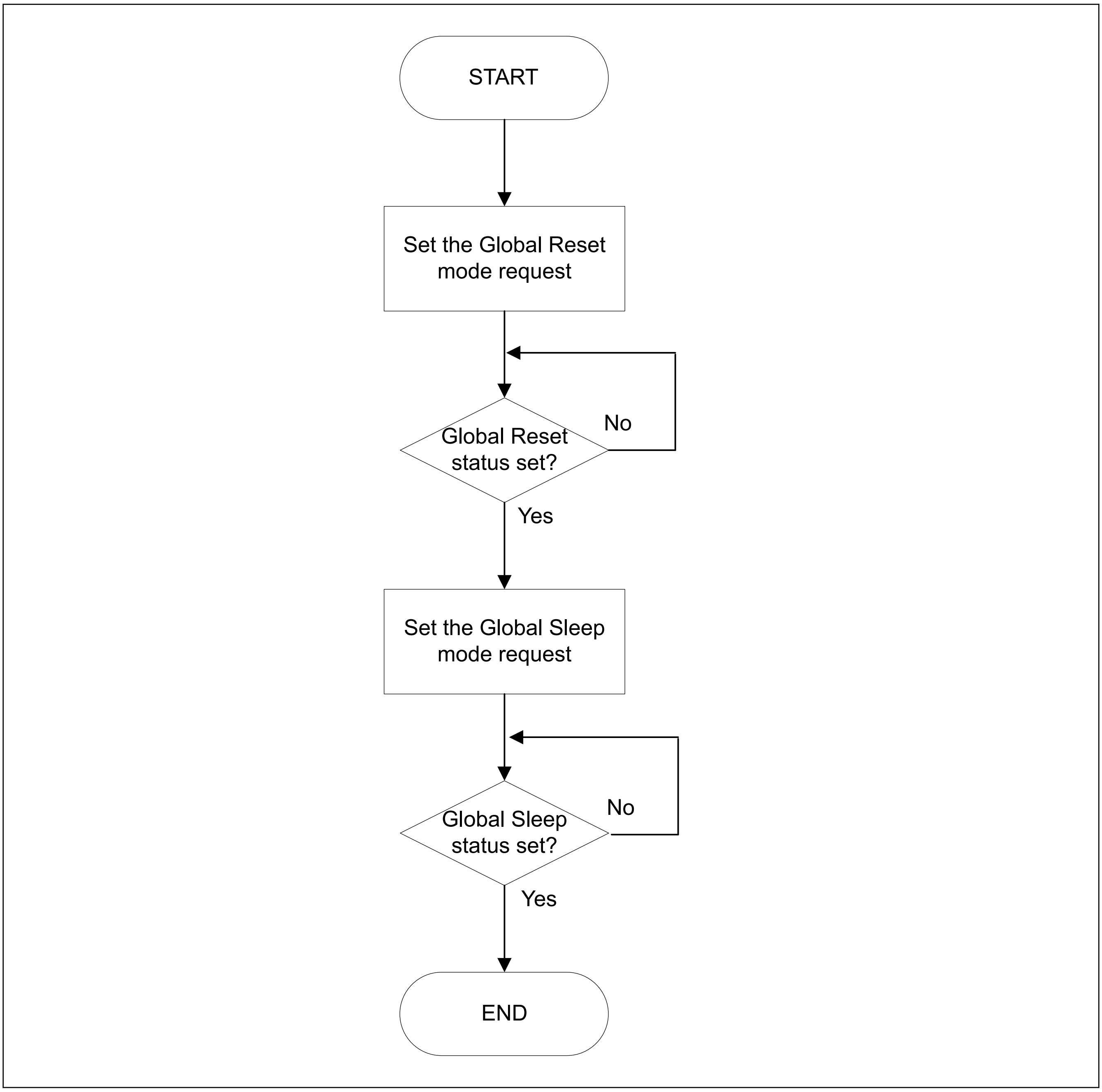


Figure 42.3 Procedure for entering Global Sleep mode

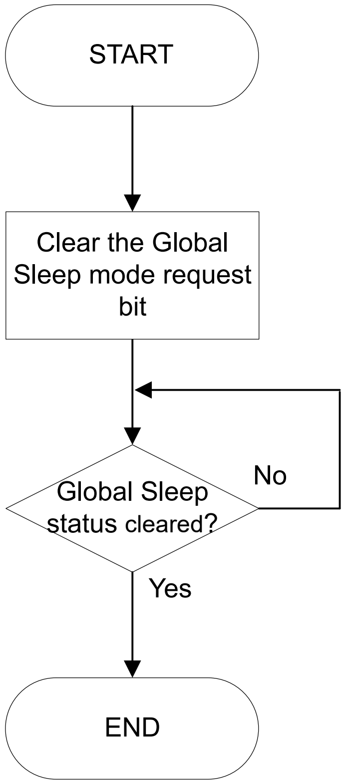


Figure 42.4 Procedure for exiting Global Sleep mode

42.3.2.2 Global Reset Mode

The CANFD module enters this mode in the following ways:

- Global Mode Control bit CFDGCTR.GMDC in the Global Control Register is configured for Global Reset mode while the CANFD module is in Global Halt or Global Operation mode
- Global Sleep Mode Request bit is cleared while CANFD module is in Global Sleep mode.

In Global Reset mode, all CANFD module functions are suspended and all status and flag registers are initialized.

Additionally, all FIFOs and TX Queues are disabled and transmission control bits are cleared.

Configuration registers (except the test mode registers) are not initialized in this mode to their MCU reset values and the CANFD module can be configured.

See section 42.3.4. Global Mode and Channel Mode Transition Interactions for a detailed description of the behavior of all registers when transition to Global Reset mode is performed.

Setting the Global mode to Reset by setting the Global Mode Control bits CFDGCTR.GMDC in the Global Control Register to 01b sets Channel Mode Control bits CFDC0CTR.CHMDC in the Channel Control Registers to 01b and forces the channel into the Channel Reset mode.

For channels that are already in Channel Reset mode or Channel Sleep mode, this automatic transition is not performed (CFDC0CTR.CHMDC of related channel already set to 01b).

After setting Global Mode Control bit CFDGCTR.GMDC to Reset mode, it is necessary to confirm that the Reset Mode Status bit CFDGSTS.GRSTSTS in the Global Status Register has been updated, indicating successful transition to Global Reset mode before CFDGCTR.GMDC can be changed again.

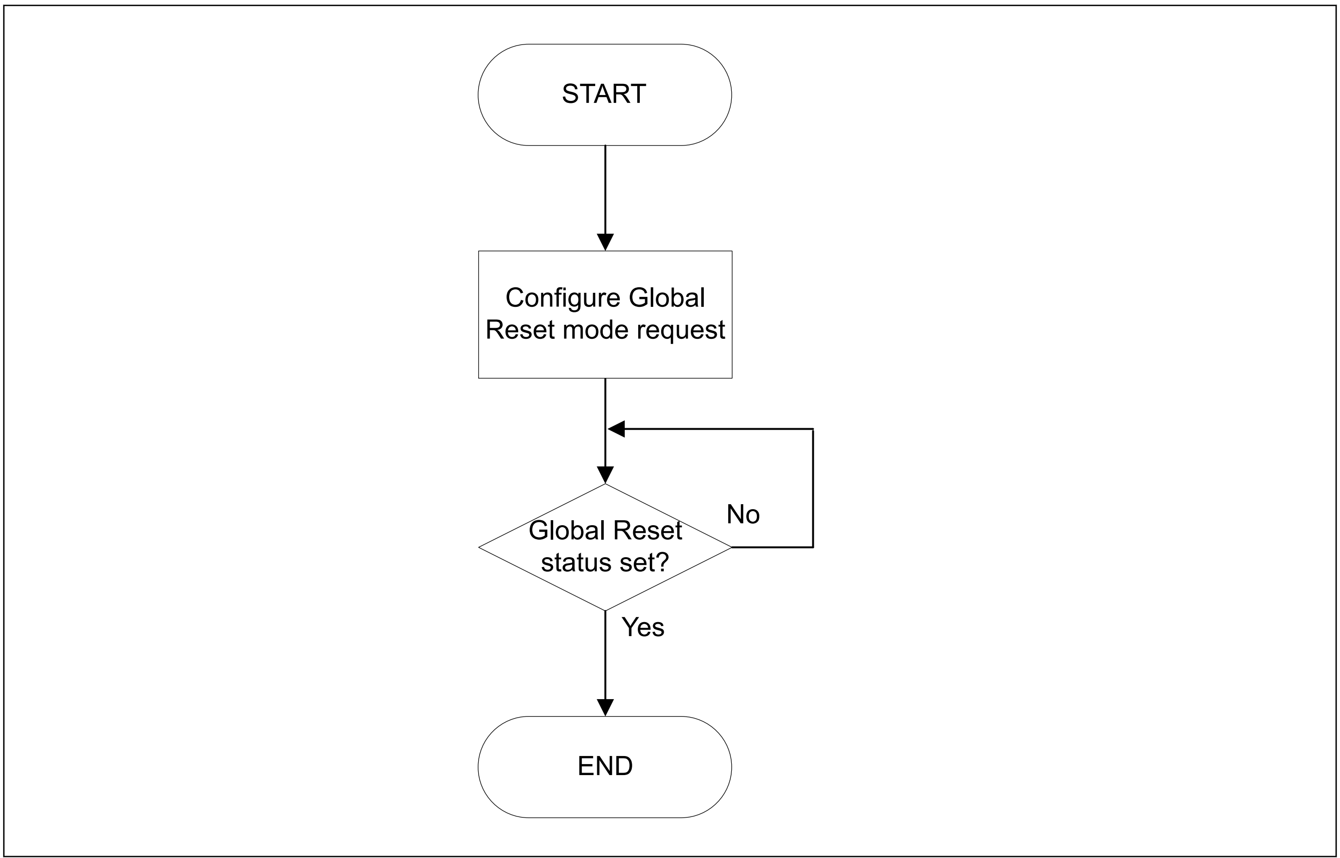


Figure 42.5 Procedure for entering Global Reset mode

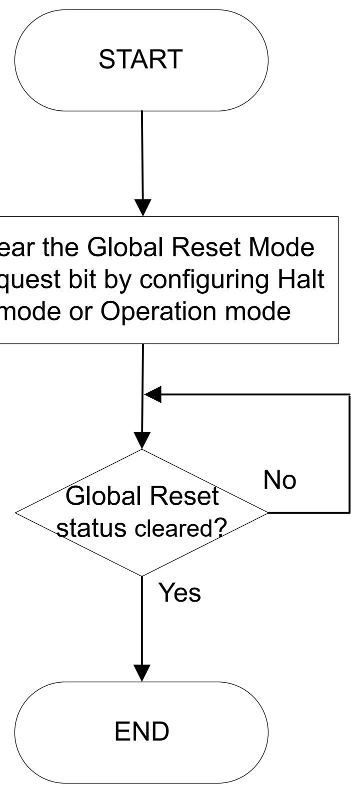


Figure 42.6 Procedure for exiting Global Reset mode

42.3.2.3 Global Halt Mode

The CANFD module enters this mode in the following ways:

- Global Mode Control bit CFDGCTR.GMDC in the Global Control Register is configured for Global Halt mode while the CANFD module is in Global Reset mode:
  - The channel in either Channel Reset or Channel Sleep mode remains in this mode
- Global Mode Control bit CFDGCTR.GMDC in the Global Control Register is configured for Global Halt mode while the CANFD module is in Global Operation mode:
  - The channel in Channel Reset, Channel Halt, or Channel Sleep mode remains in this mode
  - The channel in Channel Operation mode transitions to Channel Halt mode
  - Global Halt Mode Status bit is set when the channel has left Channel Operation mode.

If a transmission or reception is ongoing for a channel, the transition to Channel Halt mode is delayed until completion of the communication.

Similarly, if a channel is in bus-off, the full bus-off recovery sequence may be delayed depending on the channel configuration.

In Global Halt mode, all communications are suspended and CANFD logic does not cause any change to the Status and Flag registers (only when a channel is in the bus-off that its REC and TEC values are cleared). Additionally, the test mode configuration and control registers are not initialized in this mode.

The Global Halt mode should be used to configure global module test modes.

See section 42.3.4. Global Mode and Channel Mode Transition Interactions for a detailed description of the behavior of all registers when transition to Global Halt mode is performed.

Setting the Global mode to Halt by setting the Global Mode Control bit CFDGCTR.GMDC in the Global Control Register to 10b sets Channel Mode Control bits CFDC0CTR.CHMDC in the Channel Control Registers to 10b for the channel that are in Channel Operation mode and forces these channels into the Channel Halt mode.

For the channel that are already in Channel Reset, Channel Halt, or Channel Sleep mode, this automatic transition is not performed.

Therefore, the Global Halt mode request can be used to shut down all CANFD channel communications without loss of messages and disruption on the related CAN bus (no interruption of reception/transmission processes on the channels).

After setting the Global Mode Control bit CFDGCTR.GMDC to Halt mode, it is necessary to confirm that the Halt Mode Status bit CFDGSTS.GHLTSTS in the Global Status Register has been updated to indicate a successful transition to Global Halt mode. Do not specify any other SFR setting until confirming CFDGSTS.GHLTSTS is set.

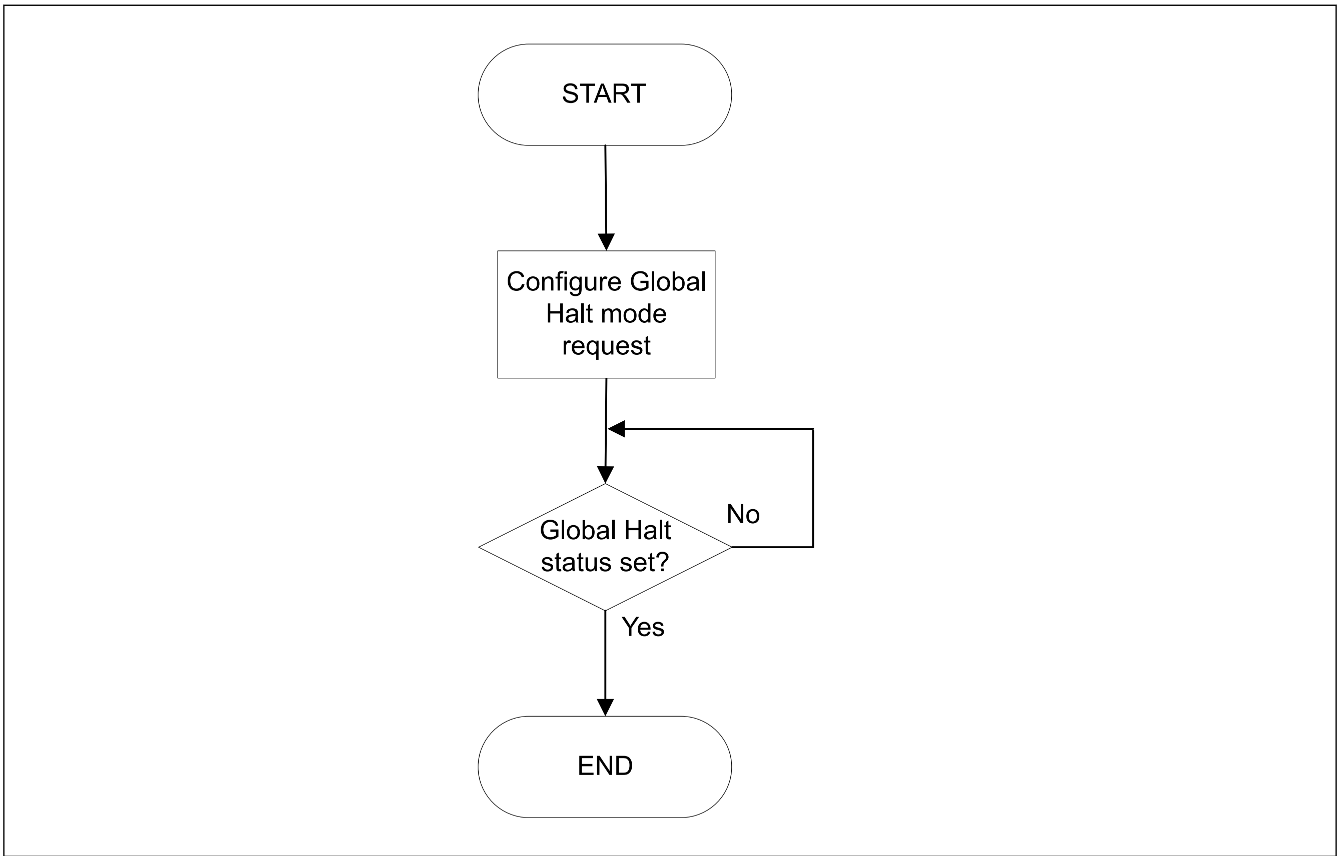


Figure 42.7 Procedure for entering Global Halt mode

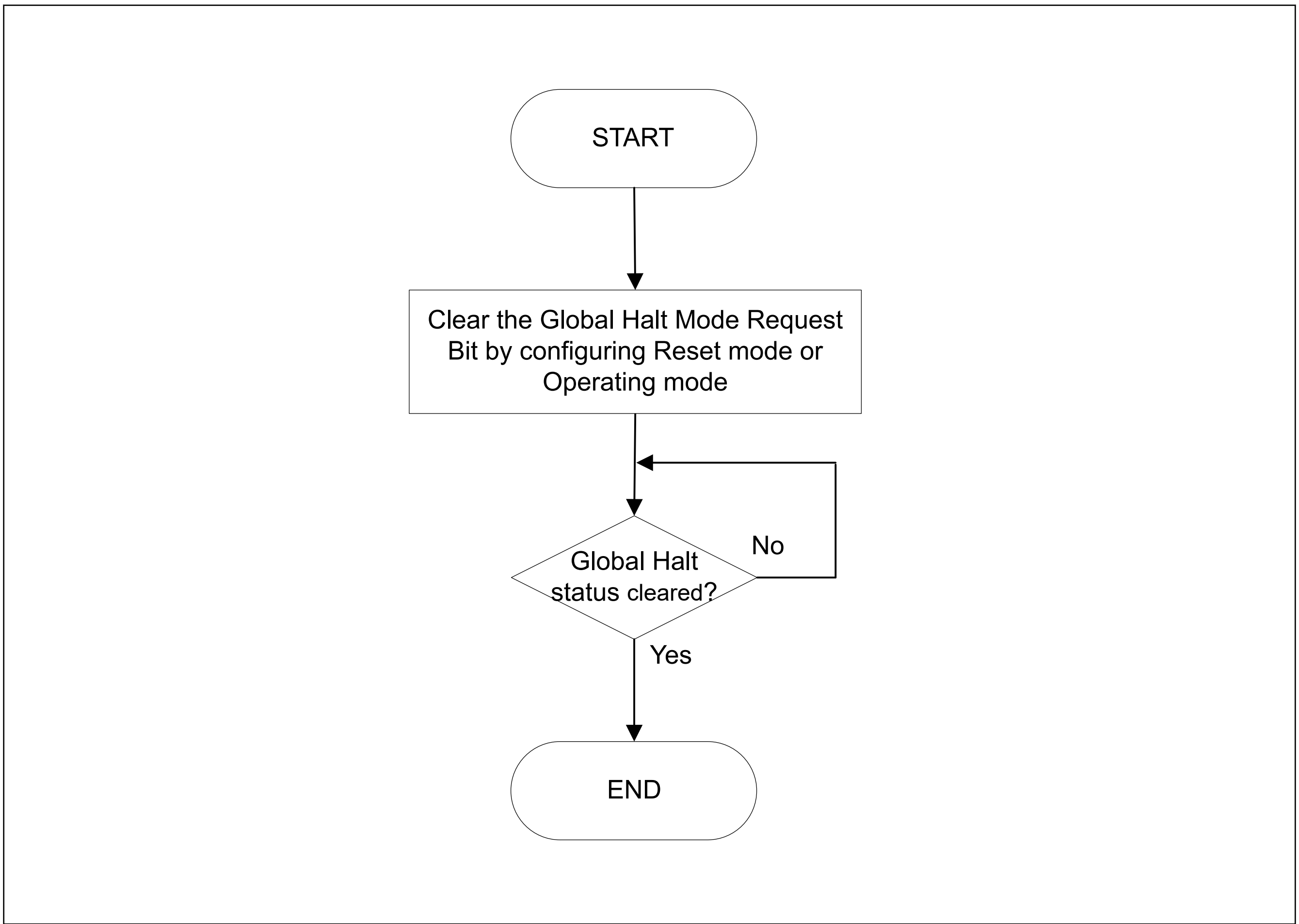


Figure 42.8 Procedure for exiting Global Halt mode

42.3.2.4 Global Operation Mode

The CANFD module enters this mode when the Global Mode Configuration bits are set to Global Operation mode.

The CANFD channel can only be set to Channel Operation mode and start CAN communication when CANFD is in Global Operation mode.

After setting the Global Mode Control bit CFDGCTR.GMDC to Global Operation mode, it is necessary to confirm that the Global Reset Mode Status bit CFDGSTS.GRSTSTS and the Global Halt Mode Status bit CFDGSTS.GHLTSTS in the Global Status Register have been cleared to indicate a successful transition to Global Operation mode before CFDGCTR.GMDC can be modified again.

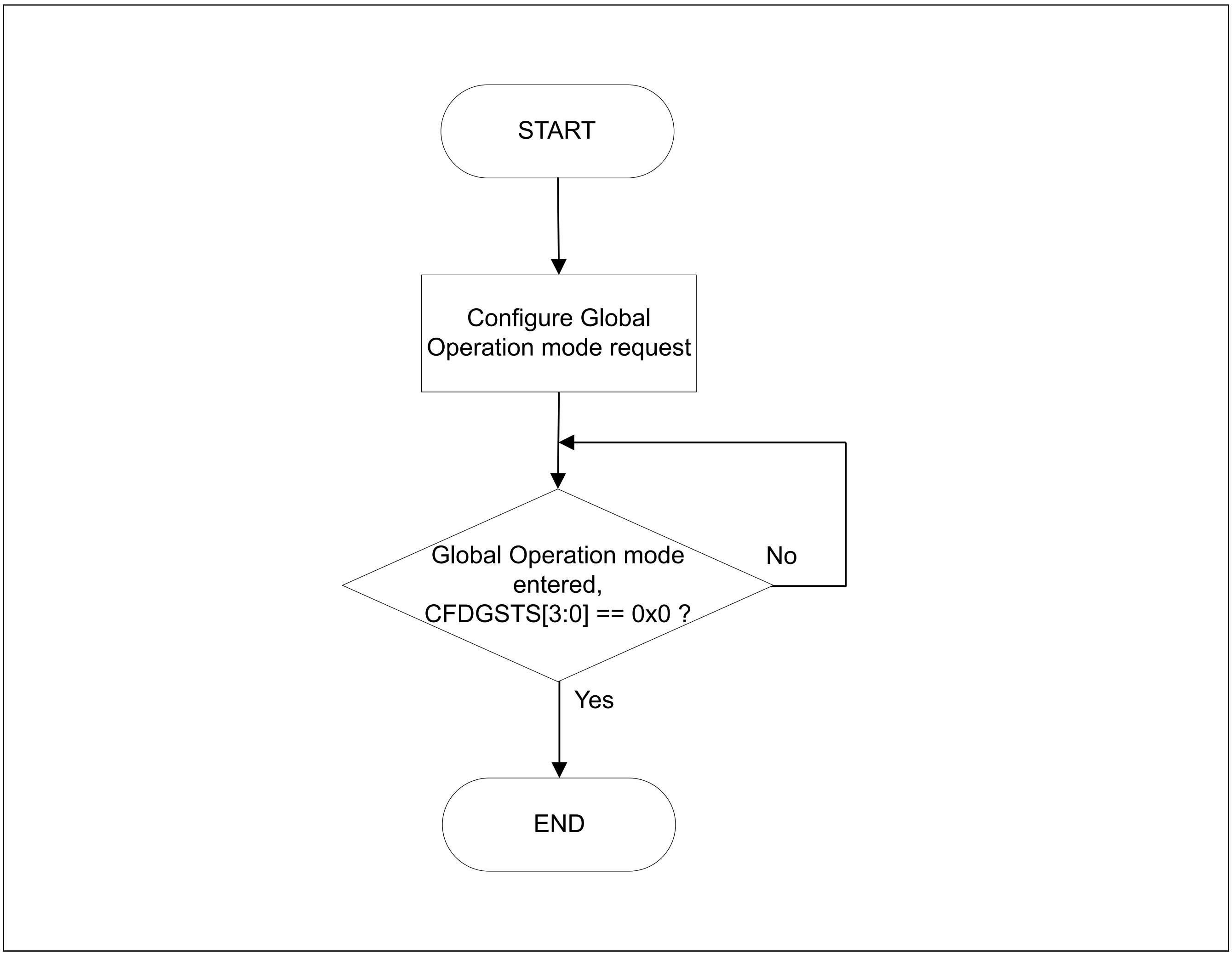


Figure 42.9 Procedure for entering Global Operation mode

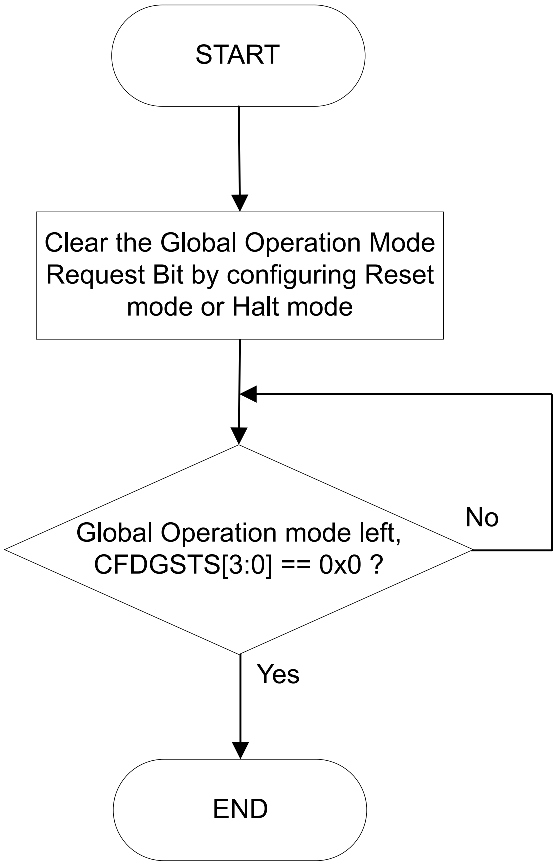


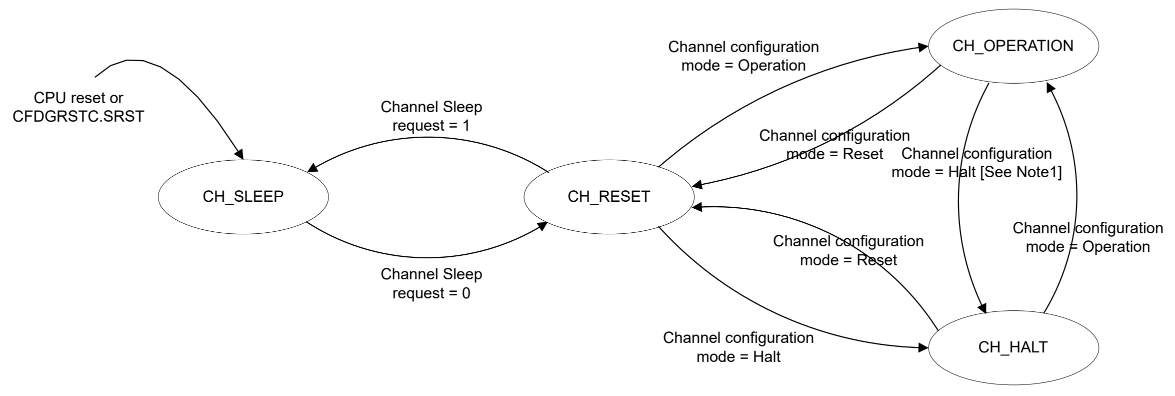
Figure 42.10 Procedure for exiting Global Operation mode

42.3.3 Channel Modes

A CAN channel can be in one of the following four channel modes:

- Reset
- Halt
- Operation
- Sleep.

Figure 42.11 shows the possible transitions between the channel modes.



Note: The transition timing depends on BOM setting in the Channel Configuration Register:

- when BOM = 01b, the state transition timing is immediate after entering bus-off state
- when BOM = 10b, the state transition timing is at the end of bus-off state
- when BOM = 11b, the state transition timing is matching with the setting of CHMDC to Halt mode.

Figure 42.11 Transition between CAN channel modes

42.3.3.1 CAN Channel Sleep Mode

After the release of a hardware reset or after setting and clearing the CFDGRSTC.SRST bit, a CAN channel of the CANFD module automatically enters Channel Sleep mode.

A CAN channel also enters Channel Sleep mode when the related Channel Sleep Mode Request bit is set while the CAN channel is in Channel Reset mode. Do not set this control bit in Channel Halt mode or Channel Operation mode.

Entering the CAN Channel Sleep mode instantly stops the clock supplied to the CAN channel unit and therefore reduces power consumption.

After setting the Channel Sleep Mode Request bit, it is necessary to confirm that the Channel Sleep mode status has been updated to indicate a successful transition to Channel Sleep mode before the Channel Sleep Mode Request bit can be cleared again.

During Channel Sleep mode, do not write to channel related registers. Read operation is still possible.

42.3.3.2 CAN Channel Reset Mode

A CANFD CAN channel enters this mode in the following ways:

- Channel Mode Control bit CFDC0CTR.CHMDC in the Channel Control Registers is configured for Channel Reset mode while the related CAN channel is in Channel Halt mode or Channel Operation mode
- Channel Sleep Mode Request bit is cleared while the related CAN channel is in Channel Sleep mode
- Global Mode Control bit CFDGCTR.GMDC is set to Global Reset mode and CAN channel is not in Channel Sleep mode or Channel Reset mode.

In Channel Reset mode, all CAN channel status and flag registers are initialized.

Additionally, all channel related transmission control bits are cleared and the channel related TX Queue is disabled.

Configuration registers (except the Channel Test Mode registers) are not initialized in this mode and the CAN channel can be configured for communication.

See section 42.3.4. Global Mode and Channel Mode Transition Interactions for a detailed description of the behavior of all registers when transition to Channel Reset mode is performed.

After setting the Channel Mode Control bit CFDC0CTR.CHMDC to Channel Reset mode, it is necessary to confirm that the Reset Mode Status bit CFDC0STS.CRSTSTS in the related Channel Status Registers has been updated to indicate a successful transition to Channel Reset mode before the related CFDC0CTR.CHMDC bit can be modified again.

See Table 42.14 for the behavior of transitioning to Channel Reset mode while CAN communication is ongoing.

42.3.3.3 CAN Channel Halt Mode

A CANFD CAN channel enters this mode in the following ways:

- Channel Mode Control bit CFDC0CTR.CHMDC in the Channel Control Registers is configured for Channel Halt mode while the related CAN channel is in Channel Reset mode or Channel Operation mode
- Global Mode Control bit CFDGCTR.GMDC is set to Global Halt mode and CAN channel is in Channel Operation mode.

In Channel Halt mode, all channel CAN communication is suspended but all status and flag registers remain unchanged during Channel Halt mode entry (except for the bus-off case where REC and TEC values are cleared for this channel).

In addition, the Channel Test Mode Configuration and Control registers are not initialized in this mode.

The Channel Halt mode should be used to configure channel test modes.

See section 42.3.4. Global Mode and Channel Mode Transition Interactions for a detailed description of the behavior of all registers when transition to Channel Halt mode is performed.

After setting the Channel Mode Control bit CFDC0CTR.CHMDC to Channel Halt mode, it is necessary to confirm that the Halt Mode Status bit CFDC0STS.CHLTSTS in the related Channel Status Register has been updated to indicate a successful transition to Channel Halt mode before the related CFDC0CTR.CHMDC can be modified again.

See Table 42.14 for the transition behavior to Channel Halt mode while CAN communication is ongoing.

Table 42.14 Transition behavior in CAN Reset mode and Halt mode

Mode	State		
	Receiver	**Transmitter**	**Bus-Off**
CAN Channel Reset mode (CFDC0CTR.CHMDC = 01b)	The CAN channel enters Channel Reset mode without waiting for the completion of the ongoing reception.*1	The CAN channel enters Channel Reset mode without waiting for the completion of the ongoing transmission.*1	The CAN channel enters Channel Reset mode without waiting for the completion of the bus-off recovery.
CAN Channel Halt mode (CFDC0CTR.CHMDC = 10b)	CAN channel enters Channel Halt mode at the end of the ongoing reception or error.*2	CAN channel enters Channel Halt mode after completion of the ongoing transmission.	When CFDC0CTR.BOM is set to 00b, a Channel Halt mode request is accepted only after the completion of the full bus-off recovery sequence. When CFDC0CTR.BOM is set to 10b, the CAN channel transits automatically to Channel Halt mode after waiting for the completion of the bus-off recovery. When CFDC0CTR.BOM is set to 01b, the CAN channel transits automatically to Channel Halt mode without waiting for the completion of the bus-off recovery. When CFDC0CTR.BOM is set to 11b, the CAN channel enters Channel Halt mode as soon as Channel Halt mode is requested (without waiting for the completion of the bus-off recovery).

Note 1. If the entry to Channel Reset mode is required only at the end of an ongoing communication, then Channel Halt mode can be requested first to prevent interruption of CAN communication by direct transition to Channel Reset mode. After the CAN channel enters Channel Halt mode, the Channel Reset mode can be requested.

Note 2. If CAN communication is locked at dominant level after an error flag, software can detect this situation by monitoring the channel related BusLock flag and resolve lock condition by setting the CAN channel to Channel Reset mode.

42.3.3.4 CAN Channel Operation Mode

The Channel Operation mode is activated by setting the CFDC0CTR.CHMDC bits to 00b. If 11 consecutive recessive bits are detected after entering the CAN Operation mode, the CFDC0STS.COMSTS bit is set and the CAN channel:

- Enables the functions of the channel communication by allowing the channel to become an active node on the CAN network
- Releases the internal fault confinement logic including receive and transmit error counters

At this point, the CAN channel can start transmission and reception of CAN messages.

Within the CAN Channel Operation mode, the channel may be in four different sub-modes, depending on which type of communication functions are performed (see Figure 42.12):

- Channel idle: The CAN channel is neither receiving nor transmitting
- Channel receives: The channel is receiving a CAN message sent by another CAN node
- Channel transmits: The channel is transmitting a CAN message

  Note: The channel may receive its own message simultaneously when Self-test mode is enabled.
- Channel is in bus-off state: The CAN channel is cut off from CAN bus communication.

After setting the Channel Mode Control bit CFDC0CTR.CHMDC to Channel Operation mode, it is necessary to confirm that the Channel Reset Mode Status bit CFDC0STS.CRSTSTS and the Channel Halt Mode Status bit CFDC0STS.CHLTSTS in the Channel Status Register have been updated to indicate a successful transition to Channel Operation mode before the related CFDC0CTR.CHMDC bit can be changed again.

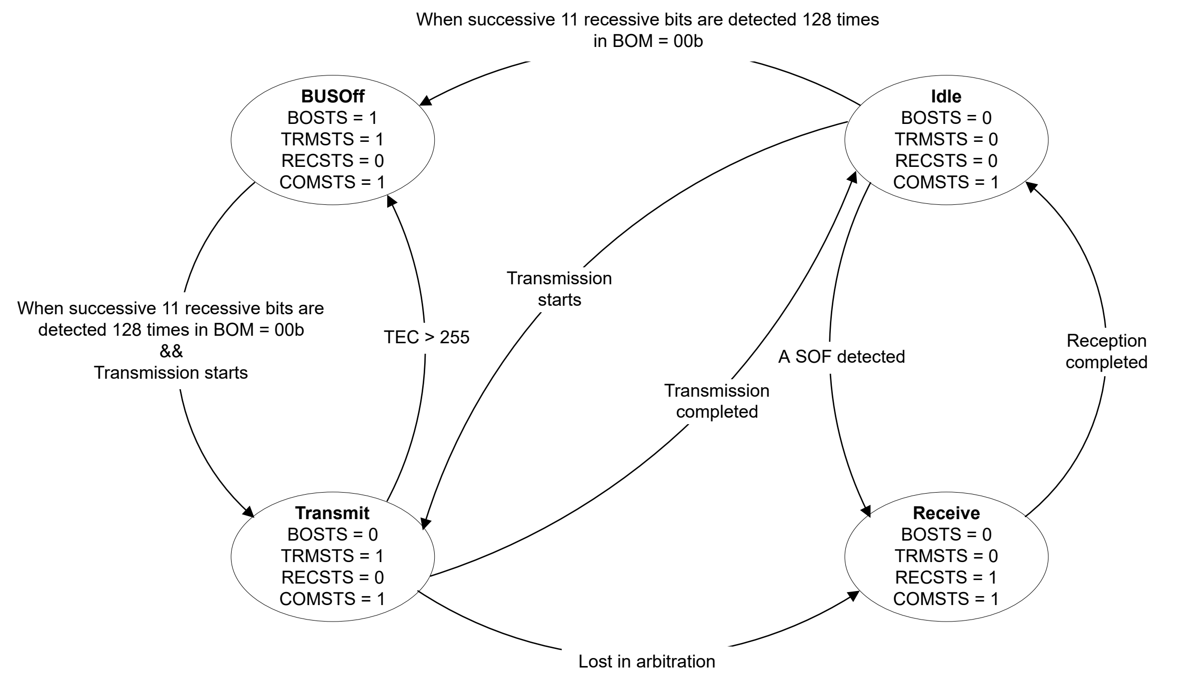


Figure 42.12 Sub-modes of CAN Channel Operation mode (only when BOM = 00b)

42.3.3.5 CAN Channel Bus-Off State

The CAN channel bus-off state is entered according to the fault confinement rules of the CAN specification. The following modes can be configured for returning to the CAN Channel Operation mode from the bus-off state:

- CFDC0CTR.BOM = 00b:
  Bus-Off recovery is compliant to ISO 11898-1, namely the CAN channel re-enters CAN communication (error active state) after 11 consecutive recessive bits are detected 128 times. TEC and REC counters are initialized to 0. The Bus-Off Recovery Flag CFDC0ERFL.BORF is set in this case.
- CFDC0CTR.BOM = 01b:

The CAN channel changes the value of the CFDC0CTR.CHMDC bits within the CAN Channel Control Register to 10b and switches immediately to Channel Halt mode automatically after entering bus-off state. TEC and REC counters are initialized to 0 and the Bus-Off Recovery Flag CFDC0ERFL.BORF is not set in this case.

- CFDC0CTR.BOM = 10b:
  The CAN channel changes the value of the CFDC0CTR.CHMDC bits within the CAN Channel Control Register to 10b as soon as it reaches bus-off state and enters Channel Halt mode automatically after the CAN channel has completed the bus-off recovery sequence (after 11 consecutive recessive bits are detected 128 times). TEC and REC counters are initialized to 0 and the Bus-Off Recovery Flag CFDC0ERFL.BORF is set in this case.
- CFDC0CTR.BOM = 11b:
  Bus-off recovery is initiated but CAN channel can immediately enter Channel Halt mode when still in bus-off state if a request is made to enter Channel Halt mode.
  TEC and REC counters are initialized to 0 and the Bus-Off Recovery Flag CFDC0ERFL.BORF is not set.
  Without setting CFDC0CTR.CHMDC [1:0] = 10b and when 11 recessive bits are detected 128 times continuously, transition conditions become the same as CFDC0CTR.BOM = 00b.

Note: If the recovery from bus-off occurs normally in this mode (after waiting for 128 sequences of 11 consecutive recessive bits), and no halt request has been generated during this period, then the Bus-Off Recovery flag CFDC0ERFL.BORF is set.

When software writes to the CFDC0CTR.CHMDC bit at the same time as the CAN channel enters Halt mode (at the start of bus-off when CFDC0CTR.BOM = 01b, or at the end of bus-off when CFDC0CTR.BOM = 10b), the software request has the highest priority.

Note: In the above case, the automatic setting of the CFDC0CTR.CHMDC bit to Channel Halt mode request is performed when the CFDC0CTR.CHMDC bit value is previously 00b (Channel Operation mode).

Additionally, it is possible to force the CAN channel to recover from the bus-off state by setting CFDC0CTR.RTBO to 1. The error state changes from bus-off state to integrating state with a maximum delay of 1 CAN bit time, and the CAN communication becomes possible again after 11 consecutive recessive bits are detected. The Bus-Off Recovery Flag is not set in this case, and the TEC and REC counters are initialized to 0.

Before setting CFDC0CTR.RTBO to 1, all pending transmissions from the TX message buffers, TX Queues and/or Common FIFO in TX mode should be disabled.

The disable of the pending transmission message buffer, TX Queue or FIFO must be confirmed by the corresponding acknowledge flags.

For the TX message buffer, the acknowledge flags are the Transmission Result Flags (CFDTMSTSj.TMTRF). For the TX Queue, it is the TX Queue Empty flag (CFDTXQSTS.TXQEMP). For the FIFO, it is the FIFO Empty flag (CFDCFSTS.CFEMP).

The CFDC0CTR.RTBO bit should be used for bus-off recovery only when CFDC0CTR.BOM is set to 00b.

Setting this bit in any state other than bus-off has no effect and the bit is cleared immediately.

Table 42.15 shows the settings for the Bus-Off Entry flag CFDC0ERFL.BOEF and the Bus-Off Recovery flag CFDC0ERFL.BORF for the different configurations of CFDC0CTR.BOM.

Table 42.15 Behavior of Bus-off Entry and Recovery flags

BOM	BOEF bit set	BORF bit set
00b	Always (on entry to bus-off)	Always (on exit from bus-off)
00b CFDC0CTR.RTBO set to 1	Always (on entry to bus-off)	Only if normal bus-off recovery occurs before software sets CFDC0CTR.RTBO to 1'
01b	Always (on entry to bus-off)	Never
10b	Always (on entry to bus-off)	Always (on exit from bus-off)
11b	Always (on entry to bus-off)	Only if normal bus-off recovery occurs before software issues a Halt request

For an efficient software procedure, it is not necessary to wait for the bus-off recovery sequence to end.

It is possible to perform a transmission re-initialization during bus-off recovery. To do this, follow the recommended software flow in Figure 42.13.

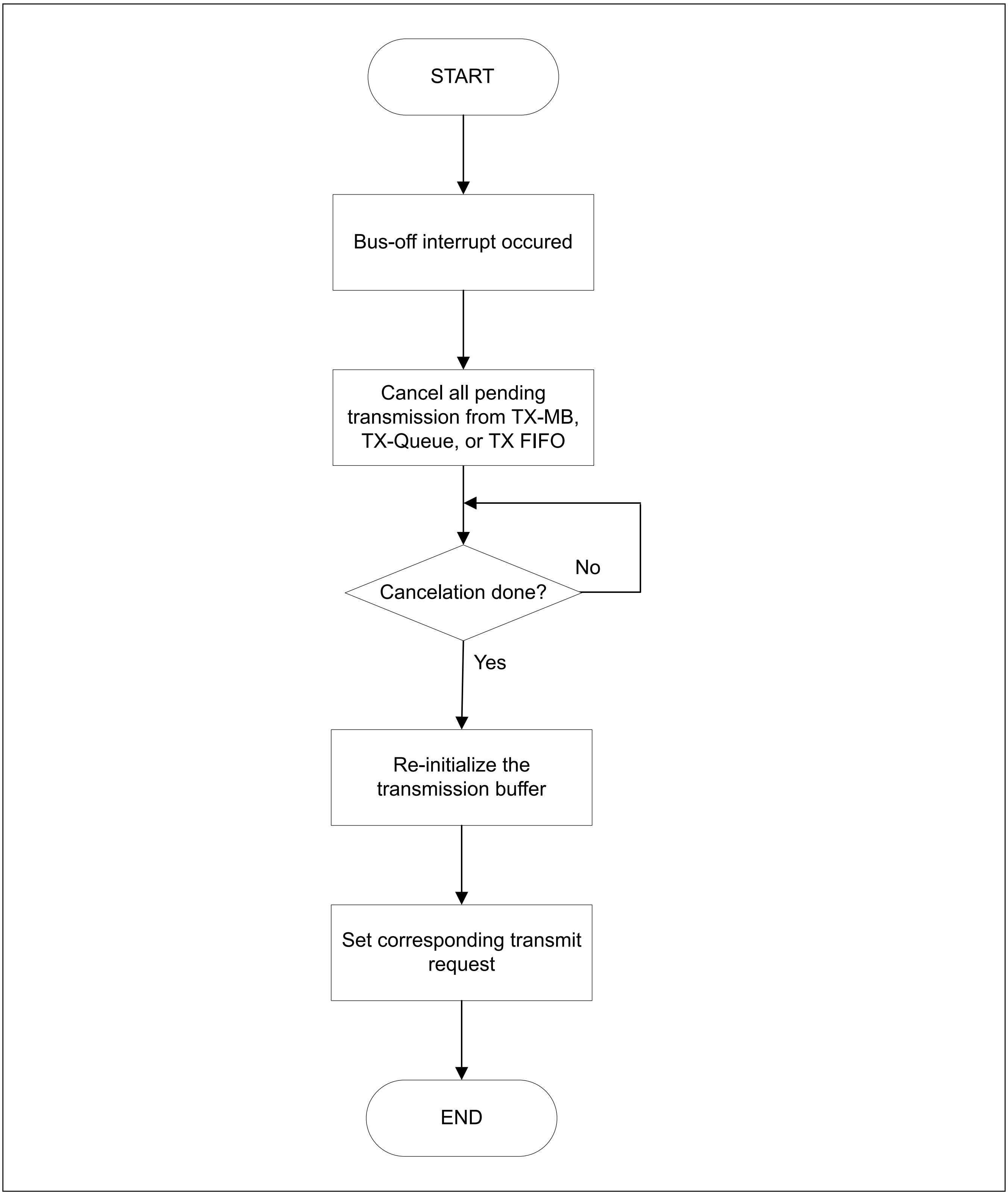


Figure 42.13 Transmission re-initialization during bus-off

42.3.4 Global Mode and Channel Mode Transition Interactions

The interaction between Global mode setting and Channel mode setting is as follows:

- Changing the Channel Mode Control bit CFDC0CTR.CHMDC in the Channel Control Registers does not affect the Global Mode Control bit CFDGCTR.GMDC.
- Changing the Global Mode Control bit CFDGCTR.GMDC affects the channel mode control as described in Table 42.16.

Table 42.16 Interaction between Global and Channel mode transition

Global mode change	Channel mode	Channel mode transition action
Sleep → Reset	Sleep	Channel remains in Sleep mode
Sleep → Halt	— (Global mode change not possible)	
Sleep → Operation	— (Global mode change not possible)	
Reset → Sleep	Sleep	Channel remains in Sleep mode
	Reset	Channel Sleep request bit is set automatically, channel enters Sleep Mode
Reset → Halt	Sleep	Channel remains in Sleep mode
	Reset	Channel remains in Reset mode
Reset → Operation	Sleep	Channel remains in Sleep mode
	Reset	Channel remains in Reset mode
Halt → Sleep	— (Global mode change not possible)	
Halt → Reset	Sleep	Channel remains in Sleep mode
	Reset	Channel remains in Reset mode
	Halt	Channel mode control is set to Reset mode, channel enters Reset mode
Halt → Operation	Sleep	Channel remains in Sleep mode
	Reset	Channel remains in Reset mode
	Halt	Channel remains in Halt mode
Operation → Sleep	— (Global mode change not possible)	
Operation → Reset	Sleep	Channel remains in Sleep mode
	Reset	Channel remains in Reset mode
	Halt	Channel mode control is set to Reset mode, channel enters Reset mode
	Operation	Channel mode control is set to Reset mode, channel enters Reset mode
Operation → Halt	Sleep	Channel remains in Sleep mode
	Reset	Channel remains in Reset mode
	Halt	Channel remains in Halt mode
	Operation	Channel mode control is set to Halt mode, channel enters Halt mode after communication finished

42.3.4.1 Timing of Global Mode Change

The transition time for the Global mode changes is shown in the following table.

Table 42.17 Maximum transition time for the global mode (1 of 2)

From	To	Maximum transition time
GL_SLEEP	GL_RESET	3 peripheral clock cycles*2
GL_RESET	GL_SLEEP	3 peripheral clock cycles
GL_RESET	GL_HALT	10 peripheral clock cycles
GL_RESET	GL_OPERATION	10 peripheral clock cycles

Table 42.17 Maximum transition time for the global mode (2 of 2)

From	To	Maximum transition time
GL_HALT	GL_RESET	2 CAN bit times
GL_HALT	GL_OPERATION	3 peripheral clock cycles
GL_OPERATION	GL_RESET	2 CAN bit times
GL_OPERATION	GL_HALT	3 CAN frames*1 *3

Note 1. The given transition time is the time without any errors on the bus. In case of an error condition, the transition time can lengthen to an uncalculated result. The transition time can also come to a stuck condition for locked RX lines or continued error conditions.
Note 2. Exit GL_SLEEP mode only when CFDGSTS.GRAMINIT is cleared.
Note 3. TQ, CAN frame and CAN bits are related to the individual channels. For the maximum transition time, the channel with the lowest baud rate must be used.

42.3.4.2 Timing of Channel Mode Change

The transition time for the Channel mode changes is shown in the following table.

Table 42.18 Maximum transition time for the channel mode

From	To	max. transition time
CH_SLEEP	CH_RESET	3 peripheral clock cycles
CH_RESET	CH_SLEEP	3 peripheral clock cycles
CH_RESET	CH_HALT	3 CAN bit times
CH_RESET	CH_OPERATION	4 CAN bit times
CH_HALT	CH_RESET	2 CAN bit times
CH_HALT	CH_OPERATION	4 CAN bit times*3
CH_OPERATION	CH_RESET	2 CAN bit times
CH_OPERATION	CH_HALT	2 CAN frames*1 *2

Note 1. The time specified for this transition does not include the case where channel enters bus-off state. For bus-off, the timing depends on the configuration of the CFDC0CTR.BOM[1:0] bits.
Note 2. The given transition time is the time without any errors on the bus. In case of an error condition, the transition time can lengthen to an uncalculated result. The transition time can also come to a stuck condition for locked RX lines or continued error conditions.
Note 3. In general, if the baud rate prescaler value CFDC0NCFG.NBRP is changed in CH_HALT mode, the transition time can deviate. The internal prescaler is a free running down counter that creates the TQ clock, and new BRP value is captured when the counter reaches the value 0.

42.4 Initialization

Before joining CAN communications, configure the following settings:

- Clock setting
- Bit timing setting (nominal and data rate)
- Baud Rate setting (nominal and data rate)
- CANFD setting
- Acceptance Filter setting (configuration of Global Acceptance Filter List)
- Reception, Transmission and GW-FIFO setting
- CAN Operation mode setting

42.4.1 Initialization of CAN Clock, Bit Timing and Baud Rate

42.4.1.1 Bit Timing Conditions

The following lines describe the composition of each segment and the restriction that apply to the segment setting.

1. Each segment setting
   SS = Fixed to 1 TQ

TSEG1 = See to (CFDC0NCFG) and (CFDC0DCFG)*1
TSEG2 = See to (CFDC0NCFG) and (CFDC0DCFG)*1
SJW = See to (CFDC0NCFG) and (CFDC0DCFG)*1
SS + TSEG1 + TSEG2 = 5 to 49 TQs for Data Bit Rate and 8 to 385 for Nominal Bit Rate

2. Restriction on TSEG1, TSEG2 and SJW
TSEG1(N) > TSEG2(N) ≥ SJW(N)
TSEG1(D) ≥ TSEG2(D) ≥ SJW(D)*1
When only classical frames are used, configure the bit fields TSEG1 and TSEG2 of CFDC0DCFG to valid values.

Note 1. This feature is not available in the classical CAN function.

Table 42.19 shows an example of how to set the bit timing to achieve the required Sample Point settings.

Table 42.19 Bit timing examples

1 bit	Set value (TQ)				Sample point*1(%)
	SS	TSEG1	TSEG2	SJW	
5TQ	1	2	2	1	60.00
8TQ	1	4	3	1	62.50
	1	5	2	1	75.00
10TQ	1	6	3	1	70.00
	1	7	2	1	80.00
12TQ	1	8	3	1	75.00
	1	9	2	1	83.33
15TQ	1	10	4	1	73.33
	1	11	3	1	80.00
16TQ	1	10	5	1	68.75
	1	11	4	1	75.00
20TQ	1	12	7	1	65.00
	1	13	6	1	70.00
24TQ	1	15	8	1	66.66
	1	16	7	1	70.83
50TQ	1	39	10	4	80.00

Note 1. Sample point (in case of 75%)

Figure 42.14 Sample point (in case of 75%)

42.4.1.2 CAN Bit Timing

In the CAN protocol, each bit in a communication frame is composed of three segments that can be configured individually for channel using the related CFDC0NCFG and CFDC0DCFG*1 registers.

Note 1. This register is not available in the classical CAN function.

Figure 42.15 shows the segment composition of a bit and the sample point in it.

Of these segments, the Time Segment 1 (TSEG1) and Time Segment 2 (TSEG2) are used to specify the position of the sample point, so that the timing at which each bit on the CAN bus is sampled can be altered by changing the values of these segments.

The minimum resolution for this timing is referred to as Time Quantum (TQ), which is determined by the clock frequency supplied to the CAN channel and the divide-by-N value of the baud rate prescaler (nominal and data rate).

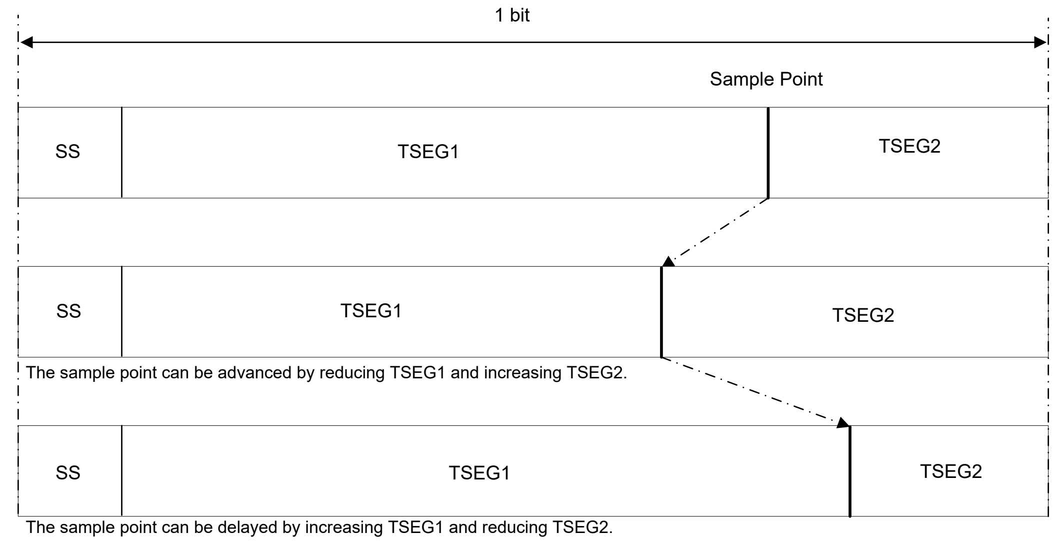


Figure 42.15 Segment composition of a bit and the sample point

1. SS: Synchronization Segment
   This segment is used to synchronize bits by monitoring a recessive-to-dominant edge during the interframe space. This consists of intermission, suspend transmission, bus idle, during bus idle, and all nodes that can start transmission.
2. TSEG1: Time Segment 1
   This segment absorbs physical delays on the CAN network. A physical delay on the network is two times the total sum of a bus delay, input comparator delay, and output driver delay. It can be lengthened by SJW.
3. TSEG2: Time Segment 2
   This segment is used to correct a phase error by performing resynchronization. It can be shortened by SJW. While sending or receiving a message, communication frames between some nodes may get out of sync due to a drift in the oscillator frequency or a delay in the transmission path. This is referred to as a phase error.
4. SJW: Resynchronization Jump Width
   This is the maximum width by which bits that have become out of sync due to a phase error may be corrected.

Figure 42.15 shows only one symbolic sample point.

42.4.1.3 Baud Rate

Either the CANFD core clock (CANFDCLK) or the external oscillator clock (CANMCLK) can be selected globally as CAN communication clock.

The transfer speed is determined by the DLL clock, the divide-by-N value of the baud rate prescaler, and the number of TQs in one bit.

$$\text{baud rate} = \frac{\text{DLL\_Clock}}{(\text{number\_of\_time\_quanta\_per\_bit}) \times (\text{BRP} + 1)}$$

Figure 42.16 shows a block diagram of the circuit that generates the CAN channel communication clock and Table 42.20 shows a baud rate examples.

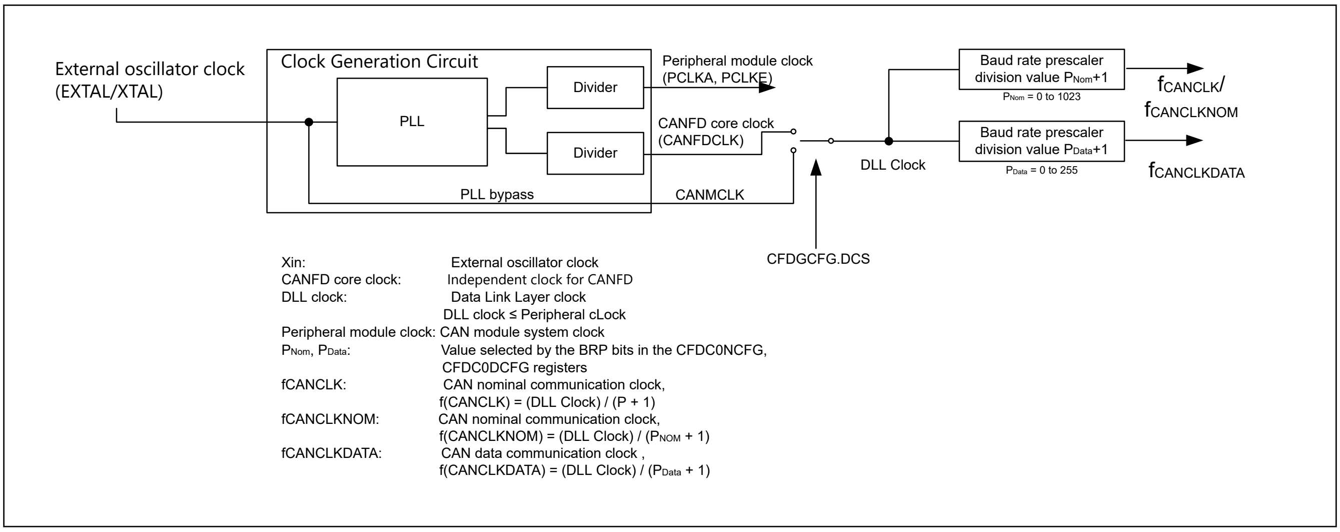


Figure 42.16 Block diagram of the circuit that generates the CAN channel communication clock

Table 42.20 Nominal baud rate calculation formula and example CAN communication configurations

Baud rate calculation formula	(DLL clock) (baud rate prescaler divide-by-N value*1) × (number of TQs in one bit)							
	40 MHz	32 MHz	30 MHz	24 MHz	20 MHz	16 MHz	10 MHz	8 MHz*2
1 Mbps	8TQ (5) 20TQ (2)	8TQ (4) 16TQ (2)	10TQ (3) 15TQ (2)	8TQ (3) 12TQ (2) 24TQ (1)	10TQ (2) 20TQ (1)	8TQ (2) 16TQ (1)	10TQ (1)	8TQ (1)
500 Kbps	8TQ (10) 20TQ (4)	8TQ (8) 16TQ (4)	10TQ (6) 15TQ (4) 20TQ (3)	8TQ (6) 12TQ (4) 24TQ (2)	10TQ (4) 20TQ (2)	8TQ (4) 16TQ (2)	10TQ (2) 20TQ (1)	8TQ (2) 16TQ (1)
250 Kbps	8TQ (20) 20TQ (8)	8TQ (16) 16TQ (8)	10TQ (12) 15TQ (8) 20TQ (6)	8TQ (12) 12TQ (8) 24TQ (4)	10TQ (8) 20TQ (4)	8TQ (8) 16TQ (4)	10TQ (4) 20TQ (2)	8TQ (4) 16TQ (2)
125 Kbps	8TQ (40) 20TQ (16)	8TQ (32) 16TQ (16)	10TQ (24) 15TQ (16) 20TQ (12)	8TQ (24) 12TQ (16) 24TQ (8)	10TQ (16) 20TQ (8)	8TQ (16) 16TQ (8)	10TQ (8) 20TQ (4)	8TQ (8) 16TQ (4)
83.3 Kbps	8TQ (60) 12TQ (40) 16TQ (30) 24TQ (20)	8TQ (48) 12TQ (32) 16TQ (24) 24TQ (16)	8TQ (45) 10TQ (36) 12TQ (30) 15TQ (24) 20TQ (18) 24TQ (15)	8TQ (36) 12TQ (24) 16TQ (18) 24TQ (12)	8TQ (30) 10TQ (24) 12TQ (20) 15TQ (16) 16TQ (15) 20TQ (12) 24TQ (10)	8TQ (24) 12TQ (16) 16TQ (12) 24TQ (8)	8TQ (15) 10TQ (12) 12TQ (10) 15TQ (8) 20TQ (6) 24TQ (5)	8TQ (12)
33.3 Kbps	8TQ (150) 12TQ (100) 16TQ (75) 20TQ (60) 24TQ (50)	8TQ (120) 10TQ (96) 12TQ (80) 15TQ (64) 16TQ (60) 20TQ (48) 24TQ (40)	10TQ (90) 12TQ (75) 15TQ (60) 20TQ (45)	8TQ (90) 10TQ (72) 12TQ (60) 15TQ (48) 16TQ (45) 20TQ (36) 24TQ (30)	8TQ (75) 10TQ (60) 12TQ (50) 15TQ (40) 20TQ (30) 24TQ (25)	8TQ (60) 10TQ (48) 12TQ (40) 15TQ (32) 16TQ (30) 20TQ (24) 24TQ (20)	10TQ (30) 12TQ (25) 15TQ (20) 20TQ (15)	8TQ (30)

Note: Shown in ( ) are the baud rate prescaler divide-by-N value.
Note 1. Baud rate prescaler divide-by-N value = P + 1 (P = 0 - 1023) P: value selected by the BRP bits in the Channel Configuration Registers.
Note 2. Minimum frequency to achieve maximum nominal baud rate of 1 Mbps.

Table 42.21 Baud rate calculation example for nominal and data bit rate CAN communication configurations

Baud rate calculation formula	(DLL clock) (baud rate prescaler divide-by-N value*1) × (number of TQs in one bit)	
	40 MHz	**20 MHz**
Nominal 1 Mbps Data 5 Mbps	40TQ (1)	20TQ (1)
	8TQ (1)	Not possible
Nominal 500 Kbps Data 2 Mbps	80TQ (1)	40TQ (1)
	20TQ (1)	10TQ (1)

Note: Shown in ( ) are the baud rate prescaler divide-by-N values and this table is not available in the classical CAN function.

Note 1. Baud rate prescaler divide-by-N value = P + 1 (P = 0 - 1023) P: value selected by the BRP bits in the Channel Configuration Registers.

For optimum clock tolerance in networks using the FD frame format, the length of the time quantum should be the same in nominal bit time and in data bit time. This means CFDC0NCFG.NBRP = CFDC0DCFG.DBRP.

Additionally, if transceiver delay compensation is used, do not program the CFDC0DCFG.DBRP bit to be greater than 1, as 1 means divide by 2.

42.4.1.4 Setting of CAN Clock, Bit Timing and Baud Rate

Figure 42.17 shows the procedure for setting the CAN clock and the baud rate for a channel.

These settings should be performed during Channel Reset mode (Configuration mode) for the CAN channels.

Before going to channel communication state, the baud rate must be configured, otherwise the mode does not switch correctly.

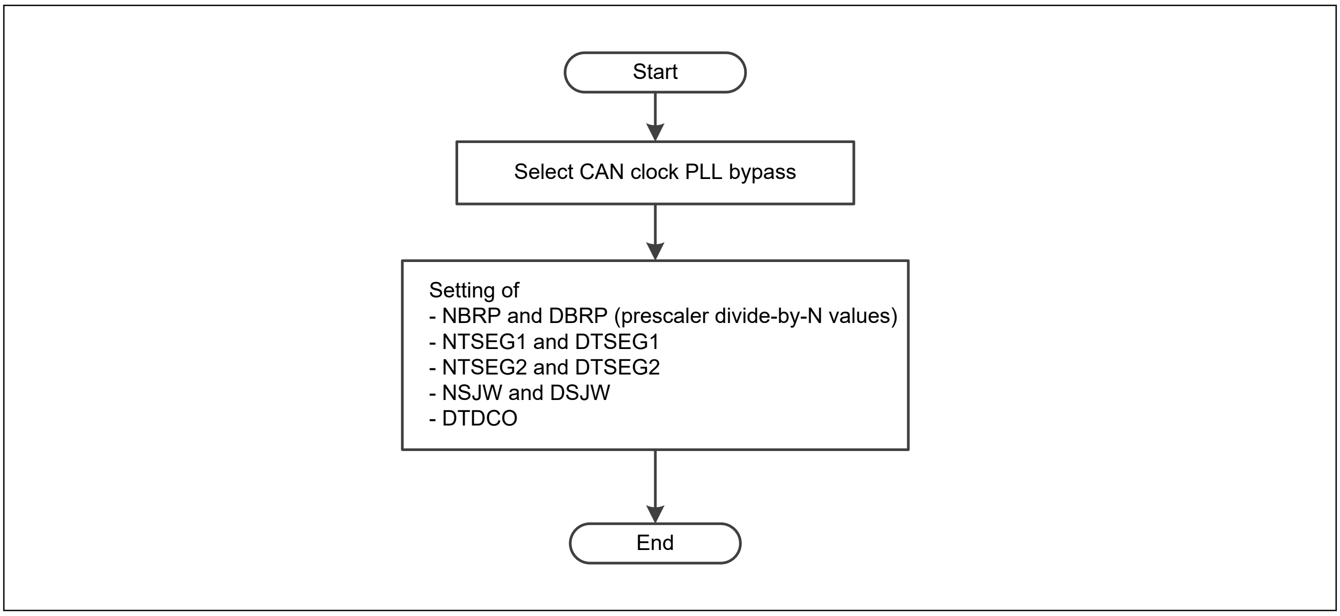


Figure 42.17 Procedure for setting the CAN bit timing and baud rate

42.4.1.5 Transmitter Delay Compensation

This chapter is not valid for classical CAN.

When a high baud rate is used such as 5 to 8 Mbps for the data phase, the transmitter delay can become greater than TSEG1. In this case, the transmitter always detects a bit-error in the data phase of the CANFD frame. The TDC compensates for the inability of the transmitter to receive its own transmitted bit at the sample point of that bit.

There is another symbolic sample point known as the Secondary Sample Point (SSP) that is used only during the data phase of CANFD frames. This is derived from the Transceiver Delay Compensation Result bit (CFDC0FDSTS.TDCR) as shown in Figure 42.18.

The resolution of the configuration, measured and offset values is based on the CAN channel DLL clock.

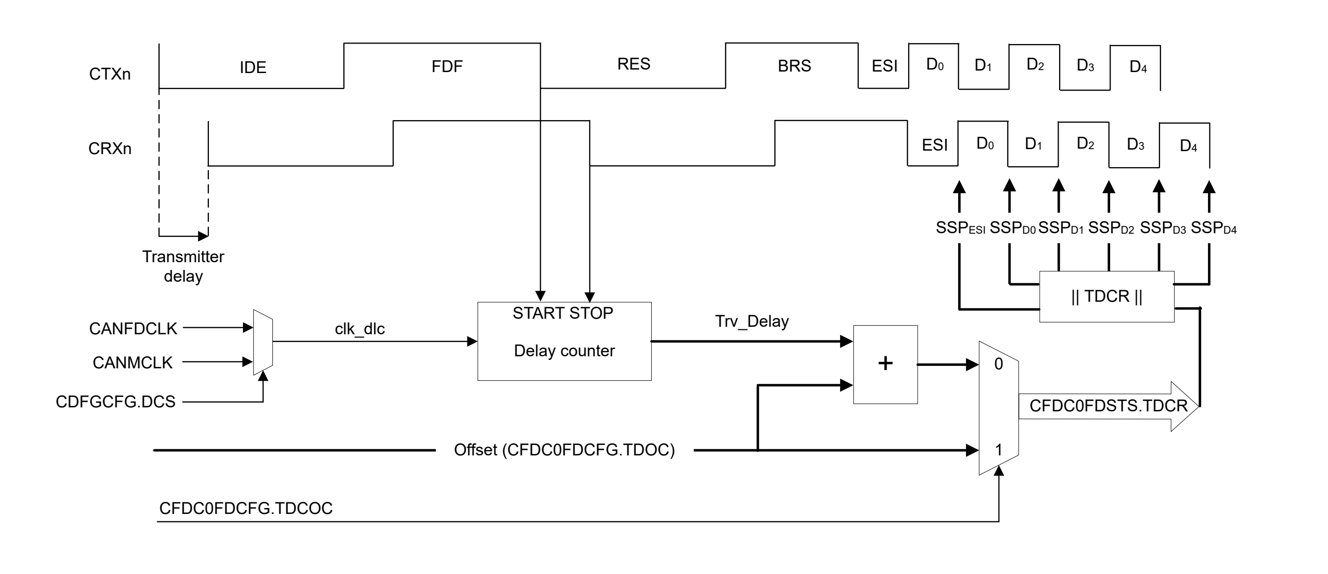


Figure 42.18 Transmitter delay compensation

The measured Trv_Delay is based on the number of clk_dlc clock cycles. The delay is counted up by one for each started clock until the dominant value is seen on CRXn. Figure 42.19 shows the measured result. Trv_Delay counted to maximum 127 with a clk_dlc clock.

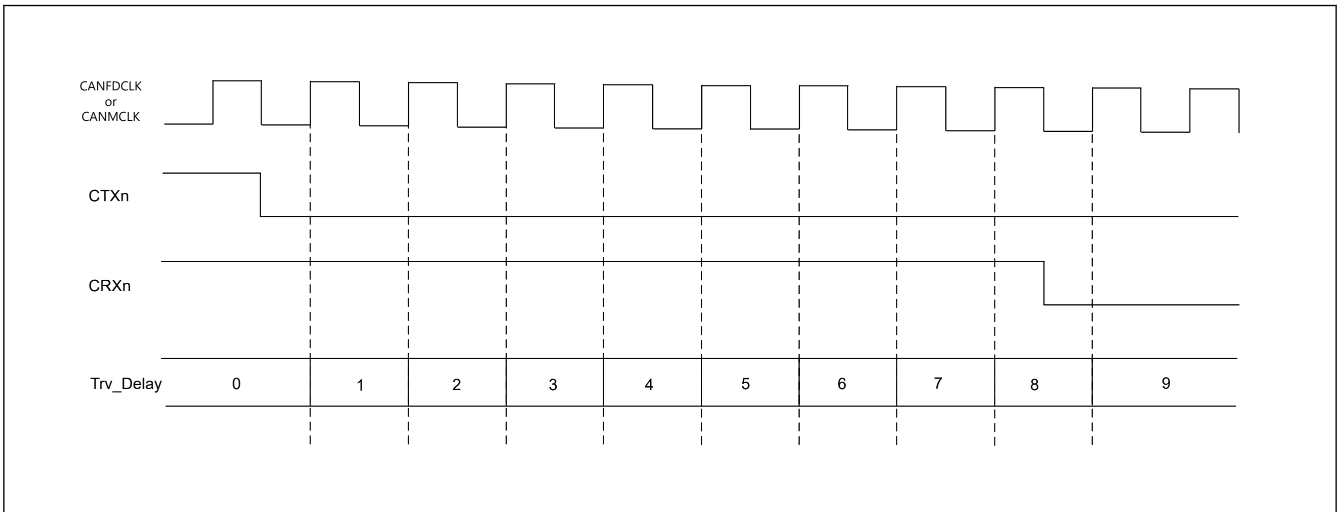


Figure 42.19 Trv_Delay measurement example

The SSP is calculated by taking the result from CFDC0FDSTS.TDCR and rounding the value down to the nearest integer number of data time quanta.

Figure 42.20 shows the positioning of the secondary sample point. When CFDC0FDCFG.TDCOC is equal to 0, the SSP is equal to the Trv_Delay (measured delay) + CFDC0FDCFG.TDCO, rounded down to the nearest integer number of time quanta. Usually, the TDCO value should have the size of (SyncSegment data + TSEG1 data) to position the SSP to a theoretical location of the sample point.

If the CFDC0FDCFG.TDCOC is equal to 1, the SSP is defined by CFDC0FDCFG.TDCO. If CFDC0DCFG.DBRP is greater than 0, the value is also rounded down to the nearest integer number of time quanta.

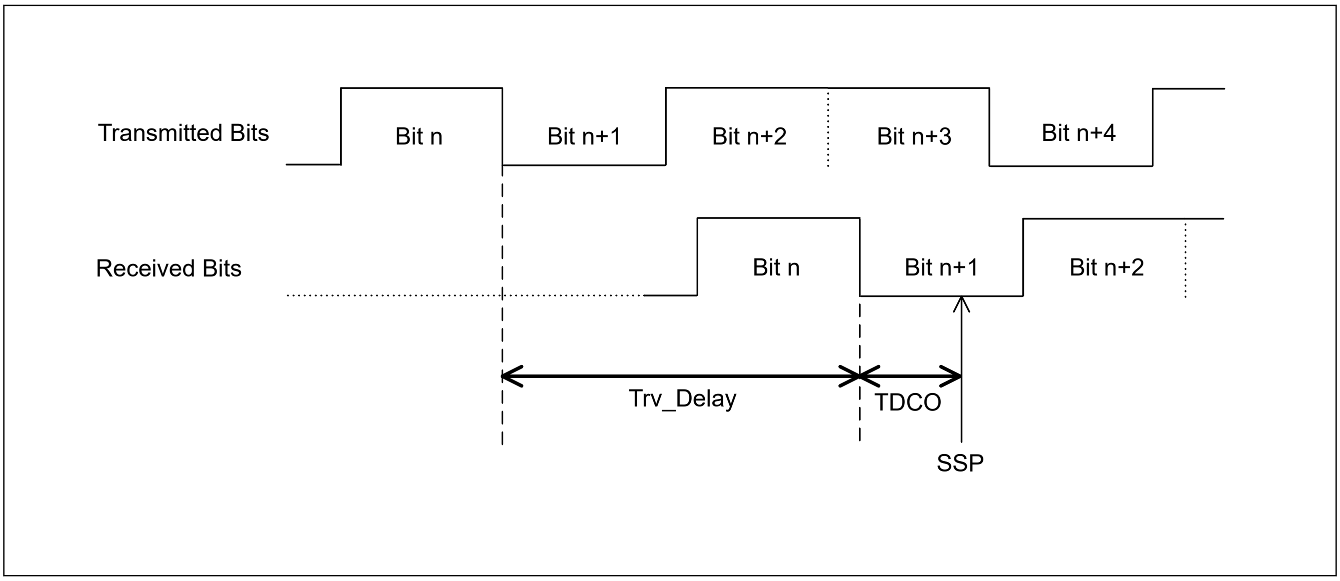


Figure 42.20 Position of the secondary sample point

The maximum delay (Trv_Delay + TDCO) which can be compensated by the CANFD module is (6 data bits - 2clk_dlc).

The ISO 11898-1 allows you to set different values for BRP_data and BRP_nom.

If different values are used for CFDC0NCFG.NBRP and CFDC0DCFG.DBRP, then two CAN nodes may be out of synchronization at the point when the bit rate changes from nominal bit rate to data bit rate after sample point of the BRS bit. This condition is shown in Figure 42.21.

The length of the time quantum should be the same in the nominal bit time and in the data bit time. This means CFDC0NCFG.NBRP = CFDC0DCFG.DBRP.

Different bit rates can be achieved by selecting different configuration values for the Time Segments. The nominal bit rate can be configured from 8 to 385 TQs and the data bit rate from 5 to 49 TQs.

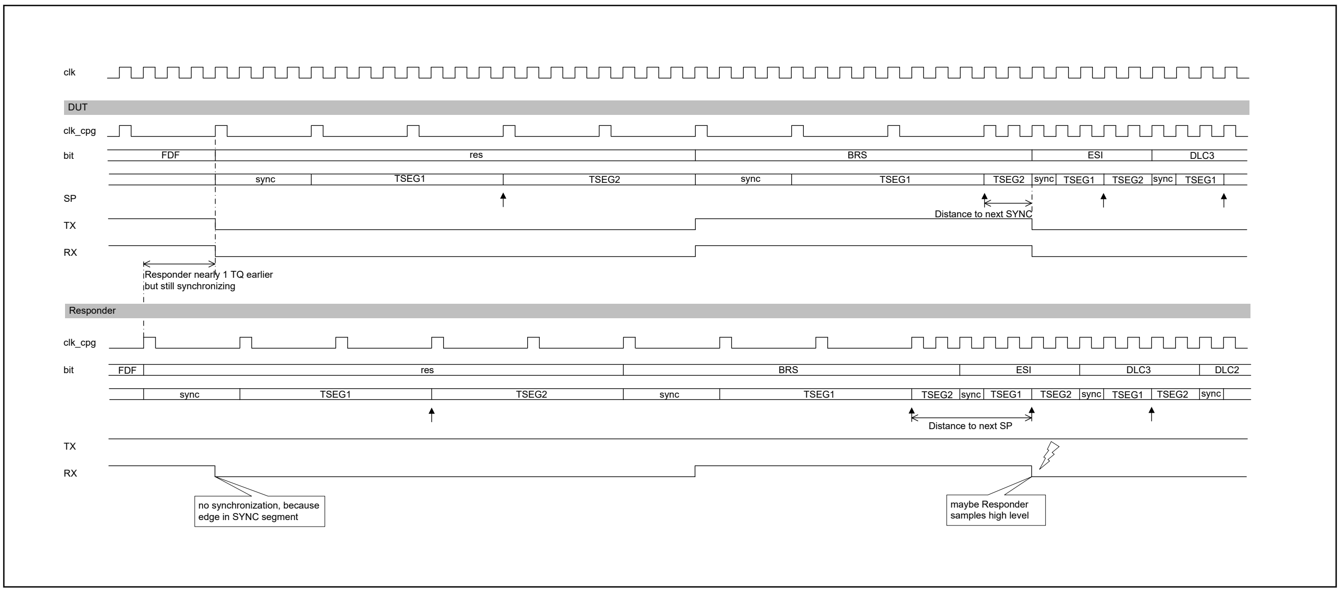


Figure 42.21 Loss of synchronization between two CAN nodes

The transmitter delay compensation measurement result is updated at the falling edge from FDF bit to RES bit when configured accordingly (CFDC0FDCFG.TDCE = 1, CFDC0FDCFG.TDCOC = 0).

Figure 42.22 shows the read flow to get the measured transmitter delay compensation result.

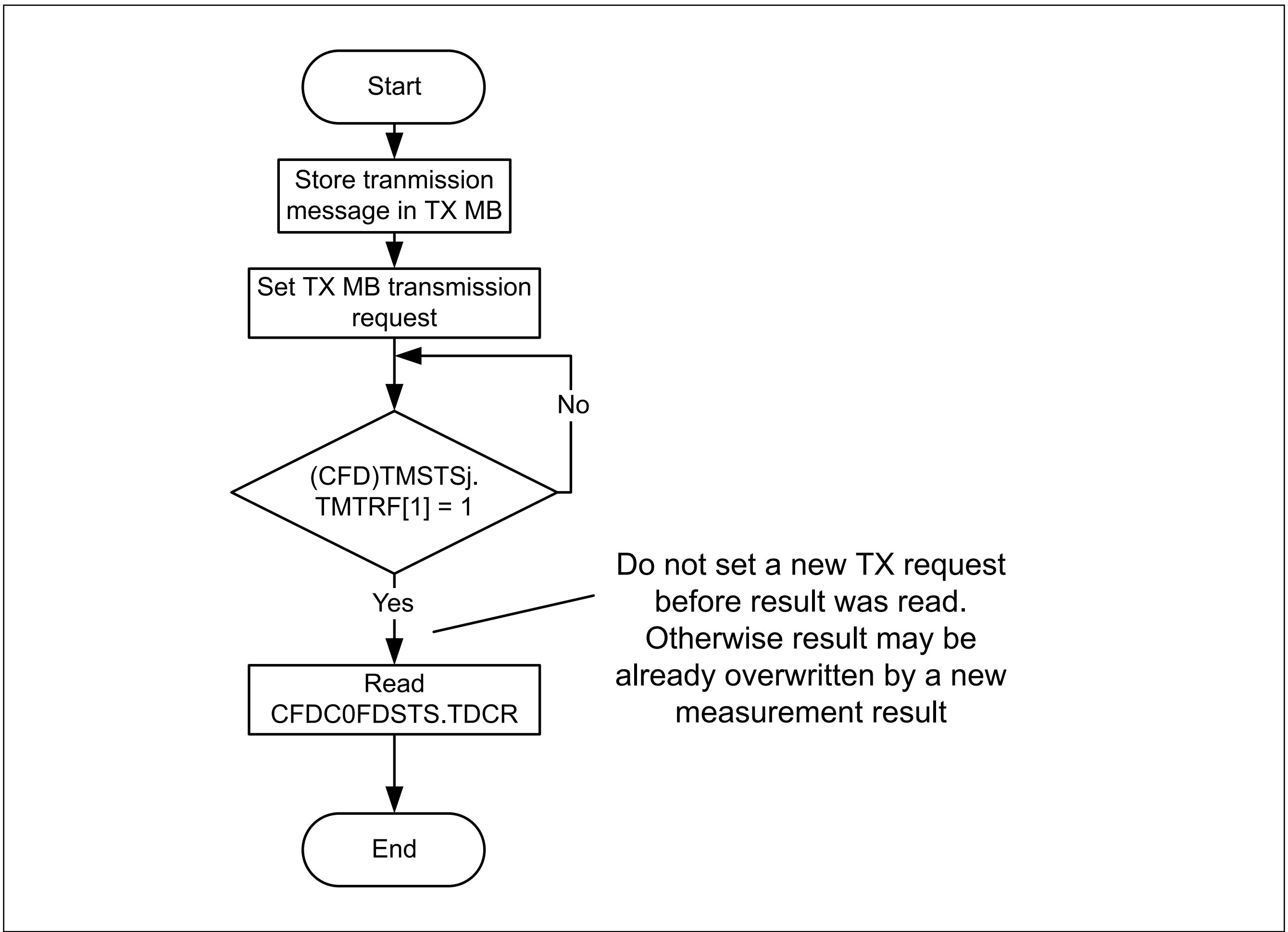


Figure 42.22 TDC result read flow

42.4.2 CAN Module Configuration after Hardware Reset

After a hardware reset (power on reset) or after setting and clearing a CFDGRSTC.SRST bit, the CANFD module enters Global Sleep mode automatically.

To enable configuration of the CANFD module, you must exit Sleep mode by clearing the Global Sleep Request bit CFDGCTR.GSLPR to 0.

After a hardware reset, the module starts RAM initialization, the CFDGSTS.GRAMINIT bit in the Global Status Register is set automatically to indicate that the CANFD logic is initializing the RAM.

After RAM initialization is complete, this bit is cleared automatically.

RAM initialization is necessary to avoid setting of false ECC error flag after hardware reset the random data presented in the RAM.

Do not access registers of CANFD in either read or write until RAM initialization is complete and the CFDGSTS.GRAMINIT bit is cleared.

Before going to communication mode, the Global Acceptance Filter List and message FIFO buffers must be configured. In addition, CAN channel must be configured such as CAN bit timing. For this configuration, CAN channel must be released from Channel Sleep mode and must be configured for communication in Channel Reset mode (Configuration mode).

Figure 42.23 shows the configuration procedure. For details about each step, see section 42.5. Acceptance Filtering Function using Global Acceptance Filter List (AFL), section 42.6. FIFO Buffers and Normal Message Buffer Configuration, section 42.7. Interrupts and DMA and section 42.4.1.3. Baud Rate.

The CANFD module does not perform the RAM initialization sequence after executing a software reset by setting CFDGRSTC.SRST.

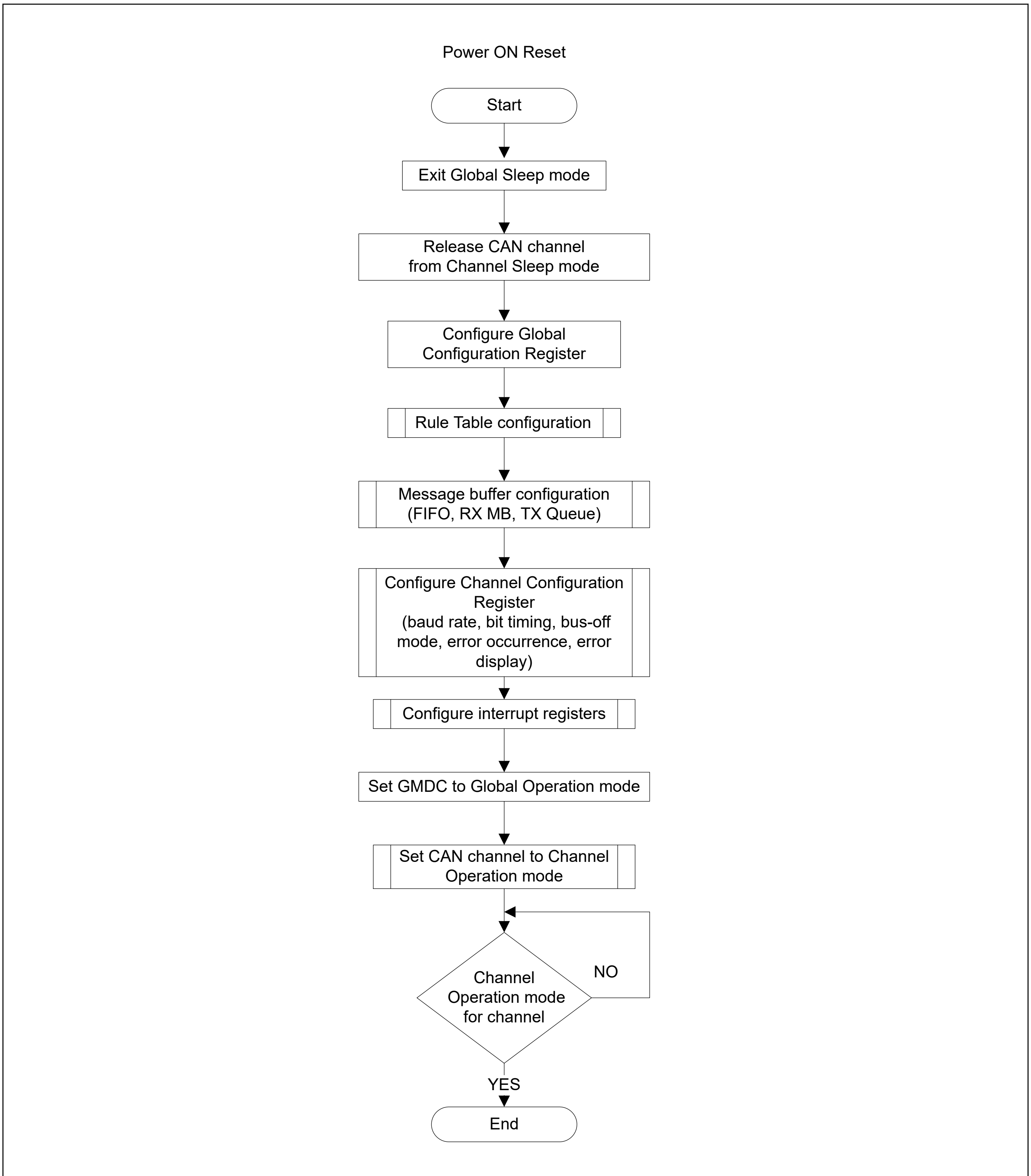


Figure 42.23 Configuration procedure after a hardware reset

42.5 Acceptance Filtering Function using Global Acceptance Filter List (AFL)

42.5.1 Overview

The CANFD module can handle message acceptance filtering with a global Acceptance Filter List (called AFL). Each element of the AFL defines a filter rule for messages received on a specific channel.

The following actions are performed based on the AFL entries:

- Acceptance filtering based on received CAN Identifier and masking
- DLC filtering based on received DLC value
- Message data payload according to the CFDGCFG.CMPOC bit*1
- Storage of accepted messages in the message buffer objects defined in the related AFL entry
- Attaching a 16-bit pointer to the stored messages defined in the related AFL entry, for example to support AUTOSAR applications
- Attaching a 2-bit information label to the stored messages defined in the related AFL entry

Note 1. This feature is not available in the classical CAN function.

The CANFD module allows a maximum of 16 AFL entries.

During acceptance filtering process, each AFL entry in a channel is checked against the received message by the acceptance filter unit. The check starts from the lowest AFL entry number for this channel.

AFL search stops when a match of the received identifier with a configured identifier/mask combination occurs or when the received identifier has been compared against all AFL entries defined for the related channel. If no match occurs, then the received message is rejected. No notification is given to the application in this case.

Additionally, an automatic DLC filtering is performed for each accepted message if DLC check is globally enabled. If the DLC value of the received message is equal to or higher than the configured DLC value in the matching AFL entry, the DLC check is passed.

If DLC replacement (CFDGCFG.DRE bit) is enabled, DLC value configured in the matching AFL entry is greater than 0x0 and DLC check passes, then the configured value of DLC in the matching AFL entry is stored in the destination RXMB or FIFO Buffer.

If the received value of DLC is greater than the configured DLC value in the matching AFL entry, then the additional data bytes received on the CAN Bus are not stored in the destination RXMB or FIFO Buffer. These additional data bytes are stored as 0x00 in the destination RXMB or FIFO Buffer.

If DLC replacement is enabled and DLC value of matching AFL entry is 0x0, then the received value of DLC is stored in the destination RX MB or FIFO Buffer.

If DLC replacement (the CFDGCFG.DRE bit) is disabled and DLC check passes, then the received value of DLC on the CAN bus is stored in the destination RXMB or FIFO buffer.

If the received value of DLC is greater than the configured DLC value in the matching AFL entry, then the additional data bytes received from the CAN bus are also stored in the destination RXMB or FIFO buffer.

If DLC value of the received message is less than the configured DLC value in the matching AFL entry, then DLC check fails. In this case, the received message is rejected and is not stored in any RXMB or FIFO buffer.

Additionally, DLC check failure is flagged by the DLC Error Flag in the Global Error Flag Register. If configured, an error interrupt is also generated. The DLC replacement configuration has no impact if the DLC check fails.

If a message has passed both acceptance filtering and DLC filtering, it is stored in a single reception message buffer and/or in FIFO buffers configured for reception function.

This message storage target information is also defined in the same AFL entry. Do not set a target at the AFL entry which is not configured.

Each accepted received message can be stored into a maximum of 2 different target destinations (single reception message buffer and/or FIFO buffers).

The programming of more than 2 target destinations is not allowed. If more destinations are programmed, then the internal timing might lead to a race condition that prevents the storage of received messages in the message RAM.. Correct configuration of the numbers of target destination is the responsibility of the application.

Additional protection mechanism is made for the case when a received message contains more data payload Bytes than possible to store in the target destination (CFDRMNB.RMPLS, CFDRFCCa.RFPLS or CFDCFCC.CFPLS).

If CFDGCFG.CMPOC = 0, the message is completely rejected and is stored in the target destination. When CFDGCFG.CMPOC = 0 and RX or Common FIFO full including the received message contains more data payload bytes than possible to store in the target destination (CFDRMNB.RMPLS, CFDRFCCa.RFPLS or CFDCFCC.CFPLS), the corresponding CFDFMSTS.RFxMLT or CFDFMSTS.CFxMLT bit is not set to 1, respectively.

When CFDGCFG.CMPOC = 1, the received data bytes greater than CFDRMNB.RMPLS is rejected. When CFDGCFG.CMPOC = 1 and RX or Common FIFO full including the received message contains more data payload bytes than possible to store in the target destination (CFDRMNB.RMPLS, CFDRFCCa.RFPLS or CFDCFCC.CFPLS), the corresponding CFDFMSTS.RFxMLT or CFDFMSTS.CFxMLT bit is set to 1, respectively.

Depending on the CFDGCFG.DRE bit, the original received DLC or the DLC value configured at the AFL entry is stored.

Regardless of the CFDGCFG.CMPOC bit setting, CFDGERFL.CMPOF is set to 1 if a payload overflow condition is detected.

The DLC filtering is performed before the payload overflow function. So for one reception frame, only one flag can be set at the same time with CFDGERFL.DEF or CFDGERFL.CMPOF*1.

Note 1. This bit is not available in the classical CAN function.

42.5.2 Allocation of AFL Entries

The number of AFL entries per channel can be configured using the dedicated field in the related Global Acceptance Filter Configuration Registers (see Figure 42.24).

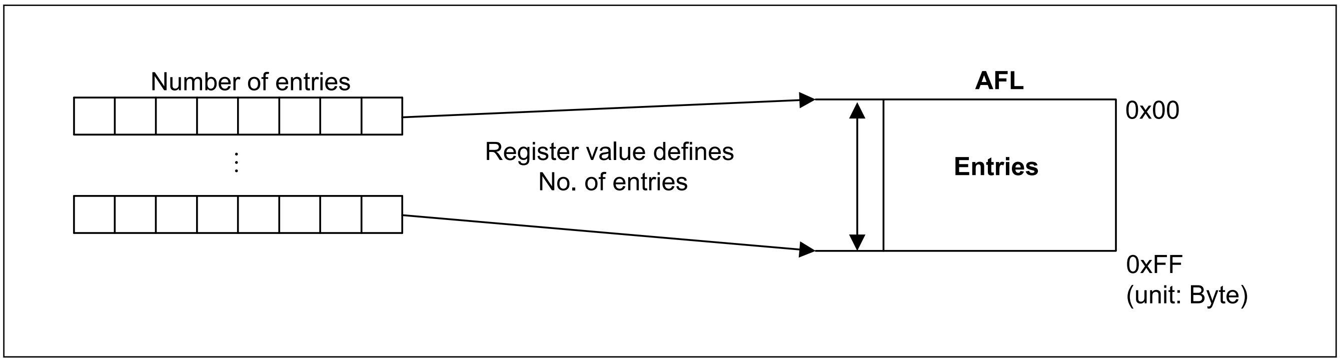


Figure 42.24 Configuration of AFL for each channel

The minimum number of entries for one channel is 0 (no entries defined for the channel) and the maximum number of entries is 16.

All entries are unique for a channel and overlapping or sharing of entries is not supported. Correct configuration of the AFL is the responsibility of the application.

The CANFD module does not flag errors related to the configuration of the AFL.

42.5.3 AFL Entry Description

Each AFL entry consists of 16 bytes. The fields in all entries are identical.

Each entry contains the following information for acceptance filtering and DLC filtering:

- Identifier (11 bits for Standard Frame format, 29 bits for Extended Frame format):
  Acceptance filter unit checks the identifier field of the received message against the identifier field of each AFL entry (full 29 bits masking of identifier bits is possible, see information that follows).
- IDE bit:
  Acceptance filter unit checks the IDE bit of the received message against this bit and selects the relevant part of the identifier field for acceptance filtering (masking of IDE bit is possible, see the information that follows).
- RTR bit:
  Acceptance filter unit only accepts data frames (RTR = 0) or remote frames (RTR = 1) according to the setting of this bit (masking of RTR bit is possible, see the information that follows).
- Loopback Configuration bit:
  This bit can enable or disable the AFL entry depending on the Loopback Configuration or Mirror mode condition.
- Mask for Identifier bits (29 bits):

Each bit in the identifier mask can mask the corresponding identifier bit in the AFL entry during acceptance filtering, see Figure 42.25.

- Mask for IDE bit:
  If this Mask bit masks the IDE bit of the AFL entry in both Standard Identifier and Extended Identifier format, messages can be accepted by this AFL entry. The identifier of the received message is compared against the Standard Identifier part of the AFL entry for Standard Identifier format messages and against the Extended Identifier part of the AFL entry for Extended Identifier format messages.

- Mask for RTR bit:
  If this Mask bit masks the RTR bit of the AFL entry in both frame formats, data frame and remote frame formats are accepted by this AFL entry.

- Pointer information (16 bits):
  This 16-bit pointer is attached to a received message accepted by the related AFL entry. The pointer is added during message storage in the message buffer area and can be used by application as support function. The pointer information can be used for example to support PDU identifier allocation for the received message in AUTOSAR systems.

- Information label (2 bits):
  This 2-bit label is attached to a received message accepted by the related AFL entry. The label is added during message storage in the message buffer area and can be used by application as support function.

- DLC value for automatic DLC filtering:
  If the DLC value of the received message is equal or higher than the configured DLC value, the DLC check is passed.

If the DLC value in this AFL entry is configured to 0, DLC filtering is effectively disabled for this entry (all accepted messages pass DLC filtering).

Each AFL entry contains the following information for the handling of received messages:

- Message buffer number of one single reception message buffer as target for received message storage
- Single reception message buffer enable bit to configure the single reception message buffer number to be valid or invalid, as target for received message storage
- FIFO direction pointer - each bit of the FIFO direction pointer configures a dedicated FIFO as possible target for a received message

There is no hardware protection against such storage of message. Therefore, the FIFO direction pointer must be configured carefully.

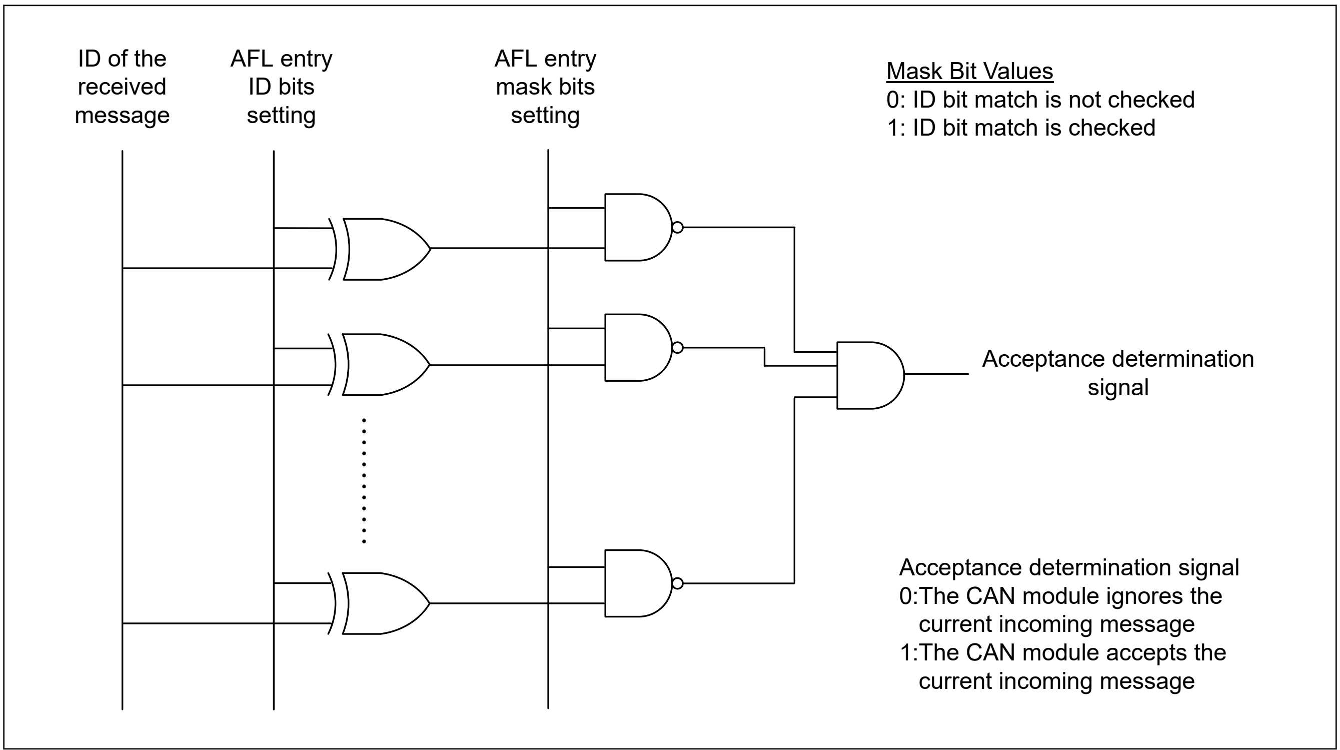


Figure 42.25 Acceptance function

42.5.4 Entering Entries in the AFL

Application software can enter one full entry into the AFL using the following registers:

- Global AFL ID Entry Register: Part 1 of the AFL entry
- Global AFL Mask Entry Register: Part 2 of the AFL entry
- Global AFL Pointer 0 Entry Register: Part 3 of the AFL entry
- Global AFL Pointer 1 Entry Register: Part 4 of the AFL entry.

16 sets of these registers form a group of AFL entries. The AFL should only be configured in CH_RESET or CH_HALT mode.

Follow the configuration shown in Figure 42.26 to program the AFL.

After entering all entries in Configuration mode, locking of the AFL access should be performed to protect unwanted write access to the AFL.

Write protection is active during all Global modes (GL_RESET, GL_HALT, and GL_OPERATION) if the lock bit is set.

Read access to AFL is still possible during all Global modes even when AFL data access is disabled (consistency check of AFL contents is possible during run time).

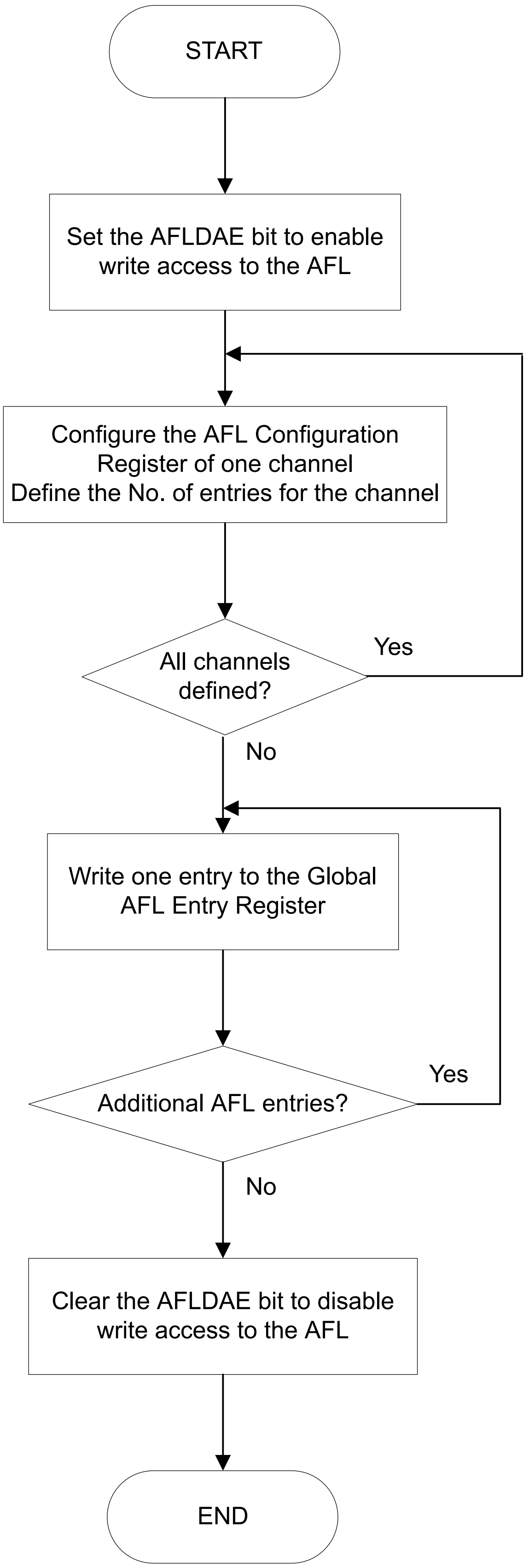


Figure 42.26 AFL configuration flow

42.5.5 Loopback Modes

If the Loopback Configuration bit is set, the AFL entry is only valid in Loopback test mode (Self-test mode 0 or Self-test mode 1) or in mirror mode when receiving messages that were transmitted by the respective CAN channel itself.

The AFL entry is not valid for received messages in loopback mode transmitted by other CAN nodes on the bus. The expression valid or invalid for the related entry means that this AFL entry is or is not compared against the received message ID respectively.

If the Loopback Configuration bit is 0, the AFL entry is only valid for:

- Received messages transmitted by other CAN nodes on the bus in normal (non-loopback mode) and mirror modes
- Received messages transmitted by other CAN nodes or the CAN channel itself in Loopback test mode.

The mirror mode can be enabled with the CFDGCFG.MME bit in the Global Configuration Register. If CFDGCFG.MME bit is set, then a successfully transmitted message can be stored back in an RX message buffer or FIFO buffer if a matching entry is configured in the AFL for that channel.

The Loopback Configuration bit in the matching AFL entry must be set to store this frame.

If Mirror mode and Loopback test mode are configured at the same time, the Loopback test mode behavior applies.

Table 42.22 shows the behavior of the acceptance filter unit depending on the setting of the related input signals.

Table 42.22 Behavior of acceptance filter based on the loopback configuration setting in AFL entry

Mirror Mode Enable (MME Configuration bit)	Loopback in test mode (Self-test mode 0 or Self-test mode 1)	Channel mode	Loopback Configuration bit in AFL entry	AFL entry
0	0	Receiver	0	Valid
			1	Invalid
		Transmitter	0	Invalid
			1	Invalid
	1	Receiver	0	Valid
			1	Invalid
		Transmitter	0	Valid
			1	Valid
1	0	Receiver	0	Valid
			1	Invalid
		Transmitter	0	Invalid
			1	Valid
	1	Receiver	0	Valid
			1	Invalid
		Transmitter	0	Valid
			1	Valid

Note: The expression valid or invalid for the related entry means that this AFL entry is or is not compared against the received message ID, respectively.

42.5.6 IDE Masking

When the GAFLIDEM bit is 0 in an AFL entry, the IDE bit configured in the AFL entry is not used for ID matching. In this case, the use of ID[10:0] or ID[28:0] matching is based on the received IDE bit.

Consider the following example:

- The ID and Mask fields of an AFL entry x is configured as follows:
  - CFDGAFLID [x] = 0xC0553A20 → IDE = 1, RTR = 1, LLB = 0, ID[10:0] = 0x220 / ID[28:0] = 0x00553A20
  - CFDGAFLMr = 0x0000FFFF → IDEM = 0. RTRM = 0, IDM[10:0] = 0x7FF / IDM[28:0] = 0x0000FFFF

- The comparison result for the four different received IDs with AFL entry x is described as follows:
  - If a frame with IDE = 0 and ID = 0x220 is received, this is considered as a match
  - If a frame with IDE = 0 and ID = 0x320 is received, this is not a match
  - If a frame with IDE = 1 and ID = 0x1FFF3A20 is received, this is considered as a match
  - If a frame with IDE = 1 and ID = 0x08803220 is received, this is not a match.

42.5.7 Updating AFL Entry During Communication

You can update the AFL entry without disabling all CAN communications. Choose the entry number to be updated by setting the AFL entry number, and ignore the enable bit.

This entry number is ignored from the AFL matching while the entry is being updated.

Figure 42.27 shows the update flow for an AFL entry.

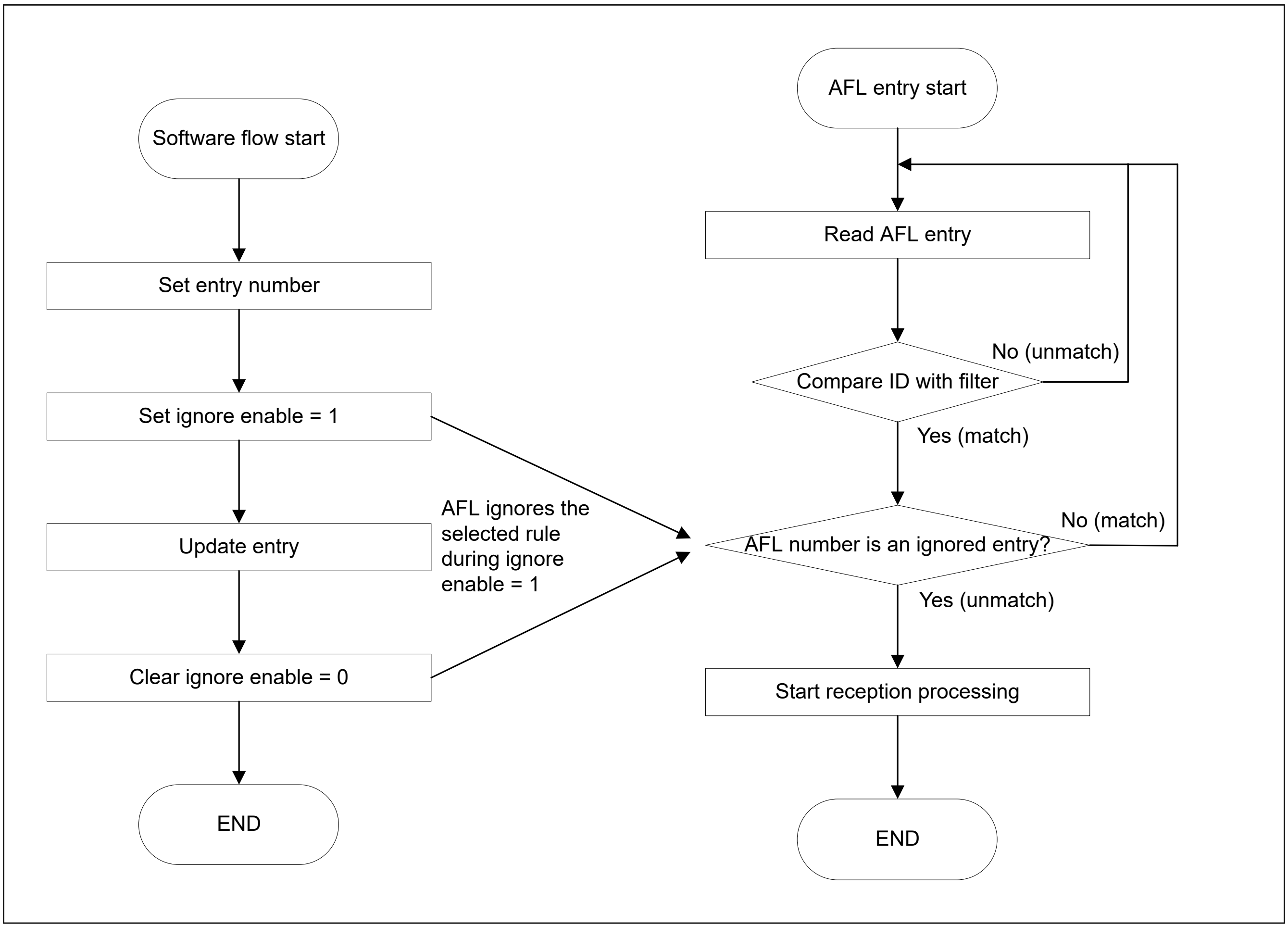


Figure 42.27 Update flow for an AFL entry

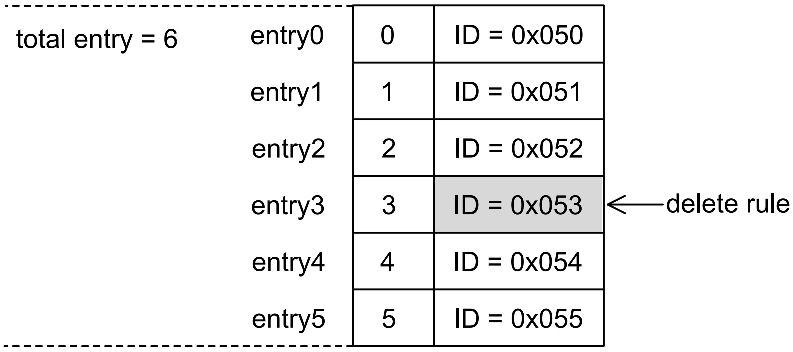
The method to update an AFL entry is as follows:

1. Set the entry number to CFDGAFLIGNENT register.
2. Set the value 0xC401 (key code and enable bit) to CFDGAFLIGNCTR register.
3. CFDGAFLECTR.AFLDAE is set to 1.
4. Set the new rule to CFDGAFLIDr, CFDGAFLMr, CFDGAFLP0r, CFDGAFLP1r registers.
5. CFDGAFLECTR.AFLDAE is cleared to 0.
6. Set the value 0xC400 (key code and clear enable bit) to CFDGAFLIGNCTR register.

Note: This entry number is ignored during the periods from (2) to (5).

(1) Example 1: Deleting an entry

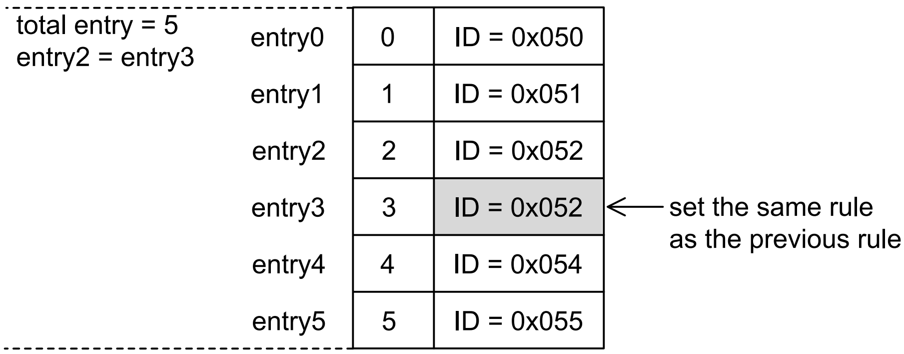
Deleting entry3 when the total number of entries is 6 channels.



How to delete an entry

1. Set 0x00000003 to CFDGAFLIGNENT register.
2. Set 0x0000C401 to CFDGAFLIGNCTR register.
3. Set 0x00000100 to CFDGAFLECTR register.
4. Set the same rule as the previous rule by accessing CFDGAFLIDr, CFDGAFLMr, CFDGAFLP0r, CFDGAFLP1r (r = 3, this is entry3).
5. Set 0x00000000 to CFDGAFLECTR register.
6. Set 0x0000C400 to CFDGAFLIGNCTR register.

Entry3 is now deleted.



(2) Example 2: Adding an entry

Adding a new entry to entry3 when the total number of entries is 6.

total entry = 5
entry2 = entry3

entry0	0	ID = 0x050	
entry1	1	ID = 0x051	
entry2	2	ID = 0x052	
entry3	3	ID = 0x052	← add new rule in this position
entry4	4	ID = 0x054	
entry5	5	ID = 0x055	

How to add an entry

1. Set 0x00000003 to CFDGAFLIGNENT register.
2. Set 0x0000C401 to CFDGAFLIGNCTR register.
3. Set 0x00000100 to CFDGAFLECTR register.
4. Set the new rule by accessing CFDGAFLIDr, CFDGAFLMr, CFDGAFLP0r, CFDGAFLP1r (r = 3, this is entry3).
5. Set 0x00000000 to CFDGAFLECTR register.
6. Set 0x0000C400 to CFDGAFLIGNCTR register.

The new entry is now added.

total entry = 6

entry0	0	ID = 0x050	
entry1	1	ID = 0x051	
entry2	2	ID = 0x052	
entry3	3	ID = 0x056	← add new rule
entry4	4	ID = 0x054	
entry5	5	ID = 0x055	

The AFL filter can be used to set CFDGAFLCFG, and addition/deletion of an entry is possible. Therefore, it is necessary to set the maximum number to be used to CFDGAFLCFG.

42.6 FIFO Buffers and Normal Message Buffer Configuration

This section describes the process for configuring the number of RX message buffers, the FIFO buffers, and the flat TX message buffers in the CANFD module. The message buffers are mapped as shown in Figure 42.28.

The RX message buffers can be accessed with the RX Message Buffer Registers.

The RX FIFO buffers and the common FIFO buffers configured in RX mode or TX mode can only be accessed with the FIFO Access Registers.

If the common FIFO is configured in TX mode, you can only write data into the FIFO buffer using the FIFO Access registers.

If the common FIFO is configured in RX mode, you can only read data from the FIFO Access Registers.

The TX message buffers can be accessed with the TX Message Buffer Registers.

If unused message buffer locations are read, the message buffer locations are read as unknown values.

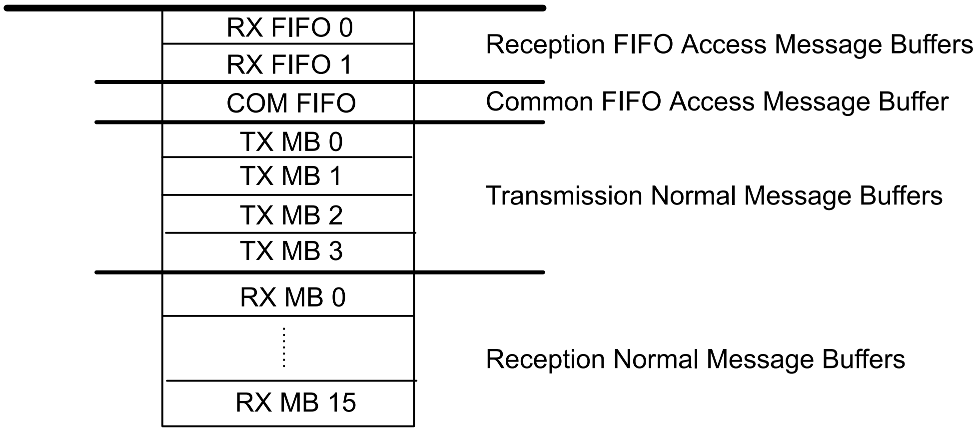


Figure 42.28 Message buffer configuration

42.6.1 Normal RX Message Buffers

In CANFD module, the frames received can be stored in normal RX message buffers based on the configuration of the AFL entries.

Additionally, the number of normal RX message buffers required in the system can be chosen up to a fixed maximum limit.

42.6.1.1 Normal RX Message Buffer Configuration

In CANFD module, the number of normal RX message buffers can be configured by writing to the RX Message Buffer Number Register.

The limiting values for the configuration of number of message buffers are:

- Minimum value = 0x00 (no normal RX MB)
- Maximum value = 0x10

Do not use values outside these limits.

The AFL entries for routing the received messages to normal RX message buffers must be configured to match the requirements of the system.

The AFL entries must also be configured properly, and an AFL entry for normal RX message buffers should not exceed the number of message buffers configured in the RX Message Buffer Number Register.

Note: There is no internal check procedure provided in CANFD module against wrong configuration of the AFL.

The data field size of the RX message buffer can be configured with the CFDRMNB.RMPLS bit. The default size is 8 bytes and the maximum data payload size is 64 bytes.

When the receiving frame exceeds the data field size, then the acceptance depends on the configuration of CFDGCFG.CMPOC (message rejecting or data payload cut).

Note: RMPLS and CMPOC bit is not available in the classical CAN function, so, these feature is not valid for classical CAN.

42.6.2 FIFO Buffers

The CANFD module provides a fixed number of FIFO buffers to support storage of frames for reception and transmission functions.

The number of reception-only FIFO buffers is fixed to 2. However, common FIFO buffer channel can be configured to store messages for transmission or reception function.

These FIFO buffers can be enabled or disabled, and the following parameters can be configured to match the system requirements:

- Size
- Interrupt structure
- Message lost mechanism
- Message overwrite mechanism of the FIFO buffers
- Location of the TX FIFO.

When the receiving frame exceeds the data field size, the acceptance depends on the configuration of the CFDGCFG.CMPOC bit (message rejecting or data payload cut).

42.6.2.1 FIFO Buffers Configuration

In CANFD module, the FIFO buffers can be configured to match the system requirements.

The total number of FIFO buffers = 2 RX FIFO buffers + 1 common FIFO buffer = 3 FIFO buffers.

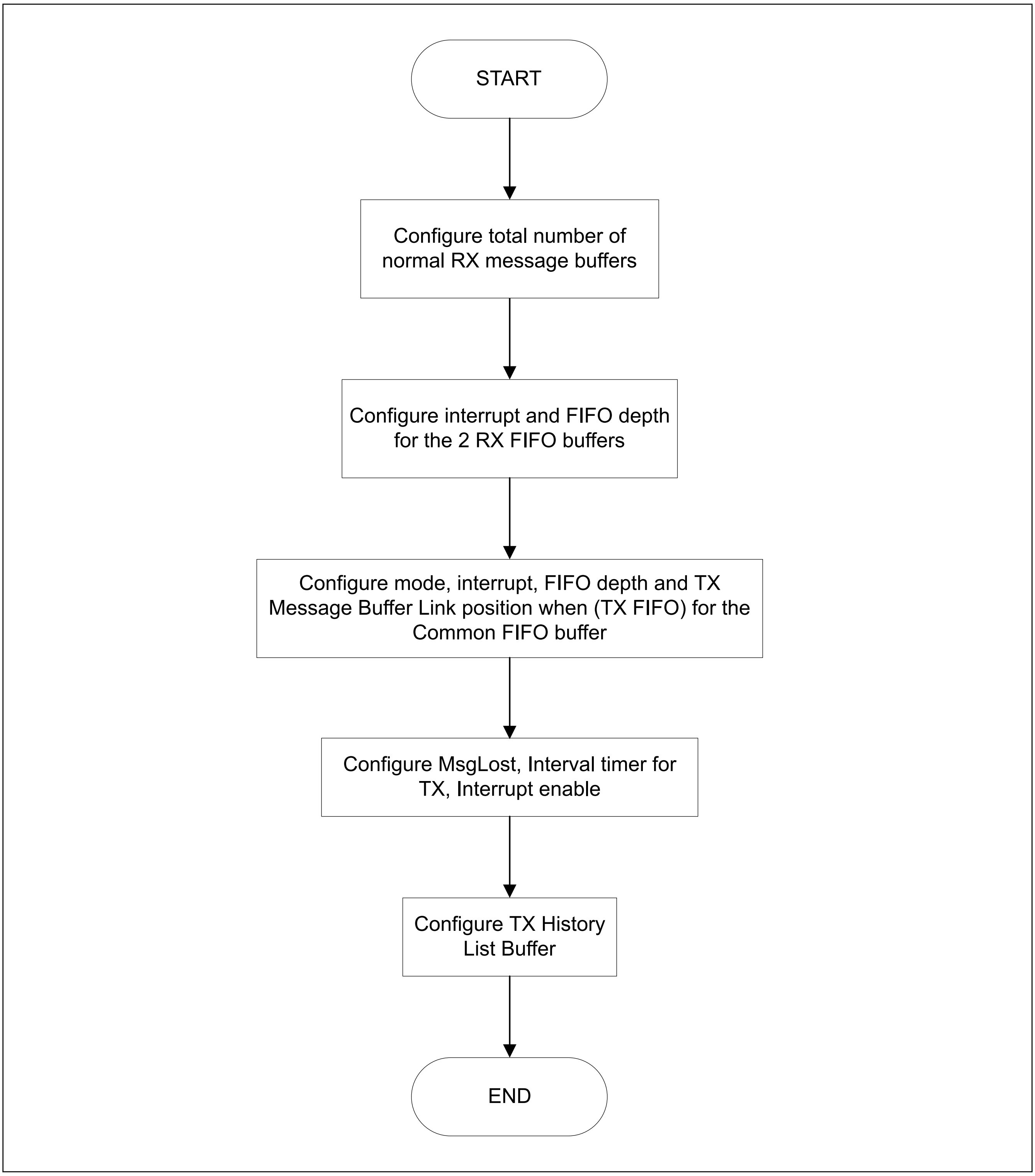


Figure 42.29 FIFO buffer configuration flow in CANFD module

As shown in Figure 42.29, the various FIFO buffers can be configured by writing to the RX FIFO Configuration/Control Registers and the Common FIFO Configuration/Control Registers.

For the 2 RX FIFO buffers, the following parameters can be configured:

- Interrupts
- FIFO depth
- FIFO payload data size.

For the common FIFO buffer, the following parameters can be configured:

- Mode
- Interrupts FIFO depth
- FIFO payload data size
- FIFO TX link position.

(1) FIFO mode configuration of Common FIFO buffer

The mode of the common FIFO buffer can be configured by writing to the CFDCFCC.CFM[1:0] bits in the Common FIFO Configuration/Control Register. The possible modes of configuration for Common FIFO buffer are:

- 0b RX mode (default mode after hardware reset)
- 1b TX mode

Messages can only be read from the RX FIFO buffers and the Common FIFO buffer configured in RX mode. Messages are stored by the CAN module in these FIFO buffers based on the AFL entries.

Messages can be read and written into the Common FIFO buffer configured in TX mode. These messages are transmitted on the appropriate CAN channel.

The pointers can only be incremented when a new message is stored in the FIFO buffer and decremented when a message is transmitted on the corresponding CAN channel by the CANFD module.

After a hardware reset, the Common FIFO buffer is configured in RX mode by default. Only enable the FIFO buffers after configuring the Common FIFO buffer in the required modes.

(2) FIFO TX message buffer link configuration

When the common FIFO is configured as TX FIFO, the FIFO buffer must be linked to a normal TX message buffer to participate in the transmission scan.

Do not write data into a TX message buffer that is linked to a Common FIFO buffer. Also, the TX message buffer linked to a Common FIFO buffer should not be a part of the TX Queue.

The TX message buffer link of each Common FIFO buffer can be configured by writing to the CFDCFCC.CFTML[1:0] bits in the Common FIFO Configuration/Control Registers. Available options for TX message buffer link configuration are:

- 0x00: TX Message Buffer 0
- 0x01: TX Message Buffer 1
- 0x10: TX Message Buffer 2
- 0x11: TX Message Buffer 3

(3) FIFO depth configuration

The depth of each FIFO buffer can be configured by writing to the CFDRFCCa.RFDC[2:0] bits and CFDCFCC.CFDC[2:0] bits in the RX FIFO Configuration/Control Registers and the Common FIFO Configuration/Control Registers. The 6 available options for depth configuration are:

- 0x000: 0 Message (FIFO buffer cannot be enabled)
- 0x001: 4 Messages
- 0x010: 8 Messages
- 0x011: 16 Messages
- 0x100: 32 Messages
- 0x101: 48 Messages

The RAM allocation for RX message buffers along with FIFO buffers is limited to 16 messages with 64 data bytes. Configuration of the RX message buffers, along with FIFO buffers, that exceeds this maximum limit should not be done.

CANFD module logic does not check the validity of the configuration.

Note: If the FIFO depth of a common FIFO is 4 messages or more (CFDCFCC.CFDC[2:0] > 000b), then the Common FIFO TX message buffer link is valid when the FIFO is disabled or enabled.

If FIFO depth is 0 messages, then the Common FIFO TX message buffer link is not valid when the FIFO is disabled or enabled.

(4) FIFO payload size configuration

The data size of each FIFO buffer can be configured by writing to the CFDRFCCa.RFPLS[2:0] bits and CFDCFCC.CFPLS[2:0] bits in the RX FIFO Configuration/Control Registers and the Common FIFO Configuration/Control Registers. The eight available options for depth configuration are:

- 000b: 8 bytes
- 001b: 12 bytes
- 010b: 16 bytes
- 011b: 20 bytes
- 100b: 24 bytes
- 101b: 32 bytes
- 110b: 48 bytes
- 111b: 64 bytes

The RAM allocation for RX message buffers along with FIFO buffers is limited to 16 messages with 64 data bytes. Configuration of the RX message buffers, along with FIFO buffers, that exceeds this maximum limit should not be done.

CANFD module logic does not check the validity of the configuration.

Note: This feature is not available in the classical CAN function.

(5) FIFO interrupt configuration

The Interrupt generation conditions for the FIFO buffers can be configured by writing to the CFDRFCCa.RFIM and CFDCFCC.CFIM bit in the RX FIFO Configuration/Control Registers and the Common FIFO Configuration/Control Registers. The two available options are:

- 0:
  - RX FIFO mode: Interrupt generated when the Common FIFO counter reaches CFDRFCCa.RFIGCV/ CFDCFCC.CFIGCV value
  - TX FIFO mode: Interrupt generated when the Common FIFO transmits the last message successfully
- 1:
  - RX FIFO mode: Interrupt generated at the end of storage of every received message
  - TX FIFO mode: Interrupt generated for every successfully transmitted message

If the Interrupt Mode bit is 0 for a RX FIFO, then interrupt is generated based on the configuration of the CFDRFCCa.RFIGCV[2:0] bits.

Similarly, if the Interrupt Mode bit is 0 for a Common FIFO configured in RX mode, then interrupt is generated based on the configuration of CFDCFCC.CFIGCV[2:0] bits.

The eight available options for configuring the FIFO counter value for generation of an interrupt are:

- 000b: Interrupt generated when FIFO is 1/8th Full
- 001b: Interrupt generated when FIFO is 1/4th Full
- 010b: Interrupt generated when FIFO is 3/8th Full
- 011b: Interrupt generated when FIFO is 1/2 Full
- 100b: Interrupt generated when FIFO is 5/8th Full
- 101b: Interrupt generated when FIFO is 3/4th Full
- 110b: Interrupt generated when FIFO is 7/8th Full
- 111b: Interrupt generated when FIFO is Full.

In this case, an interrupt is generated when the message count matches the configured value.

However, there are some limitations on the configuration of the CFDRFCCa.RFIGCV[2:0] and CFDCFCC.CFIGCV[2:0] bits depending on the FDC[2:0] bits (FIFO Depth Configuration), see Table 42.23.

Table 42.23 FIFO interrupt generation counter and FIFO depth configuration

RFDC[2:0] (CFDC[2:0])	RFIGCV[2:0] (CFIGCV[2:0])							
	111b	110b	101b	100b	011b	010b	001b	000b
000b	Don’t care (FIFO cannot be enabled)							
001b	Allowed	Not allowed	Allowed	Not allowed	Allowed	Not allowed	Allowed	Not allowed
010b	Allowed							
011b	Allowed							
100b	Allowed							
101b	Allowed							
110b	Allowed							
111b	Allowed							

42.6.2.2 FIFO Buffers Control

The FIFO interrupt must be enabled by setting any one of the following bits in the RX FIFO Configuration/Control Registers:

- CFDRFCCa.RFIE

In addition, the FIFO interrupt must be enabled by setting any one of the following bits in the Common FIFO Configuration/Control Register:

- CFDCFCC.CFRXIE
- CFDCFCC.CFTXIE

After configuration is complete, each FIFO can be enabled by setting the CFDRFCCa.RFE and CFDCFCC.CFE bits in the RX FIFO Configuration/Control Registers and the Common FIFO Configuration/Control Register to allow transmission and reception of messages.

42.7 Interrupts and DMA

42.7.1 Interrupts

The CANFD module generates several interrupts. The interrupt output, which is connected to the Interrupt Controller Unit (ICU), can be controlled by the corresponding interrupt enable bit.

The status flag is set independent of this enable bit.

The channel transmission interrupt has an additional status flag register. The status bits are set when the corresponding interrupt enables are set.

The status flag register supports the identification of the interrupt source for the channel transmission, as this interrupt is driven by several trigger sources.

The interrupts in the CANFD module can be classified into two groups, global interrupts and channel interrupts:

- Global interrupts:
  The CANFD module can generate 3 global interrupts:
  - Global interrupt for successful reception into the 2 RX FIFO buffers
  - Global error interrupt.
  - Global Interrupt for successful reception into the 16 RX message buffers
- Channel interrupts:
  Channel of the CANFD module can generate 3 channel interrupts:

1. Channel transmission
   - Transmission completion from channel
   - Transmission abort from channel
   - Transmission from TX Queue for a channel
   - Channel THL interrupt
   - Successful transmission from a Common FIFO in TX mode for a channel.
2. Channel error interrupt
3. Successful reception in a Common FIFO in RX mode for a channel.

The interrupts are cleared when the corresponding flag bits are cleared or the Interrupt enable bits are cleared.

Table 42.24 gives an overview of interrupt sources for the different interrupt outputs. The interrupt outputs are active-high.

Table 42.24 Interrupt source overview

Parameter	Interrupt	Name	Interrupt source	Interrupt clearing
Global Interrupts	Successful reception into at least one RX FIFO	CAN_RXF	Interrupt flag of corresponding RX FIFO for which interrupt is enabled	Clear the interrupt flag of corresponding RX FIFO buffer for which interrupt is enabled
	Global Error	CAN_GLERR	Any of the following: ● DLC Error flag ● Message Lost Status bit ● TX History Entry Lost Status bit ● CANFD Message Payload overflow flag	Clear all of : ● DLC Error flag ● Message Lost flags in all of the FIFO Status Registers ● TX History List Entry Lost flag ● CANFD Message Payload overflow flag
	Successful reception into at least one RXMB	CANn_RXMB (n = 0, 1)	Interrupt flag of corresponding RXMB for which interrupt is enabled	Clear the interrupt flag of corresponding RXMB buffer for which interrupt is enabled
Channel Transmission Interrupts	Channel successful transmission	CANn_TX (n = 0, 1)	Any channel related TXMB Successful flag when interrupt is enabled*1	Clear all channel related TXMB Result status bits for which the interrupt is enabled
	Channel Abort		Any channel related TXMB Abort flag when interrupt is enabled*1	Clear all channel related TXMB Result Status bits for which the interrupt is enabled globally
	Channel transmission from TX Queue		Related channel TX Queue Interrupt flag	Clear related channel TX Queue Interrupt flag
	Channel THL Interrupt		Channel THL Interrupt status flag	Clear the relevant THL Interrupt status flag
	Channel COM FIFO TX Interrupt		Interrupt Flag for Common FIFOs in TX mode belonging to the related channel	Clear the interrupt flags of Common FIFOs in TX mode belonging to the related channel
Channel Error Interrupt	Channel Error	CANn_CHERR (n = 0, 1)	Any channel related error flag in the Channel Error Flag Register for which interrupt is enabled in the Channel Error Interrupt Enable Register	Clear all channel related error flags in the Channel Error Flag Register for which interrupt is enabled in the Channel Error Interrupt Enable Register
Channel COM RX FIFO Interrupt	Channel COM FIFO RX Interrupt	CANn_COMFRX (n = 0, 1)	Interrupt flag for Common FIFOs in RX mode belonging to the related channel	Clear the interrupt flags of Common FIFOs in RX mode belonging to the related channel

Note 1. These interrupts are only set for TX Message Buffers that do not belong to an enabled TX Queue and are not pointing to a common FIFO.
Separate interrupts are provided for common FIFO buffers and TX Queue.

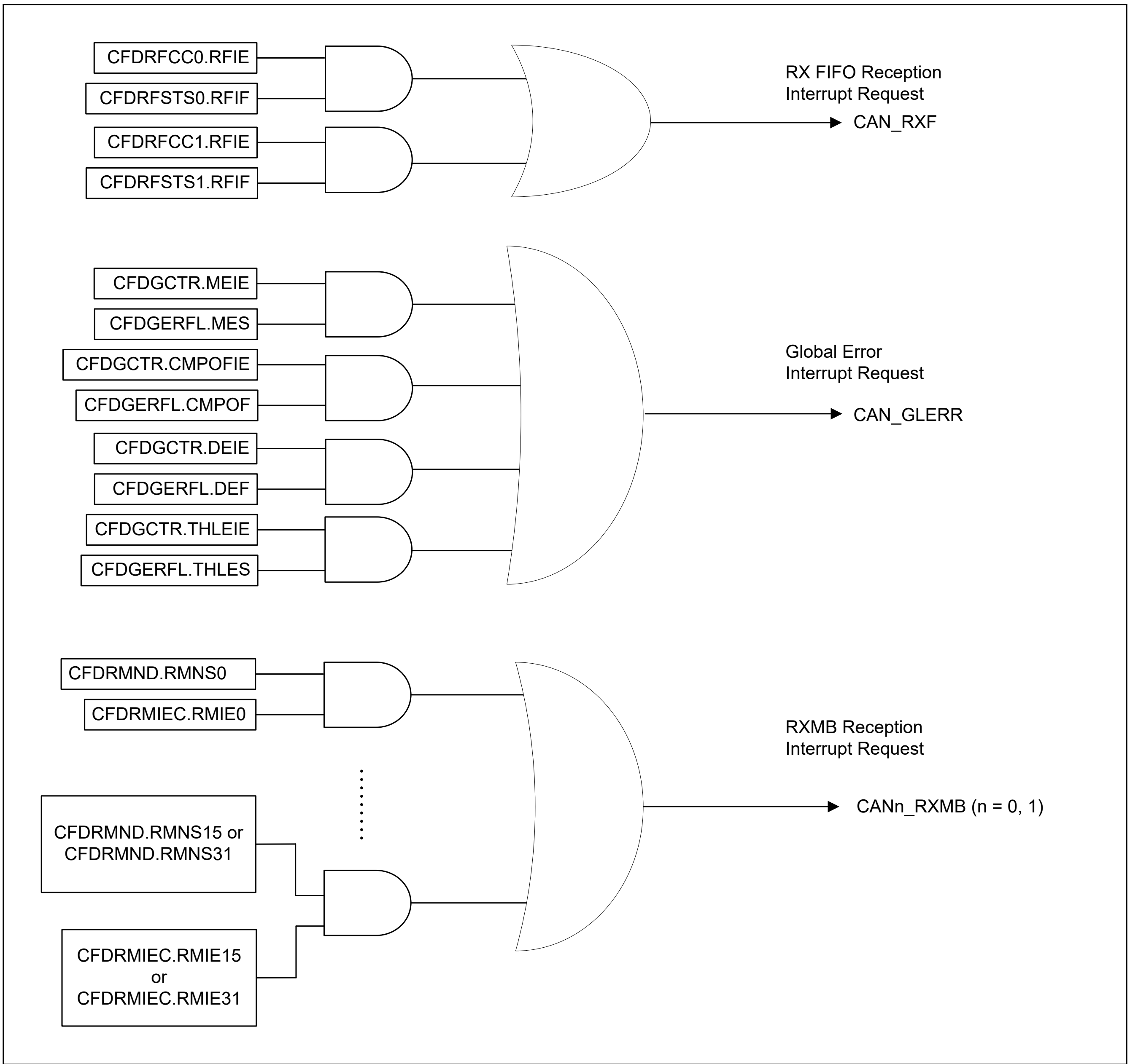


Figure 42.30 Global interrupt block diagram

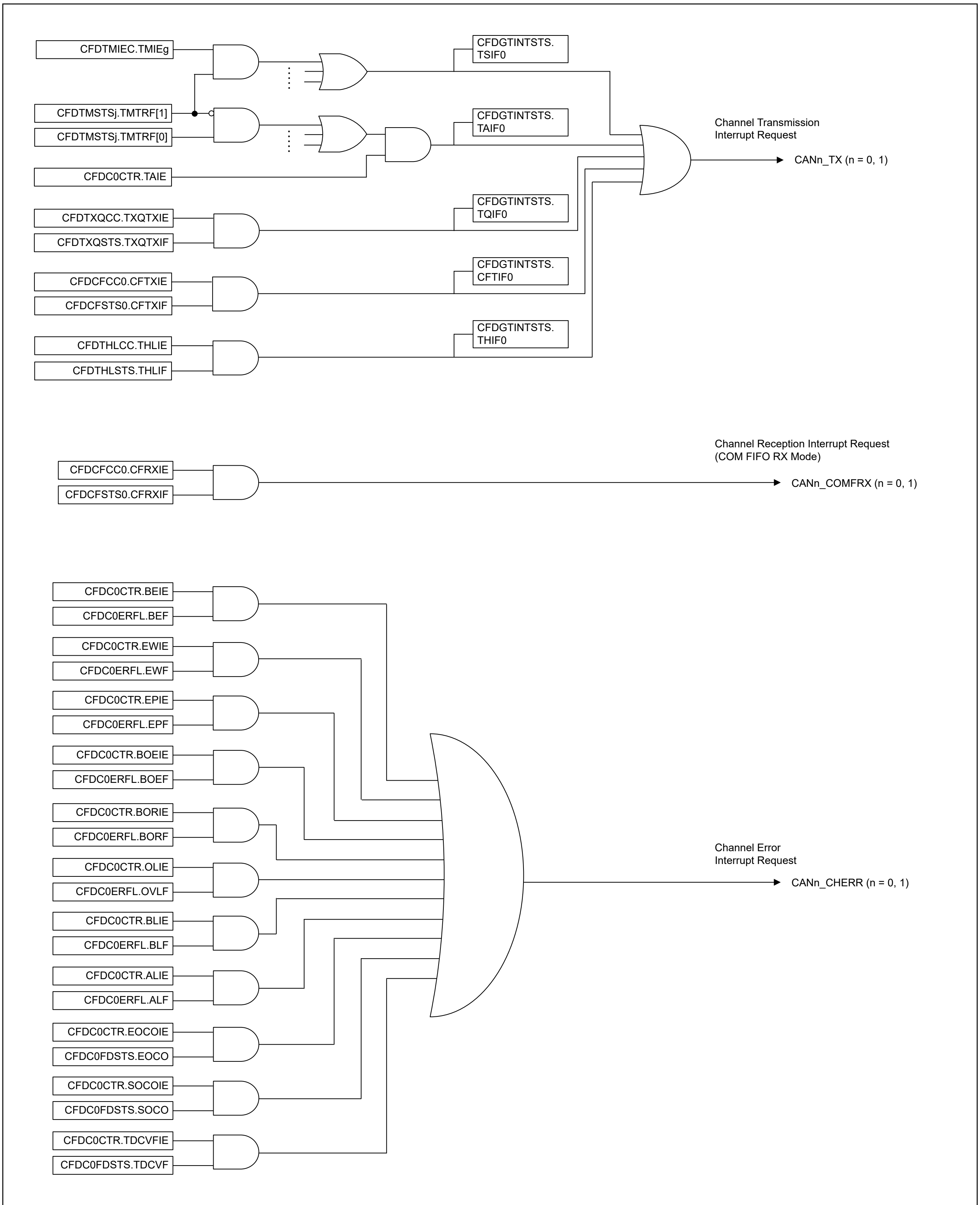


Figure 42.31 Channel interrupt block diagram

42.7.2 DMA Transfer

The CANFD module has message buffers that can be associated with a DMA channel:

- Reception DMA

– 2 RX FIFO message buffers
– Common FIFO Message Buffer

Figure 42.32 shows the potential DMA channels.

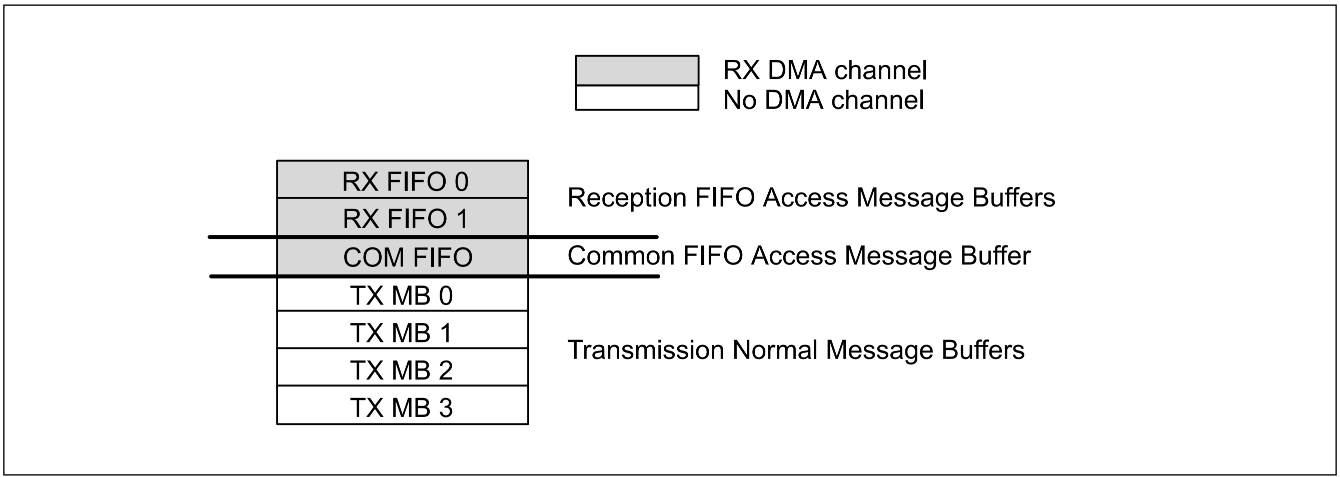


Figure 42.32 Message buffer connectable to a DMA channel

A DMA channel transfer request is generated for each FIFO entry to the DMAC when the related CFDCDTCT.RFDMAE or CFDCDTCT.CFDMAE is set to 1 and the belonging FIFO is not empty.

Reception FIFO Interrupt should be disabled for this particular FIFO (CFDRFCCa.RFIE or CFDCFCC.CFRXIE)

Use the regular start address for the DMA access window address. See Figure 42.33.

Table 42.25 DMA channel access window address

b = Message buffer component index	Message Buffer Component	Register	P	Regular Start Address
b = [0…1]	RFMBCPb[0]	CFDRFIDb	x	0x0520 + b × 0x004C
		CFDRFPTRb	x	0x0524 + b × 0x004C
		CFDRFFDSTSb	x	0x0528 + b × 0x004C
		CFDRFDFbp	[0…15]	0x052C + p × 0x0004 + b × 0x004C
—	CFMBCP0[0]	CFDCFID	x	0x05B8
		CFDCFPTR	x	0x05BC
		CFDCFFDCSTS	x	0x05C0
		CFDCFDFp	[0…15]	0x05C4 + p × 0x0004

DMA FIFO pointer decrement is done automatically by reading the last configured data payload byte (CFDRFCCa.RFPLS or CFDCFCC.CFPLS).

Note: The DMA must read the exact length of the configured data payload size (CFDRFCCa.RFPLS or CFDCFCC.CFPLS).

Note: This feature is not available for classical CAN function because CFDRFCCa.RFPLS and CFDCFCC.CFPLS are not in classical CAN.

Do not write to the FIFO control registers when DMA is enabled. The DMA enable of the particular DMA FIFO (CFDCDTCT.RFDMAE or CFDCDTCT.CFDMAE) can be set at any time. Figure 42.33 shows a configuration flow for an initial setup.

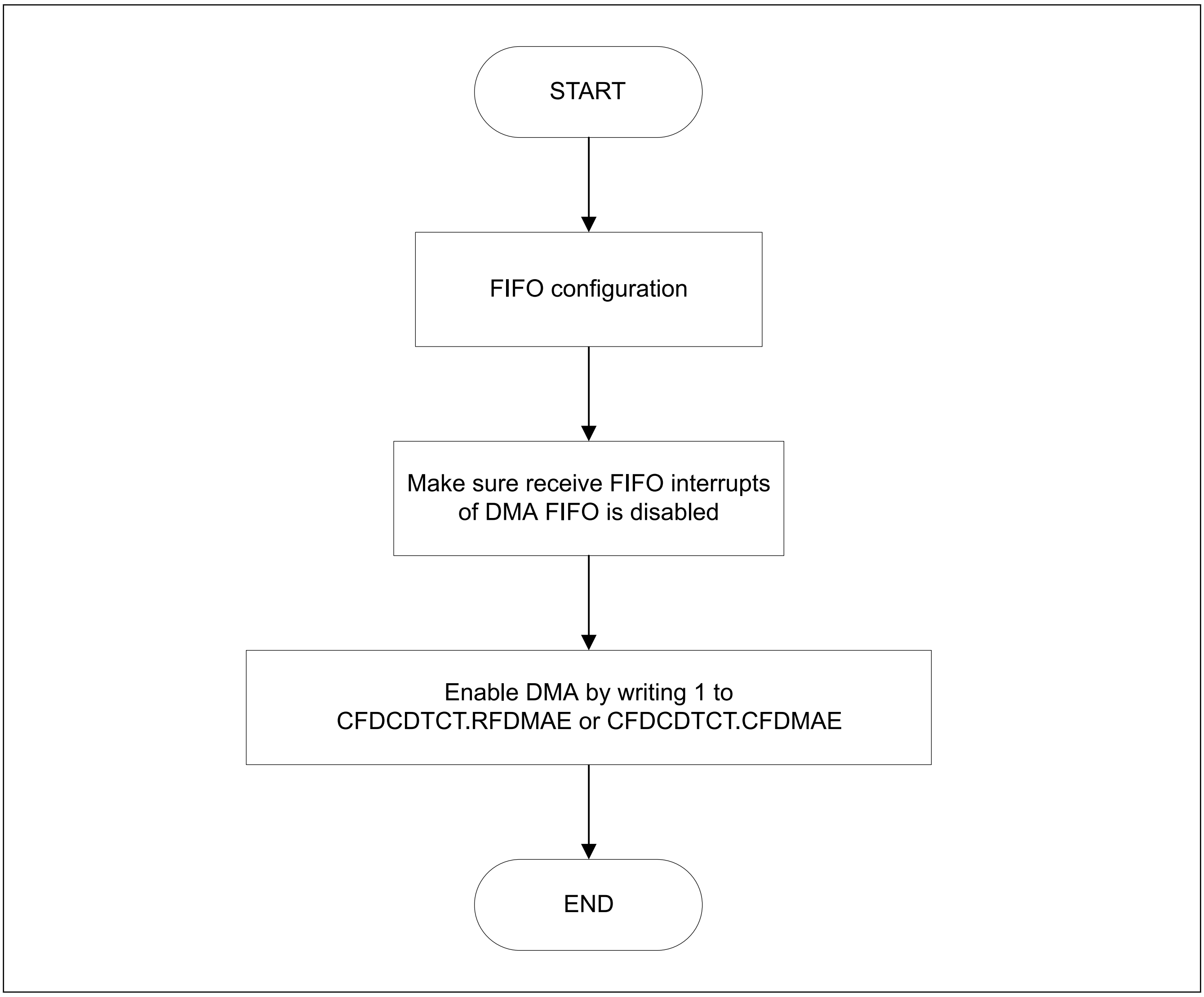


Figure 42.33 DMA enable flow

To disable a DMA transfer request, you must disable the particular DMA enable bit (CFDCDTCT.RFDMAE or CFDCDTCT.CFDMAE). If the disable is made during an ongoing transfer, then the transfer must be completed first before further action can be taken. The transfer status can be identified by the CFDCDTSTS.RFDMASTS or CFDCDTSTS.CFDMASTS bit. See Figure 42.34 for the DMA disable flow. When the DMA is disabled, consideration should be made for the remaining or new incoming messages to this particular reception FIFO.

When the FIFO is not disabled, reception to the FIFO continues.

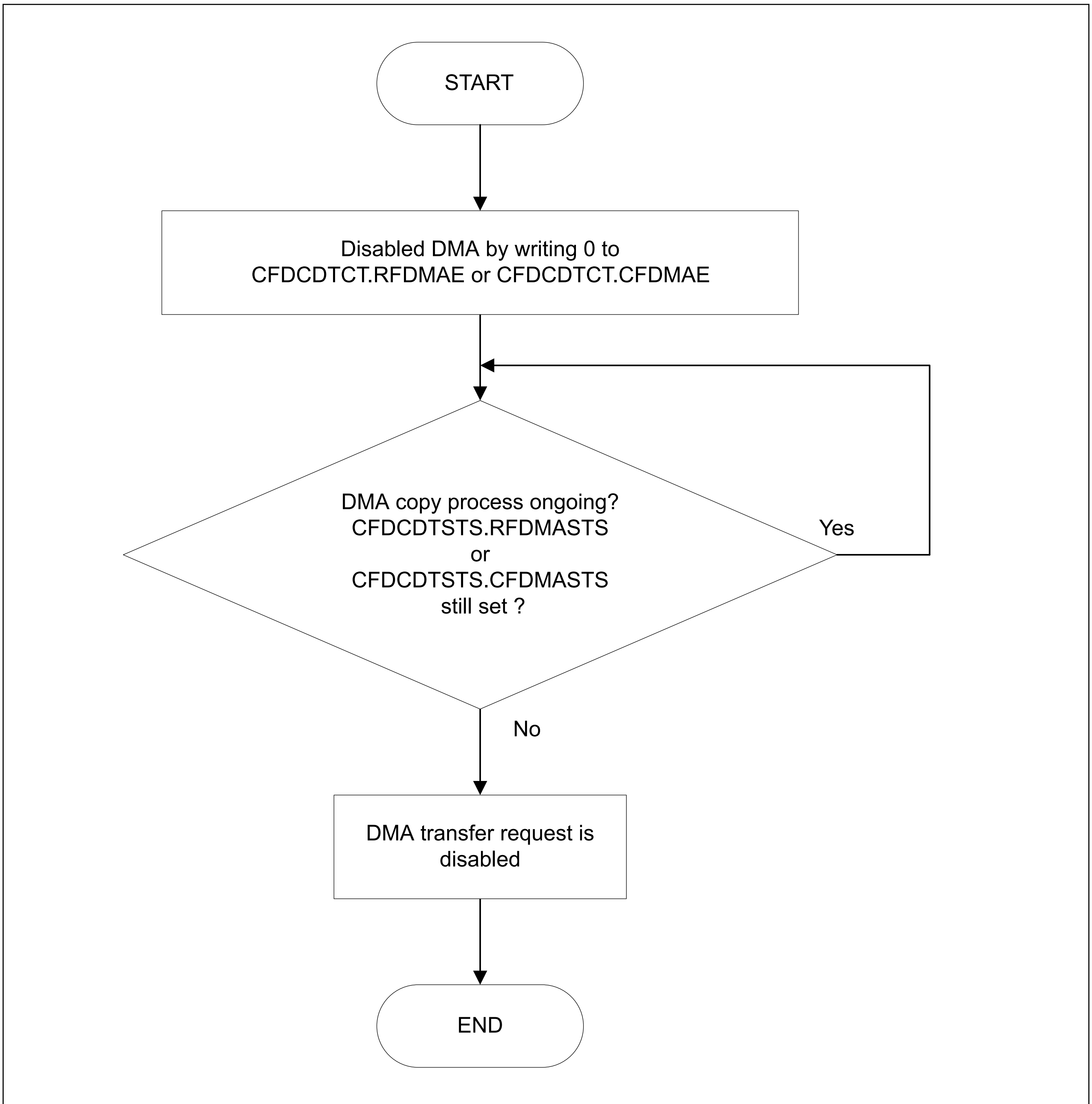


Figure 42.34 DMA disable flow

42.8 Reception and Transmission

42.8.1 Reception

In the CANFD module, CAN messages received on any of the channels are stored in RX message buffers, RX FIFO buffers, or Common FIFO buffers configured in RX mode depending on the Acceptance Filter List entries.

- Up to 16 RX message buffers can be configured
- 2 RX FIFO buffers available
- 1 Common FIFO Buffer can be configured in RX mode

42.8.1.1 Message Storage in RX Message Buffers

When a message is successfully received and stored in a RX message buffer, the corresponding New Data flag is set in the RX Message Buffer New Data Register.

The CAN message can be read from the corresponding RX message buffer.

If a new message is stored into a RX message buffer before the previous message in this message buffer can be read, then the original message is overwritten. There is no mechanism for preventing a new message from overwriting the current message in the RX message buffer. If such a loss of messages is not acceptable, then RX FIFO should be used for storing related messages.

Note: Users should do the same processing as the existing software flow also when using interrupt. (see Figure 42.36)

Note: Unused data bytes are filled with 0x00 depending on the DLC value.

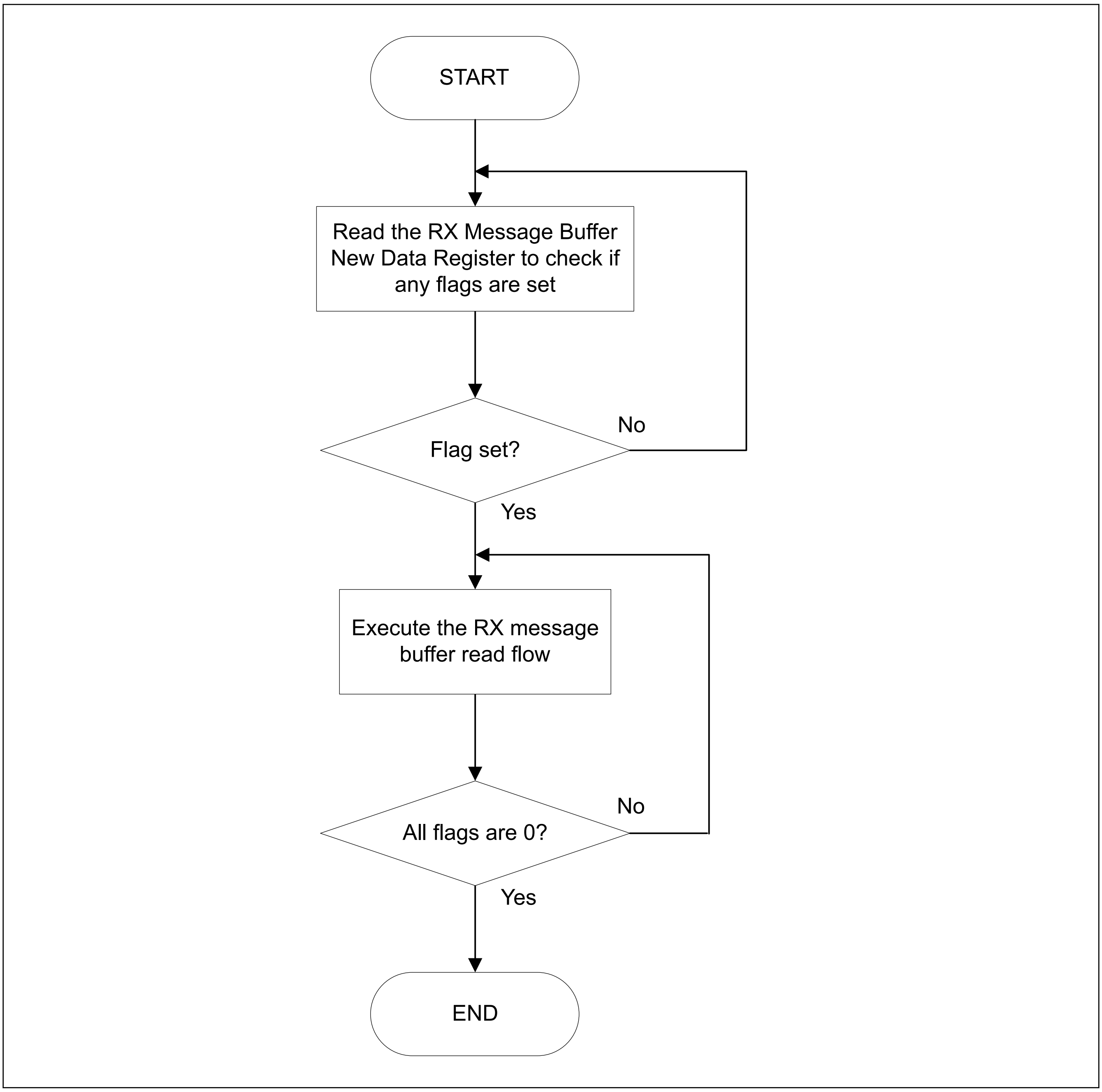


Figure 42.35 Access flow of RX message buffer (Polling)

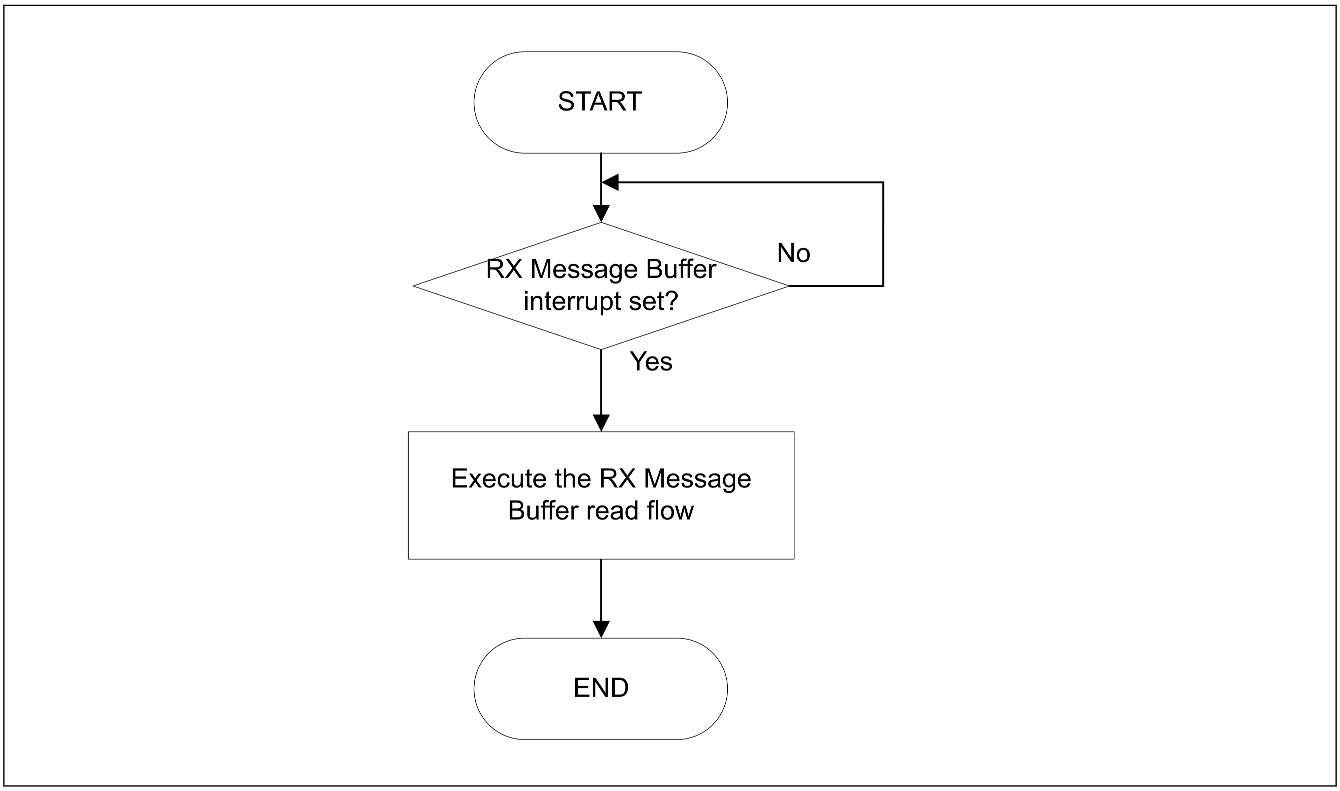


Figure 42.36 RX Message Buffer Message Access Flow (interrupt)

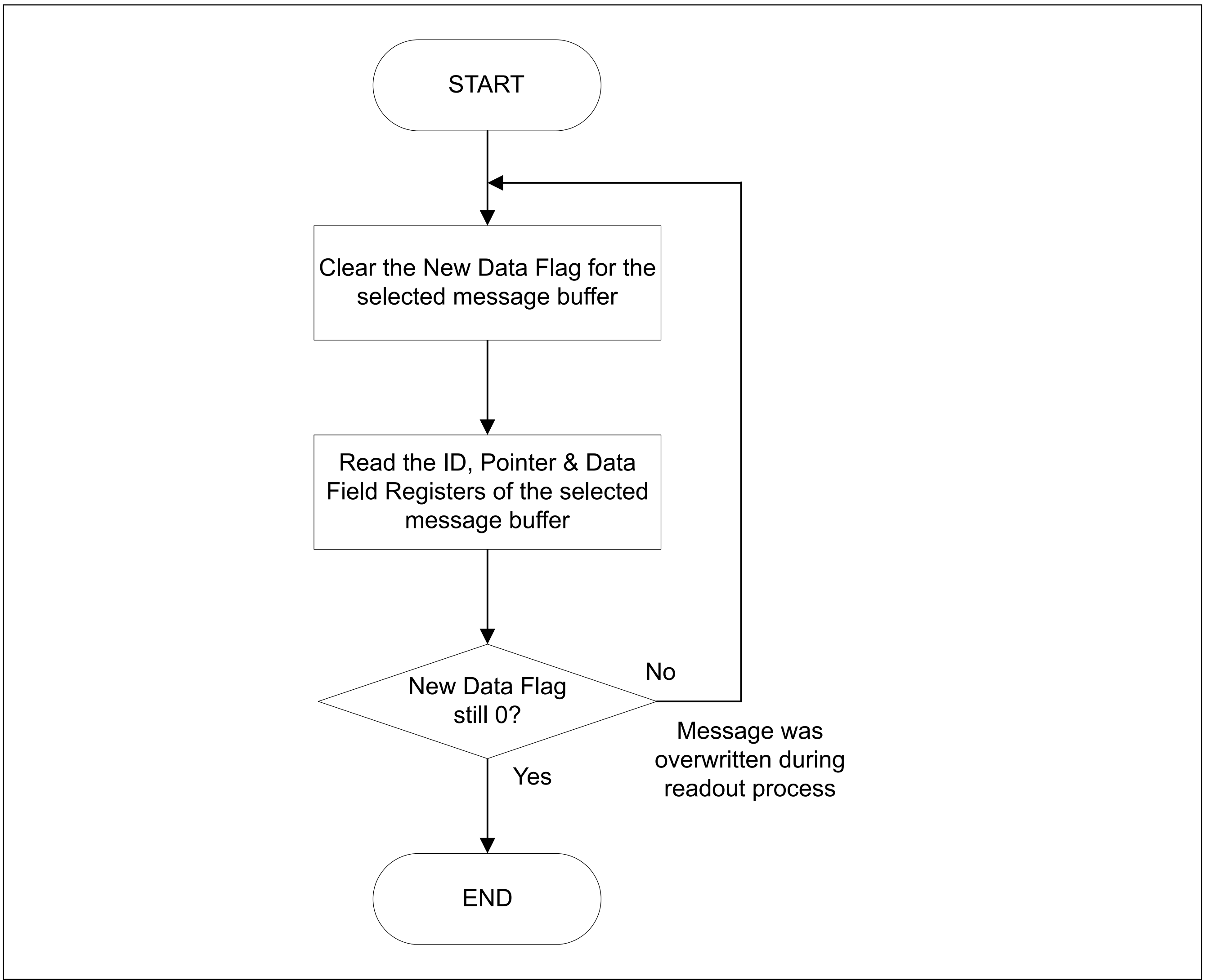


Figure 42.37 Read flow of RX message buffer

42.8.1.2 Message Storage in FIFO Buffers

The AFL entries for routing the received messages to RX FIFO buffers or Common FIFO buffer configured in RX mode should be configured based on system requirements.

The CFDGAFLP1r.GAFLFDP[8,1:0] field in the matching AFL entry selects the FIFO buffers to which the related reception message is stored.

When the received message is stored in one or more RX FIFO buffers or Common FIFO buffer configured in RX mode, the message counter value is incremented in the corresponding RX FIFO Status Registers or Common FIFO Status Register.

Depending on the configuration of the FIFO buffers, an interrupt might also be generated.

The message can be read from the corresponding FIFO Access registers.

Note: Because many messages can be stored in the FIFO buffers, reading more than one message may be required to read the latest message stored in a FIFO buffer.

If the message count value matches the FIFO depth, the FIFO Full flag is set.

When the value 0xFF is written to the corresponding FIFO Pointer Control Register, the message count is decremented by 1.

Only write 0xFF to the FIFO Pointer Control register after reading the complete message from the FIFO Access registers of the corresponding FIFO.

When all the messages stored in the FIFO are read, the FIFO Empty flag is set.

If a new message is stored into the FIFO when the FIFO message count matches the FIFO depth (FIFO full condition), the FIFO Message Lost flag is set and the new message is lost (no overwrite of already stored messages takes place).

An appropriate value can be configured as warning level to generate an interrupt before the FIFO full condition occurs to avoid loss of a message due to an overrun condition.

Note: The message lost can be set only in RX mode by CAN, and the flag is not set when the CPU is overloading the FIFO buffers.

The RX FIFO buffers and the Common FIFO buffers configured in RX mode can be disabled at any time by clearing the CFDRFCCa.RFE or CFDCFCC.CFE bit in the RX FIFO Configuration/Control Registers and the Common FIFO Configuration/Control Registers.

When the CFDRFCCa.RFE or CFDCFCC.CFE bit is cleared, the message read and write pointers of the FIFO are cleared and are no longer active. Therefore, all messages in the FIFO buffers are lost and no further messages can be stored into the FIFO.

When the RX FIFO buffers or Common FIFO buffer configured in RX mode is assigned as a DMA channel, software should not access the FIFO Access Register of this FIFO buffer or write 0xFF to the FIFO Pointer Control Register (CFDCFPCTR.CFPC or CFDRFPCTRa.RFPC). This can lead to unintended FIFO message decrement. The DMA channel controls the FIFO decrement automatically.

Note: If the interrupt flag is set for a FIFO buffer and then the FIFO is disabled, the interrupt flag is not cleared automatically. The interrupt flag should be cleared before disabling the FIFO.

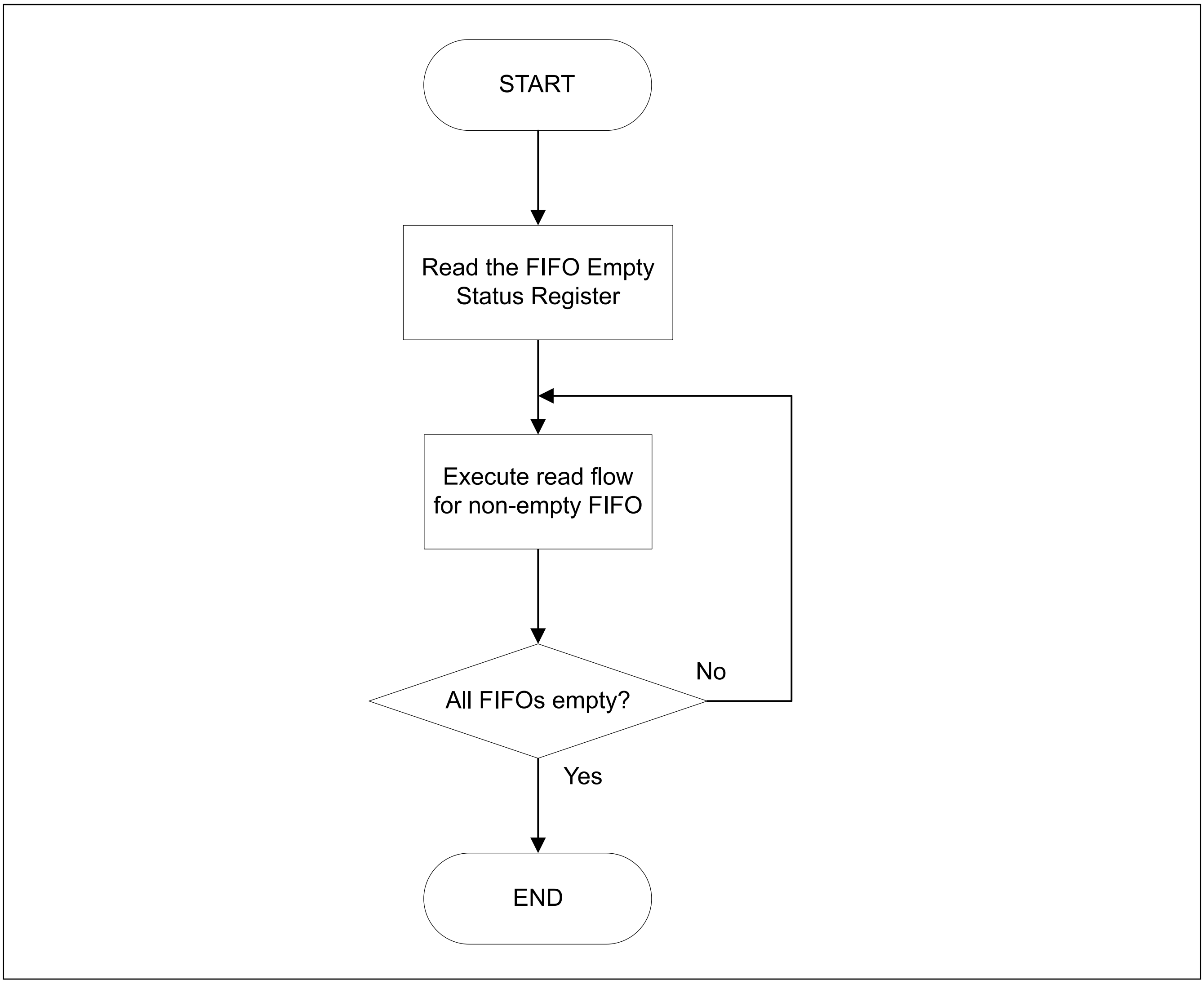


Figure 42.38 Access flow of FIFO buffer message (example for polling case)

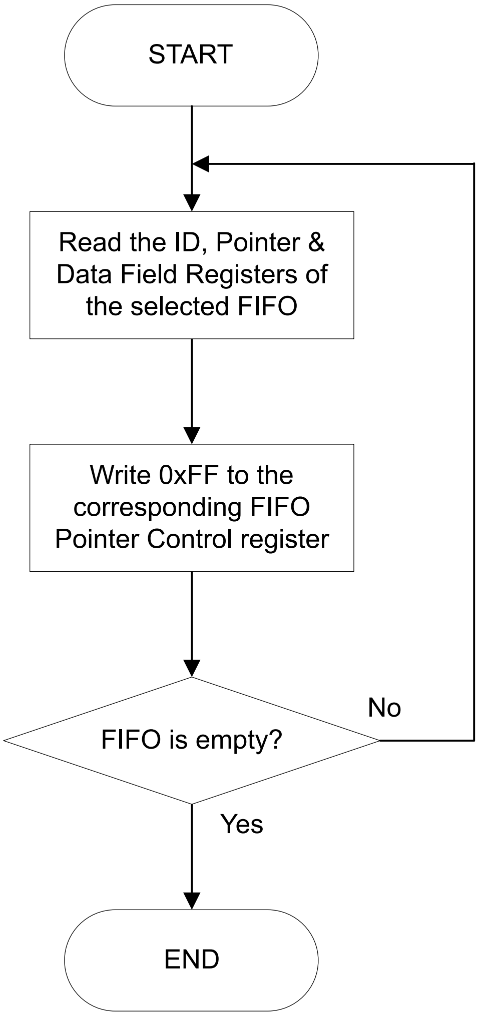


Figure 42.39 Read flow of RX FIFO buffer message (example for polling case)

Note: When the next frame is received before clearing the completion interrupt flag of reception, the completion interrupt of reception is not set again.

Even when an interruption flag is cleared after the completion processing of reception, the already received interrupt flag is not set.

It is necessary to perform the completion processing of reception even before the next completion of frame reception, and to clear an interruption flag.

When processing does not meet the condition, after checking that receiving data is empty, interrupt flag is cleared, and it checks that receiving data is empty again.

42.8.1.3 Timestamp

The timestamp counter is a free-running counter that can be used to check reception time of an incoming message or transmission time of successful transmitted messages. The Timestamp counter value is captured based on the CFDGFDCFG.TSCCFG[1:0] configuration (at the sample point of start of frame, point in time when the frame is valid, or for CANFD frames also at the sample point of the RES bit). For reception, it is stored together with the message ID and data into the target RX message buffer or RX FIFO.

For transmit message, the timestamp counter value is stored as part of the TX History List entry.

The counter can be clocked with the peripheral clock or with the CAN channel bit timing clock. The counter source clock can be configured with the CFDGCFG.TSSS bit of the Global Configuration Register. If this bit is 0, the peripheral clock is used. If the bit is 1, the selected CAN channel bit time clock is used.

The channel selection is performed with the CFDGCFG.TSBTCS bit of the Global Configuration Register.

Care must be taken when using selected CAN channel bit time clock as the clock source. When entering Channel Halt mode or Channel Reset node, for this channel, the timestamp counter is stopped. For other CAN channels, the timestamp counter value is not updated.

If peripheral clock is selected as the timestamp counter clock source, Channel modes do not affect the timestamp counter function.

The source clock for the timestamp counter can be divided by a factor defined by the CFDGCFG.TSP bits (timestamp prescaler) in the Global Configuration Register.

The timestamp counter can be reset to 0x0000 with the CFDGCTR.TSRST bit (timestamp reset).

42.8.2 Transmission

There are several possible transmission configurations:

- Normal transmission
- FIFO transmission
- TX Queue transmission
  A fixed number of transmission message buffers (4 TX message buffers) are dedicated. These message buffers are only used for transmission and cannot be configured for reception.
  Additionally, transmission from TX Queue or Common FIFO in TX mode can be configured in the following way (see Figure 42.40):
  - TX Queue: Up to four transmission message buffers can be grouped to form a TX Queue with a common access window.
    Upper transmission message buffers are used to form the TXQ.
    TXQ has an access window.
    - TXQ is transmission Message Buffer 0.
- Common FIFO (TX mode): Common FIFO in TX mode is linked to a dedicated channel.
  Channel has a fixed number of one Common FIFO assigned to it. Within the channel, a Common FIFO configured in TX mode, can be freely linked (assigned) between 0 and 3 transmission message buffers (only one FIFO to one transmission message buffer).
  The Common FIFO buffer then replaces the transmission message buffer linked to it.
  Transmission Control and Status registers of these transmission message buffers should not be used.

See Figure 42.28 for information about Common FIFO buffer assignment to related channel.

Note: Common FIFO buffer should not be linked to TX message buffers that are already part of a TX Queue.

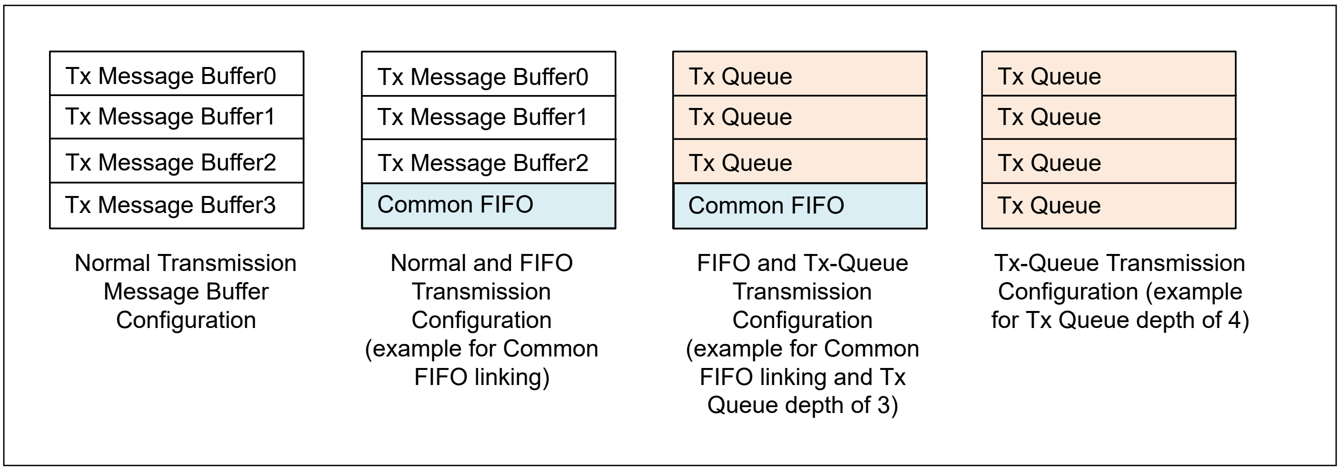


Figure 42.40 Configuration of channel transmission message buffer

42.8.2.1 Transmission Priority

If two or more transmission message buffers of a channel are configured for transmission, then the transmission priority in the CANFD module can be selected from the following two modes:

- CAN ID priority
- Message buffer number priority.

The transmission priority mode is common for all message buffers. It can be configured with the CFDGCFG.TPRI bit in the Global Configuration Register.

For message buffer number priority transmission, the smallest message buffer number with transmission request has the highest priority for transmission. This also includes the TX message buffers linked to the Common FIFO buffer configured in TX mode.

However, message buffer number priority should not be used if TX Queue is enabled.

For CAN ID priority transmission, ID priority complies with the CAN bus arbitration rule (as specified in ISO 11898-1 specification). All TX message buffers can enter the ID priority comparison for message buffers configured for transmission. This also includes the TX message buffers linked to the Common FIFO buffer configured in TX mode and includes the TX Queue message buffers.

If the ID of two or more message buffers is the same, then the smaller message buffer number has higher priority for transmission.

Note: For Common FIFO buffer configured in TX mode, only the message currently being pointed to by the FIFO read pointer can be included in the transmission arbitration.

If the message is being transmitted from the FIFO, then the next pending message within the same FIFO is considered in the transmission arbitration.

In contrast to this, all transmission message buffers of a TX Queue participate in internal transmission arbitration.

Figure 42.41 shows the transmission configuration flow.

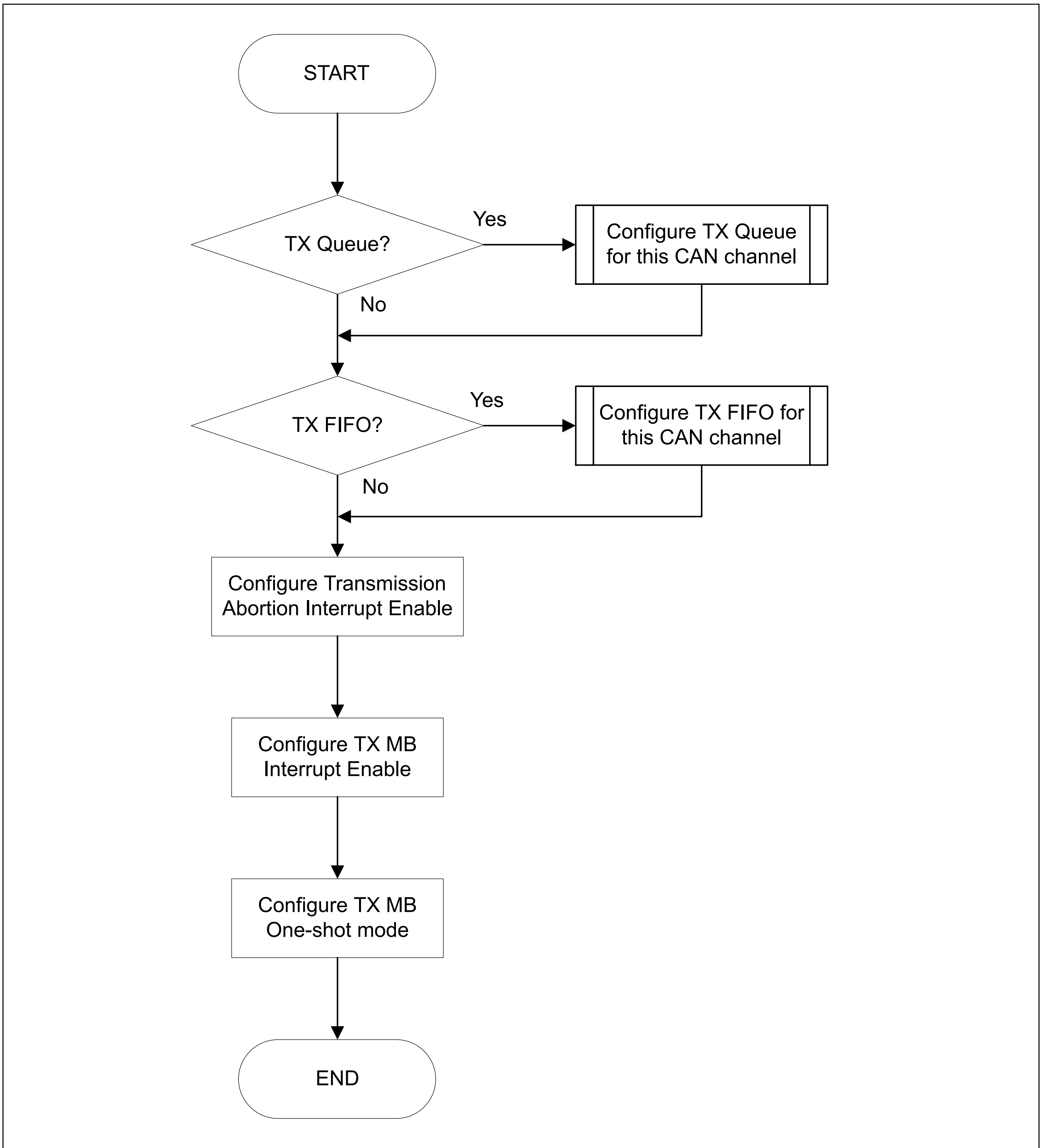


Figure 42.41 Flow for transmission configuration

42.8.2.2 Normal Transmission

Each transmission message buffer has two modes of message transmission:

1. Regular transmission mode
   If the message buffer is placed in regular transmission mode, the data frame or remote frame set in that message buffer can be transmitted.
   Completion of regular transmission can be checked through the related TX Message Buffer Transmission Result flag bits (CFDTMSTSj.TMTRF) in the TX Message Buffer Status Registers. These bits are set to 10b or 11b when the regular transmission is successful.

When arbitration is lost or an error occurs, message transmission is further attempted if no transmission abort request is set for this transmission message buffer.
New internal transmission arbitration for this channel is performed for all message buffers with transmission request.

2. One-shot transmission mode

When the CFDTMCi.TMOM bit of the TX Message Buffer Control Registers is set for a transmission message buffer, the message buffer is placed in One-shot transmission mode and attempts to transmit a message only once.

Completion of One shot transmission can be checked through the related TX Message Buffer Transmission Result Flag bits (CFDTMSTSj.TMTRF) in the TX Message Buffer Status Registers. The CFDTMSTSj.TMTRF bits are set to 10b or 11b when One-shot transmission is successful.

The CFDTMSTSj.TMTRF bits are set to 01b when arbitration is lost or an error occurs during transmission of the related message buffer.

Additional message transmission is not attempted in this case.

The regular transmission request procedure after a configuration is shown in Figure 42.42.

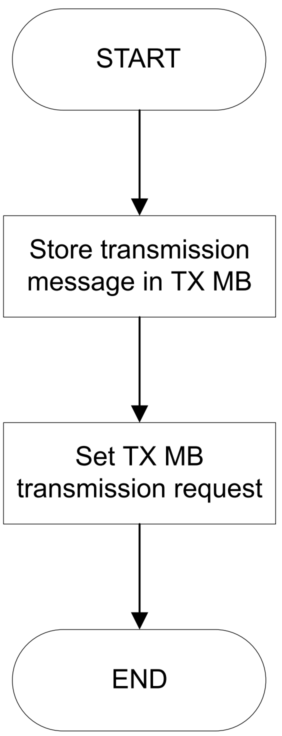


Figure 42.42 Transmission request procedure using normal TX Message Buffer mode

(1) Setting for TX Message Buffer Control Register

Table 42.26 shows configuration of a normal CAN transmission mode.

Table 42.26 Configuration of CAN transmission mode (1 of 2)

Transmission request CFDTMCi.TMTR	Transmission abort request CFDTMCi.TMTAR	One-shot enable CFDTMCi.TMOM	Communication activity
0	0	0	Message buffer disabled
0	0	1	Message buffer disabled

Table 42.26 Configuration of CAN transmission mode (2 of 2)

Transmission request CFDTMCi.TMTR	Transmission abort request CFDTMCi.TMTAR	One-shot enable CFDTMCi.TMOM	Communication activity
1	0	0	Configured as a transmission message buffer for a data frame or a remote frame
1	0	1	Configured as a one-shot transmission message buffer for a data frame or a remote frame
1	1	0	Transmission abort requested
1	1	1	One-shot transmission abort requested

The configuration bits can be configured in the TX Message Buffer Control Registers.

Figure 42.43 shows timings for successful transmission for two message buffers.

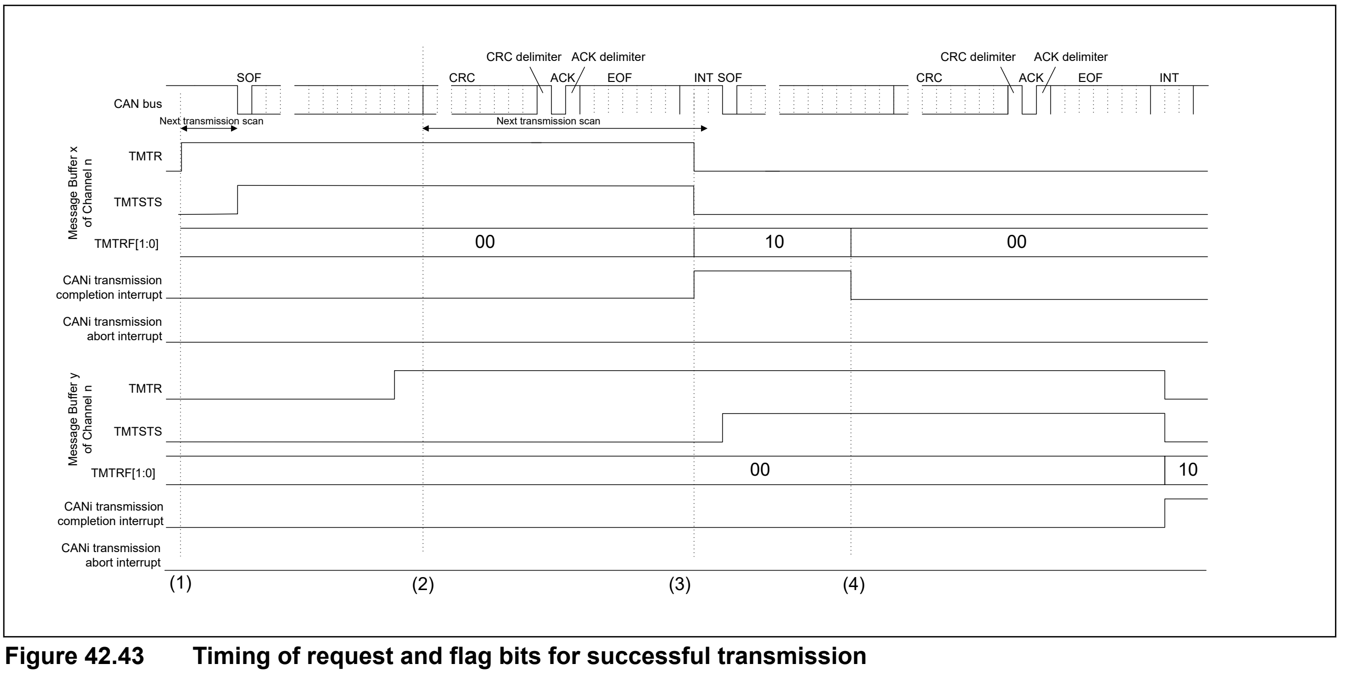


Figure 42.43 Timing of request and flag bits for successful transmission

1. If the CFDTMCi.TMTR bit in the TX Message Buffer Control Registers is set in the bus idle state, the message buffer scanning procedure determines the highest priority message buffer for transmission.
   When the transmission message buffer is determined, the CFDTMSTSj.TMTSTS bit in the related TX Message Buffer Status Registers is set (transmitting/transmitter), and CAN channel starts the transmission *1.
2. At the first bit of CRC, the transmission scanning procedure starts for the next possible transmission when pending transmission requests exist.
3. If the message has been successfully transmitted, the CFDTMSTSj.TMTRF[1:0] bits in the corresponding TX Message Buffer Status Registers are set to 10b and CFDTMSTSj.TMTSTS and the CFDTMCi.TMTR bits are cleared.
   When the TMIE bit in the TX Message Buffer Interrupt Enable Configuration Registers is set (interrupt enabled), the CAN successful transmission interrupt request is generated.
   To clear the related interrupt line, clear the CFDTMSTSj.TMTRF flag bits.
4. Before starting the next transmission, clear the CFDTMSTSj.TMTRF bits. Load the next message in the transmission message buffer and set the CFDTMCi.TMTR bit again.
   CFDTMCi.TMTR bit cannot be set again before CFDTMSTSj.TMTRF[1:0] bits are cleared.

Note 1. If arbitration is lost after the CAN channel starts the transmission, the CFDTMSTSj.TMTSTS bit is cleared.

The transmission scanning procedure is performed again to search for the highest priority transmission message buffer from the beginning of the first CRC bit.

If an error occurs either during transmission or following the loss of arbitration, then during the error frame, the transmission scanning procedure is performed again to search for the highest priority transmission message buffer.

Note: The setting point of CFDTMSTSj.TMTSTS is not always fixed at the start of the SOF. It may be delayed up to the start of the standard ID due to the synchronization logic implemented for the PLL bypass.

Figure 42.44 shows timings for transmission abort for two message buffers.

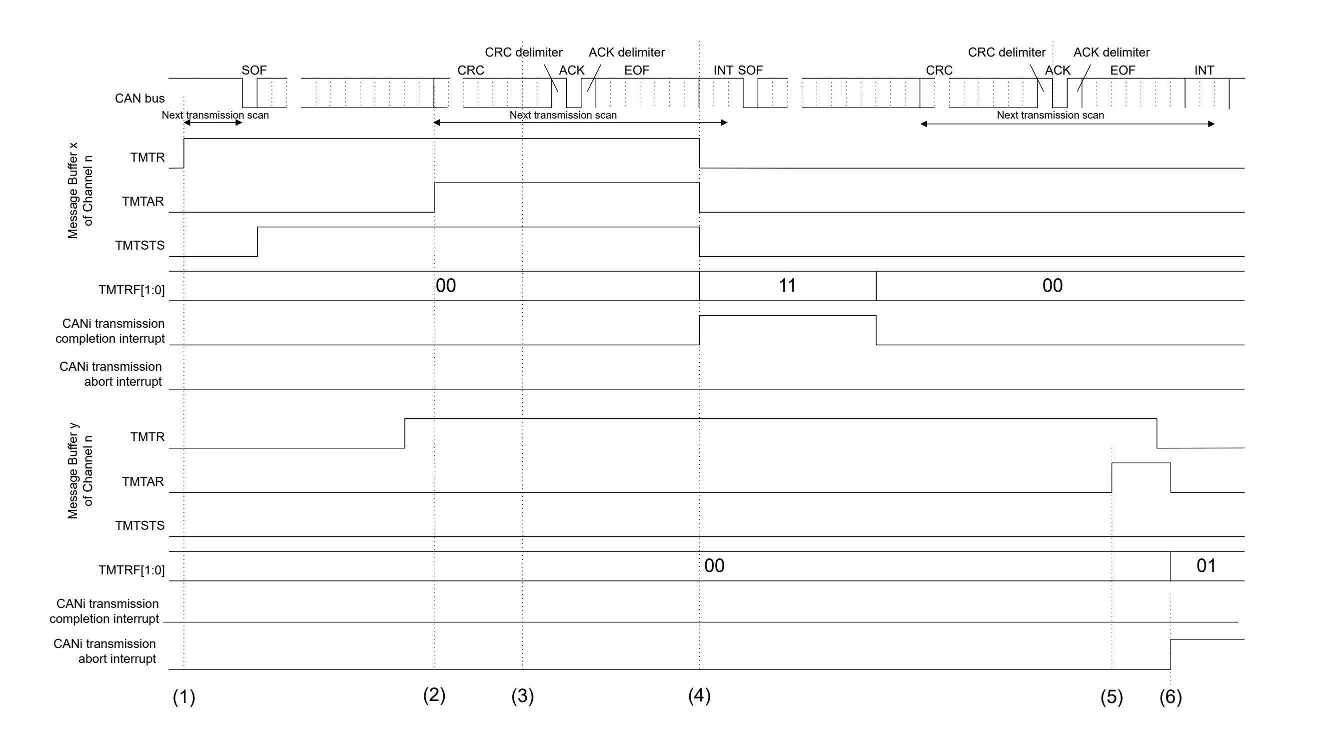


Figure 42.44 Timing of request and flag bits for transmission abort

1. If the CFDTMCi.TMTR bit in the TX Message Buffer Control Registers is set in the bus idle state, the message buffer scanning procedure determines the highest priority message buffer for transmission.
   When the transmission message buffer is determined, the CFDTMSTSj.TMTSTS bit in the TX Message Buffer Status Registers is set (transmitting/transmitter), and CAN channel starts the transmission*1.
2. If the CFDTMCi.TMTAR bit is set when the related message buffer is already selected for transmission or currently transmitting, the message is not aborted, if no error occurs or arbitration is lost.
3. At the first CRC bit, the transmission scanning procedure starts for the next transmission. In this example, timing chart message buffer y is not selected as the next transmission message buffer.
4. If the message has been successfully transmitted, the CFDTMSTSj.TMTRF[1:0] bits in the corresponding TX Message Buffer Status Registers are set to 11b and the CFDTMSTSj.TMTSTS and CFDTMCi.TMTR bits are cleared.
   When the TMIE bit in the TX Message Buffer Interrupt Enable Configuration Registers is set (interrupt enabled), the CAN successful transmission interrupt request is generated.
   To clear the related interrupt line, clear the CFDTMSTSj.TMTRF[1:0] bits.
5. Another CAN node is transmitting on the CAN bus (CFDTMSTSj.TMTSTS is not set). If the CFDTMCi.TMTAR bit is set when the related channel is under transmission scan, the transmission request cannot be cleared.
6. After internal processing time, the transmission is aborted and the CFDTMSTSj.TMTRF[1:0] bits are set to 01b.
   If the message buffer is not transmitting or selected as the next transmission message buffer or under transmit scan, then the abort is immediately accepted and the corresponding CFDTMSTSj.TMTRF[1:0] bits in the TX Message Buffer Status Registers are set to 01b.
   In addition, CFDTMCi.TMTR, and CFDTMCi.TMTAR bits are cleared automatically.
   When the transmission abort interrupt enable TAIE bit of the related Channel Control Register is set then an interrupt is generated for successful transmission abort.
   To clear the related interrupt line the CFDTMSTSj.TMTRF[1:0] bits have to be cleared.

Note 1. If arbitration is lost after the CAN channel starts the transmission, the CFDTMSTSj.TMTSTS bit is cleared.

The transmission scanning procedure is performed again to search for the highest priority transmission message buffer from the beginning of the first CRC bit.

If an error occurs, either during transmission, or following the loss of arbitration, then during the error frame, the transmission scanning procedure is performed again to search for the highest priority transmission message buffer.

42.8.2.3 TX FIFO Transmission

One common FIFO buffer is assigned to CANFD module. The FIFO buffer can be linked to any normal TX message buffer position for this channel with the CFDCFCC.CFTML bits in the Common FIFO Configuration/Control Register if configured in TX mode.

When the transmission scan starts and the FIFO buffer corresponding to this TX message buffer is enabled, the relevant message in the FIFO buffer participates in the transmission scan.

Configuration of a TX message buffer linked to a FIFO buffer configured in TX mode should not be done.

(1) TX FIFO Operation

CAN messages can be written into the TX FIFO by writing to the corresponding FIFO Access registers.

When the value 0xFF is written into the corresponding FIFO Pointer Control Register, the message count of the related FIFO is incremented by 1.

Only write to the FIFO Pointer Control register after writing the complete message to the corresponding FIFO Access registers. If the message count matches the FIFO depth, the FIFO Full flag is set.

The oldest message in the TX FIFO is included in the scan for transmission by the corresponding CANFD module channel logic.

When a message is successfully transmitted from the TX FIFO, the message count value is decremented by 1. When all the messages from the FIFO are transmitted, the FIFO Empty flag is set.

The interrupt generation conditions for the TX FIFO buffer can be configured by configuring the CFDCFCC.CFIM bit in the corresponding Common FIFO Configuration/Control Register.

If CFDCFCC.CFIM bit is 0, then interrupt is generated when the last message is successfully transmitted from the TX FIFO buffer.

If CFDCFCC.CFIM bit is 1, then interrupt is generated for every successfully transmitted message from the TX FIFO buffer.

The Common FIFO can set interrupt when CAN frame transmission is complete.

The Common FIFO buffer configured in TX Mode can be disabled by clearing the CFDCFCC.CFE bit in the Common FIFO Configuration/Control Register. If this bit is cleared to 0, the FIFO Empty flag is set as follows:

- Immediately if the message from the TX FIFO is neither scheduled for the next transmission nor in transmission
- Following the transmission completion, the detection of an error on the CAN bus, loss of arbitration or transition to Channel or Global Halt mode if the transmission from the TX FIFO is already scheduled for transmission or already in transmission.

Note: The Common FIFO buffer is considered as disabled after clearing the CFDCFCC.CFE bit only when the Empty flag is set for the corresponding Common FIFO buffer.

Other possible messages pending from the TX FIFO are lost and their transmission must be requested again. Before CFDCFCC.CFE is set again, ensure that CFDCFSTS.CFEMP bit is set and that there are no pending aborts from the TX FIFO.

When the CFDCFCC.CFE bit is cleared, the message read and write pointers of the FIFO are cleared and are no longer active. Therefore, all messages in the FIFO buffers are lost and no further message can be stored into the FIFO.

The FIFO transmission request procedure after configuration is shown in Figure 42.45.

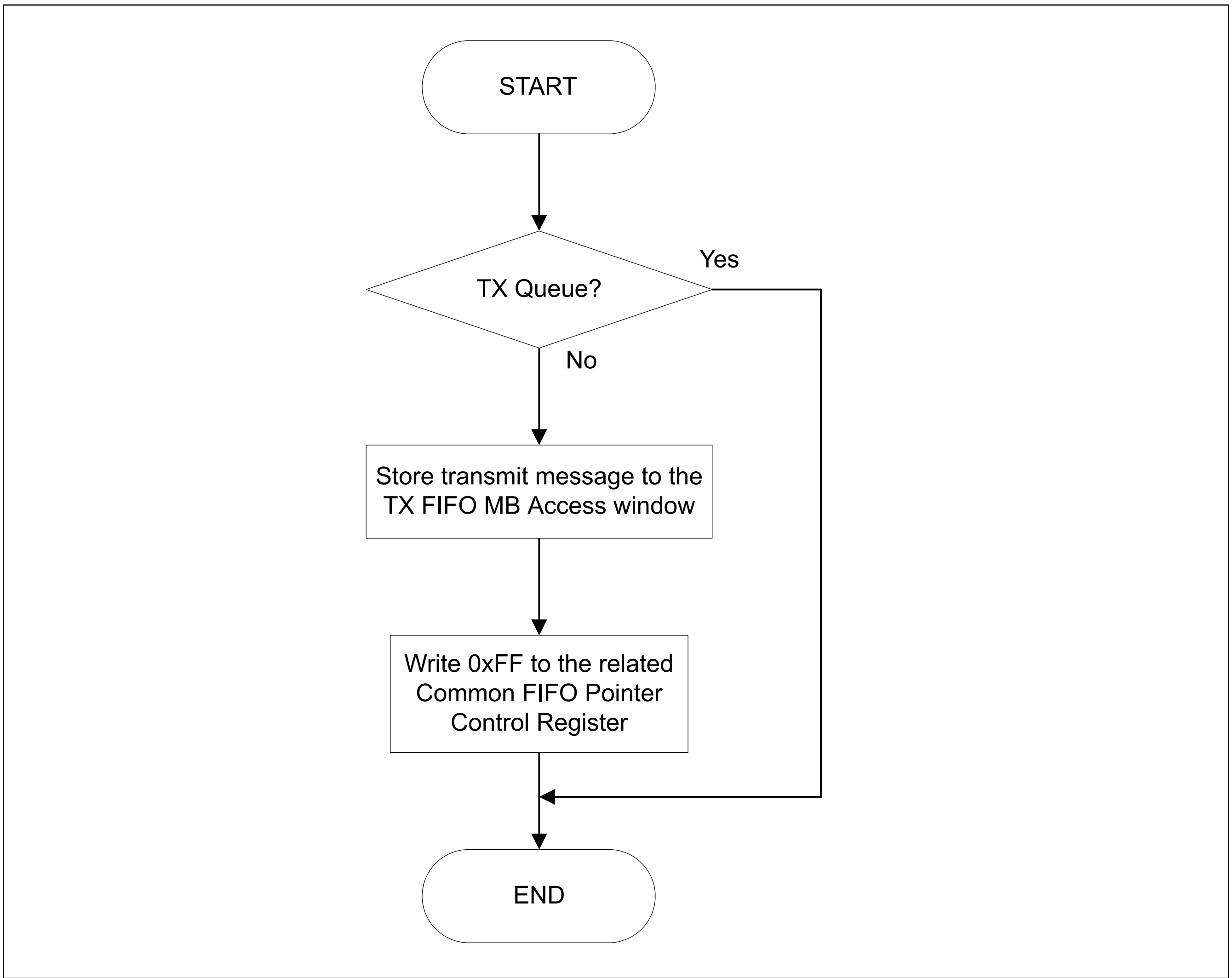


Figure 42.45 Request procedure for TX FIFO transmission

(2) Interval Timer for FIFO Transmission

For each Common FIFO in TX mode, it is possible to specify a delay between two consecutive messages that are configured for transmission from the same FIFO buffer. This delay is called interval time. This interval time starts after the first message has been successfully transmitted from the FIFO buffer after the CFDCFCC.CFE bit is set.

When the Common FIFO in TX mode is enabled, the first message is transmitted without considering this interval time.

The interval timer stops counting when:

- FIFO is disabled by clearing the CFDCFCC.CFE bit.
- CAN channel is in CH_RESET mode.

The interval time is specified by the CFDCFCC.CFITT value from 0 to 255 timer units in the Common FIFO Configuration/ Control Register.

The timer unit can be defined based on two different source clocks for the interval timer. To disable the interval timer for FIFO transmission, select a value of 0.

The timer source can be selected with the configuration bit CFITSS in the Common FIFO Configuration/Control Register.

If CAN channel bit time clock is configured as the clock source, and the CAN channel enters CH_HALT, CH_RESET, or CH_SLEEP mode, the interval timer is stopped for that channel.

If peripheral clock is selected as the interval timer clock source, the interval timer is stopped only when the CAN channel is in CH_RESET or CH_SLEEP mode.

The reference clock can be used to configure the interval time in fixed time units. It is based on the peripheral clock. The reference clock prescaler value CFDGCFG.ITRCP in the Global Configuration Register defines the relation between the peripheral clock frequency/period and the reference clock period.

See Table 42.27 for CFDGCFG.ITRCP configuration values to achieve different reference clock periods based on the peripheral clock frequency and period.

Table 42.27 Configuration example for the reference clock of the FIFO interval timer

Reference clock/Peripheral clock	1 µs	100 µs	500 µs
16 MHz/62.5 ns	16	1600	8000
20 MHz/50 ns	20	2000	10000
32 MHz/31.25 ns	32	3200	16000
50 MHz/20 ns	50	5000	25000

The reference clock resolution can be specified by the interval timer reference clock resolution value CFDCFCC.CFITR in the Common FIFO Configuration/Control Register.

The interval time is based on the reference clock period multiplied by the configured value (x1 or x10). The reference clock based interval timer can be used to satisfy the requirements of the ISO 15765-2 Separation Time. The whole range for the separation time from 100 µs to 127 ms can be covered.

The specified interval time starts after successful transmission event (after EOF7 state of the CAN protocol).

When the interval time has elapsed, the next transmission request is raised by the related TX FIFO. Therefore, the interval time defines the minimum time between two messages transmitted from one FIFO.

The next message is sent at earliest after this interval time. Figure 42.46 shows an example timing of the internal processing.

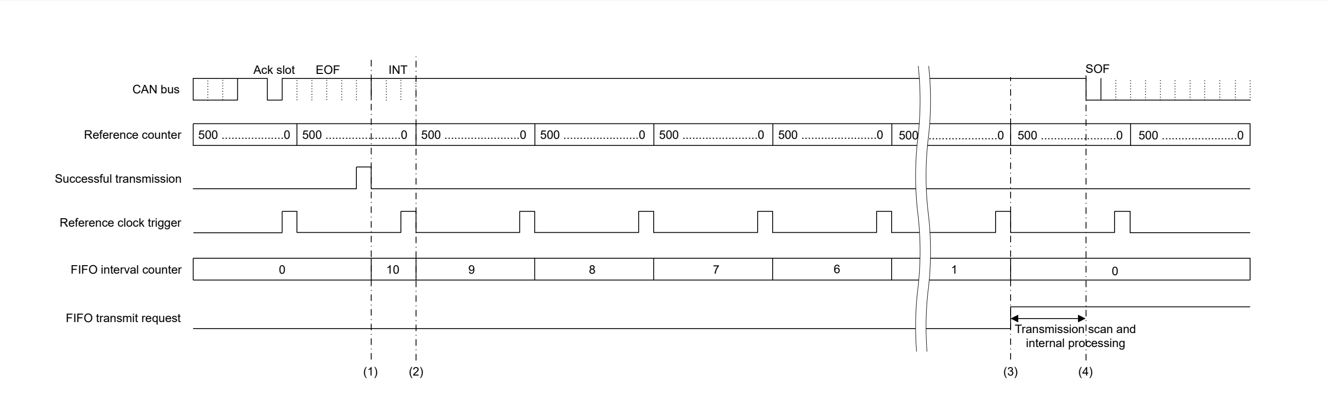


Figure 42.46 Example for interval processing time

The configuration for the timing in Figure 42.46 is as follows:

- Peripheral clock frequency = 50 MHz
- Interval timer reference clock (CFDGCFG.ITRCP) = 500 times
- Reference clock from the settings in Figure 42.46 = 10 µs
- Common FIFO interval timer source selection (CFDCFCC.CFITSS) = 0
- Common FIFO interval timer resolution (CFDCFCC.CFITR) = 0
- Common FIFO interval transmission time (CFDCFCC.CFITT) = 10 times
- Theoretical message separation interval = 100 µs

1. Internal FIFO interval timer is restarted with the occurrence of successful transmission result. This restart is not synchronized to the reference clock trigger. Therefore, the first interval is counting less or equal to 1 reference clock interval.
2. With the next reference clock trigger the FIFO interval timer is decremented.

3. When the FIFO interval timer reached the value 0, the FIFO transmit request is set.
4. When the FIFO is selected for transmission, the transmission starts. Due to internal processing, this usually takes less than 3 CAN bit time, between the internal FIFO transmit request set in step 3. and the actual transmission.

In the worst case when multiple events such as a reception scan, an internal message routing, a transmit scan on all channels occur, it can take up to 120 peripheral clock cycles.

As shown in Figure 42.46, it is not guaranteed that the minimum interval is always equal to the configured value. If a minimum time must never be breached, configure CFDCFCC.CFITT to the required minimum value plus 1.

If additional TX message buffers or TX FIFO are configured for transmission of the same channel, the real delay between two messages transmitted from a TX FIFO can be much longer than specified by the interval time. This is due to higher priority message transmission from these TX message buffers or TX FIFO.

Figure 42.47 shows a block diagram of the FIFO interval time generation circuit.

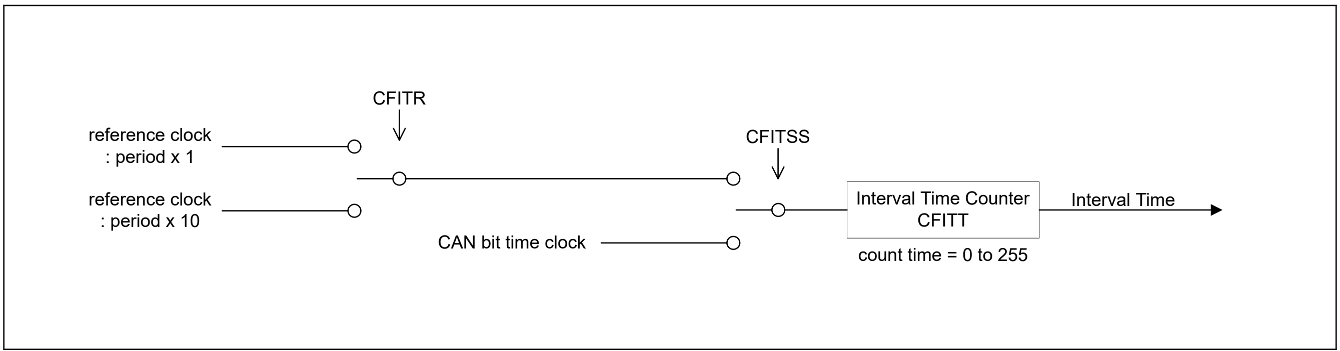


Figure 42.47 Block diagram of FIFO interval timer

42.8.2.4 TX Queue

Each enabled TX Queue for a specific channel consists of 3 to 4 TX message buffers, which are accessed through one access window.

- The first TX Queue can be configured with a depth of three up to four buffers and uses TX Message Buffer No. 0 as access window (referred to as TXQ)

All the TXQ messages enter the priority comparison for the transmission, which should be only ID Priority (CFDGCFG.TPRI = 0).

The registers for TXQ are:

- CFDTXQCC
- CFDTXQSTS
- CFDTXQPCTR

See related access registers TX Message Buffer ID Registers (TMID[m]), TX Message Buffer Pointer Registers (TMPTR[m]), TX Message Buffer Data Field 0 Registers, and TX Message Buffer Data Field 1 Registers (TMDF[0:1][m]) when access window TXQ0 is used.

The depth of each TXQ buffer can be configured by writing to the CFDTXQCC.TXQDC[1:0] bits of the TX Queue Configuration/Control Register. TXQ can be set from TXMB0 to TXMB3 as a queue buffer at the maximum.

The 4 available options for the depth configuration of TXQ buffer are:

- 0x00: TX Queue disabled
- 0x01: reserved
- 0x10: 3 Messages
- 0x11: 4 Messages

Do not access all the TX message buffers forming the TX Queue directly (except TX Message Buffer No. 0, which act as TX Queue access window).

When a system writes in TXQ, it writes in send data, after checking the state of TXQ.

Do not access or configure the related TX Message Buffer Control Registers.

The messages stored to the TX Queue access window are internally stored to a free buffer of the TX Queue.

When the buffer is full, no further access should be done to the queue, until it is no longer full. If access is a software write when the buffer of TXQ is full, send data is overwritten.

The TX Queue can be disabled by clearing the TXQE bit in the TX Queue Configuration/Control Register. If this bit is cleared, the TX Queue Empty flag is set as follows:

- Immediately if the message from the TX Queue is neither scheduled for the next transmission nor in transmission
- Following the transmission completion, the detection of an error on the CAN bus, loss of arbitration or transition to Channel or Global Halt mode if the transmission from the TX Queue is already scheduled for transmission or already in transmission.

Note: The TX Queue is disabled only when the Empty flag is set after clearing the TXQE bit for the corresponding TX Queue.

Other possible messages pending from the TX Queue are lost and their transmission must be requested again.

Before TXQE is set again, ensure that the CFDTXQSTS.TXQEMP bit is set and that there is no pending abort from the TX Queue.

When the TXQE bit is cleared, all messages in the TX Queue buffers are lost and no further message should be stored in the TX Queue.

When a message has been stored to the TX Queue, write 0xFF in the TX Queue Pointer Control Register. This sets the transmit request automatically and changes the internal message buffer pointer to the next free message buffer location of the TX Queue.

Note: If two messages with the same ID are stored in the TX Queue, the order of transmission of these messages can be different from the order in which they were stored in the TX Queue.

To avoid this condition, it is important to confirm that the previous message with the same ID was successfully transmitted before a new message with the same ID is stored in the TX Queue.

For the TX Queue, a dedicated interrupt can be enabled by setting the TXQIE bit of the TX Queue Configuration/Control Register.

The interrupt mode can be configured with the CFDTXQCC.TXQIM bit of the same register either to generate an interrupt for every transmitted message or for the last transmitted message.

The TX Queue transmission request procedure after configuration is shown in Figure 42.48.

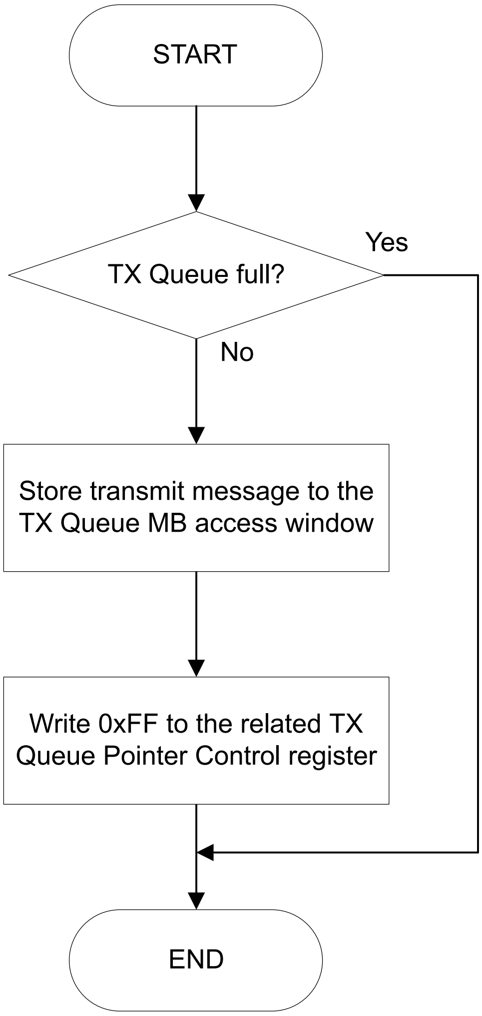


Figure 42.48 TX Queue transmission request

42.8.2.5 TX History List

The TX History List function records the information of the successfully transmitted message in the TX History List Buffers. Two TX History List buffers are provided and THL buffer can store up to 8 TX History List entries.

The CFDTHLCC.THLDTE bit of the TX History List Configuration/Control Register can be used to configure if only message information from TX FIFO or TX Queue is stored, or if all transmit message information from TX Queue, TX FIFO, or normal TX message buffers is stored in the TX History List.

Each transmit message can be individually configured for acceptance to the TX History List with the CFDCFID.THLEN bit in the Message Buffer Pointer Register.

The message information is stored to the TX History List Buffer of a CAN channel after the message is successfully transmitted.

Storing to the list is not synchronized with the status of CFDTMSTSj.TMTRF[1:0] bits in the TX Message Buffer Status Register.

Due to internal processing, the storage to the list can happen with a delay after the successful transmission indication.

Storing the TX History List data can be recognized by the condition that the THLIF is set to 1 when the THLIE bit is configured to 1 or when the TX History List counter CFDTHLSTS.THLMC[5:0] is increased.

In the worst case when multi events like reception scan, internal message routing on happen.

- Maximum delay time from setting the CFDTMSTSj.TMTRF to store the TX History List data is 70 peripheral bus clock cycles.

The History list records the following information of a transmitted message:

- Buffer type:
  - 001: TX Message Buffer
  - 010: TX FIFO
  - 100: TX Queue
- Buffer number:
  TX message buffer, TX Queue message buffer or TX message buffer link for the Common FIFO buffer from which transmission occurred. The number depends on the buffer type. See Table 42.28.
- Transmission ID:
  Transmission pointer stored in the transmission message
- Transmit timestamp:
  Message timestamp captured at capture point as configured by CFDGFDCFG.TSCCFG.
- Transmission information label:
  Transmission information label stored in the transmission message.

Table 42.28 TX History List Buffer number entry

Buffer Number	BT[2:0] Buffer Type		
	001b TX Message Buffer	101b TX FIFO	100b TX Queue
00b	Message Buffer 0	Number shown corresponds to the common FIFO. TX Message Buffer Link CFTML of the related Common FIFO configuration	Number shown corresponds to the Message Buffer belonging to the TX Queue which the frame was transmitted
01b	Message Buffer 1		
10b	Message Buffer 2		
11b	Message Buffer 3		

The Transmission ID entry is used to identify which message of a TX FIFO or TX Queue has been successfully transmitted because the TX FIFO or TX Queue number alone is not sufficient.

Therefore, a unique number can be attached to each transmission message stored in a TX FIFO or TX Queue. This unique identification number should be written to the CFDCFFDCSTS.CFPTR[15:0] part of the Common FIFO Access Pointer Register for a TX FIFO or to the CFDTMFDCTRb.TMPTR[15:0] part of the TX Message Buffer Pointer Register of the TX Queue access window message buffer.

When the message is successfully transmitted, this identification number is stored together with the other message related information to the TX History List and can be read using the Transmission ID (TID) of the TX History List Access Register.

Also, for normal TX message buffers, the CFDTMFDCTRb.TMPTR[15:0] part of the TX Message Buffer Pointer Register is stored in the Transmission History List and the information label is the same.

Figure 42.49 shows a transmission preparation flow when TX History List is used.

Read access to the TX History List Access Register is done for every single entry.

After reading one entry, 0xFF must be written to the corresponding TX History List Pointer Control Register to be able to access the next entry until TX History List is empty.

Figure 42.50 shows an example flow for processing the TX History List information.

The TX History Lists have dedicated interrupts, which can be configured with the CFDTHLCC.THLIM bit of the corresponding TX History List Configuration/Control Register and enabled with the CFDTHLCC.THLIE bit of the same registers, either to generate an interrupt when the History List reached a filling level of 75% or for every new TX History List entry.

An entry lost indication is flagged by the CFDTHLSTS.THLELT bit in the TX History List Status Register. The status of this bit is also shown by the THLES bit in the Global Error Flag Register.

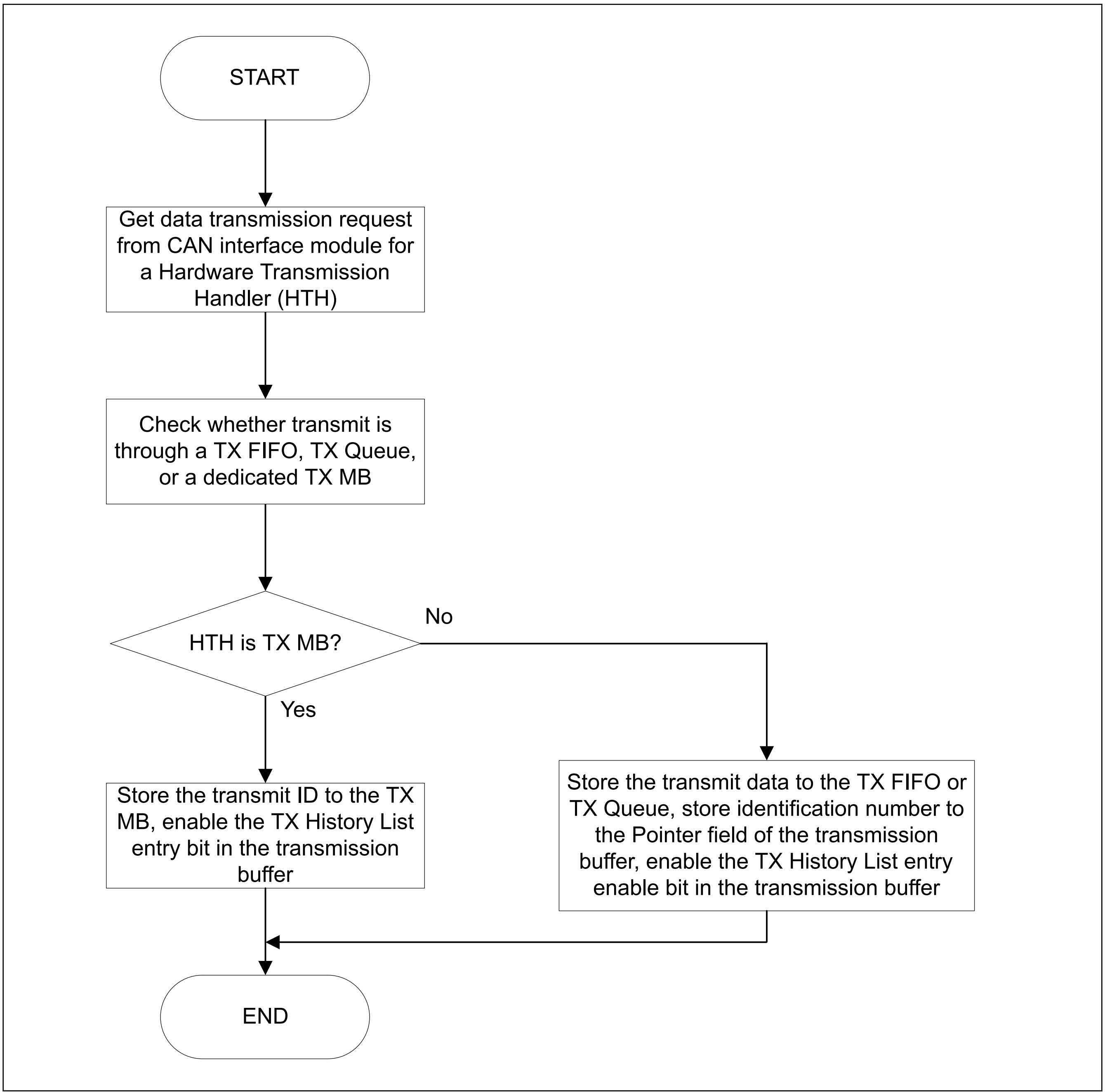


Figure 42.49 TX History List preparation flow

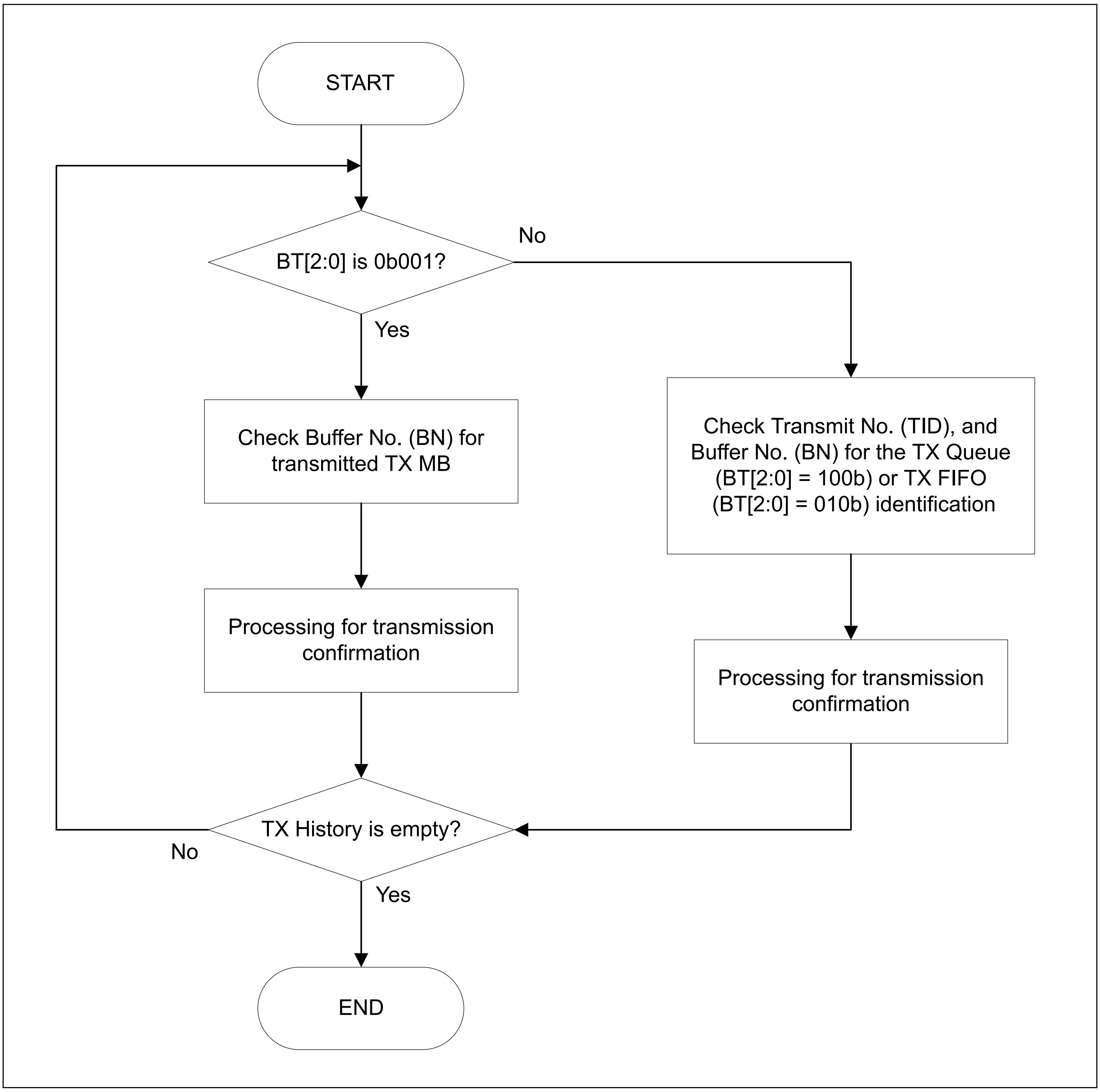


Figure 42.50 TX History List processing flow

42.8.2.6 TX Data Padding

This chapter is not valid for classical CAN.

If the data length code (DLC) of the transmitting message has a higher number of data bytes than the buffer size, the data bytes beyond the restricted range are replaced by bytes with the value of 0xCC.

This can happen for Common FIFO configured as (TX mode) when the transmit message DLC is higher than the CFDCFCC.CFPLS.

This can also happen in FD only mode, if a Classical frame is configured with a DLC bigger than 8.

42.9 Test Mode

The CANFD module can be configured into test modes to allow testing of certain features. These features are provided only for special purposes and care must be taken when configuring the CANFD module in test modes.

Note: All test modes are mutually exclusive unless it is explicitly stated that some functions can be enabled across other test modes.

Do not enable any combination of the various test modes specified in this section.

The test modes can be broadly split into 2 groups:

- Channel specific test modes
- Global test modes.

42.9.1 Channel Specific Test Modes

CAN channel can be configured into the following test modes:

- Basic test mode
- Listen-only mode
- Self-test mode 0 (External loop back mode)
- Self-test mode 1 (Internal loop back mode)
- Restricted operation mode.

42.9.1.1 Basic Test Mode

The basic test mode should be used when there is requirement for a particular test setting to be enabled other than when in Listen-only and Self-test modes.

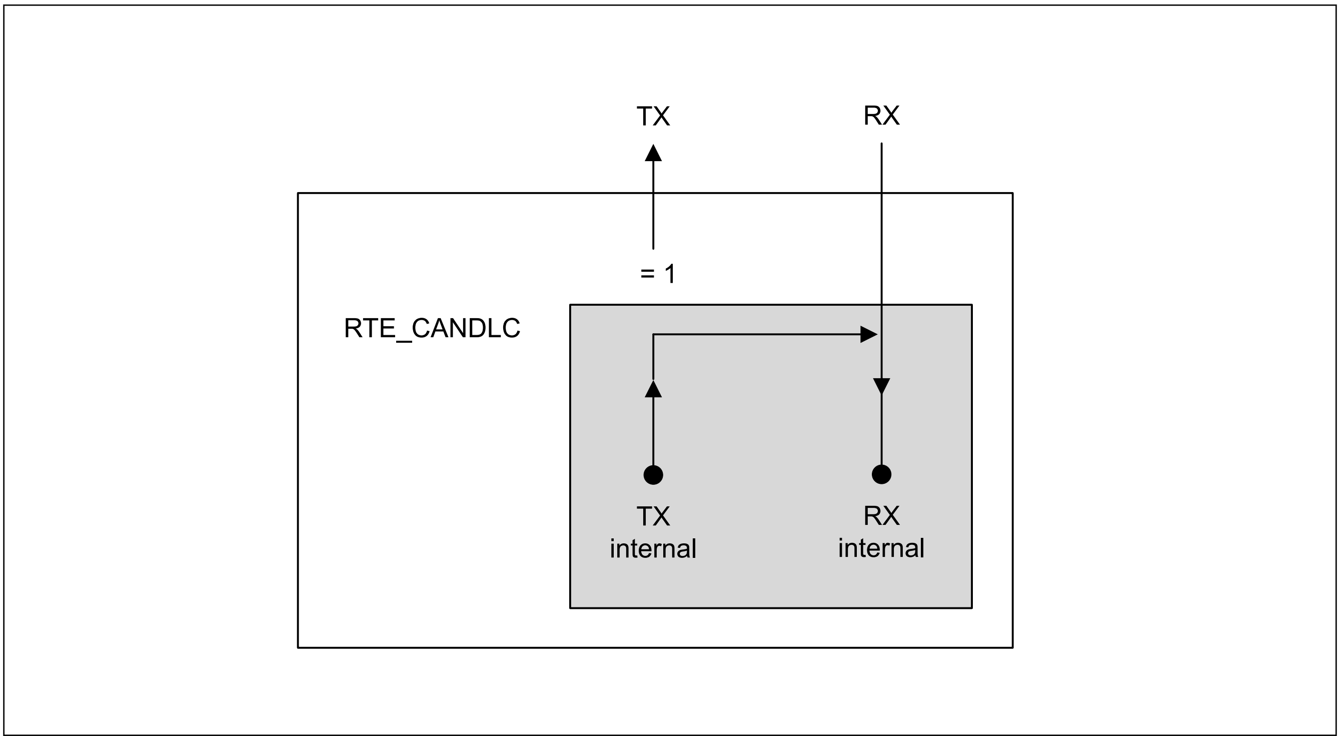
42.9.1.2 Listen-only Mode

The ISO 11898-1 recommends an optional bus-monitoring mode. In this mode, the CAN channel is able to receive valid data frames and valid remote frames. However, it sends only recessive bits on the CAN bus and is not allowed to transmit.

If the CAN engine is required to send a dominant bit (ACK bit, overload flag, active error flag), the bit is routed internally so that the CAN engine monitors this as dominant. The external TX pin remains in recessive state.

This mode can be used for baud rate detection. In this mode, an error interrupt is generated if a bus error occurs and the interrupt is enabled.

In this mode, it is not permitted to request transmission from any normal TX message buffer or TX FIFO.

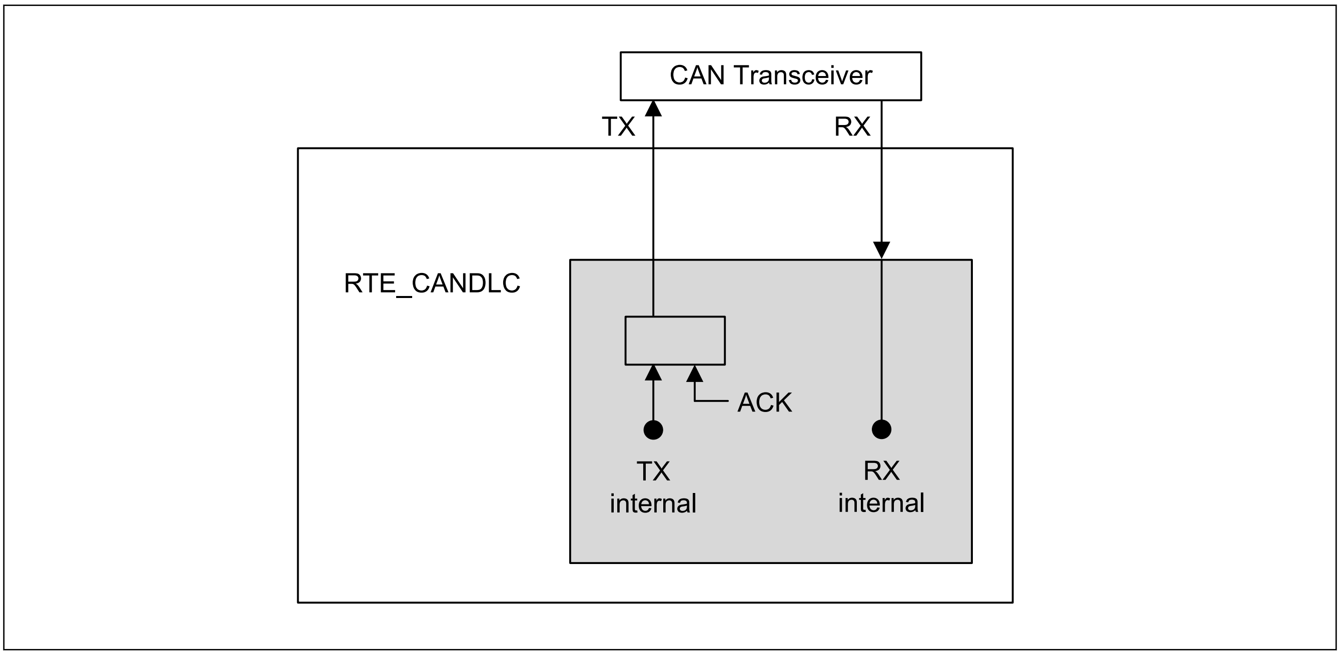


42.9.1.3 Self-test Mode 0 (External loopback mode)

In Self-test mode 0, the CAN engine treats its own transmitted messages as received messages through the CAN transceiver and stores them into its receive message buffers.

To be independent of external stimulation, the engine generates its own Acknowledge bit.

This test can be used for CAN transceiver tests and the RX/TX pins should be connected to the transceiver.

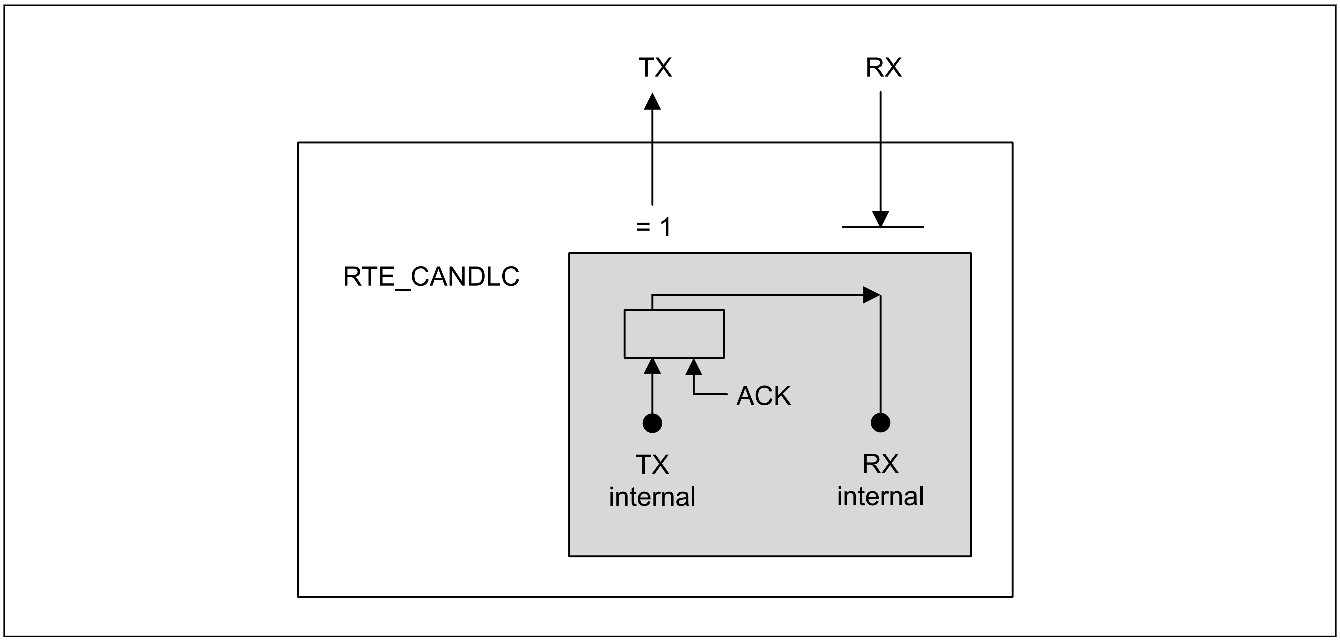


42.9.1.4 Self-test Mode 1 (Internal loopback mode)

In Self-test mode 1, the CAN engine treats its own transmitted messages as received messages and stores them into the receive buffer. This mode is provided for self-test functions. To be independent of external stimulation, the CAN engine generates its own Acknowledge bit. In this mode the CAN engine performs an internal feedback from TX internal to RX internal. The actual value of the external RX input is disregarded by the CAN engine.

The external TX pin outputs only recessive bits. The RX/TX pins do not need to be connected to the CAN bus or any external device.

Note: The channel pins are also disconnected from the internal CAN bus communication line.



42.9.1.5 Restricted Operation Mode

This chapter is not valid for classical CAN.

In Restricted operation mode, the CAN node is able to receive valid data and remote frames generating the Acknowledge bit.

Active error or overload frames cannot be transmitted, instead it waits for the occurrence of bus idle condition to resynchronize itself to the CAN communication after an error or overload condition occurs.

Additionally, the Receive and Transmit Error Counter (REC and TEC) are frozen independently of the occurrence of errors. The mode is specified in ISO 11898-1 and the setting of transmit request is permitted.

42.9.2 Global Test Modes

The CANFD module can be configured into the following test modes:

- RAM test mode
- Bit Flip Test

The test modes in the following table are protected by a special software procedure to enable the mode. This software procedure enables write access to the test mode by a specific unlock key as shown in the table.

Test mode	Unlock key 1	Unlock key 2
RAM test mode	0x7575	0x8A8A

If the software sequence of the two consecutive unlock key write accesses (half-word or word access) is interrupted by any other write access to the register or if incorrect data is written to the Global Unlock Key Register, the corresponding test mode cannot be set and the sequence must be restarted.

After the two unlock key write accesses, the next write access should be to set the corresponding test mode enable bit. If this is not followed, the unlock mechanism reset and the test mode enable bit cannot be set and the unlock sequence must be restarted.

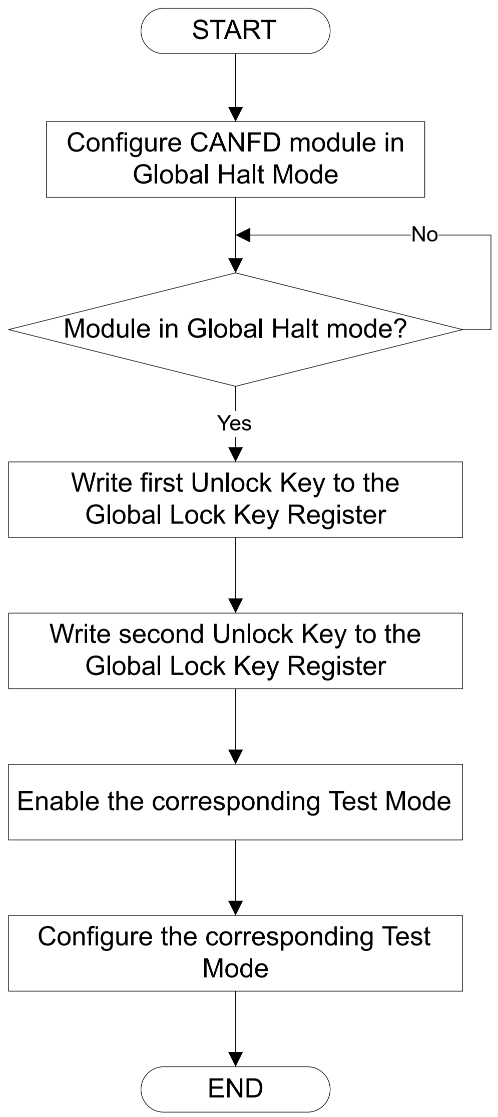


Figure 42.51 Unlock software protection routine

42.9.2.1 RAM Test Mode

The CANFD module can be configured in RAM test mode by setting the CFDGTSTCTR.RTME bit in the Global Test Control Register when the corresponding lock key is previously written. This is a special test mode, in which, the complete RAM area can be accessed.

Note: The actual RAM size is bigger than the RAM area initialized after a hardware reset. Therefore, ECC error flag (of the ECC macro) may be set if CPU reads data from this uninitialized RAM area while CANFD module is in RAM test mode.

In this mode, the RAM area is split into number of pages (pn) of 256 bytes, each which can be accessed with the CFDRPGACCk register.

The page should be selected for read/write access by writing to the CFDGTSTCFG.RTMPS[3:0] bits in the Global Test Control Register. Data can then be read from or written in to the RAM Test Page Access Registers.

Figure 42.52 shows the structure of the pages in the RAM when performing a RAM test mode.

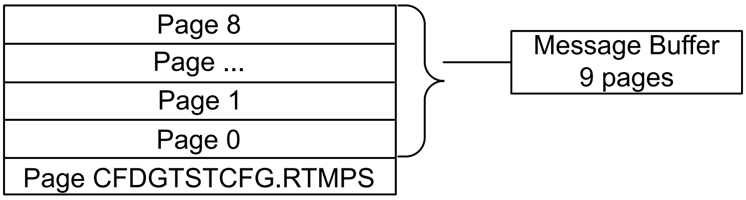


Figure 42.52 RAM page structure

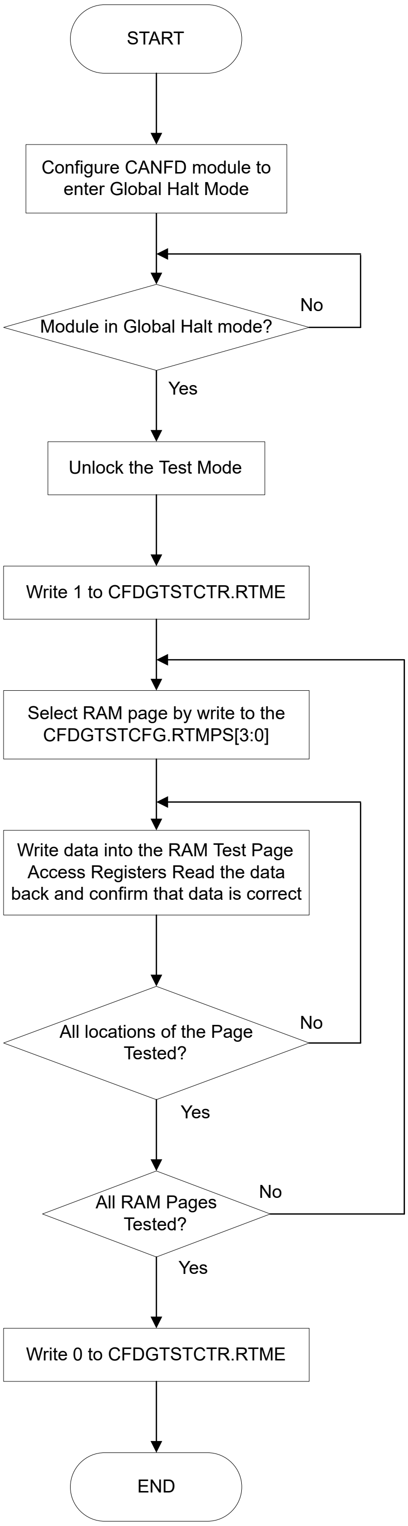
The total available RAM size is 2072 bytes for the Message Buffer RAM.

The pn and CFDGTSTCFG.RTMPS[3:0] values for the MB RAMs are calculated in the following way:

pn = ceil (total RAM size in bytes / number of bytes per page)

- MB RAM:
  pn = ceil (2072 / 256) = 9 pages
  CFDGTSTCFG.RTMPS[3:0] = 0 to 8 inclusive

Figure 42.53 shows the software flow for RAM test mode.



Note: Change into the following status before changing to RAMTEST.

- Cancel of a request for transmission.
- Disable of all the FIFO and TXQ.
- Clear of the receiving flag for a receiving buffer.

Figure 42.53 Software flow for RAM test mode

To exit this test mode, the CFDGTSTCTR.RTME bit must be cleared. The CFDGTSTCTR.RTME bit is cleared by writing 0 to it.

The CFDGTSTCTR.RTME bit is cleared automatically when the CANFD module enters Global Reset mode from the test mode.

42.9.2.2 Bit Flip Test

Bit Flip Test can invert the bit (the 1st bit of ID) of the beginning of the bit stream to receive.

If this function is used by a transmitting node, a bit error or an arbitration lost will occur.

If this function is used by a receiving node, a CRC error or a stuff error will occur.

Users should refer to the bit stuffing rule when using this feature, as there is the possibility of receiving a stuff error (due to the inversion) rather than a CRC error.

The following sequence should be used to perform CRC Error testing. In the sequence below CANFD module is the receiver.

1. Set the CFDC0CTR.BFT bit to 1’b1, in order to invert the first bit of the incoming bit stream from sending node
2. Wait for the CANn_CHERR (n = 0, 1) output signal to set to 1’b1
3. Read either the CFDC0ERFL.CRCREG or the CFDC0FDCRC.CRCREG (depending on the received frame type: Classical or FD). The value should be different from the received CRC value of the reference message from sending node.
4. Check that CFDC0ERFL.CERR is 1’b1

As the CRC generator logic is shared for RX and TX there is no need to create a separate TX CRC Error test.

42.10 Usage Notes

42.10.1 Enter Software Standby Mode

The following steps are required, before the MCU enters Software Standby mode.

1. Set the global halt mode request (CFDGCTR.GMDC = 10b).
2. Wait for completion of transition to global halt mode (CFDGSTS.GHLTSTS = 1b).
3. Set the global reset mode request (CFDGCTR.GMDC = 01b).
4. Wait for completion of transition to global reset mode (CFDGSTS.GRSTSTS = 1b).
5. Set the global sleep mode request (CFDGCTR.GSLPR = 1b).
6. Wait for completion of transition to global sleep mode (CFDGSTS.GSLPSTS = 1b).
7. Read the CFDGLOCKK register.

42.10.2 Return from Software Standby Mode

The procedures described in section 42.4.2. CAN Module Configuration after Hardware Reset are required, after the MCU returns from Software Standby mode.

42.10.3 Module-Stop Function Setting

CANFD operation can be disabled or enabled using Module Stop Control Register C (MSTPCRC). The CANFD module is initially stopped after reset. Releasing the module-stop state enables access to the registers.